Massachusetts Institute of Technology
Department of Physics
8.962 Spring 2024

Lecture 1
Spacetime and geometry part I

1.1 Goal of this course

General relativity is Albert Einstein's relativistically covariant theory of gravity. It famously makes gravity a consequence of the geometry of spacetime. The goal of the version of 8.962 taught by Scott Hughes is for you to learn the fundamentals of general relativity (roughly the first half, up to spring break) and to explore some of its most important consequences, particularly as they pertain to our universe as we observe it today, and some of the most interesting and important gravitating objects that we encounter (the second half, from spring break to the end of term).

We will assume in this course that you are familiar with spacetime as described by special relativity. Special relativity turns out to have geometric notions embedded in it, but it is possible to do a lot in special relativity without explicitly thinking about geometry. However, to go from special relativity to general relativity, we need to understand the geometry. So the start of 8.962 will largely consist of a review and re-presentation of special relativity using mathematical language that describes it geometrically, doing so in a way that carries over naturally to general relativity. The fundamental observation is that special relativity is the framework in which measurements and observations are subject to Lorentz symmetry, and thus measurements and observations made by different inertial observers can be related by Lorentz transformations.

1.2 Spacetime: Critical definitions and properties

Spacetime is a *manifold* of *events* that is endowed with a *metric*. Several terms in this opening sentence require definitions:

- *Manifold*: For our purposes, a manifold is a set of points with well-understood connectedness properties (i.e., with a defined topology). We require that it have certain well-defined, "nice" smoothness properties, and that the neighborhood of any point should look like Euclidean space (or like Lorentzian space if it includes timelike directions). See Carroll pages 54–62 (plus additional discussion in Appendix A) for very careful and rigorous discussion. The mathematical machinery that Carroll introduces goes beyond what we need in 8.962, but it is very much worth it for you to be familiar with this terminology (and, depending on your interests, you might find that this level of rigor is useful for you). Wald's textbook, chapter 2 (particularly section 2.1), is also likely to be of interest to students who desire a more mathematical formulation.

- *Event*: When and where something happens. We label events with coordinates, but the event itself is a geometric object whose meaning is independent of the coordinate representation. For example, if "event A" corresponds to a piece of chalk hitting the floor at a moment during lecture, I might assign this event the coordinates $(t, x, y, z) =$ (Feb 6 2:37 pm, 0, 0, 0); a student in the front row might assign it the coordinates (Feb 6 2:37pm, 2 meters, 0, -0.5 meters); a student in the back row whose watch is set for a different time zone might assign it the coordinates (Feb 6 7:37pm, 4 meters, 1 meter, -1 meter). We all use different coordinates to represent this event, but our different representations nonetheless describe the same point in spacetime. Many of the most important concepts and structures we will develop in this

class can be thought of as "geometric objects": entities describing some aspect of physics whose nature is independent of coordinates. Although the physical meaning of these objects "transcend" their coordinate description, we will nonetheless use coordinates for many of our calculations, and so it will be important to understand how to convert between different observers' various coordinates.

- *Metric*: A notion of distance between events in spacetime. This comes from the physics attached to these events. A manifold on its own does not include a notion of distance between its constituent points; we will develop the concept of metric in more detail in the coming discussion. Key for us is that the metric is what puts geometry into the manifold.

SR is essentially the simplest theory of spacetime consistent with most of the physics that we observe; it turns out to correspond to the zero gravity limit of general relativity (GR). A key concept for us in this discussion is the notion of an *inertial reference frame*, or IRF. Following Blandford and Thorne, we carefully define this frame as *a (conceptual) lattice of clocks and measuring rods that allows us to assign coordinates to (i.e., to label) spacetime events.* The IRF and this lattice have the following properties:

- The lattice moves freely through spacetime — no forces act on it, and it does not rotate relative to distant beacons.
- The measuring rods are orthogonal and uniformly ticked, forming an orthonormal, Cartesian coordinate system.
- All of the clocks tick uniformly.
- The clocks are synchronized using the "Einstein synchronization procedure": clock 1 emits a pulse of light at $t = t_e$. It bounces off a mirror on clock 2, and is received back at clock 1 at $t = t_r$. Clock 2 is synchronized with clock 1 such that the time of bounce is $t_b = (t_e + t_r)/2$. This synchronization is done between every pair of clocks in the lattice.

⌈Notice that the Einstein synchronization procedure takes advantage of the fact that the speed of light c is a relativistic invariant: it is a "good" synchronization procedure because the light travel time between various events (emission of light, bouncing of light, reception of light) depends upon a quantity whose value is agreed upon by all observers. Never forget the important role that the propagation of light has in understanding the way that time is measured! Later in the course, we will encounter some situations in which time seems to be behaving oddly, at least according to certain observers. The reason for this is going to be intimately tied up with the way that light propagates through spacetime.⌋

An event is represented in this IRF by the coordinates: $\mathcal{P} \doteq (t_\mathcal{P}, x_\mathcal{P}, y_\mathcal{P}, z_\mathcal{P})$. (Throughout these notes, I use the symbol "$\underset{\mathcal{O}}{\doteq}$" to mean "the object on the left-hand side is represented by the set on the right-hand side in the IRF of observer $\mathcal{O}$." If there is no ambiguity about or need to carefully specify the observer, I may simply write this "$\doteq$".) This is a good moment for a brief aside on the system of units we use: If our clocks tick once per second, then we choose our tickmarks such that the basic unit of length is 1 light second. In these units, the speed of light is then

$$c = \frac{1\,\text{lightsecond}}{\text{second}} \equiv 1\ . \tag{1.1}$$

In this system, the speed of light $c = 1$, and all speeds are dimensionless. To convert to "normal" units, think of all speeds and velocities as being scaled to c: $v = v_\text{normal}/c$.

Let $\mathcal{O}$ be an observer who uses this IRF. We can now define a *displacement vector* between 2 spacetime events, $\mathcal{Q}$ and $\mathcal{P}$:

$$\begin{aligned} \Delta\vec{x} &\doteq_{\mathcal{O}} (t_{\mathcal{Q}} - t_{\mathcal{P}}, x_{\mathcal{Q}} - x_{\mathcal{P}}, y_{\mathcal{Q}} - y_{\mathcal{P}}, z_{\mathcal{Q}} - z_{\mathcal{P}}) \\ &\doteq_{\mathcal{O}} (\Delta t, \Delta x, \Delta y, \Delta z) \ . \end{aligned} \tag{1.2}$$

We will denote the components of such a "4-vector" (so-called because it has 4 components, one for each spacetime direction) as Δx^μ (or using another letter from the Greek alphabet, depending on the needs of the calculation). When Greek letters are used to label an index, they are taken from the set $[0, 1, 2, 3]$, with direction 0 being along the "time" axis in that frame, and indices 1, 2, and 3 the space indices (typically ordered in a right-hand fashion, e.g., $1 \to x$, $2 \to y$, $3 \to z$ in a Cartesian coordinate system). We will sometimes only want the spatial pieces (at least, the spatial pieces according to some specified observer):

$$\Delta\mathbf{x} = \Delta\underset{\sim}{x} \doteq_{\mathcal{O}} (x_{\mathcal{Q}} - x_{\mathcal{P}}, y_{\mathcal{Q}} - y_{\mathcal{P}}, z_{\mathcal{Q}} - z_{\mathcal{P}}) \ . \tag{1.3}$$

We will tend to use the boldface notation for spatial quantities that we typeset, and the undertilde when writing by hand. Components of a spatial vector are denoted using lowercase Latin letters, and range over 1, 2, 3. So, in a right-handed Cartesian coordinate system, Δx^i is shorthand for $(\Delta x^1, \Delta x^2, \Delta x^3) = (\Delta x, \Delta y, \Delta z)$.

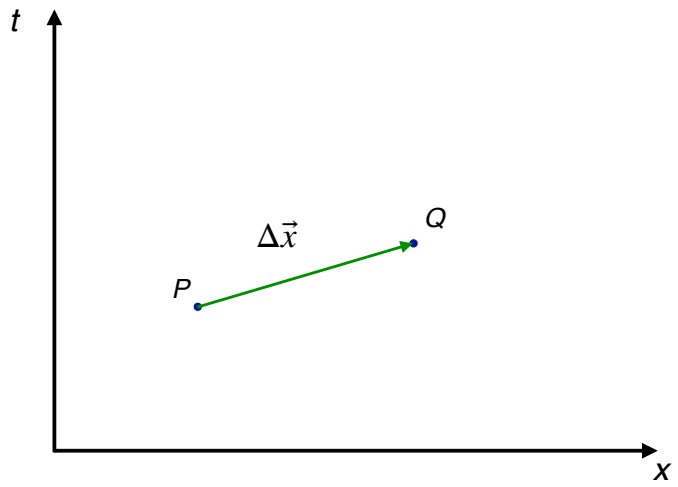


Figure 1: The displacement $\Delta\vec{x}$ between two spacetime events $\mathcal{P}$ and $\mathcal{Q}$, along with coordinate axis used by an observer in some IRF.

1.3 Lorentz transformations

We have formulated everything so far in terms of measurements made by a particular observer $\mathcal{O}$. Let us consider the events $\mathcal{P}$ and $\mathcal{Q}$ and the displacement vector $\Delta\vec{x}$ that connects them, but now as viewed by an observer $\bar{\mathcal{O}}$. The key principle we bear in mind is that $\mathcal{P}$, $\mathcal{Q}$, and $\Delta\vec{x}$ *do not change*, since they are simply geometric objects in the manifold of spacetime. However, *their representation according to $\bar{\mathcal{O}}$ is not the same as their representation according to $\mathcal{O}$.*

Because we take our starting point to be the special theory of relativity, the representations used by these two observers are related by the Lorentz transformation. Let us suppose that the IRF of $\bar{\mathcal{O}}$ moves with speed v along the "1" (i.e., x) direction as seen by IRF $\mathcal{O}$. Then, denoting

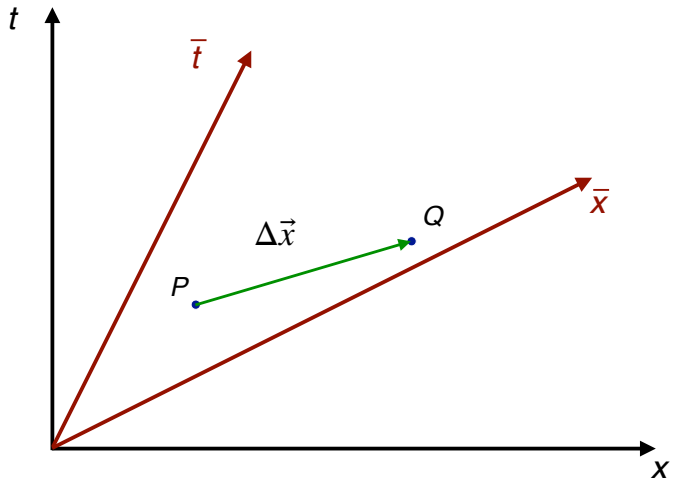


Figure 2: The displacement $\Delta\vec{x}$ between two spacetime events $\mathcal{P}$ and $\mathcal{Q}$, along with coordinate axis used by two observers in two different IRFs. The events and the displacement are the same geometric objects, though the two observers label these objects differently.

by $x^{\bar{\mu}}$ the components in $\bar{\mathcal{O}}$'s representation, we have

$$\Delta x^{\bar{0}} = \gamma\Delta x^0 - \gamma v\Delta x^1 \tag{1.4}$$
$$\Delta x^{\bar{1}} = -\gamma v\Delta x^0 + \gamma\Delta x^1 \tag{1.5}$$
$$\Delta x^{\bar{2}} = \Delta x^2 \tag{1.6}$$
$$\Delta x^{\bar{3}} = \Delta x^3 \, . \tag{1.7}$$

This can be written more compactly

$$\Delta x^{\bar{\mu}} = \sum_{\nu=0}^{3} \Lambda^{\bar{\mu}}{}_{\nu}\Delta x^{\nu} \equiv \Lambda^{\bar{\mu}}{}_{\nu}\Delta x^{\nu} \, . \tag{1.8}$$

Here, we have introduced the Lorentz transformation matrix,

$$\Lambda^{\bar{\mu}}{}_{\nu} \doteq \begin{pmatrix} \gamma & -v\gamma & 0 & 0 \\ -v\gamma & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \, , \tag{1.9}$$

as well as the Einstein summation convention: indices on the same side of the equal sign that are repeated, with one in the "upstairs" position and the other "downstairs," are taken to be summed from 0 to 3. The quantity $\gamma = 1/\sqrt{1-v^2}$.

Notice that one can write

$$\Lambda^{\bar{\mu}}{}_{\nu} = \frac{\partial x^{\bar{\mu}}}{\partial x^{\nu}} \, . \tag{1.10}$$

The right-hand side of Eq. (1.10) can be regarded as defining a more general notion of transforming between different representations of the spacetime displacement, with the Lorentz transformation as a particularly important special case.

In Eq. (1.8), it is irrelevant what label we use for the index corresponding to $\mathcal{O}$'s coordinates, as long as this label is distinct from all other index labels in the problem. Because we sum over the full range of the index being summed upon,

$$\Delta x^{\bar{\mu}} = \Lambda^{\bar{\mu}}{}_{\nu}x^{\nu} = \Lambda^{\bar{\mu}}{}_{\alpha}x^{\alpha} = \Lambda^{\bar{\mu}}{}_{\beta}x^{\beta} = \ldots \tag{1.11}$$

are all equivalent and equally good ways of representing the components $\Delta x^{\bar{\mu}}$ used by $\bar{\mathcal{O}}$ to the components used by $\mathcal{O}$. The index which is summed over is known as a *dummy index*. We will occasionally exploit the fact that the particular label we use for it is irrelevant to re-organize and simplify certain calculations which involve multiple index sums. On the other hand, we do not have freedom to relabel the index $\bar{\mu}$, which is not summed over and appears on both sides of the equation. It is often called a "free index."

1.4 Vectors more generally

The spacetime displacement $\Delta\vec{x} \doteq (\Delta t, \Delta x, \Delta y, \Delta z)$ is our prototype for a vector in spacetime, or 4-vector. We will generalize the notion by defining any quartet of numbers — "components" — that transforms between IRFs like the displacement vector to be a vector:

$$\vec{A} \underset{\mathcal{O}}{\doteq} (A^0, A^1, A^2, A^3) \,, \tag{1.12}$$

$$A^{\bar{\mu}} = \Lambda^{\bar{\mu}}{}_{\alpha} A^{\alpha} \,. \tag{1.13}$$

That this quartet of numbers is subject to this transformation law is what makes them vector components, and not just some set of numbers. We also require that these components obey the linearity laws which go into a vector space:

- If $\vec{A}$ is a vector (with components A^{α}) and a is a scalar[1], then $a\vec{A}$ is a vector (with components aA^{α}).
- If $\vec{A}$ and $\vec{B}$ are both vectors (with components A^{α} and B^{α}, respectively), then $\vec{C} = \vec{A} + \vec{B}$ is a vector (with components $C^{\alpha} = A^{\alpha} + B^{\alpha}$).

Within frame $\mathcal{O}$, we can define 4 clearly very special vectors:

$$\begin{aligned} \vec{e}_0 &\doteq (1,0,0,0) \,, & \vec{e}_1 &\doteq (0,1,0,0) \,, \\ \vec{e}_2 &\doteq (0,0,1,0) \,, & \vec{e}_3 &\doteq (0,0,0,1) \,. \end{aligned} \tag{1.14}$$

We call these *basis vectors*, since they pick out 4 principle directions associated with the coordinates used in $\mathcal{O}$. (Note that we cannot call the basis vectors "unit" vectors, at least not yet, since we have not ascertained how to compute a "magnitude" associated with each one. Hold that thought for another page or so.) Note that the index on each vector is not a component; instead, it labels which member of the set is being referenced. The components of the basis vectors can be written compactly using the Kronecker delta:

$$(\vec{e}_{\alpha})^{\beta} = \delta_{\alpha}{}^{\beta} \,. \tag{1.15}$$

Using the basis vectors, we can write a vector in terms of its components:

$$\vec{A} = A^{\alpha}\vec{e}_{\alpha} \,. \tag{1.16}$$

Notice that this is a "real" equality, not the $\doteq$ that we use to indicate that a representation holds in some specified reference frame. This means that the equality holds in all IRFs, a fact we can exploit to deduce how the basis vectors transform between different IRFs:

$$\begin{aligned} \vec{A} = A^{\alpha}\vec{e}_{\alpha} &= A^{\bar{\mu}}\vec{e}_{\bar{\mu}} \quad \text{(Equality holds in all IRFs)} \\ &= \left(\Lambda^{\bar{\mu}}{}_{\beta} A^{\beta}\right)\vec{e}_{\bar{\mu}} \quad \text{(Lorentz transformation of components)} \\ &= \left(A^{\beta}\Lambda^{\bar{\mu}}{}_{\beta}\right)\vec{e}_{\bar{\mu}} \quad \text{(Order does not matter in index notation)} \\ &= \left(A^{\alpha}\Lambda^{\bar{\mu}}{}_{\alpha}\right)\vec{e}_{\bar{\mu}} \quad \text{(Relabel dummy indices)} \end{aligned} \tag{1.17}$$

[1]Note, we haven't actually discussed what a scalar is! Hold that thought for a page or two.

With this we rearrange the equation as follows:

$$A^{\alpha}\left(\vec{e}_{\alpha} - \Lambda^{\bar{\mu}}{}_{\alpha}\vec{e}_{\bar{\mu}}\right) = 0\ , \tag{1.18}$$

from which we see that the rule for transforming basis vectors is given by

$$\boxed{\boxed{\vec{e}_{\alpha} = \Lambda^{\bar{\mu}}{}_{\alpha}\vec{e}_{\bar{\mu}}}} \tag{1.19}$$

Notice that this expresses the basis vectors in frame $\mathcal{O}$ in terms of the basis vectors in frame $\bar{\mathcal{O}}$ — quite different from the rule that we use to transform the components:

$$\boxed{\boxed{A^{\bar{\mu}} = \Lambda^{\bar{\mu}}{}_{\alpha}A^{\alpha}}} \tag{1.20}$$

In both cases, there's a simple algorithm: "Line up the indices." However, if your goal is to understand how to write down $\bar{\mathcal{O}}$'s basis vectors in terms of the basis vectors in $\mathcal{O}$, then we need the *inverse* Lorentz transformation.

Thanks to special relativity, it is physically clear what the inverse Lorentz transformation must be: the Lorentz transformation depends upon the relative velocity of the frames, $\mathbf{v}$, so if we want the inverse transformation, we just flip the sign of all velocity terms. I.e., if

$$\vec{e}_{\alpha} = \Lambda^{\bar{\mu}}{}_{\alpha}(\mathbf{v})\vec{e}_{\bar{\mu}} \tag{1.21}$$

(writing the transformation as an explicit function of $\mathbf{v}$ for clarity), then

$$\vec{e}_{\bar{\mu}} = \Lambda^{\alpha}{}_{\bar{\mu}}(-\mathbf{v})\vec{e}_{\alpha}\ . \tag{1.22}$$

The different components of $\Lambda^{\alpha}{}_{\bar{\mu}}(-\mathbf{v})$ are identical to the components of $\Lambda^{\bar{\mu}}{}_{\alpha}(\mathbf{v})$, but with the sign of $\mathbf{v}$ flipped.

To be perfectly blunt, keeping track of these different indices and the sign associated with the velocity is something of a pain in a region a bit below the lower back. My personal mnemonic is to keep in my head the matrix that we use for transforming the displacement vector's components. When we do that, the velocity which enters the components is that of the IRF associated with the top index according to observers in the IRF associated with the bottom index.

One last transformation exercise: let's combine the Lorentz transformation with its inverse.

$$\begin{aligned}
\vec{e}_{\alpha} &= \Lambda^{\bar{\mu}}{}_{\alpha}(\mathbf{v})\vec{e}_{\bar{\mu}} \\
&= \Lambda^{\bar{\mu}}{}_{\alpha}(\mathbf{v})\left[\Lambda^{\beta}{}_{\bar{\mu}}(-\mathbf{v})\vec{e}_{\beta}\right] \\
&= \left[\Lambda^{\bar{\mu}}{}_{\alpha}(\mathbf{v})\Lambda^{\beta}{}_{\bar{\mu}}(-\mathbf{v})\right]\vec{e}_{\beta}\ .
\end{aligned} \tag{1.23}$$

This equality requires that

$$\boxed{\boxed{\Lambda^{\bar{\mu}}{}_{\alpha}\Lambda^{\beta}{}_{\bar{\mu}} = \delta^{\beta}{}_{\alpha}}} \tag{1.24}$$

This is simply another way of stating that these are elements of matrices which have an inverse relation to each other. One can likewise show

$$\boxed{\boxed{\Lambda^{\alpha}{}_{\bar{\mu}}\Lambda^{\bar{\nu}}{}_{\alpha} = \delta^{\bar{\nu}}{}_{\bar{\mu}}}} \tag{1.25}$$

1.5 Invariant product

Let us go back to the displacement vector $\Delta\vec{x} \doteq (\Delta t, \Delta x, \Delta y, \Delta z)$ for a moment. A very important result from special relativity[2] is that

$$\Delta s^2 = -\Delta t^2 + \Delta x^2 + \Delta y^2 + \Delta z^2 \tag{1.26}$$

is a *Lorentz invariant*: although different inertial observers will measure different values for Δt, Δx, etc., they will *all agree* on the value of Δs^2. Note that, because of this, we could use either the true equals "=" or the IRF representation symbol "$\doteq$"; the right-hand side is written using quantities that are defined in a particularly IRF, but the sum is independent of this frame. We will typically use = to emphasize the fact that all observers agree on the result, though one should remember that the terms on the right-hand vary from frame to frame.) We will define this to be a *Lorentz scalar*, or "scalar" for short: A quantity whose value is the same in all IRFs. (Incidentally, Δs^2 is often called the invariant interval.)

We will use this notion of an invariant to define a scalar or invariant product (I will use the terms interchangeably). Let us define

$$\Delta s^2 \equiv \Delta\vec{x} \cdot \Delta\vec{x} \, . \tag{1.27}$$

Another way of saying this is that

$$\Delta\vec{x} \cdot \Delta\vec{x} = -(\Delta x^0)^2 + (\Delta x^1)^2 + (\Delta x^2)^2 + (\Delta x^3)^2 \, . \tag{1.28}$$

This looks just like the Euclidean dot product with which you are surely familiar, but the time piece enters with a minus sign. Why? That's what physics demands in order to construct an invariant. The best I can say is that time is not on the same footing as the other dimensions: you can easily move forward and backward, left and right, up and down; but you cannot easily move to the past and the future. Physics tells the math to treat this direction differently than the others.

Since we define vectors as objects which transform like the displacement, we can develop scalar products with them in just the same way. For example,

$$\vec{A} \cdot \vec{A} = -(A^0)^2 + (A^1)^2 + (A^2)^2 + (A^3)^2 \, . \tag{1.29}$$

This is a good point to introduce some terminology:

- If $\vec{A} \cdot \vec{A} < 0$, we say that $\vec{A}$ is "timelike." Intuitively, the time component dominates the properties of this vector. More rigorously, you can find a Lorentz transformation into an IRF where $\vec{A}$ *only* has a timelike component — there exists a frame in which the spatial components of $\vec{A}$ are zero. (Actually, you can find infinitely many such frames: Once you have found one such frame, you can perform rotations to any number of additional frames in which $\vec{A}$ only has a timelike component.)

- If $\vec{A} \cdot \vec{A} > 0$, we say that $\vec{A}$ is "spacelike." Intuitively, the spatial components dominate this vector's properties. More rigorously, you can find Lorentz transformations such that $\vec{A}$ only has spatial components; there are frames that completely eliminate the timelike component of spacelike 4-vectors.

- If $\vec{A} \cdot \vec{A} = 0$, we say that $\vec{A}$ is "null" (meaning hopefully obvious) or "lightlike." To understand why we call this lightlike, suppose that $\vec{A} = \Delta\vec{x}$ — our 4-vector is a spacetime displacement vector. If $\Delta\vec{x} \cdot \Delta\vec{x} = 0$, then the events which define the endpoints of this displacement vector can be connected by a pulse of light. (Write it out and check if this is not clear; it may be helpful to restore factors of c.) This means that a lightlike or null vector is oriented parallel to the path that light follows in spacetime.

[2] If this looks slightly funny to you, remember that our unit system has $c = 1$.

A more general notion of the scalar product is

$$\vec{A} \cdot \vec{B} = -A^0 B^0 + A^1 B^1 + A^2 B^2 + A^3 B^3 \; . \tag{1.30}$$

It is simple to see that this is also invariant: consider $\vec{C} = \vec{A} + \vec{B}$. Then,

$$\vec{C} \cdot \vec{C} = \vec{A} \cdot \vec{A} + \vec{B} \cdot \vec{B} + 2\vec{A} \cdot \vec{B} \; . \tag{1.31}$$

The left-hand side and the first two terms on the right-hand side are manifestly invariant; the final term on the right-hand side therefore must be as well.

Another way of writing this final notion of scalar product is

$$\begin{aligned} \vec{A} \cdot \vec{B} &= (A^\alpha \vec{e}_\alpha) \cdot (B^\beta \vec{e}_\beta) \\ &= A^\alpha B^\beta (\vec{e}_\alpha \cdot \vec{e}_\beta) \\ &\equiv A^\alpha B^\beta \eta_{\alpha\beta} \; . \end{aligned} \tag{1.32}$$

On the last line we have defined $\vec{e}_\alpha \cdot \vec{e}_\beta \equiv \eta_{\alpha\beta}$. This is known as the *metric tensor*. Its components can be written in matrix form

$$\eta_{\alpha\beta} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \; . \tag{1.33}$$

Strictly speaking, (1.33) should be written using $\doteq$ rather than $=$; however, this tensor turns out to be represented by these components in *all* IRFs.

Using the components of $\eta_{\alpha\beta}$, we can now assess whether the basis vectors are unit vectors:

$$\vec{e}_0 \cdot \vec{e}_0 = -1 \; , \tag{1.34}$$

$$\vec{e}_1 \cdot \vec{e}_1 = \vec{e}_2 \cdot \vec{e}_2 = \vec{e}_3 \cdot \vec{e}_3 = 1 \; , \tag{1.35}$$

$$\vec{e}_\alpha \cdot \vec{e}_\beta = 0 \quad \text{if } \alpha \neq \beta. \tag{1.36}$$

These basis vectors indeed are unit vectors, with the twist that $\vec{e}_0$ is a *timelike* unit vector (logically enough), and $\vec{e}_{1,2,3}$ are all *spacelike* unit vectors (again, logically enough). They are also orthogonal to each other (we continue to call two vectors whose scalar product is zero "orthogonal," even in spacetime), so this is in fact an orthonormal basis.

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8.962 Spring 2024

Lecture 2
Spacetime and geometry part II

2.1 More on the metric and basis vectors

The metric is the mathematical entity which attaches a notion of distance to the manifold: Δs^2 in the relation

$$\Delta s^2 = \Delta\vec{x}\cdot\Delta\vec{x} = \eta_{\alpha\beta}\Delta x^\alpha \Delta x^\beta \tag{2.1}$$

provides a lot of information about the distances between events that different observers measure. If $\Delta\vec{x}$ is spacelike, then we can perform a Lorentz transformation into a frame such that $\Delta\vec{x}$ has no timelike components; $\Delta s = \sqrt{\Delta s^2}$ is then the distance an observer in this frame measures between the two events which define the endpoints of $\Delta\vec{x}$. If $\Delta\vec{x}$ is timelike, we can perform a Lorentz transformation into a frame such that $\Delta\vec{x}$ has no spatial components; $\Delta\tau = \sqrt{-\Delta s^2}$ is the elapsed time an observer in this frame measures between the events which define the endpoints of $\Delta\vec{x}$.

A differential displacement vector is given by $d\vec{x} = dx^\alpha \vec{e}_\alpha$. This relationship defines $\vec{e}_\alpha$ as a *coordinate basis* vector. For rectilinear coordinates, this term doesn't mean very much. However, the distinction between "coordinate bases" and other kinds of bases becomes very important when we begin to examine curvilinear coordinates. Consider for example two ways of writing the spatial interval $d\mathbf{x}$ in spherical polar coordinates:

$$d\mathbf{x} = dr\,\mathbf{e}_{\hat{r}} + r\,d\theta\,\mathbf{e}_{\hat{\theta}} + r\sin\theta d\phi\,\mathbf{e}_{\hat{\phi}} \tag{2.2}$$

$$= dr\,\mathbf{e}_r + d\theta\,\mathbf{e}_\theta + d\phi\,\mathbf{e}_\phi\ . \tag{2.3}$$

The form (2.2) is surely familiar to you if have studied electrodynamics; it uses what are called orthonormal polar basis vectors. The form (2.3) is perhaps new, and perhaps confusing. On first sight, you might wonder how this can possibly be dimensionally consistent — shouldn't every term on the right-hand side have the units of length? Indeed, each term on the right-hand side *does* have the units of length: the "missing" factors which make the units right are in the basis vectors. In a coordinate basis, the basis vectors are *not necessarily unit vectors*. We must in fact have $\mathbf{e}_\theta\cdot\mathbf{e}_\theta = r^2$, $\mathbf{e}_\phi\cdot\mathbf{e}_\phi = r^2\sin^2\theta$ in order for Eq. (2.3) to define a sensible displacement vector. Although Eqs. (2.2) and (2.3) use somewhat different forms, they say exactly the same thing.

We will use coordinate basis vectors exclusively in this class. The reason is that as we move to more general transformations (going beyond Lorentz transformations), it will remain the case that to change the representation we use a matrix whose elements are $\partial x^{\bar{\mu}}/\partial x^\alpha$. That this works can be seen quite simply: $d\vec{x}$ is a geometric object that is the same in all representations, so

$$d\vec{x} = dx^\alpha\vec{e}_\alpha = dx^{\bar{\mu}}\vec{e}_{\bar{\mu}}\ . \tag{2.4}$$

It follows that the matrix elements $\partial x^{\bar{\mu}}/\partial x^\alpha$ produce the transformation of the components of $d\vec{x}$ — and all other 4-vector components — from those used in frame $\mathcal{O}$ to those used in frame $\bar{\mathcal{O}}$.

The transformation we need to use in the case of orthonormal bases is a little more complicated. Though not significantly so; interested students are strongly encouraged to read about the mathematical structure introduced to describe "noncoordinate" bases in Appendix J of Carroll. But this is enough of a diversion that in a single semester class, it is one of those interesting topics that we must unfortunately not include.

2.2 Other important 4-vectors

For us, a major purpose of constructing geometric objects like vectors is that they describe aspects of the physics that "lives" in the manifold of spacetime. Perhaps the simplest example is the 4-velocity. Suppose observer $\mathcal{O}$ follows a trajectory $x^\alpha(\tau)$, with the coordinates specified in some IRF (not necessarily the rest frame of $\mathcal{O}$; indeed, perhaps $\mathcal{O}$ is accelerating, and has different rest frames at different moments). A body's trajectory through spacetime is known as its *worldline*. We will define the parameter τ more precisely in a moment; for now, just imagine that it increases in some fashion as $\mathcal{O}$ moves along this worldline. Then, we define the 4-velocity as

$$\vec{u} = \frac{d\vec{x}}{d\tau} \, . \tag{2.5}$$

The vector interval $d\vec{x}$ is bracketed by $\mathcal{O}$'s location in spacetime at parameter τ and at parameter $\tau + d\tau$. For this parameter we use the *proper time* of $\mathcal{O}$ — the time that $\mathcal{O}$ measures on their own clock[1]. Different inertial observers will report different intervals of time for $\mathcal{O}$ to move from event $\mathcal{P}$ to event $\mathcal{Q}$, but all will agree that $\mathcal{O}$'s own clock records an interval $\Delta\tau$ for this interval. The proper time is thus an excellent way of defining a time interval that everyone can agree on. Notice that the 4-velocity is the tangent vector in spacetime to the worldline that $\mathcal{O}$ traces out.

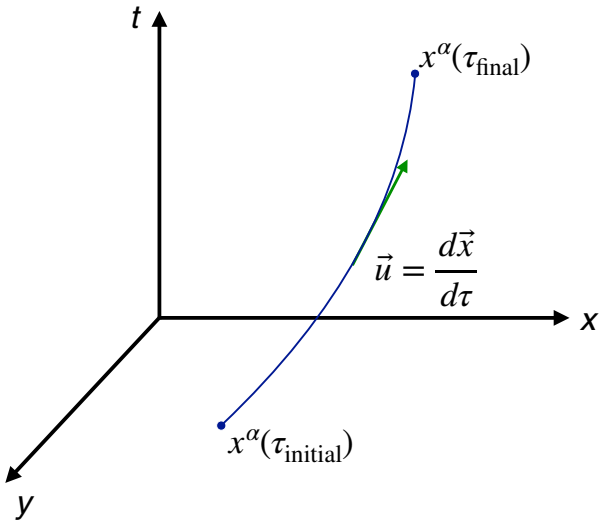


Figure 1: Illustration of the 4-velocity $\vec{u} = d\vec{x}/d\tau$ along a trajectory in spacetime.

In special relativity, we learn that the components of 4-velocity as measured by an observer who sees $\mathcal{O}$ move past with 3-velocity $\mathbf{v}$ is given by

$$\vec{u} \doteq (\gamma, \gamma\mathbf{v}) \, , \tag{2.6}$$

with $\gamma = 1/\sqrt{1 - |\mathbf{v}|^2}$. Notice that if $\mathcal{O}$ happens to be at rest with respect to that observer, then $\vec{u} \doteq (1, \mathbf{0})$. Notice also that

$$\vec{u} \cdot \vec{u} = -\gamma^2 + |\mathbf{v}|^2\gamma^2 = -1 \, . \tag{2.7}$$

The 4-velocity is *always* a timelike vector, with unit magnitude.

An important quantity we can construct from the 4-velocity is the 4-momentum:

$$\vec{p} = m\vec{u} \tag{2.8}$$

$$\doteq (E, \mathbf{p}) \, . \tag{2.9}$$

[1]Sometimes people ask what is "proper" about proper time. This term is taken from French or Latin, and really means "$\mathcal{O}$'s own time." I like the German term eigenzeit, but it doesn't seem to have caught on in English.

The form (2.8) is a definition of $\vec{p}$ that describes any object with mass[2] m and 4-velocity $\vec{u}$; the form (2.9) describes the timelike and spacelike components of $\vec{p}$ as measured in some particular IRF. The timelike component is the *energy* measured in that IRF; the spacelike components are the *3-momentum* (often simply "momentum") which that IRF measures. A few remarks about this very important object:

- All observers agree on $\vec{p}\cdot\vec{p}$; using the fact that $\vec{p} = m\vec{u}$ and $\vec{u}\cdot\vec{u} = -1$, we see that

$$\vec{p}\cdot\vec{p} = -m^2 \ . \tag{2.10}$$

  Because this is true in all IRFs, we can use it to understand the relation between an object's energy and 3-momentum as measured in some IRF:

$$\begin{aligned} & \vec{p}\cdot\vec{p} = -m^2 = -(p^0)^2 + |\mathbf{p}|^2 \\ \longrightarrow \quad & E^2 - |\mathbf{p}|^2 = m^2 \qquad \text{or} \qquad E^2 = |\mathbf{p}|^2 + m^2 \ . \end{aligned} \tag{2.11}$$

  This final form should be familiar from studies of special relativity (albeit perhaps with factors of c hidden by our choice of units).

- A "body" can have a mass of zero; photons are well-known examples of this limit. In this case, the body still has energy and momentum, and thus has a well-defined 4-momentum. From the above relation, we see that in this case, any observer measures this zero-mass body to have energy equal to the magnitude of its 3-momentum: $E = |\mathbf{p}|$. It's worth noting one cannot assign a 4-velocity to such a body. If a body has a 4-velocity, then there exists a rest frame for that body at each moment on its worldline. You are hopefully familiar with the fact that zero mass objects cannot be at rest, consistent with the idea that they do not have a well-defined 4-velocity.

- 4-momentum is conserved in any interaction. If we have N particles that interact, then

$$\vec{p}_{\text{tot}} \equiv \sum_{i=1}^{N} \vec{p}_i \tag{2.12}$$

  is conserved in the interaction. This is true even if N changes during the interaction (e.g., a particle decays, or two bodies collide inelastically to form a new body).

- Many calculations based on conservation of 4-momentum can be expedited by working in a special IRF in which

$$\vec{p}_{\text{tot}} \doteq (E, \mathbf{0}) \ . \tag{2.13}$$

  This special IRF is known as the *center of momentum* frame. In this frame, the interacting system is at rest as a whole (although members of the system could very well be zipping around quite speedily).

2.3 Exploiting the invariant product

The invariance of the scalar product is often a way to rapidly deduce a result that is of great utility. The first example we'll look at concerns the energy of a body as measured by some particular observer. Let $\vec{p}_A$ be the 4-momentum of body A, and let $\vec{u}_{\mathcal{O}}$ be the 4-velocity of observer $\mathcal{O}$. What does $\mathcal{O}$ measure for the energy of body A?

[2]Notice that the mass is a Lorentz scalar — all observers agree on its value. It is often called the "rest mass," in recognition of the fact that it describes the object's energy when at rest (i.e., $E = mc^2$, restoring the c^2 hidden by our choice of units).

One way to do this is to go into the rest frame of $\mathcal{O}$. In that frame, $\vec{u}_{\mathcal{O}} \doteq (1, \mathbf{0})$. Also in that frame, $\vec{p} \doteq (E_{\mathcal{O}}, \mathbf{p}_{\mathcal{O}})$, where $E_{\mathcal{O}}$ and $\mathbf{p}_{\mathcal{O}}$ are the energy and 3-momentum of body A as measured by $\mathcal{O}$. This suggests that we should figure out the Lorentz transformation that puts us into the rest frame of $\mathcal{O}$, apply this transformation to $\vec{p}_A$, and from the resulting components read out $E_{\mathcal{O}}$.

That's not wrong, but the invariant product enables a tremendously effective shortcut. Because $\vec{u}_{\mathcal{O}} \doteq (1, \mathbf{0})$ in the rest frame of $\mathcal{O}$, it is clear that $E_{\mathcal{O}} = -\vec{p}_A \cdot \vec{u}_{\mathcal{O}}$. But this result involves an invariant *that holds in all IRFs*. Thus, no matter what frame one uses to construct $\vec{p}_A$ and $\vec{u}_{\mathcal{O}}$, the energy that $\mathcal{O}$ measures for the energy of body A is given by

$$\boxed{\boxed{E_{\mathcal{O}} = -\vec{p}_A \cdot \vec{u}_{\mathcal{O}}}} \tag{2.14}$$

This very powerful result gives us a totally frame-independent way to express what the energy of a body is according to some IRF. We are going to use this result several times at points in this class.

Another result concerns the 4-acceleration of a body. We described the fact that the 4-velocity $\vec{u} = d\vec{x}/d\tau$ is essentially the tangent vector to the worldline of a body moving through spacetime. If the body is not inertial, then it is accelerating, and we can define an acceleration 4-vector

$$\vec{a} = \frac{d\vec{u}}{d\tau} \, . \tag{2.15}$$

The 4-acceleration has a very important constrained relationship with respect to the 4-velocity. To see this, we use the fact that $\vec{u} \cdot \vec{u} = -1$. Differentiating this relationship, we have

$$\vec{a} \cdot \vec{u} = 0 \, . \tag{2.16}$$

The 4-acceleration is always orthogonal to the 4-velocity in spacetime. This is in stark contrast to physics based on 3-velocity and 3-acceleration, where these quantities need not have any particular relationship to one another at any moment. Equation (2.16) is an important sanity check when doing relativistic kinematics; a friend of mine was able to instantly reject a paper he was refereeing for a journal by noting that the author's proposed kinematics did not respect this relationship.

More importantly for us, Eq. (2.16) helps us understand the meaning of 4-acceleration. Let us go into the rest frame of the observer whose 4-velocity is $\vec{u}$. (If the body is accelerating, that frame will not be its rest frame for long. As such, this is often called its "instantaneous rest frame.") As mentioned many times, in this frame $\vec{u} \doteq (1, \mathbf{0})$. In order for $\vec{a} \cdot \vec{u} = 0$, we must have $a \doteq (0, \mathbf{a})$ in this frame. Here, $\mathbf{a}$ is the ordinary 3-acceleration in this frame. Further, we see from this that $\vec{a} \cdot \vec{a} = |\mathbf{a}|^2$ — a frame-independent statement about the 4-acceleration that nonetheless involves the 3-acceleration in the body's instantaneous rest frame. Putting all this together, we see that a body experiencing a 4-acceleration $\vec{a}$ feels a 3-acceleration of magnitude $|\mathbf{a}| = \sqrt{\vec{a} \cdot \vec{a}}$ in their own rest frame.

2.4 Tensors more generally

So far, we have introduced one object that we have labeled a "tensor," the metric $\eta_{\alpha\beta} = \vec{e}_\alpha \cdot \vec{e}_\beta$. More generally, tensors are geometric objects which act like functions which enact linear operations on vectors. Let us introduce a precise definition:

A tensor of type $\binom{0}{N}$ is a geometric object which enacts a functional mapping of N vectors into a Lorentz-invariant scalar, and which is linear in each of its N arguments.

(You might wonder: why the zero in $\binom{0}{N}$? Can't there be something else there? Hold that thought a bit: we're walking up to a fully general definition, but need to lay some groundwork first.)

Using this definition, we see that the invariant scalar product is actually a $\binom{0}{2}$ tensor:

$$\begin{aligned}
\vec{A}\cdot\vec{B} &= a && \text{2 vectors map to a Lorentz scalar.} && (2.17)\\
(\alpha\vec{A})\cdot\vec{B} &= \alpha a && \text{Linear in the 1st argument (part 1 of linearity).} && (2.18)\\
(\vec{A}+\vec{C})\cdot\vec{B} &= \vec{A}\cdot\vec{B}+\vec{C}\cdot\vec{B} && \text{Linear in the 1st argument (part 2 of linearity).} && (2.19)\\
\vec{A}\cdot(\beta\vec{B}) &= \beta a && \text{Linear in the 2nd argument (part 1 of linearity).} && (2.20)\\
\vec{A}\cdot(\vec{B}+\vec{C}) &= \vec{A}\cdot\vec{B}+\vec{A}\cdot\vec{C} && \text{Linear in the 2nd argument (part 2 of linearity).} && (2.21)
\end{aligned}$$

The "scalar product" tensor is in fact the metric. In very abstract form, we define the tensor $\boldsymbol{\eta}$ as a "2 slot" function that takes vectors as inputs, and yields the scalar product as output:

$$\boldsymbol{\eta}(\vec{A},\vec{B}) = \vec{A}\cdot\vec{B} = \eta_{\alpha\beta}A^\alpha B^\beta \ . \tag{2.22}$$

The usual linearity rules apply: $\boldsymbol{\eta}(\alpha\vec{A}+\vec{B},\vec{C}) = \alpha\boldsymbol{\eta}(\vec{A},\vec{C})+\boldsymbol{\eta}(\vec{B},\vec{C})$, etc.

Because tensors take vectors — frame-independent, geometric objects — as input, and yield Lorentz invariant scalars as output, they are themselves a kind of geometric object. We have intuition about vectors as "directed line segments" pointing in space or spacetime. It's harder to build such intuition about tensors; some can be thought of as an object that, in a meaningful way, points in multiple directions at once, but other tensors are harder to characterize. Because the tensors are themselves frame-independent geometric objects, they will have different components — i.e., different representations — in different reference frames. The scalar we get out when we fill up all their "slots" with vectors will, however, be invariant.

To get the components out, we simply put basis vectors in the slots. Our lone example so far is the metric:

$$\boldsymbol{\eta}(\vec{e}_\alpha,\vec{e}_\beta) = \vec{e}_\alpha\cdot\vec{e}_\beta = \eta_{\alpha\beta} \ . \tag{2.23}$$

Using this plus the linearity rules, we can see how to get tensor components in another frame of reference. Let's work this out: using $\vec{e}_{\bar\mu} = \Lambda^\alpha{}_{\bar\mu}\vec{e}_\alpha$ plus $\eta_{\bar\mu\bar\nu} = \boldsymbol{\eta}(\vec{e}_{\bar\mu},\vec{e}_{\bar\nu})$, we have

$$\begin{aligned}
\eta_{\bar\mu\bar\nu} &= \boldsymbol{\eta}(\Lambda^\alpha{}_{\bar\mu}\vec{e}_\alpha,\Lambda^\beta{}_{\bar\nu}\vec{e}_\beta)\\
&= \Lambda^\alpha{}_{\bar\mu}\Lambda^\beta{}_{\bar\nu}\boldsymbol{\eta}(\vec{e}_\alpha,\vec{e}_\beta)\\
&= \Lambda^\alpha{}_{\bar\mu}\Lambda^\beta{}_{\bar\nu}\eta_{\alpha\beta} \ . && (2.24)
\end{aligned}$$

The final rule we find aligns with a slogan introduced previously: "Just line up the indices." It should be noted that this is a fairly silly example, because it turns out that the components of the metric are just $\mathrm{diag}(-1,1,1,1)$ in all IRFs[3], at least as long as we use rectilinear coordinates. Nonetheless, the principle holds even in this silly case, and the need to enact this transformation law is far less trivial on all other tensors we will encounter in this class.

2.5 $\binom{0}{1}$ tensors: 1-forms

A 1-form is a mapping from a single vector to Lorentz invariant scalars:

$$\tilde{p}(\vec{A}) = \alpha \ . \tag{2.25}$$

Here, $\tilde{p}$ is a 1-form, and α is a Lorentz scalar. By inheriting tensor properties, it is easy to see that 1-forms satisfy all the rules needed to define a vector space: if $\tilde{s} = \tilde{p}+\tilde{q}$ and $\tilde{r} = \beta\tilde{p}$ then $\tilde{s}$ and $\tilde{r}$ are also 1-forms; $\tilde{s}(\vec{A}) = \tilde{p}(\vec{A})+\tilde{q}(\vec{A})$, $\tilde{r}(\vec{A}) = \beta\tilde{p}(\vec{A})$. We get their components with the basis vectors:

$$\tilde{p}(\vec{e}_\alpha) = p_\alpha \ . \tag{2.26}$$

[3] $\mathrm{diag}(a,b,c,d)$ means the matrix whose entries are a, b, c, d down the diagonal, and zero everywhere else.

This gives us a very nice way of understanding the Lorentz scalar that we get when we apply a 1-form to a vector:

$$\begin{aligned} \tilde{p}(\vec{A}) &= \tilde{p}(A^\alpha \vec{e}_\alpha) \\ &= A^\alpha \, \tilde{p}(\vec{e}_\alpha) \\ &= A^\alpha p_\alpha \, . \end{aligned} \tag{2.27}$$

The resulting Lorentz scalar is just what we get when we *contract* the (upstairs) vector indices on the components of $\vec{A}$ with the (downstairs) 1-form indices on the components of $\tilde{p}$. This operation is called *contraction*. Using a similar operation, it is straightforward to show that 1-form components transform by the "line up the indices" rule:

$$\begin{aligned} p_{\bar{\mu}} &= \tilde{p}(\vec{e}_{\bar{\mu}}) \\ &= \tilde{p}(\Lambda^\alpha{}_{\bar{\mu}} \vec{e}_\alpha) \\ &= \Lambda^\alpha{}_{\bar{\mu}} \tilde{p}(\vec{e}_\alpha) \\ &= \Lambda^\alpha{}_{\bar{\mu}} p_\alpha \, . \end{aligned} \tag{2.28}$$

In our discussion of basis vectors, we introduced a set of special vectors which were oriented along the 4 directions of spacetime. We similarly will find it useful to define a set of special 1-forms oriented (in a way to be described later) to pick out special directions. Let us define a set of 1-forms $\tilde{\omega}^\alpha$ such that any 1-form can be written using them and components as follows:

$$\tilde{p} = p_\alpha \tilde{\omega}^\alpha \, . \tag{2.29}$$

To actually define what these quantities are, we use the fact that $\tilde{p}(\vec{A}) = p_\alpha A^\alpha$ in the following way:

$$\begin{aligned} p_\alpha A^\alpha &= \tilde{p}(\vec{A}) \\ &= p_\alpha \tilde{\omega}^\alpha (A^\beta \vec{e}_\beta) \\ &= p_\alpha A^\beta \tilde{\omega}^\alpha (\vec{e}_\beta) \, . \end{aligned} \tag{2.30}$$

This last equation leads us to require that

$$\boxed{\boxed{\tilde{\omega}^\alpha(\vec{e}_\beta) = \delta^\alpha{}_\beta}} \tag{2.31}$$

An example representation that satisfies this is the very simple set

$$\tilde{\omega}^0 \doteq (1,0,0,0) \, , \tag{2.32}$$

$$\tilde{\omega}^1 \doteq (0,1,0,0) \, , \tag{2.33}$$

$$\tilde{\omega}^2 \doteq (0,0,1,0) \, , \tag{2.34}$$

$$\tilde{\omega}^3 \doteq (0,0,0,1) \, . \tag{2.35}$$

These look quite a lot like the basis vectors. They are not; the difference is akin to the difference between row vectors and column vectors in matrix manipulations. In this vein, it's worth remarking that 1-forms are sometimes called "dual vectors," especially in older texts. We are going to see very soon that 1-forms can be constructed from vectors, and vice versa, in a very simple way, making it clear that they are very closely related geometric entities.

We conclude this discussion with an example of a 1-form. Imagine that some observer follows a trajectory through spacetime with 4-velocity $\vec{u} = d\vec{x}/d\tau$ along their worldline, and that all of spacetime is filled with some scalar field which varies from event to event $\Phi(t,x,y,z)$. What is the rate of change of Φ along the worldline?

If this were simple Euclidean 3-space, you would surely assert that the answer is

$$\left.\frac{d\Phi}{dt}\right|_{\text{along traj}} = \frac{\partial\Phi}{\partial t} + \frac{dx}{dt}\frac{\partial\Phi}{\partial x} + \frac{dy}{dt}\frac{\partial\Phi}{\partial y} + \frac{dz}{dt}\frac{\partial\Phi}{\partial z}$$
$$= \frac{\partial\Phi}{\partial t} + \mathbf{v}\cdot\boldsymbol{\nabla}\Phi \ . \tag{2.36}$$

Generalizing this to spacetime is easy: we change per unit time to per unit proper time, and "upgrade" certain other concepts in a similar manner:

$$\left.\frac{d\Phi}{d\tau}\right|_{\text{along traj}} = \frac{dt}{d\tau}\frac{\partial\Phi}{\partial t} + \frac{dx}{d\tau}\frac{\partial\Phi}{\partial x} + \frac{dy}{d\tau}\frac{\partial\Phi}{\partial y} + \frac{dz}{d\tau}\frac{\partial\Phi}{\partial z}$$
$$= u^\alpha \frac{\partial\Phi}{\partial x^\alpha}$$
$$= u^\alpha \partial_\alpha \Phi \ . \tag{2.37}$$

On the last line, we have introduced the notation $\partial/\partial x^\alpha = \partial_\alpha$. We see that the components of the spacetime gradient are the components of a 1-form:

$$\tilde{d}\Phi \equiv \partial_\alpha \Phi \, \tilde{\omega}^\alpha \ . \tag{2.38}$$

We will use this in the next lecture to get some intuition about the geometry associated with 1-forms.

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Lecture 3
Spacetime and geometry part III

3.1 Wrapping up the geometric nature of 1-forms

The notion of the gradient as a 1-form gives us a very nice alternative way of writing down the basis 1-forms. Recall that basis 1-forms are subject to the condition that

$$\tilde{\omega}^{\alpha}(\vec{e}_{\beta}) = \delta^{\alpha}{}_{\beta} \ . \tag{3.1}$$

We also know that $\partial_{\beta} x^{\alpha} = \delta^{\alpha}{}_{\beta}$. This tells us that basis 1-form components can be "generated" by computing the gradient of our coordinate system. Since the gradient is a 1-form, we recognize the basis 1-forms as what we get by taking the gradient of the coordinates:

$$\boxed{\tilde{\omega}^{a} = \tilde{d}x^{\alpha}} \tag{3.2}$$

This is all well and good, but what mental picture should we carry in our heads as we think about the 1-forms? We have a mental picture of basis vectors as indicating a magnitude and a direction, with the basis vectors picking out the principle directions associated with our coordinate system — i.e., a spacetime trajectory that is everywhere parallel to $\vec{e}_0$ is one on which all of the coordinates except x^0 are constant, and x^0 steadily changes. The basis vector $\vec{e}_0$ defines a trajectory parallel to the x^0 axis, and likewise for a trajectory along any of the other basis vectors.

To try to develop a similar kind of picture about the geometry of 1-forms, imagine a 2-dimensional space and some scalar field that fills this space. For instance, suppose that our scalar field is height h as a function of coordinate location (x, y):

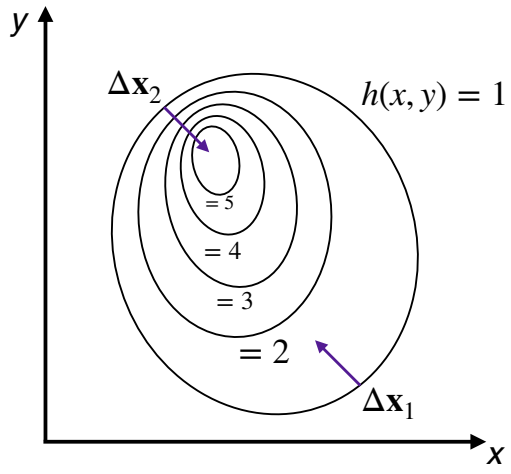


Figure 1: Level surfaces of a scalar function ("height" h) that depends on x and y, with two displacement vectors $\Delta\mathbf{x}_{1,2}$ that pierce several of the function's contours.

You are presumably familiar with the idea that the gradient of a function illustrates where the function is most rapidly changing. In a figure like Fig. 3.1, we would say that the gradient is largest

where the level surfaces of h are closest together. This tells us that the 1-form $\tilde{dh}$ is, in some meaningful sense, "largest" in the top left of the nested ovals shown in Fig. 3.1, and "smallest" in the bottom right of these ovals.

Consider some displacement vector $\Delta\mathbf{x}$ in this 2-dimensional plane. What do get if we contract the gradient of h with this displacement vector?

$$\tilde{dh}(\Delta\mathbf{x}) = \partial_i h\, \Delta x^i \equiv \Delta h(\text{along } \Delta\mathbf{x}) \,. \tag{3.3}$$

Contracting the gradient 1-form along a displacement vector tells us how much the function h changes along that vector[1]. Figure 3.1 illustrates this with two examples: $\Delta\mathbf{x}_1$ is in a place where the level surfaces are far apart, so the total Δh along this displacement vector is small. This means that the 1-form's magnitude is small here. By constrast, $\Delta\mathbf{x}_2$ is located where the surfaces are quite close together, and Δh along this vector is much larger. The 1-form's magnitude must be much larger there. In both cases, you can think of the "net Δh" found by the contraction as something like the number of h contours that are pierced by these vectors.

To tie together all these pieces, the geometric picture that has been developed to "visualize" 1-forms is that the 1-form basically *is* the set of level surfaces of the function. When those level surfaces are packed very densely, the 1-form is "large"; when the level surfaces are spread far out, the 1-form is "small."

In this way of thinking about 1-forms, the basis 1-forms are essentially level surfaces of constant coordinate. Whereas the basis vector $\vec{e}_0$ points along a line with only x^0 changing, the basis 1-form $\tilde{dx}^0$ is the set of 3-dimensional surfaces ("3-surfaces") in which x^0 is constant; they fill all of the rest of space. This is a way in which the 1-forms can be regarded as dual to vectors: the basis vector $\vec{e}_\alpha$ points along x^α and only x^α; the basis 1-form $\tilde{dx}^\alpha$ doesn't "point" along a direction in spacetime, but instead spans the 3-surface normal to that direction. This is a geometric notion that will prove particularly useful when we start to examine fluxes.

The textbook by Misner, Thorne, and Wheeler has some nice illustrations of 1-forms and basis 1-forms. Scans of these figures are posted to the 8.962 canvas page along with these lecture notes.

3.2 From vectors to 1-forms and back; tensors in generality

So far, we have only discussed in detail one tensor, the metric: a $\binom{0}{2}$ tensor that maps a pair of vectors to a Lorentz scalar:

$$\boldsymbol{\eta}(\vec{A}, \vec{B}) = \vec{A} \cdot \vec{B} \,. \tag{3.4}$$

What happens if we apply the metric to just a single vector?

$$\begin{aligned} \boldsymbol{\eta}(\vec{A}, -) &= \text{something that maps a single vector to a Lorentz scalar} \\ &= \text{a 1-form} \end{aligned} \tag{3.5}$$

We therefore define the 1-form that is dual to $\vec{A}$ as

$$\boldsymbol{\eta}(\vec{A}, -) = \tilde{A} \,. \tag{3.6}$$

Let's get its components in the usual way:

$$\begin{aligned} A_\alpha &= \tilde{A}(\vec{e}_\alpha) \\ &= \boldsymbol{\eta}(\vec{A}, \vec{e}_\alpha) \\ &= \boldsymbol{\eta}(A^\beta \vec{e}_\beta, \vec{e}_\alpha) \end{aligned} \tag{3.7}$$

[1] Strictly speaking, we should only use a differential displacement, and integrate up the change in h along the full vector. This does not affect the point we are emphasizing here.

Sorting through the various definitions and using the fact that $\eta_{\beta\alpha} = \eta_{\alpha\beta}$, we see that

$$\boxed{\boxed{A_\alpha = \eta_{\alpha\beta} A^\beta}} \tag{3.8}$$

We see that the metric converts the vector to a 1-form by "lowering" its components. We can make this an invertible procedure by defining the "metric inverse": we seek the tensor $\boldsymbol{\eta}^{-1}$ whose components $\eta^{\alpha\beta}$ are related to the components $\eta_{\alpha\beta}$ so that

$$\eta^{\alpha\gamma}\eta_{\gamma\beta} = \delta^\alpha{}_\beta \ . \tag{3.9}$$

(Hopefully it is simple to see the components $\eta^{\alpha\beta}$ have exactly the same matrix representation as $\eta_{\alpha\beta}$; this will not be the case for the metric and its inverse in general, but holds in special relativity in "inertial" coordinates.) Using this, it is a straightforward exercise to show that given the 1-form components A_α the vector components A^α are simply

$$A^\alpha = \eta^{\alpha\beta} A_\beta \ . \tag{3.10}$$

Using these notions, we now have a plethora of ways of writing the invariant scalar product between two vectors:

$$\begin{aligned} \vec{A}\cdot\vec{B} &= \boldsymbol{\eta}(\vec{A},\vec{B}) \\ &= \tilde{A}(\vec{B}) \\ &= \tilde{B}(\vec{A}) \\ &= \eta_{\alpha\beta}A^\alpha B^\beta \\ &= \eta^{\alpha\beta}A_\alpha B_\beta \\ &= A_\alpha B^\alpha \\ &= A^\alpha B_\alpha \ . \end{aligned} \tag{3.11}$$

These are all completely equivalent ways of writing the invariant scalar product. The distinction between "vectors" and "1-forms" is becoming nugatory: both mathematical objects really encode the same information, just packaged a bit differently.

When we introduced 1-forms, we treated them as $\binom{0}{1}$ tensors — geometric objects in the manifold which take a single vector as input and yield a Lorentz scalar as output. But we could just as well have said that the *vectors* are tensors — geometric objects which take a single 1-form as input, yielding a Lorentz scalar as output:

$$\tilde{p}(\vec{A}) = p_\alpha A^\alpha = p^\alpha A_\alpha = \vec{p}(\tilde{A}) = \vec{A}(\tilde{p}) = \tilde{A}(\vec{p}) \ . \tag{3.12}$$

All of the forms in the above line are totally equivalent ways of writing a particular Lorentz invariant scalar. By now, the distinction between the function which enacts the linear map (originally the 1-form, now extended to include the vector) and the object which that function acted upon has become rather irrelevant. The index notation really highlights the fundamental equivalence of these various "species" of geometric objects; thanks to the fact that we can raise and lower indices with the metric, the distinction is not so important.

This motivates us to redefine and expand our definition of a tensor:

A tensor of type $\binom{M}{N}$ is a geometric object which enacts a functional mapping of N vectors and M 1-forms into a Lorentz-invariant scalar, and which is linear in each of its (M, N) arguments.

Such an object is represented in a specified IRF by an object with M "upstairs" and N "downstairs" indices. One can easily adjust this, however, using the metric:

$$\binom{M}{N} \underset{\text{lower}}{\rightarrow} \binom{M-1}{N+1} \qquad \text{Example:} \quad R_{\alpha\beta\gamma\delta} = \eta_{\alpha\mu} R^\mu{}_{\beta\gamma\delta} \tag{3.13}$$

$$\binom{M}{N} \underset{\text{raise}}{\rightarrow} \binom{M+1}{N-1} \qquad \text{Example:} \quad S^{\alpha}{}_{\beta\gamma} = \eta^{\alpha\mu} S_{\mu\beta\gamma} \; . \tag{3.14}$$

When we discussed vectors and 1-forms, we introduced basis objects which attached to the components and allowed us to write down frame-invariant expressions for these objects in terms of components that hold only in a particular frame. Do we need to do the same thing for tensors? For example, do we need objects $\boldsymbol{\omega}^{\alpha\beta}$ and $\mathbf{e}_{\alpha\beta}$ such that

$$\boldsymbol{\eta} = \eta_{\alpha\beta}\, \boldsymbol{\omega}^{\alpha\beta} \quad \text{and} \quad \boldsymbol{\eta}^{-1} = \eta^{\alpha\beta}\, \mathbf{e}_{\alpha\beta} \qquad ? \tag{3.15}$$

Let's assume that the answer to this question is yes, and examine the consequences of this assumption. We know that

$$\boldsymbol{\eta}(\vec{e}_\mu, \vec{e}_\nu) = \eta_{\mu\nu} \; . \tag{3.16}$$

According to our assumption that we need a basis tensor, we can further write this

$$\eta_{\mu\nu} = \eta_{\alpha\beta}\, \boldsymbol{\omega}^{\alpha\beta}(\vec{e}_\mu, \vec{e}_\nu) \; . \tag{3.17}$$

For this equality to hold, we must have

$$\boldsymbol{\omega}^{\alpha\beta}(\vec{e}_\mu, \vec{e}_\nu) = \delta^{\alpha}{}_{\mu} \delta^{\beta}{}_{\nu} \; . \tag{3.18}$$

However, we already know that these Kronecker deltas are exactly what we get when we contract the basis 1-forms with basis vectors:

$$\delta^{\alpha}{}_{\mu} = \tilde{\omega}^{\alpha}(\vec{e}_\mu) \; , \qquad \delta^{\beta}{}_{\nu} = \tilde{\omega}^{\beta}(\vec{e}_\nu) \; . \tag{3.19}$$

This tells that we don't need a new category of objects to make the tensor basis object $\boldsymbol{\omega}^{\alpha\beta}$, we can simply make it by attaching a pair of basis 1-forms to the tensor components. We often denote the tensor produced by combining together vectors and 1-forms using the symbol $\otimes$, meaning "outer product" or "tensor product":

$$\boldsymbol{\eta} = \eta_{\alpha\beta}\, \tilde{\omega}^{\alpha} \otimes \tilde{\omega}^{\beta} \; . \tag{3.20}$$

Likewise, can write

$$\boldsymbol{\eta}^{-1} = \eta^{\alpha\beta}\, \vec{e}_\alpha \otimes \vec{e}_\beta \; . \tag{3.21}$$

We can generalize this to any rank of tensor:

$$\mathbf{R} = R^{\alpha}{}_{\beta\gamma\delta}\, \vec{e}_\alpha \otimes \tilde{\omega}^{\beta} \otimes \tilde{\omega}^{\gamma} \otimes \tilde{\omega}^{\delta} \; . \tag{3.22}$$

It's worth noting that carrying around all these basis objects is a lot of baggage. When we actually work with equations that have lots of tensors and indices, we will generally just work with expressions involving the components, with the understanding that for the equation to hold in all reference frames, basis objects must be attached and are (essentially) coming along for the ride[2]. There are two places where bearing in mind the accompanying basis objects becomes particularly useful. The first is in thinking about transformations. IRF $\mathcal{O}$ writes tensor $\mathbf{R}$ as

$$\mathbf{R} = R^{\alpha}{}_{\beta\gamma\delta}\, \vec{e}_\alpha \otimes \tilde{\omega}^{\beta} \otimes \tilde{\omega}^{\gamma} \otimes \tilde{\omega}^{\delta} \; . \tag{3.23}$$

IRF $\bar{\mathcal{O}}$ writes this tensor as

$$\mathbf{R} = R^{\bar{\mu}}{}_{\bar{\nu}\bar{\rho}\bar{\sigma}}\, \vec{e}_{\bar{\mu}} \otimes \tilde{\omega}^{\bar{\nu}} \otimes \tilde{\omega}^{\bar{\rho}} \otimes \tilde{\omega}^{\bar{\sigma}} \; . \tag{3.24}$$

[2] An alternate notation, introduced in Wald's textbook (Sec. 2.4), is particularly elegant and compact. Wald's notation looks like the index notation we use for components, but in fact with careful definitions and by carefully using a subset of lower-case Latin letters, he essentially "hides" the basis objects in a way that allows them to "come along for the ride" much more compactly than the notation I have introduced.

How do we relate the components $R^{\bar{\mu}}{}_{\bar{\nu}\bar{\rho}\bar{\sigma}}$ to the components $R^{\alpha}{}_{\beta\gamma\delta}$? We already know how to transform the basis objects:

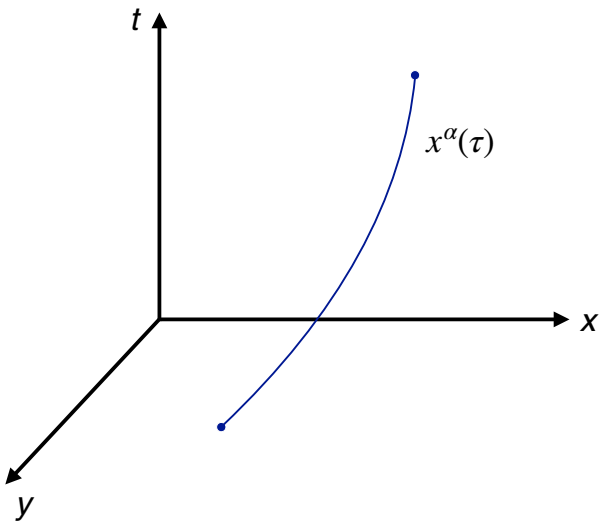
$$\vec{e}_{\bar{\mu}} = \Lambda^{\alpha}{}_{\bar{\mu}}\vec{e}_{\alpha} \ , \qquad \tilde{\omega}^{\bar{\nu}} = \Lambda^{\bar{\nu}}{}_{\beta}\tilde{\omega}^{\beta} \ , \qquad \text{etc.} \tag{3.25}$$

By using these rules for transforming the basis objects, using the fact that $\Lambda^{\alpha}{}_{\bar{\mu}}\Lambda^{\bar{\mu}}{}_{\beta} = \delta^{\alpha}{}_{\beta}$, and finally using the fact that the tensor $\mathbf{R}$ is a geometric object which must be the same in all frames, with a little algebra we can show that

$$R^{\bar{\mu}}{}_{\bar{\nu}\bar{\rho}\bar{\sigma}} = R^{\alpha}{}_{\beta\gamma\delta}\Lambda^{\bar{\mu}}{}_{\alpha}\Lambda^{\beta}{}_{\bar{\nu}}\Lambda^{\gamma}{}_{\bar{\sigma}}\Lambda^{\delta}{}_{\bar{\sigma}} \ . \tag{3.26}$$

At the end of the day, the mantra "line up the indices" holds as usual.

The second place where it is useful to remember that basis objects are coming along for the ride is in defining the derivatives of tensors. Imagine that some observer follows a trajectory $x^{\alpha}(\tau)$ through spacetime,



and that a tensor field $\mathbf{T} = T^{\alpha}{}_{\beta}\,\vec{e}_{\alpha}\otimes\tilde{\omega}^{\beta}$ fills this region. How does this tensor field vary along this trajectory? The logic of basic calculus holds:

$$\begin{aligned} \frac{d\mathbf{T}}{d\tau} &= \lim_{\Delta\tau\to 0}\frac{\mathbf{T}(\tau+\Delta\tau)-\mathbf{T}(\tau)}{\Delta\tau} \\ &= \frac{dT^{\alpha}{}_{\beta}}{d\tau}\,\vec{e}_{\alpha}\otimes\tilde{\omega}^{\beta} \qquad \text{(for now!)} \end{aligned} \tag{3.27}$$

This is quite simple *for now* because our basis objects are constant — they do not vary as functions of spacetime coordinate. *This will not always be true.* You have presumably seen in other contexts basis vectors which are non-constant, such as the unit vectors used in polar coordinate systems. When we begin to include such cases, we'll need to take into account how the basis objects themselves change in spacetime. For now, with the basis objects constant, then we simply need to compute the derivative of the tensor's components:

$$\frac{dT^{\alpha}{}_{\beta}}{d\tau} = u^{\gamma}\partial_{\gamma}T^{\alpha}{}_{\beta} \ . \tag{3.28}$$

The terms $\partial_{\gamma}T^{\alpha}{}_{\beta}$ define the components of a new tensor, the gradient of $\mathbf{T}$:

$$\tilde{d}\mathbf{T} = \partial_{\gamma}T^{\alpha}{}_{\beta}\,\tilde{\omega}^{\gamma}\otimes\vec{e}_{\alpha}\otimes\tilde{\omega}^{\beta} \ . \tag{3.29}$$

4 Characterizing matter in spacetime

Let's go back to thinking about how we describe things that, as physicists, we will want to understand. So far, we have introduced two important vectors which tell us about the kinematics of particles:

$$\text{The 4-velocity:} \qquad \vec{u} \doteq (\gamma, \gamma\mathbf{v}) \qquad \vec{u}\cdot\vec{u} = -1 \ , \tag{4.30}$$

$$\text{The 4-momentum:} \qquad \vec{p} = m\vec{u} \doteq (E, \mathbf{p}) \qquad \vec{p}\cdot\vec{p} = -m^2 \ . \tag{4.31}$$

As we discussed previously, the 4-momentum works perfectly well for a zero-mass "particle." In this case, it is often convenient to write $\vec{p} \doteq (E, E\mathbf{e}_k)$, where $\mathbf{e}_k$ describes the direction of propagation.

This is fine for single particles, but much of what we want to describe in physics is more complicated. How do we describe more interesting bunches of matter? We begin with the simplest continuum form of matter, "dust." We define dust as a collection of non-interacting particles (meaning that the collection does not generate any pressure), for which there exists a rest frame for each infinitesimal element[3].

Let's begin by going into such a rest frame. The simplest question we can ask is: How many dust particles do we have per unit of volume? Let us define

$$n_0 \equiv \text{Number density of dust in rest frame of an element.} \tag{4.32}$$

Let's now move into a new IRF. What is the number density of dust in this frame? The total number of dust particles in the volume must be the same; we cannot create or destroy matter simply by changing frames of reference. But, the volume of the element is smaller in this frame due to the length contraction that goes with the change of frame, so the density (number per volume) is concomitantly larger:

$$\begin{aligned} n &= \text{Number density of rest in some other frame} \\ &= \gamma n_0 \ . \end{aligned} \tag{4.33}$$

As usual, $\gamma = 1/\sqrt{1-|\mathbf{v}|^2}$, where $\mathbf{v}$ is the velocity of the dust particles in this IRF. In addition to this non-rest density, there is a current of dust flowing through spacetime thanks to the fact that it is not at rest in this frame. We define a "number current" by

$$\mathbf{n} = n\mathbf{v} = \gamma n_0 \mathbf{v} \ , \tag{4.34}$$

where $\mathbf{v}$ is the velocity of each dust particle in this IRF. Notice the dimensions of this quantity: n is number per unit volume, so $\mathbf{n}$ is number per unit area per unit time[4].

If you stare at the number density n and the number current $\mathbf{n}$ for a few moments, you can see that they very naturally combine to form a 4-vector. Let us define the *number flux 4-vector*:

$$\begin{aligned} \vec{N} &\doteq (n, \mathbf{n}) \\ &= n_0 \vec{u} \ . \end{aligned} \tag{4.35}$$

The 4-velocity $\vec{u}$ that enters this relation is the 4-velocity that a particular element of the dust distribution follows through spacetime. Notice that $n_0 = \sqrt{-\vec{N}\cdot\vec{N}}$: the rest density determines the "magnitude" or normalization of this timelike 4-vector.

[3] Perhaps a large collection of dust shares a rest frame. If not, you can imagine picking out some fiducial small volume dV and going to the rest frame of that element.

[4] In units with $c = 1$, per unit area per unit time has the same dimensions as per unit volume.

We can use our basis 1-forms to pick out how much dust flows across a particular surface: going back to the notation in which we treat 1-forms as linear functional operators on vectors, we have

$$\begin{aligned} \tilde{dx}^\alpha(\vec{N}) &= \text{The flux of } \vec{N} \text{ along the } x^\alpha \text{ direction} \\ &= N^\alpha \, . \end{aligned} \tag{4.36}$$

Choosing the $\alpha = 0$ component, we see that the number density is just the flux of $\vec{N}$ in the time direction! A pile of dust on your desk, apparently doing nothing, is zipping through spacetime in the timelike direction at all moments. The number density characterizes motion through time.

Because the number of dust particles can be neither created nor destroyed, it is subject to a conservation law. In a particular reference frame, we can look at a fiducial volume element, count the change of particles in this volume from t to $t + \Delta t$, and count how the particles cross over the boundaries of the element. The result (which hopefully looks very familiar) is that the rate of change of the volume in the element is closely related to the divergence of the number current in that volume:

$$\frac{\partial n}{\partial t} = -\frac{\partial n^x}{\partial x} - \frac{\partial n^y}{\partial y} - \frac{\partial n^z}{\partial z} = -\boldsymbol{\nabla} \cdot \mathbf{n} \, . \tag{4.37}$$

(This calculation is identical to the analysis of charge conservation in electrodynamics: you just replace electric charge — which is also Lorentz invariant, as it is also a quantity that cannot be created or destroyed by changing frames — with number of dust particles.) It is hopefully now clear that this equation can be rewritten in the manifestly Lorentz invariant form

$$\boxed{\boxed{\partial_\alpha N^\alpha = 0}} \tag{4.38}$$

When we have encountered differential conservation laws like this in previous physics courses, we have found that they can be converted into an integral form. To remind ourselves of how this works, let's choose some particular IRF. In this IRF, conservation of number is governed by the equation

$$\frac{\partial}{\partial t} \int_{V^3} n \, dV = -\int_{\partial V^3} \mathbf{n} \cdot d\mathbf{a} \, , \tag{4.39}$$

where

$$\begin{aligned} V^3 &= \text{some 3-volume at moment } t \\ \partial V^3 &= \text{the boundary of } V^3, \text{ with} \\ d\mathbf{a} &= \text{an outward-directed area element on that boundary.} \end{aligned} \tag{4.40}$$

Such an equation can be quite useful, but note that it is entirely frame-dependent — it relies on a particular observer's notion of the time direction, and of the space coordinates that this observer uses. Is it possible to formulate an integral conservation law in language that is frame-independent? In other words, is there a *covariant* way to put together something akin to Eq. (4.39)?

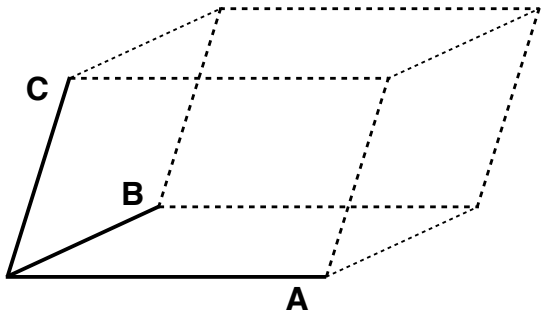
Massachusetts Institute of Technology
Department of Physics
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Lecture 4
Spacetime and geometry part IV

4.1 Volumes and volume integrals

The previous lecture ended wondering whether we could define a notion of a volume integral that would be useful for us, motivated by the idea that we'd like to be able to "integrate up" a differential expression of a conservation principle to describe the way some quantity (number, charge; perhaps energy and momentum) are conserved in some specified volume. The usual framework is manifestly *not* Lorentz covariant: conservation laws tend to balance the rate at which the amount of some conserved "stuff" in a volume changes against the flow of that "stuff" through the volume's sides. This explicitly depends on defining a time (in order that rates have useful meaning) and space (to define the geometry of the volume and its sides).

Let's set aside concerns about Lorentz covariance for a few moments and just think about how to formulate these integrals using a framework that is as general and coordinate-independent as possible. For intuition, think about a 3-volume: consider the parallelepiped whose sides are defined by the 3-vectors **A**, **B**, and **C**:



A really nice way to compute the volume of this figure is by using the dot product and cross product associated with 3-vectors:

$$\text{3-volume} = \mathbf{A}\cdot(\mathbf{B}\times\mathbf{C}) = \mathbf{B}\cdot(\mathbf{C}\times\mathbf{A}) = \mathbf{C}\cdot(\mathbf{A}\times\mathbf{B}) \ . \tag{4.1}$$

These three forms are totally equivalent, and can be expressed very nicely using a matrix determinant, which in turn can be expressed quite neatly using the Levi-Civita symbol:

$$\text{3-volume} = \begin{vmatrix} A^1 & A^2 & A^3 \\ B^1 & B^2 & B^3 \\ C^1 & C^2 & C^3 \end{vmatrix} = \epsilon_{ijk} A^i B^j C^k \ . \tag{4.2}$$

The Levi-Civita symbol is the totally antisymmetric symbol, defined as

$$\begin{aligned} \epsilon_{123} = \epsilon_{231} = \epsilon_{312} &= +1 \quad \text{(All even permutations of 123)} \\ \epsilon_{132} = \epsilon_{321} = \epsilon_{213} &= -1 \quad \text{(All odd permutations of 123)} \\ \epsilon_{112} &= 0 \quad \text{(Any index repeated)} \end{aligned} \tag{4.3}$$

In Euclidean 3-space, the Levi-Civita symbol can be regarded as the components of a type $\binom{0}{3}$ tensor. When its 3 "slots" are filled by vectors, the output is the volume of the parallelepiped whose sides are defined by those vectors:

$$V^3(\mathbf{A},\mathbf{B},\mathbf{C}) = \boldsymbol{\epsilon}(\mathbf{A},\mathbf{B},\mathbf{C}) \; . \tag{4.4}$$

Notice that we only fill two of the slots, what we get is a 1-form:

$$\tilde{\Sigma}(\mathbf{B},\mathbf{C}) \equiv \boldsymbol{\epsilon}(-,\mathbf{B},\mathbf{C}) \tag{4.5}$$

or

$$\Sigma_i(\mathbf{B},\mathbf{C}) = \epsilon_{ijk} B^j C^k \; . \tag{4.6}$$

The magnitude of this 1-form is the surface area corresponding to the face spanned by the two vectors this tensor acts upon. Bearing in mind the idea that a 1-form corresponds to the level surface of a function, we can see that the idea that a volume's face is described geometrically by a 1-form makes a lot of sense. (We can always construct the "normal" vector with which you may be more familiar by raising an index with the metric. In Euclidean 3-space with Cartesian coordinates, the metric's components are just the identity matrix, so this is a rather trivial operation.)

With these notions, we can rigorously spell out Gauss's theorem in geometric language: for some 3-vector field $\mathbf{F}$,

$$\int_{V^3} (\boldsymbol{\nabla}\cdot\mathbf{F})\, dV = \oint_{\partial V^3} \mathbf{F}(d\tilde{\Sigma}) \; . \tag{4.7}$$

Here, V^3 is some 3-volume, and ∂V^3 is the boundary of that 3-volume. To construct the volume element dV and the area element 1-form $d\tilde{\Sigma}$, we need to imagine that at each point we integrate over a differential triple $d\mathbf{x}_1$, $d\mathbf{x}_2$, $d\mathbf{x}_3$. Then,

$$\begin{aligned} dV &= \epsilon_{ijk} dx_1^i dx_2^j dx_3^k \; , \\ d\Sigma_i &= \epsilon_{ijk} dx_i^j dx_2^k \; . \end{aligned} \tag{4.8}$$

Note, the particular differential legs that go into $d\Sigma_i$ will depend on the orientation of ∂V^3. For example, a particularly "nice" choice of triple could be $d\mathbf{x}_1 = dx\,\mathbf{e}_x$, $d\mathbf{x}_2 = dy\,\mathbf{e}_y$, $d\mathbf{x}_3 = dz\,\mathbf{e}_z$. With this choice, $dV = dx\,dy\,dz$, and $d\Sigma_i$ will have magnitude $dx\,dy$, $dy\,dz$, $-dx\,dz$, etc, depending upon the detailed geometry of each surface element.

The spacetime generalization of all this follows quite naturally. Consider the *4-volume* enclosed by a parallelepiped in spacetime with sides $\vec{A}$, $\vec{B}$, $\vec{C}$, $\vec{D}$: it is simply given by

$$\text{4-volume} \equiv V^4 = \epsilon_{\alpha\beta\gamma\delta} A^\alpha B^\beta C^\gamma D^\delta \; . \tag{4.9}$$

We are using the 4-dimensional Levi-Civita symbol, for which $\epsilon_{0123} = +1$, and likewise for all even permutations of 0123; $\epsilon_{1023} = -1$, and likewise for all odd permutations of 0123; and $\epsilon_{0023} = 0$, and likewise whenever an index is repeated.

The surface "area" of each face of the parallelepiped has the dimensions of 3-volume, and can be regarded as a spacetime 1-form. For example, two of the faces have the form

$$\Sigma_\alpha = \epsilon_{\alpha\beta\gamma\delta} B^\beta C^\gamma D^\delta \; ; \tag{4.10}$$

the other six faces involve different combinations of 3 of the 4 defining vectors. Gauss's theorem then carries over to the generalization

$$\int_{V^4} (\partial_\alpha F^\alpha)\, dV^4 = \oint_{\partial V^4} F^\alpha\, d\Sigma_\alpha \; , \tag{4.11}$$

where

$$\begin{aligned} dV^4 &= \epsilon_{\alpha\beta\gamma\delta} dx_0^\alpha \, dx_1^\beta \, dx_2^\gamma \, dx_3^\delta \, , \\ d\Sigma_\alpha &= \epsilon_{\alpha\beta\gamma\delta} dx_1^\beta \, dx_2^\gamma \, dx_3^\delta \, . \end{aligned} \tag{4.12}$$

Similar to the 3-space example, the particular differential legs that go into $d\Sigma_\alpha$ will depend on the orientation of ∂V^4.

4.2 Integral formulation of conservation laws in spacetime

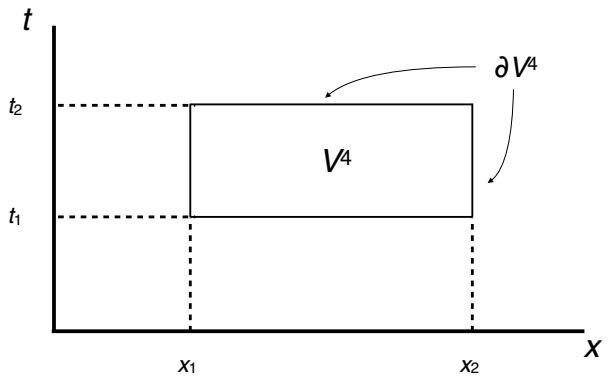
This has all been fairly abstract up to this point. Let's make it concrete by considering the number flux 4-vector for dust:

$$\int_{V^4} (\partial_\alpha N^\alpha) \, dV^4 = \oint_{\partial V^4} N^\alpha \, d\Sigma_\alpha \, . \tag{4.13}$$

The differential conservation law we deduced for this quantities was $\partial_\alpha N^\alpha = 0$, so the covariant integral conservation law is

$$\boxed{\oint_{\partial V^4} N^\alpha d\Sigma_\alpha = 0} \tag{4.14}$$

We emphasize that Eq. (4.14) holds in *all* reference frames: it is a completely covariant integral formulation of the conservation of number in some spacetime volume. However, specific inertial observers will split up the volume V^4 in different ways. Let's look at this equation from the perspective of a specific frame, for which the volume looks as follows:



(This is obviously just a 2-dimensional slice of the 4-volume.) When we evaluate the integral (4.14), there will be 8 contributions, corresponding to the 8 different faces of the 4-cube:

$$\begin{aligned} 0 = \oint_{\partial V^4} N^\alpha d\Sigma_\alpha &= \int_{t=t_2} N^0 \, dx \, dy \, dz - \int_{t=t_1} N^0 \, dx \, dy \, dz \\ &+ \int_{x=x_2} N^1 \, dt \, dy \, dz - \int_{x=x_1} N^1 \, dt \, dy \, dz + \ldots \end{aligned} \tag{4.15}$$

Now consider the limit $t_2 \to t_1 + dt$, and rearrange:

$$\int_{t=t_1+dt} N^0 \, dx \, dy \, dz - \int_{t=t_1} N^0 \, dx \, dy \, dz = -dt \left[\int_{x=x_2} N^1 \, dy \, dz - \int_{x=x_1} N^1 \, dy \, dz + \ldots \right] \, . \tag{4.16}$$

Recalling that, in this frame $N^0 = n$, the number density of dust, and $N^{1,2,3}$ are the components of $\mathbf{n}$, the number flux 3-vector, the above equation becomes

$$\frac{\partial}{\partial t} \int_{V^3} n \, dx \, dy \, dz = - \oint_{\partial V^3} \mathbf{n} \cdot d\mathbf{a} \, . \tag{4.17}$$

This is exactly the rule that we wrote down earlier, demonstrating that Eq. (4.14) is entirely equivalent to this rule, but is written in a *covariant* manner — meaning that it is a formulation that works in all IRFs.

4.3 Electrodynamics: Field equations and equation of motion

A quantity which shares some of the mathematical structure of the number-flux 4-vector is the electric current:

$$\vec{J} \doteq (\rho, \mathbf{J}) \; . \tag{4.18}$$

The continuity equation of electrodynamics takes the form $\partial_\alpha J^\alpha = 0$ in our language. Integrating this over some V^4 and applying Gauss's theorem tells us that

$$\oint_{\partial V^4} J^\alpha d\Sigma_\alpha = 0 \; . \tag{4.19}$$

Examining Eq. (4.19) in some specified IRF, this equation tells us how the charge contained in a volume changes in response to currents that flow into and out of that volume.

Part of what makes the electric current so important is that it acts as the source for electric and magnetic fields. Working in Cartesian coordinates in 3-space (so that whether indices are "upstairs" or "downstairs" is irrelevant — the metric which raises and lowers is just the identity), the field equations in component form are given by

$$\begin{aligned} \partial_i E_i = 4\pi\rho &= 4\pi J^0 & (4.20)\\ \epsilon_{ijk}\partial_j B_k - \partial_0 E_i &= 4\pi J^i & (4.21)\\ \epsilon_{ijk}\partial_j E_k + \partial_0 B_i &= 0 & (4.22)\\ \partial_i B_i &= 0 \; . & (4.23) \end{aligned}$$

You presumably learned about this in another course, perhaps without the 4πs but with factors of ϵ_0 and μ_0; the difference is just an uninteresting choice of units. The 6 components of the electromagnetic field are perfectly suited to be packaged into an antisymmetric 2nd-rank tensor; we define components in a particular frame according to

$$F^{0i} = E^i = -F^{i0} \; , \qquad F^{ij} = \epsilon^{ijk} B_k \; . \tag{4.24}$$

This object has the matrix representation

$$F^{\mu\nu} \doteq \begin{pmatrix} 0 & E_x & E_y & E_z \\ -E_x & 0 & B_z & -B_y \\ -E_y & -B_z & 0 & B_x \\ -E_z & B_y & -B_x & 0 \end{pmatrix} \; . \tag{4.25}$$

It's not hard to show that the field equations are given by

$$\begin{aligned} \partial_\nu F^{\mu\nu} &= 4\pi J^\mu \; , & (4.26)\\ \partial_\mu F_{\nu\lambda} + \partial_\nu F_{\lambda\mu} + \partial_\lambda F_{\mu\nu} &= 0 \; . & (4.27) \end{aligned}$$

Just expand these out and you'll see that Eq. (4.26) is equivalent to Eqs. (4.20) and (4.21), and (4.27) is equivalent to (4.22) and (4.23). An alternate way of writing (4.27) is to define a "dual" electromagnetic field tensor,

$$G_{\alpha\beta} = \epsilon_{\alpha\beta\gamma\delta} F^{\gamma\delta} \; ; \tag{4.28}$$

using this, the source-free part of the Maxwell equations becomes

$$\partial_\alpha G^{\alpha\beta} = 0 \; . \tag{4.29}$$

This is quite pretty, though one should bear in mind that we are soon going to use the symbol $G^{\alpha\beta}$ for a totally different (and extremely important) tensor in our relativistic theory of gravity. It's also worth remarking that when one lowers or raises indices, various minus signs come into play thanks to the negative component of the "time-time" piece of the metric. For example,

$$F_{\mu\nu} \doteq \begin{pmatrix} 0 & -E_x & -E_y & -E_z \\ E_x & 0 & B_z & -B_y \\ E_y & -B_z & 0 & B_x \\ E_z & B_y & -B_x & 0 \end{pmatrix} . \tag{4.30}$$

Be careful and double check all quantities when working with these tensors; it's easy to introduce stray minus signs.

One of the beautiful features of the electromagnetic field equation as written in this tensorial index notation is that conservation of the source is automatically built into the formalism. To see this, take the divergence of the sourced field equation (4.26):

$$\begin{aligned} 4\pi\partial_\mu J^\mu &= \partial_\mu\partial_\nu F^{\mu\nu} & & \\ &= \partial_\nu\partial_\mu F^{\nu\mu} & \text{(Dummy indices can be relabeled, so switch } \mu \text{ and } \nu) & (4.31) \\ &= -\partial_\nu\partial_\mu F^{\mu\nu} & (F^{\mu\nu} \text{ is antisymmetric}) & (4.32) \\ &= -\partial_\mu\partial_\nu F^{\mu\nu} & (\partial_\mu\partial_\nu \text{ is symmetric}) & (4.33) \\ &= 0 . & & (4.34) \end{aligned}$$

The final equality comes from comparing the first line with the fourth line: the right-hand side is equal to its negative, which is only true if it is zero. The field equations thus automatically enforce conservation of the current vector J^μ.

The trick that the above calculation introduced is often called "symmetry-antisymmetry": whenever an object that is symmetric on exchange of indices is contracted onto an object that is antisymmetric on index exchange, then the result must be zero. Another place where this result comes up is in the equation of motion a charge feels due to an electromagnetic field. In previous physics cources, you probably learned the Lorentz force law $\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$. This is not wrong, but it is not written in covariant form: we have selected notions of "time" and "space," so this is written for some observer's IRF. The covariant version of this is

$$\frac{dp^\mu}{d\tau} = qF^{\mu\nu}u_\nu , \tag{4.35}$$

or

$$a^\mu = \frac{q}{m}F^{\mu\nu}u_\nu . \tag{4.36}$$

A lovely feature of our formalism is that this equation automatically builds in the idea that the 4-acceleration is spacetime orthogonal to the 4-velocity. To see this, contract u_μ into both sides of this equation. On the left, we have $a^\mu u_\mu$, something we hope is zero. On the right, we have something proportional to $F^{\mu\nu}u_\mu u_\nu$. The tensor $F^{\mu\nu}$ is antisymmetric on exchange of indices; the combination $u_\mu u_\nu$ is symmetric. Thanks to "symmetry-antisymmetry," the right-hand side is indeed zero; our force law is one allowed by relativistic kinematics.

4.4 The stress-energy tensor

Let's come back to our dust, and now look at something more interesting than just the number of particles per unit volume. Suppose each particle has a mass m; in "normal" units, we would say that each particle has a rest energy mc^2. In the rest frame (and going back to $c = 1$ units), the rest energy density of the dust is given by

$$\rho \equiv \rho_0 = mn_0 . \tag{4.37}$$

How about in another frame? Suppose that in frame $\mathcal{O}$, the dust is moving with $\mathbf{v}$ relative to the rest frame. The number density goes up by a factor of γ due to the length contraction of the fiducial volume. The energy per particle also goes up by γ, since in this frame the particles are not at rest. So,

$$\rho = (\gamma m)(\gamma n_0) = \gamma^2 \rho_0 \ . \tag{4.38}$$

This is neither the transformation law of a scalar, nor of a 4-vector component. In fact, m is a component of $\vec{p}$ (it is the energy, at least in one particular frame), and n_0 is a component of $\vec{N}$ (it is the number density in that same frame), so the density that we originally assembled in fact comes out of the tensor product of two vectors. What we've developed is one component of a quantity known as the *stress-energy tensor* for dust:

$$\mathbf{T} = \vec{N} \otimes \vec{p} = n_0 m \, \vec{u} \otimes \vec{u} = \rho_0 \, \vec{u} \otimes \vec{u} \ , \tag{4.39}$$

or

$$T^{\alpha\beta} = \rho_0 u^\alpha u^\beta \ . \tag{4.40}$$

The physical meaning of the stress-energy tensor is simple: $T^{\alpha\beta}$ describes the flux of 4-momentum component p^α in the x^β direction. This holds *in general*, not just for dust (which are using because it is simple and illustrative). Let's think about what the components mean the four principle groupings of space and time in some IRF:

$$T^{00} = \text{flux of } p^t \text{ in the } t \text{ direction} = \text{energy density for observers in this IRF} \tag{4.41}$$

$$T^{0i} = \text{flux of } p^t \text{ in the } x^i \text{ direction} = \text{energy flux along } x^i \tag{4.42}$$

$$T^{i0} = \text{flux of } p^i \text{ in the } t \text{ direction} = \text{density of momentum } p^i \tag{4.43}$$

$$T^{ij} = \text{flux of } p^i \text{ in the } x^j \text{ direction} \ . \tag{4.44}$$

Much of what we encounter in physics is not dust. To understand $T^{\alpha\beta}$ for whatever situation we are studying, it is fairly common to understand what the components mean in some IRF, and then "promote" terms to a covariant form.

An important example for the study of gravitational physics is the "perfect" fluid. This is a fluid in which there is no energy flow in the rest frame (meaning, for example, that it does not transport heat) and for which it exerts no lateral stresses in that frame (i.e., T^{ij} is diagonal — there is no viscosity to transport p^i in a direction other than x^i). These simplifications mean that this is a rather fictional fluid; it has been derided as a mathematical description of "dry" water. Nonetheless, for certain applications (especially ones where the goal is to understand gravity), this approximation includes all the terms which make a significant contribution to the physics we wish to study. Section 4.5 in the textbook by Schutz does a very nice job laying out what is needed to go beyond this basic picture; for us, the basic picture is sufficient. A perfect fluid is characterized by two numbers, the energy density in the fluid's rest frame ρ and the *pressure* P which describes momentum flux. In a perfect fluid's rest frame, the stress-energy tensor's components are

$$T^{\alpha\beta} \doteq \begin{pmatrix} \rho & 0 & 0 & 0 \\ 0 & P & 0 & 0 \\ 0 & 0 & P & 0 \\ 0 & 0 & 0 & P \end{pmatrix} \doteq \text{diag}(\rho, P, P, P) \ . \tag{4.45}$$

This tensor can be written in a totally covariant manner by recognizing that the ρ attaches to the fluid 4-velocity of each element (recall that the components above are in the fluid's rest frame), and that the P attaches to the tensor which projects quantities orthogonal to those velocities (which you worked out on a problem set):

$$\mathbf{T} = \rho \, \vec{u} \otimes \vec{u} + P \, (\boldsymbol{\eta} + \vec{u} \otimes \vec{u}) \tag{4.46}$$

or

$$T_{\alpha\beta} = \rho \, u_\alpha u_\beta + P(\eta_{\alpha\beta} + u_\alpha u_\beta) = (\rho + P) u_\alpha u_\beta + P \eta_{\alpha\beta} \ . \tag{4.47}$$

4.5 Conservation of stress-energy

The stress-energy tensor is the tool that we use to describe how energy and momentum are distributed in spacetime; as such, it is the tool that we use to describe how energy and momentum are conserved for matter and fields that are so distributed. Because energy and momentum are connected into a single object (indeed, what is energy to one observer will be a mixture of energy and momentum to another), we do not treat their conservation laws separately, but combine them into the following form:

$$\boxed{\partial_\alpha T^{\alpha\beta} = 0} \tag{4.48}$$

To really see that this corresponds to conservation of energy and conservation of momentum, we need to look at it in some particular observer's reference frame. Let's choose a frame, and begin with the timelike component of Eq. (4.48):

$$\partial_\alpha T^{\alpha 0} = 0 \; ; \tag{4.49}$$

or,

$$\frac{\partial T^{00}}{\partial t} = -\frac{\partial T^{i0}}{\partial x^i} \; . \tag{4.50}$$

In words, the rate of change of energy density T^{00} measured by our observer is equal to the (minus) divergence of the energy flux. This is exactly the form of energy conservation we should expect on intuitive grounds! Integrating both sides over a 3-volume and invoking Gauss[1], we come to an integral formulation of energy conservation:

$$\frac{\partial}{\partial t}\int_{V^3} T^{00} dV^3 = -\int_{\partial V^3} T^{0i} d\Sigma_i \; . \tag{4.52}$$

The rate of change of energy in V^3 is balanced by the flux of energy across the boundary ∂V^3. (An equivalent way to do this would be to integrate

The spatial component of our general law gives us momentum conservation in this frame:

$$\partial_\alpha T^{\alpha j} = 0 \quad \longrightarrow \quad \frac{\partial T^{0j}}{\partial t} = -\frac{\partial T^{ij}}{\partial x^i} \; , \tag{4.53}$$

$$\frac{\partial}{\partial t}\int_{V^3} T^{0j} dV^3 = -\int_{\partial V^3} T^{ij} d\Sigma_i \; . \tag{4.54}$$

The rate of change of momentum component j in V^3 is balanced by the flux of momentum component j across the boundary ∂V^3.

It's worth re-emphasizing that Eq. (4.48) is, in "spacetime thinking," the fundamental conservation law. Energy and momentum conservation, expressed by Eqs. (4.52) and (4.54) follow from that only after splitting spacetime into a particular observer's space and time. They are 100% correct, and very useful in a particular frame, but have not been expressed in a fully covariant way.

[1] Or, we could integrate (4.48) over a 4-volume, apply Gauss to that and deduce the law

$$\int_{\partial V^4} T^{\alpha\beta} d\Sigma_\beta = 0 \; . \tag{4.51}$$

Nothing changes if we do that — analyzing the $\alpha = 0$ component reproduces our result for conservation of energy, and $\alpha = i$ reproduces that for conservation of momentum. This is simply another route to the same results.

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Lecture 5
More on the stress-energy tensor; curvilinear coordinates

5.1 Stress-energy symmetry

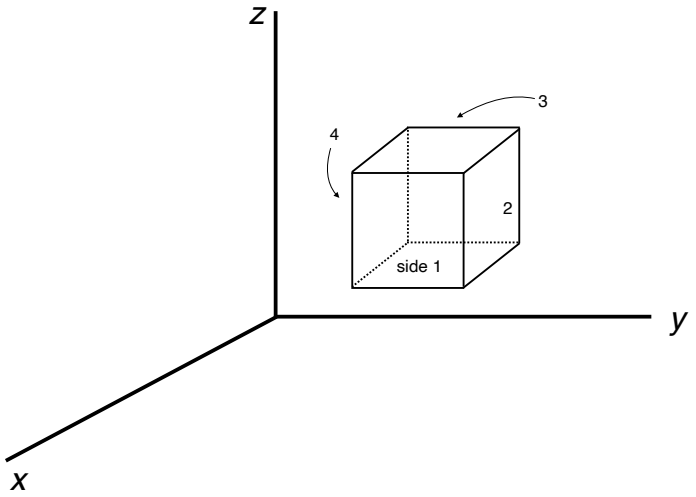
In the previous lecture, we introduced the stress-energy tensor, a quantity $T^{\alpha\beta}$ which describes the flux of p^α in the x^β direction. Let us look at it component by component for the case of energy and momentum of dust moving with 4-velocity $\vec{u} \doteq (\gamma, \gamma\mathbf{v})$ in some specified IRF:

$$T^{00} = \text{Energy density} = \gamma^2\rho_0 \ , \tag{5.1}$$
$$T^{0i} = \text{Energy flux} = \gamma^2\rho_0 v^i \ , \tag{5.2}$$
$$T^{i0} = \text{Momentum density} = \gamma^2\rho_0 v^i \ , \tag{5.3}$$
$$T^{ij} = \text{Momentum flux or stress} = \gamma^2\rho_0 v^i v^j \ . \tag{5.4}$$

(Here, ρ_0 is the energy density of the dust in its rest frame, i.e., the dust's rest-energy density.) Notice that $T^{\alpha\beta} = T^{\beta\alpha}$: the stress-energy tensor is symmetric under exchange of indices.

This symmetry is manifestly clear for the case of dust, but in fact holds much more generally. Picking a particular IRF, the symmetry $T^{0i} = T^{i0}$ follows from the fact that energy and momentum are the same thing reviewed in different reference frames. In other words, if energy is flowing, we must have momentum as well[1]. Energy flux is thus equivalent to momentum density. The symmetry $T^{ij} = T^{ji}$ is less obvious. However, one can quickly show that if this symmetry were violated, then a physically absurd situation could develop.

Consider a small cube with sides l embedded in some $T^{\alpha\beta}$:



Let's compute the 3-forces that act due to the stress-energy tensor upon faces 1 through 4 as illustrated here, and then use those forces to compute the torque about the z axis.

[1]It is worth noting that in "normal" units, we may need to insert factors of c to connect T^{0i} and T^{i0} with energy flux and momentum density, respectively.

The total 3-force that acts on face 1 is given by

$$F^i_{\text{face 1}} = \int_{\text{face 1}} T^{ix} dy\, dz \simeq T^{ix}\, l^2 \ . \tag{5.5}$$

The approximate equality follows because we assume l is very small (smaller than the scale over which the stress-energy tensor varies in this region). Repeating this calculation for the other three faces, we find

$$F^i_{\text{face 2}} \simeq T^{iy}\, l^2 \ , \qquad F^i_{\text{face 3}} \simeq -T^{ix}\, l^2 \ , \qquad F^i_{\text{face 4}} \simeq -T^{iy}\, l^2 \ . \tag{5.6}$$

(Our assumption of l small means that we assume the components T^{ij} are approximately constant over the cube.) Note that no net force acts on the cube, so our stress-energy tensor does not lead to weird accelerations.

Let us next examine the torques that these forces exert about an axis through the center of the cube, parallel to the z axis:

$$\tau^z_1 = (\text{torque due to } \mathbf{F}_{\text{face 1}})^z = -x F^y_{\text{face 1}} = -\frac{1}{2} T^{yx}\, l^3 \ . \tag{5.7}$$

The other 3-force components generate the following contributions to the torque:

$$\tau^z_3 = x F^y_{\text{face 3}} = -\frac{1}{2} T^{yx}\, l^3 \ , \tag{5.8}$$

$$\tau^z_2 = y F^x_{\text{face 2}} = \frac{1}{2} T^{xy}\, l^3 \ , \tag{5.9}$$

$$\tau^z_4 = -y F^x_{\text{face 4}} = \frac{1}{2} T^{xy}\, l^3 \ . \tag{5.10}$$

The total torque about the axis parallel to z exerted on this cube is thus

$$\tau^z = l^3 \left(T^{xy} - T^{yx} \right) \ . \tag{5.11}$$

By itself, there's nothing very interesting here. We would like to understand how this behaves as $l \to 0$. Equation (5.11) seems to indicate that the torque strictly vanishes as $l \to 0$, independent of the properties of T^{xy} and T^{yx}. This is not wrong, but we should dig in a little deeper before concluding that everything is OK. Imagine that the cube is made of some material of density ρ, and thus has a momentum of inertia

$$I = \alpha(\rho l^3) l^2 \ . \tag{5.12}$$

(The quantity α is a number less than or of order unity; if ρ is constant, $\alpha = 1/6$.) The cube feels an angular acceleration, tending to make it spin about this axis, whose value is

$$\frac{d^2\theta}{dt^2} = \frac{\tau^z}{I} \propto \frac{(T^{xy} - T^{yx})}{l^2} \ . \tag{5.13}$$

Unless this is strictly zero — i.e., unless $T^{xy} = T^{yx}$ — then the source of our stress-energy would tend to cause small regions to unstably rotate. Considering torques about the other axes of the cube leads us to conclude we need $T^{ij} = T^{ji}$ in order to avoid "weird spin ups" in small regions of our stress-energy tensor.

We emphasize that this discussion is *not* a rigorous proof that $T^{ij} = T^{ji}$; it is just a physical motivation that if this symmetry were violated, then weird and unphysical behavior would result. Later in the course, we will see a way to extract the stress-energy tensor from a Lagrangian density, and will see that the procedure for doing this guarantees that $T^{\alpha\beta} = T^{\beta\alpha}$.

5.2 Examples of stress-energy tensors

So far, we have discussed two examples of stress-energy tensors: that of dust, $T^{\alpha\beta} = \rho_0 u^\alpha u^\beta$; and that of the perfect fluid, $T^{\alpha\beta} = (\rho + P)u^\alpha u^\beta + P\eta^{\alpha\beta}$. Arguably, this is actually *one* example, since dust can be considered a perfect fluid with $P = 0$. This example is particularly important in gravity studies, but it is worth knowing of other examples:

5.2.1 A point particle moving through spacetime

A point body of mass m moving on some worldline $z^\mu(\tau)$ has a stress-energy tensor given by

$$T^{\mu\nu} = m \int d\tau \, u^\mu u^\nu \, \delta^{(4)} \left[\vec{x} - \vec{z}(\tau)\right] \ . \tag{5.14}$$

We have introduced here the 4-dimensional Dirac delta function; it can be decomposed into a product of 4 1-dimensional deltas:

$$\delta^{(4)} \left[\vec{x} - \vec{z}(\tau)\right] = \delta\left[t - z^0(\tau)\right] \delta\left[x - z^1(\tau)\right] \delta\left[y - z^2(\tau)\right] \delta\left[z - z^3(\tau)\right] \ . \tag{5.15}$$

Using the rule

$$\int f(x)\delta[g(x)]dx = \frac{f(x_0)}{|dg/dx|_{x=x_0}} \ , \tag{5.16}$$

where x_0 is the[2] zero of $g(x)$ in the domain of the integral, we can integrate up to find

$$T^{\mu\nu} = m\frac{u^\mu u^\nu}{u^t}\delta^{(3)} \left[\mathbf{x} - \mathbf{z}(\tau)\right] \ . \tag{5.17}$$

This form is used quite a bit in modern research as a tool for modeling the stress-energy of "small" bodies (i.e., bodies sufficiently pointlike that their internal structure can be ignored).

5.2.2 An electromagnetic field

If a region is filled with an electromagnetic field with field tensor $F^{\mu\nu}$, then

$$T^{\mu\nu} = \frac{1}{4\pi}\left[F^{\mu\lambda}F^{\nu}{}_{\lambda} - \frac{1}{4}\eta^{\mu\nu}F^{\lambda\sigma}F_{\lambda\sigma}\right] \ . \tag{5.18}$$

Going into a particular frame and using the association of components $F^{\mu\nu}$ with electric and magnetic field components, it is not hard to see that

$$T^{00} = \frac{\mathbf{E}\cdot\mathbf{E} + \mathbf{B}\cdot\mathbf{B}}{8\pi} \ , \tag{5.19}$$

$$T^{0i} = \frac{(\mathbf{E}\times\mathbf{B})^i}{4\pi} \ , \tag{5.20}$$

$$= T^{i0} \ , \tag{5.21}$$

$$T^{ij} = \frac{1}{8\pi}\left[(\mathbf{E}\cdot\mathbf{E} + \mathbf{B}\cdot\mathbf{B})\,\delta^{ij} - 2\left(E^iE^j + B^iB^j\right)\right] \ . \tag{5.22}$$

The terms for T^{00} and T^{0i} are hopefully familiar: they are the usual electromagnetic energy density (perhaps in funny units) and the Poynting vector, respectively. The result for T^{ij} is perhaps a little

[2] If there is more than one zero, then one must sum over all the zeros. This does not pertain to our analysis.

less familiar; it expresses a both an attractive tension associated with the field as well as a pressure that the field generates. A simple example can be found by putting $\mathbf{E} = E^x \mathbf{e}_x$, for which

$$T^{\mu\nu} \doteq \frac{(E^x)^2}{8\pi} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} . \tag{5.23}$$

The term T^{xx} indicates that this electromagnetic field generates an attractive stress along the x direction (think of the plates of a capacitor being pulled toward one another), but also a pressure in the y and z directions.

5.3 Flat spacetime in curvilinear coordinates

Everything we have done so far has been done in *inertial* coordinates — that is, coordinates for which an inertial observer's worldline can be expressed as a line. These coordinates are spatially Cartesian, and really easy to work with. However, they are limiting. To prepare us for the calculations we will need to do when we start thinking about spacetimes that are curved, let us start developing tools to do flat spacetime physics in curvilinear coordinate systems. Let's begin with the simplest example beyond Cartesian/inertial coordinates: plane polar coordinates (t, r, ϕ, z), related to our inertial coordinate system by $x = r\cos\phi$, $y = r\sin\phi$.

We continue to work in a coordinate basis, so it remains the case that

$$d\vec{x} = dx^\alpha \vec{e}_\alpha = dt\,\vec{e}_t + dr\,\vec{e}_r + d\phi\,\vec{e}_\phi + dz\,\vec{e}_z \ . \tag{5.24}$$

Since $d\phi$ has dimensions of angle, $\vec{e}_\phi$ must have dimensions of length. This is not a "normal" basis: $\vec{e}_\phi \cdot \vec{e}_\phi \neq 1$. We transform between these bases using the transformation matrix $L^{\alpha}{}_{\bar{\mu}} \equiv \partial x^\alpha / \partial x^{\bar{\mu}}$ (reserving from now on the symbol Λ for transformation matrices that are just Lorentz transformations). Let us write x^α for the inertial coordinates, and $x^{\bar{\mu}}$ for the curvilinear coordinates. The components of this transformation matrix are then given by

$$L^{\alpha}{}_{\bar{\mu}} = \frac{\partial x^\alpha}{\partial x^{\bar{\mu}}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\phi & \sin\phi & 0 \\ 0 & -r\sin\phi & r\cos\phi & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} . \tag{5.25}$$

The inverse transformation involves the matrix whose components are

$$L^{\bar{\mu}}{}_{\alpha} = \frac{\partial x^{\bar{\mu}}}{\partial x^\alpha} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\phi & -\sin\phi/r & 0 \\ 0 & \sin\phi & \cos\phi/r & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} . \tag{5.26}$$

You should be able to show straightforwardly that these matrices are inverses of each other.

Moving forward, we will define objects as tensors if their components transform between different coordinate representations using the L matrix which expresses the Jacobian between those coordinates:

$$V^{\bar{\mu}} = L^{\bar{\mu}}{}_{\alpha}\, V^\alpha = \frac{\partial x^{\bar{\mu}}}{\partial x^\alpha}\, V^\alpha \ , \tag{5.27}$$

$$\vec{e}_{\bar{\mu}} = L^{\alpha}{}_{\bar{\mu}}\, \vec{e}_\alpha = \frac{\partial x^\alpha}{\partial x^{\bar{\mu}}}\, \vec{e}_\alpha \ , \tag{5.28}$$

$$T^{\bar{\mu}\bar{\nu}}{}_{\bar{\sigma}} = L^{\bar{\mu}}{}_{\alpha} L^{\bar{\nu}}{}_{\beta} L^{\gamma}{}_{\bar{\sigma}}\, T^{\alpha\beta}{}_{\gamma} = \left(\frac{\partial x^{\bar{\mu}}}{\partial x^\alpha}\right)\left(\frac{\partial x^{\bar{\nu}}}{\partial x^\beta}\right)\left(\frac{\partial x^\gamma}{\partial x^{\bar{\sigma}}}\right) T^{\alpha\beta}{}_{\gamma} \ . \tag{5.29}$$

These forms reduce to our previous rule when the two coordinates are inertial representations of special relativity. However, there is now in general a significant difference: $L^{\bar{\mu}}{}_{\alpha}$ is now a function, and can vary from point to point. Our previous transformation only involved matrices whose components $\Lambda^{\bar{\mu}}{}_{\alpha}$ are constants.

Let's briefly examine several important geometric entities in the new coordinate representation. Two of the basis vectors are different:

$$\vec{e}_r = L^{\alpha}{}_r \, \vec{e}_{\alpha} = \cos\phi \, \vec{e}_x + \sin\phi \, \vec{e}_y \ , \tag{5.30}$$

$$\vec{e}_{\phi} = L^{\alpha}{}_{\phi} \, \vec{e}_{\alpha} = -r \sin\phi \, \vec{e}_x + r \cos\phi \, \vec{e}_y \ , \tag{5.31}$$

Notice that $\vec{e}_{\phi}$ grows with distance. You can think of this because the basis vector subtends a particular amount of fixed angle.

We find the metric in the usual way, though we now introduce a new symbol:

$$g_{\alpha\beta} = \vec{e}_{\alpha} \cdot \vec{e}_{\beta} = \mathrm{diag}(-1, 1, r^2, 1) \ . \tag{5.32}$$

From now on, we reserve the symbol $\eta_{\alpha\beta}$ strictly for $\mathrm{diag}(-1, 1, 1, 1)$, the inertial coordinate representation of the metric of special relativity. Notice that the components of the inverse metric are no longer the same as the components of the metric:

$$g^{\alpha\beta} = \mathrm{diag}(-1, 1, 1/r^2, 1) \ . \tag{5.33}$$

The line element we find with this metric is

$$ds^2 = g_{\alpha\beta} dx^{\alpha} \, dx^{\beta} = -dt^2 + dr^2 + r^2 \, d\phi^2 + dz^2 \ . \tag{5.34}$$

As with the basis vectors, two of the basis 1-forms are different:

$$\tilde{dr} = L^{r}{}_{\alpha} \, \tilde{dx}^{\alpha} = \cos\phi \, \tilde{dx} + \sin\phi \, \tilde{dy} \ , \tag{5.35}$$

$$\tilde{d\phi} = L^{\phi}{}_{\alpha} \, \tilde{dx}^{\alpha} = -\frac{\sin\phi}{r} \tilde{dx} + \frac{\cos\phi}{r} \tilde{dy} \ . \tag{5.36}$$

5.4 Christoffel symbols

A major reason why this detail matters is in calculating derivatives of tensorial objects: our basis objects are no longer constant, and we need to include how they vary when we take derivatives. For the case of plane polar coordinates,

$$\begin{aligned} \frac{\partial \vec{e}_r}{\partial r} &= 0 \ , & \frac{\partial \vec{e}_r}{\partial \phi} &= \frac{\vec{e}_{\phi}}{r} \ , \\ \frac{\partial \vec{e}_{\phi}}{\partial r} &= \frac{\vec{e}_{\phi}}{r} \ , & \frac{\partial \vec{e}_{\phi}}{\partial \phi} &= -r\vec{e}_r \ . \end{aligned} \tag{5.37}$$

Let's think about this in the context of the derivative of a simple tensor object, a vector $\vec{V} = V^{\alpha}\vec{e}_{\alpha}$. We write its gradient in fully tensorial form using the basis 1-forms as follows:

$$\boldsymbol{\nabla}\vec{V} = \frac{\partial}{\partial x^{\beta}} \left[(V^{\alpha}\vec{e}_{\alpha}) \right] \tilde{dx}^{\beta} \equiv \partial_{\beta} \left[(V^{\alpha}\vec{e}_{\alpha}) \right] \tilde{dx}^{\beta} \ . \tag{5.38}$$

The derivative we need to evaluate here is

$$\partial_{\beta} \left(V^{\alpha}\vec{e}_{\alpha} \right) = \left(\partial_{\beta} V^{\alpha} \right) \vec{e}_{\alpha} + V^{\alpha} \partial_{\beta} \vec{e}_{\alpha} \ . \tag{5.39}$$

The last term involves the derivatives of the basis vectors themselves. They can be organized in a very helpful way as follows:

$$\partial_{\beta}\vec{e}_{\alpha} = \Gamma^{\mu}_{\beta\alpha}\vec{e}_{\mu} \ . \tag{5.40}$$

The quantity we introduced, $\Gamma^{\mu}_{\beta\alpha}$, is known as the *Christoffel symbol*.

Caution: Slightly different conventions exist for how the Christoffel symbol is defined. The way we have defined it (the x^β derivative of $\vec{e}_\alpha$) is how it is done in, for example, the textbooks by Carroll and Schutz. Some other sources (for example, Schutz and MTW) define this quantity to be $\Gamma^{\mu}_{\alpha\beta}$. This difference turns out to be irrelevant as long as we work in a coordinate basis, since the Christoffel symbol is symmetric on the lower two indices in this case. (We will soon prove this.)

For the plane polar coordinates we have introduced, there are three non-zero Christoffel symbols (two of which are equal due to this symmetry):

$$\Gamma^{\phi}_{r\phi} = \Gamma^{\phi}_{\phi r} = 1/r \ , \qquad \Gamma^{r}_{\phi\phi} = -r \ . \tag{5.41}$$

All the other Christoffel symbols are zero in this case.

Let us return to the derivative of the vector:

$$\begin{aligned} \partial_\beta \vec{V} &= [(\partial_\beta V^\alpha)\vec{e}_\alpha + V^\alpha(\partial_\beta \vec{e}_\alpha)] \\ &= \left[(\partial_\beta V^\alpha)\vec{e}_\alpha + V^\alpha \Gamma^{\mu}_{\beta\alpha}\vec{e}_\mu\right] \\ &= \left[(\partial_\beta V^\alpha)\vec{e}_\alpha + V^\mu \Gamma^{\alpha}_{\beta\mu}\vec{e}_\alpha\right] \\ &= \left(\partial_\beta V^\alpha + V^\mu \Gamma^{\alpha}_{\beta\mu}\right)\vec{e}_\alpha \\ &\equiv (\nabla_\beta V^\alpha)\,\vec{e}_\alpha \ . \end{aligned} \tag{5.42}$$

On the last line, we introduced the *covariant derivative* of the vector components V^α:

$$\nabla_\beta V^\alpha = \partial_\beta V^\alpha + V^\mu \Gamma^{\alpha}_{\beta\mu} \ . \tag{5.43}$$

The importance of the covariant derivative is that $\nabla_\beta V^\alpha$ represents the components of a tensor: if we change our coordinate representation, then the components change in the usual way:

$$\nabla_{\bar{\nu}} V^{\bar{\mu}} = \left(\frac{\partial x^{\bar{\mu}}}{\partial x^\alpha}\right)\left(\frac{\partial x^\beta}{\partial x^{\bar{\nu}}}\right)\nabla_\beta V^\alpha \ . \tag{5.44}$$

A corollary of this statement is that in general coordinates, the components $\partial_\beta V^\alpha$ do *not* represent the components of a tensor. If these components did in fact represent tensor components, then we would expect that

$$\partial_{\bar{\nu}} V^{\bar{\mu}} \stackrel{?}{=} \left(\frac{\partial x^{\bar{\mu}}}{\partial x^\alpha}\right)\left(\frac{\partial x^\beta}{\partial x^{\bar{\nu}}}\right)\partial_\beta V^\alpha \ . \tag{5.45}$$

Let's see what happens if we enact this transformation:

$$\begin{aligned} \partial_{\bar{\nu}} V^{\bar{\mu}} &= \left(\frac{\partial x^\beta}{\partial x^{\bar{\nu}}}\right)\partial_\beta\left[\left(\frac{\partial x^{\bar{\mu}}}{\partial x^\alpha}\right)V^\alpha\right] \\ &= \left(\frac{\partial x^\beta}{\partial x^{\bar{\nu}}}\right)\left(\frac{\partial x^{\bar{\mu}}}{\partial x^\alpha}\right)\partial_\beta V^\alpha + \left(\frac{\partial x^\beta}{\partial x^{\bar{\nu}}}\right)\left(\frac{\partial^2 x^{\bar{\mu}}}{\partial x^\alpha \partial x^\beta}\right)V^\alpha \ . \end{aligned} \tag{5.46}$$

The second term in the above equation spoils the tensorial nature of $\partial_\beta V^\alpha$, showing that the $\stackrel{?}{=}$ in (5.45) is not in fact an equality. On an upcoming problem set, you will show that that the Christoffel symbol $\Gamma^{\alpha}_{\beta\mu}$ is also not tensorial; but, that the manner in which it "fails" to transform as a tensor exactly compensates for how the partial derivative fails. As a consequence, the sum of the partial derivative and the Christoffel term transforms exactly as tensor components need to transform.

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Lecture 6
Wrap up of the covariant derivative; introduction to curved spacetimes

6.1 More on the covariant derivative

The covariant derivative is one of the most important tools one uses when studying physics in non-inertial coordinate systems:

$$\nabla_\beta V^\alpha = \partial_\beta V^\alpha + V^\mu \Gamma^\alpha{}_{\beta\mu} \; . \tag{6.1}$$

In (6.1), neither of the terms on the right transform between representations as components of a tensor; but, their sum — and thus the left-hand side — *does* transform in such a way.

Let's take a look at a concrete example, the "covariant divergence" of a 4-vector, and let's do it using the Christoffel symbols we worked out in the previous lecture:

$$\begin{aligned} \nabla_\alpha V^\alpha &= \partial_\alpha V^\alpha + V^\mu \Gamma^\alpha{}_{\alpha\mu} \\ &= \partial_t V^t + \partial_r V^r + \partial_\phi V^\phi + \partial_z V^z + V^r \Gamma^\phi{}_{\phi r} \\ &= \partial_t V^t + \partial_r V^r + \frac{V^r}{r} + \partial_\phi V^\phi + \partial_z V^z \; . \end{aligned} \tag{6.2}$$

From your study of electrodynamics, you may recall that terms like V^r/r enter the divergence in plane polar coordinates, so that term at least should not be a surprise. However, the term $\partial_\phi V^\phi$ should inspire us to pause and check some details: *Does this term have the right dimensions to contribute to the divergence?* Remember that we work in a coordinate basis, meaning that the differential displacement $d\vec{x}$ is related to its components as $d\vec{x} = dx^\alpha \vec{e}_\alpha$. Since $d\vec{x}$ must have the dimensions of length but $dx^\phi = d\phi$ is an angle, we infer that $\vec{e}_\phi$ must have the dimensions of length.

Suppose $\vec{V}$ has dimensions $[V]$. Then, the dimensions of the components V^t, V^r, and V^z will all be $[V]$. However, the component V^ϕ must have dimensions $[V]$/length in order for $V^\phi \vec{e}_\phi$ to have dimension $[V]$. Bearing this in mind, we can see that each term on the right-hand side of Eq. (6.2) in fact has dimension $[V]$/length — exactly as they should in order to represent a divergence.

What about other tensorial quantities? Let's start with the simplest "tensors" of all — scalars. Since they involve no basis objects, their covariant derivatives are the simplest possible:

$$\nabla_\alpha \Phi = \partial_\alpha \Phi \; . \tag{6.3}$$

The components of the covariant derivative of a scalar are just the partial derivative.

How about 1-forms? Let's deduce what the covariant derivative of a 1-form is by considering derivatives of the scalar $p_\alpha A^\alpha$:

$$\nabla_\beta \left(p_\alpha A^\alpha \right) = \partial_\beta \left(p_\alpha A^\alpha \right) = A^\alpha \partial_\beta p_\alpha + p_\alpha \partial_\beta A^\alpha \; . \tag{6.4}$$

Now, let us write $\partial_\beta A^\alpha = \nabla_\beta A^\alpha - \Gamma^\alpha{}_{\beta\mu} A^\mu$:

$$\begin{aligned} \nabla_\beta \left(p_\alpha A^\alpha \right) &= A^\alpha \partial_\beta p_\alpha + p_\alpha \nabla_\beta A^\alpha - p_\alpha \Gamma^\alpha{}_{\beta\mu} A^\mu \\ &= p_\alpha \nabla_\beta A^\alpha + A^\alpha \left(\partial_\beta p_\alpha - p_\mu \Gamma^\mu{}_{\beta\alpha} \right) \; . \end{aligned} \tag{6.5}$$

On the second line, we have used our freedom to relabel dummy indices to exchange α and μ, allowing us to pull out a common A^α factor in two of the terms. We now demand that the covariant derivative obey the Leibniz rule for the derivative of the product of quantities:

$$\nabla_\beta \left(p_\alpha A^\alpha\right) = p_\alpha \nabla_\beta A^\alpha + A^\alpha \nabla_\beta p_\alpha \ . \tag{6.6}$$

Comparing Eqs. (6.5) and (6.6), we read off

$$\boxed{\boxed{\nabla_\beta p_\alpha = \partial_\beta p_\alpha - \Gamma^\mu{}_{\beta\alpha} p_\mu}} \tag{6.7}$$

From this, it is simple to find a rule expressing the derivatives of the basis 1-forms:

$$\partial_\beta \tilde{\omega}^\alpha = -\Gamma^\alpha{}_{\beta\mu} \tilde{\omega}^\mu \ . \tag{6.8}$$

The minus sign in this expression enforces the rule that $\tilde{\omega}^\alpha(\vec{e}_\beta) = \delta^\alpha{}_\beta$. Indeed, we could have started with this requirement and derived 1-form derivatives from this.

The pattern we see for vector components and 1-form components generalizes to any tensor:

$$\nabla_\beta T^{\mu\nu} = \partial_\beta T^{\mu\nu} + \Gamma^\mu{}_{\beta\alpha} T^{\alpha\nu} + \Gamma^\nu{}_{\beta\alpha} T^{\mu\alpha} \ , \tag{6.9}$$

$$\nabla_\beta T_{\mu\nu} = \partial_\beta T_{\mu\nu} - \Gamma^\alpha{}_{\beta\mu} T_{\alpha\nu} - \Gamma^\alpha{}_{\beta\nu} T_{\mu\alpha} \ , \tag{6.10}$$

$$\begin{aligned} \nabla_\beta T^{\mu\nu\cdots}{}_{\rho\sigma\cdots} \ &= \ \partial_\beta T^{\mu\nu\cdots}{}_{\rho\sigma\cdots} \\ &\quad + \Gamma^\mu{}_{\beta\alpha} T^{\alpha\nu\cdots}{}_{\rho\sigma\cdots} + \Gamma^\nu{}_{\beta\alpha} T^{\alpha\nu\cdots}{}_{\rho\sigma\cdots} + \ldots \\ &\quad - \Gamma^\alpha{}_{\beta\rho} T^{\mu\nu\cdots}{}_{\alpha\sigma\cdots} - \Gamma^\alpha{}_{\beta\sigma} T^{\mu\nu\cdots}{}_{\rho\alpha\cdots} + \ldots \end{aligned} \tag{6.11}$$

There's a $\Gamma \times$ tensor term included in the sum for each index of the tensor, entering with $+$ sign for upstairs indices, and with $-$ for downstairs. When "correcting" an upstairs tensor index, the upper index of the Γ takes the value of the index it is correcting; the left-lower index takes the value of the derivative index; and the right-lower index is a dummy index that is contracted with the tensor index being corrected. When "correcting" a downstairs index, the upstairs index is the dummy index that is contracted with the tensor index being corrected; the left-lower index is again the value of the derivative index; and the right-lower index takes the value of the index it is correcting.

6.2 A property of tensor equations; how to compute Christoffel symbols

The definition $\partial_\beta \vec{e}_\alpha = \Gamma^\mu{}_{\beta\alpha} \vec{e}_\mu$ suggests that we are going to generate all of our Christoffel symbols by studying derivatives of basis vectors. This would be as tedious and prone to error as you imagine. Fortunately, there is a better way. Our derivation of this better way allows us to introduce a key property of tensor relationships: A tensorial equation that holds in one representation must hold in **all** represenations. *Changing our coordinates cannot change the equation.*

The application of this we will examine first is the "double gradient" of a scalar function: Let us first do this exercise in inertial (i.e., spatially Cartesian) coordinates:

$$\boldsymbol{\nabla}\boldsymbol{\nabla}\Phi = \partial_\alpha \partial_\beta \Phi \, \tilde{\omega}^\alpha \otimes \tilde{\omega}^\beta \ . \tag{6.12}$$

Written using inertial coordinates, it is clear that the components of the double gradient are symmetric, since $\partial_\alpha \partial_\beta \Phi = \partial_\beta \partial_\alpha \Phi$. Because this is a tensor equation, this symmetry property must hold in all coordinate systems:

$$\boldsymbol{\nabla}\boldsymbol{\nabla}\Phi = \nabla_\alpha \nabla_\beta \Phi \, \tilde{\omega}^\alpha \otimes \tilde{\omega}^\beta \ . \tag{6.13}$$

In other words, we demand that $\nabla_\alpha \nabla_\beta \Phi = \nabla_\beta \nabla_\alpha \Phi$. Let's expand this out:

$$\begin{aligned} \nabla_\alpha \nabla_\beta \Phi &= \nabla_\beta \nabla_\alpha \Phi \\ \partial_\alpha \partial_\beta \Phi - \Gamma^\mu{}_{\alpha\beta} \partial_\mu \Phi &= \partial_\beta \partial_\alpha \Phi - \Gamma^\mu{}_{\beta\alpha} \partial_\mu \Phi \ . \end{aligned} \tag{6.14}$$

The final line simplifies to

$$\left(\Gamma^\mu{}_{\alpha\beta} - \Gamma^\mu{}_{\beta\alpha} \right) \partial_\mu \Phi = 0 \ . \tag{6.15}$$

This is only true if $\Gamma^\mu{}_{\alpha\beta} = \Gamma^\mu{}_{\beta\alpha}$, proving that the Christoffel symbol is in fact symmetric on its lower indices.

A particularly important application of this principle is the gradient of the metric. In an inertial reference frame, we have

$$\boldsymbol{\nabla \eta} = \partial_\gamma \eta_{\alpha\beta} \, \tilde{\omega}^\alpha \otimes \tilde{\omega}^\beta \otimes \tilde{\omega}^\gamma = 0 \ . \tag{6.16}$$

This clearly vanishes since the metric is a constant in inertial reference frames. In general, we have

$$\boldsymbol{\nabla g} = \nabla_\gamma g_{\alpha\beta} \, \tilde{\omega}^\alpha \otimes \tilde{\omega}^\beta \otimes \tilde{\omega}^\gamma \ . \tag{6.17}$$

Although it is only obvious that the gradient of the metric is zero in inertial coordinates, it must in fact be zero in all representations. We thus demand that $\nabla_\gamma g_{\alpha\beta} = 0$.

Let us write this gradient in three different ways, looking at multiple permutations of the indices:

$$\nabla_\gamma g_{\alpha\beta} = \partial_\gamma g_{\alpha\beta} - \Gamma^\mu{}_{\gamma\alpha} g_{\mu\beta} - \Gamma^\mu{}_{\gamma\beta} g_{\alpha\mu} = 0 \ , \tag{6.18}$$

$$\nabla_\alpha g_{\beta\gamma} = \partial_\alpha g_{\beta\gamma} - \Gamma^\mu{}_{\alpha\beta} g_{\mu\gamma} - \Gamma^\mu{}_{\alpha\gamma} g_{\beta\mu} = 0 \ , \tag{6.19}$$

$$\nabla_\beta g_{\gamma\alpha} = \partial_\beta g_{\gamma\alpha} - \Gamma^\mu{}_{\beta\gamma} g_{\mu\alpha} - \Gamma^\mu{}_{\beta\alpha} g_{\gamma\mu} = 0 \ . \tag{6.20}$$

Let us look at Eq. (6.18) minus (6.19) minus (6.20):

$$\partial_\gamma g_{\alpha\beta} - \partial_\alpha g_{\beta\gamma} - \partial_\beta g_{\gamma\alpha} - g_{\mu\beta} \left(\Gamma^\mu{}_{\gamma\alpha} - \Gamma^\mu{}_{\alpha\gamma} \right) + g_{\mu\gamma} \left(\Gamma^\mu{}_{\alpha\beta} + \Gamma^\mu{}_{\beta\alpha} \right) + g_{\mu\alpha} \left(\Gamma^\mu{}_{\beta\gamma} - \Gamma^\mu{}_{\gamma\beta} \right) = 0 \ . \tag{6.21}$$

We've used the fact that the metric is symmetric to organize some of these terms. Using the symmetry of the Christoffels on their lower indices, this further simplifies considerably:

$$g_{\mu\gamma} \Gamma^\mu{}_{\alpha\beta} = \frac{1}{2} \left(\partial_\alpha g_{\beta\gamma} + \partial_\beta g_{\gamma\alpha} - \partial_\gamma g_{\alpha\beta} \right) \equiv \Gamma_{\gamma\alpha\beta} \ . \tag{6.22}$$

Using that final definition, we see that the Christoffel symbols can be written very simply in terms of partial derivatives of the metric:

$$\Gamma^\mu{}_{\alpha\beta} = g^{\mu\gamma} \Gamma_{\gamma\alpha\beta} = \frac{1}{2} g^{\mu\gamma} \left(\partial_\alpha g_{\beta\gamma} + \partial_\beta g_{\gamma\alpha} - \partial_\gamma g_{\alpha\beta} \right) \ . \tag{6.23}$$

6.3 Summary and synopsis of the Christoffel symbols

Here are the major properties of the Christoffel symbol that we will need moving forward:

$$\partial_\beta \vec{e}_\alpha = \Gamma^\mu{}_{\alpha\beta} \vec{e}_\mu \ , \tag{6.24}$$

$$\Gamma^\mu{}_{\alpha\beta} = g^{\mu\gamma} \Gamma_{\gamma\alpha\beta} \ , \tag{6.25}$$

$$\Gamma_{\gamma\alpha\beta} = \frac{1}{2} \left(\partial_\alpha g_{\beta\gamma} + \partial_\beta g_{\gamma\alpha} - \partial_\gamma g_{\alpha\beta} \right) \ . \tag{6.26}$$

The Christoffels are symmetric on the lower indices in a coordinate basis:

$$\Gamma^\mu{}_{\alpha\beta} = \Gamma^\mu{}_{\beta\alpha} = \Gamma^\mu{}_{(\alpha\beta)} \tag{6.27}$$

$$\Gamma^{\mu}{}_{(\alpha\beta)} \equiv \frac{1}{2}\left(\Gamma^{\mu}{}_{\alpha\beta} + \Gamma^{\mu}{}_{\beta\alpha}\right) \ , \tag{6.28}$$

$$\Gamma^{\mu}{}_{[\alpha\beta]} \equiv \frac{1}{2}\left(\Gamma^{\mu}{}_{\alpha\beta} - \Gamma^{\mu}{}_{\beta\alpha}\right) = 0 \ . \tag{6.29}$$

The major and important application of this symbol is to evaluate the covariant derivative:

$$\nabla_{\alpha} T^{\beta}{}_{\gamma} = \partial_{\alpha} T^{\beta}{}_{\gamma} + \Gamma^{\beta}{}_{\alpha\mu} T^{\mu}{}_{\gamma} - \Gamma^{\mu}{}_{\alpha\gamma} T^{\beta}{}_{\mu} \ . \tag{6.30}$$

None of the individual terms on the right-hand side transform between representations as components of a tensor; but, their sum does, and thus the left-hand side transforms as the components of a tensor.

6.4 First considerations on general relativity: Existence of *locally* Lorentz frames

Everything we have discussed so far has pertained to special relativity. In special relativity, the entire manifold of spacetime can be "covered" be a single inertial reference frame. What is "special" about special relativity is the existence of such *global* Lorentz frames.

Gravity breaks this. When we include gravity, global inertial frames cannot exist if gravity acts on light. (Notes from the undergraduate relativity course which describe this are posted with this lecture as an addendum; see those notes for the physical argument which lays this out.) Indeed, it is now a well-established physical fact that gravity acts on light, and so we are forced to conclude that global Lorentz frames cannot exist. We can, however, find *local* Lorentz frames. In other words, given a metric which describes spacetime, we can find a change of representation which covers some finite region of spacetime with inertial coordinates. Let us now go through this analysis.

Let x^{α} describe our starting coordinate system, in which the metric of spacetime is $g_{\alpha\beta}$. Let $x^{\bar{\mu}}$ be coordinates in which spacetime has the metric of special relativity, at least in the vicinity of some event $\mathcal{P} \doteq x^{\bar{\mu}}_{\mathcal{P}}$. We assume that there exists some mapping between these coordinate representations, $x^{\alpha} = x^{\alpha}(x^{\bar{\mu}})$, with a matrix

$$L^{\alpha}{}_{\bar{\mu}} = \frac{\partial x^{\alpha}}{\partial x^{\bar{\mu}}} \tag{6.31}$$

which transforms tensor components between these representations. Our goal is to show that a transformation exists that allows us to put

$$g_{\bar{\mu}\bar{\nu}} = L^{\alpha}{}_{\bar{\mu}} L^{\beta}{}_{\bar{\nu}} g_{\alpha\beta} = \eta_{\bar{\mu}\bar{\nu}} \tag{6.32}$$

over some region of spacetime centered on the event $\mathcal{P}$. We will do this by expanding all the relevant quantities in a Taylor series about this event. At each order in the expansion, we will compare the number of *degrees of freedom* offered by the coordinate transformation to the number of *constraints* imposed by the metric and its derivatives. The degrees of freedom are free for us to adjust in order to meet the constraints; the constraints are imposed by the original metric (and its derivatives).

Let us do the expansions in the barred-frame coordinates:

$$g_{\alpha\beta} = g_{\alpha\beta}\big|_{\mathcal{P}} + \left(x^{\bar{\gamma}} - x^{\bar{\gamma}}_{\mathcal{P}}\right)\left(\partial_{\bar{\gamma}} g_{\alpha\beta}\right)\big|_{\mathcal{P}} + \frac{1}{2}\left(x^{\bar{\gamma}} - x^{\bar{\gamma}}_{\mathcal{P}}\right)\left(x^{\bar{\delta}} - x^{\bar{\delta}}_{\mathcal{P}}\right)\left(\partial_{\bar{\gamma}}\partial_{\bar{\delta}} g_{\alpha\beta}\right)\big|_{\mathcal{P}} + \ldots \tag{6.33}$$

$$L^{\alpha}{}_{\bar{\mu}} = L^{\alpha}{}_{\bar{\mu}}\big|_{\mathcal{P}} + \left(x^{\bar{\gamma}} - x^{\bar{\gamma}}_{\mathcal{P}}\right)\left(\partial_{\bar{\gamma}} L^{\alpha}{}_{\bar{\mu}}\right)\big|_{\mathcal{P}} + \frac{1}{2}\left(x^{\bar{\gamma}} - x^{\bar{\gamma}}_{\mathcal{P}}\right)\left(x^{\bar{\delta}} - x^{\bar{\delta}}_{\mathcal{P}}\right)\left(\partial_{\bar{\gamma}}\partial_{\bar{\delta}} L^{\alpha}{}_{\bar{\mu}}\right)\big|_{\mathcal{P}} + \ldots \tag{6.34}$$

The metric and its two derivatives provide the constraints at the event $\mathcal{P}$; the coordinate transformation matrix $L^{\alpha}{}_{\bar{\mu}}$ and its derivatives are freely specifiable at this point, and provide our degrees of freedom.

The logic of the calculation is to make

$$\begin{aligned} L^{\alpha}{}_{\bar{\mu}} L^{\beta}{}_{\bar{\nu}} g_{\alpha\beta} &= (L^{\alpha}{}_{\bar{\mu}})_{\mathcal{P}} (L^{\beta}{}_{\bar{\nu}})_{\mathcal{P}} (g_{\alpha\beta})_{\mathcal{P}} \\ &\quad + \left(x^{\bar{\gamma}} - x^{\bar{\gamma}}_{\mathcal{P}}\right) (\text{Terms involving } (\partial_{\bar{\gamma}} g_{\alpha\beta})_{\mathcal{P}} \text{ and } (\partial_{\bar{\gamma}} L^{\alpha}{}_{\bar{\mu}})_{\mathcal{P}}) \\ &\quad + \frac{1}{2}\left(x^{\bar{\gamma}} - x^{\bar{\gamma}}_{\mathcal{P}}\right)\left(x^{\bar{\delta}} - x^{\bar{\delta}}_{\mathcal{P}}\right) (\text{Terms involving } (\partial_{\bar{\gamma}}\partial_{\bar{\delta}} g_{\alpha\beta})_{\mathcal{P}} \text{ and } (\partial_{\bar{\gamma}}\partial_{\bar{\delta}} L^{\alpha}{}_{\bar{\mu}})_{\mathcal{P}}) + \dots \end{aligned} \tag{6.35}$$

as close to $\eta_{\bar{\mu}\bar{\nu}}$ as possible. Let's examine Eq. (6.35) order by order in spacetime separation from the event $\mathcal{P}$. At zeroth order (i.e., right at $\mathcal{P}$) we have 10 constraints imposed by the spacetime metric $g_{\alpha\beta}$ at $\mathcal{P}$ (due to the fact that the metric is represented by a symmetric 4×4 matrix). But, we also have 16 degrees of freedom available to effect this transformation, since $L^{\alpha}{}_{\bar{\mu}}$ is a *non*-symmetric 4×4 matrix. We thus have enough functional freedom to set the spacetime metric to that of special relativity in inertial coordinates. We in fact have 6 degrees of freedom left over. These "extra" degrees of freedom correspond to 3 boosts and 3 rotations at the event $\mathcal{P}$.

At first order in separation, our constraints are due to the derivatives $\partial_{\bar{\gamma}} g_{\alpha\beta}$ at $\mathcal{P}$. This is a total of 40 constraints, arising from the 4 derivatives of the 10 metric components. Our degrees of freedom at first order come from the derivative of the coordinate transformation matrix,

$$(\partial_{\bar{\gamma}} L^{\alpha}{}_{\bar{\mu}})_{\mathcal{P}} = \left(\frac{\partial^2 x^{\alpha}}{\partial x^{\bar{\gamma}} \partial x^{\bar{\mu}}}\right)_{\mathcal{P}} . \tag{6.36}$$

We can see that there are in fact exactly 40 degrees of freedom: 4 choices for the index α, and 10 symmetric combinations of derivatives with respect to $x^{\bar{\gamma}}$ and $x^{\bar{\delta}}$. At first order in separation, our degrees of freedom exactly match our constraints: We can set the metric to $\eta_{\bar{\mu}\bar{\nu}}$ exactly, at linear order in separation from $\mathcal{P}$.

Let's look at second order. Our constraints here arise from $\partial_{\bar{\gamma}}\partial_{\bar{\delta}} g_{\alpha\beta}$ at $\mathcal{P}$. This provides 100 constraints since we have 10 (symmetric) second derivatives acting on the 10 (symmetric) unique metric components. How many degrees of freedom do we have? The degrees of freedom come from the second derivative of the coordinate transformation matrix,

$$(\partial_{\bar{\gamma}}\partial_{\bar{\delta}} L^{\alpha}{}_{\bar{\mu}})_{\mathcal{P}} = \left(\frac{\partial^3 x^{\alpha}}{\partial x^{\bar{\gamma}} \partial x^{\bar{\delta}} \partial x^{\bar{\mu}}}\right)_{\mathcal{P}} . \tag{6.37}$$

The number of independent combinations here is 4 times the number of totally symmetric combinations of a 3 index object, with each index allowed to take 4 different values:

$$4 \cdot \left(\frac{(n(n+1)(n+2)}{3!}\right)_{n=4} = 80 . \tag{6.38}$$

We cannot satisfy the 100 constraints on the transformation at second order in separation. Instead, there are 20 components "left over" that cannot be zeroed.

What we find is that our freedom to adjust our coordinate representation lets us put any spacetime metric in the form

$$g_{\bar{\mu}\bar{\nu}} = \eta_{\bar{\mu}\bar{\nu}} + \mathcal{O}\left[(\partial^2 g)(\delta x)^2\right] . \tag{6.39}$$

This means that the metric can be put into the inertial, special relativity form at some event in spacetime, and the form accurately describes spacetime over a region with corrections scaling as the second derivative of the metric, times the spacetime separation squared. We call such a region a "local Lorentz frame," or LLF. As we'll see when we begin to understand how to bring gravity into our framework, observers in a LLF are in fact freely falling due to gravity, and so "freely falling frame" or FFF tends to be used synonymously with LLF.

A few key points to conclude this discussion:

- Notice that the corrections to the "inertial" form scale with the second derivative of the metric. In Newtonian theory, the second derivative of the gravitational potential tells us about tides. Hold that thought!

- The second derivative of any function describes the *curvature* of that function. We expect to develop a more rigorous notion of curvature that "looks like" two derivatives of the metric and with 20 independent components.

- We can see that the corrections to the inertial form of the metric are small until $\delta x \sim 1/\sqrt{\partial^2 g}$. In other words, the size of the region scales is set by the second derivative of the metric. If the derivatives are "small," the metric is accurately described by this inertial form over a large region; and vice versa.

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Lecture 7
Introduction to the principle of equivalence; transport on a curved manifold

7.1 Freely falling frames

A key foundational concept for us in special relativity was the notion of an inertial reference frame, a frame in which objects are unaccelerated if no forces act on them. We appear to lose this idea once we consider gravity: *everything* accelerates due to the gravitational force. However, focusing for the moment on the non-relativistic motion of ordinary matter, the gravitational force $\mathbf{F}_g$ that everything feels is proportional to its mass. As such, the *acceleration* imparted by gravity is universal: all objects in the same "gravitational field" experience the same acceleration. This means that *in the absence of non-gravitational forces, objects maintain their* **relative** *velocities.*

These concepts allow us to define a reference frame that captures the essence of the IRF: the *freely falling frame*, or FFF, is the frame that "falls" under gravity at the same rate as all objects within it. If we define all of our bodies' kinematics relative to the acceleration of the FFF, then the effect of gravity is essentially canceled out. Since all objects maintain their relative velocities, the FFF ends up playing a role when gravity is important much like the role that the IRF played in special relativity.

A corollary of these ideas is a rule known as the *Weak Equivalence Principle*, or WEP: *Over sufficiently small regions, the motion of freely falling bodies due to gravity cannot be distinguished from uniform acceleration.* The restriction to "sufficiently small regions" is because in any real situation, gravity will change over a large region. These changes are known as *tides*, and they set the size of the small region over which the WEP is effective. The WEP tells us that the freely falling frames are in fact exactly the local Lorentz frames that we worked out in the previous lecture.

Is this enough for us? Can we essentially promote the IRFs of special relativity to FFFs and develop a description of gravity that works with relativity? Because of tides, the answer is no. FFFs are not the same at all points, and gravity is non uniform. In special relativity, we could define an IRF and use it to describe all of spacetime; a FFF will only cover some finite portion of the spacetime manifold. The acceleration of a FFF near the surface of the Earth is very different from the acceleration of a FFF in geostationary orbit.

We can cancel out the local acceleration by changing frames, but we cannot transform away tides; that is a lesson of our derivation of local Lorentz frames, which showed that we can only put our spacetime into a special-relativity-like representation over a finite region. Tides cause trajectories which start out parallel to one another in spacetime to become non-parallel. This means that motion in spacetime violates one of Euclid's postulates. It's worth recalling that, in retrospect, we learned that this postulate of Euclid (the "parallelism" axiom) only applies when our manifold is "flat" (in a manner to be made more precise later). Tides mean that the geometry of spacetime must be curved.

Suppose we go into a FFF, within which the spacetime metric has the form $g_{\bar{\mu}\bar{\nu}} = \eta_{\bar{\mu}\bar{\nu}} + \mathcal{O}[(\delta x)^2(\partial^2 g)]$. In that frame, we can define another version of the equivalence principle: *In a sufficiently small region of spacetime, we can find a representation such that the laws of physics reduce to those of special relativity.* This formulation is knows as the *Einstein Equivalence Principle*, or EEP. It serves to guide us to understand how to formulate the laws of physics to situations in which gravity is important[1].

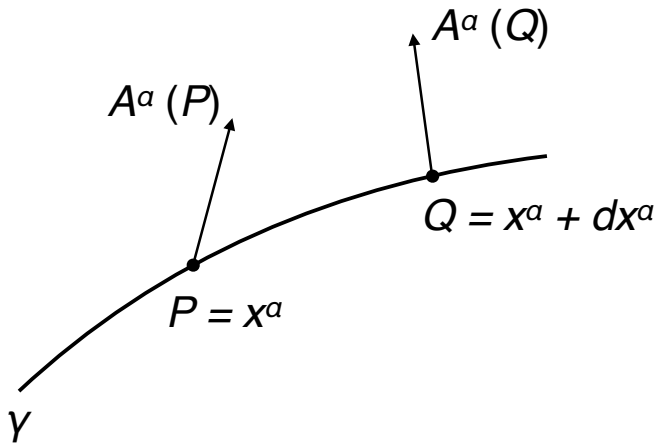
[1]It's worth noting that one other equivalence principle is often discussed: the Strong Equivalence Principle, or

7.2 Physics in a curved manifold: Parallel transport

A manifold with curvature is one on which initially parallel trajectories do not remain parallel. An example is the surface of a sphere: two north-bound trajectories at the equator will cross at the pole — the sphere's curvature caused these initial parallel paths to cross (think of lines of longitude on a globe, which all intersection the equator at 90°, but cross each other at the poles). *Not* an example is the surface of a cylinder: initial parallel paths on a surface remain parallel forever. A cylinder's surface is in fact "intrinsically" flat, though it appears to have *extrinsic* curvature due to the way we have embedded that 2-dimensional surface in 3-dimensional space.

As we think about doing physics in curved manifolds, we need to develop tools for putting things like vectors and tensors into the manifold. A useful concept for us will be the notion of a *tangent space*: the space that, at a particular point in a manifold, contains all the trajectories which pass tangent to the manifold at this point. Useful intuition is to think of the 2-dimensional surface of a sphere; the tangent space at each point on that surface is the plane which just touches the sphere at that point. The basis vectors lie in the tangent space to the manifold, and herein lies a big complication associated with doing physics in a curved manifold: *different points in the manifold have different tangent spaces.* This makes it difficult to compare vectors at different points — because the tangent spaces at event $\mathcal{P}$ and event $\mathcal{Q}$ are not the same, we must think very carefully about to relate these two vectors to each other.

In particular, how do we take derivatives of vector fields? Suppose that some curve γ lies in our manifold; the tangent vector to this curve is $\vec{u}$. We want to differentiate a vector field $\vec{A}$ that is defined in this manifold:



Our first guess for the derivative of $\vec{A}$ might be to just do the usual calculation thing:

$$\frac{\partial A^\alpha}{\partial x^\beta} = \frac{A^\alpha(\mathcal{Q}) - A^\alpha(\mathcal{P})}{dx^\beta} \ . \tag{7.1}$$

The problem with this is that the events $\mathcal{P}$ and $\mathcal{Q}$ don't have the same tangent space, so this derivative misses how the basis vectors vary from $\mathcal{P}$ to $\mathcal{Q}$. The resulting derivative is non-tensorial: though we hope to find

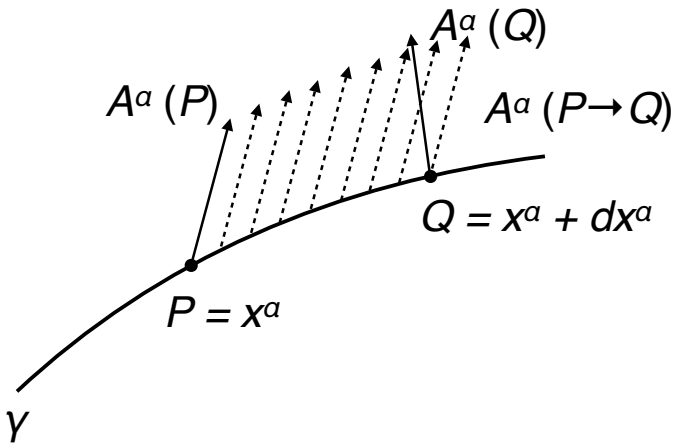
$$\partial_{\bar{\nu}} A^{\bar{\mu}} \stackrel{?}{=} \left(\frac{\partial x^\alpha}{\partial x^{\bar{\nu}}}\right)\left(\frac{\partial x^{\bar{\mu}}}{\partial x^\beta}\right)\partial_\alpha A^\beta \ , \tag{7.2}$$

we instead find

$$\partial_{\bar{\nu}} A^{\bar{\mu}} = \left(\frac{\partial x^\alpha}{\partial x^{\bar{\nu}}}\right)\left(\frac{\partial x^{\bar{\mu}}}{\partial x^\beta}\partial_\alpha A^\beta + \frac{\partial^2 x^{\bar{\mu}}}{\partial x^\beta \partial x^{\bar{\nu}}} A^\beta\right) \ . \tag{7.3}$$

SEP. We need to develop a lot more before we can use this one, which relates to the manner in which gravity itself gravitates.

To address this, we need to introduce a notion of *transporting* the field $\vec{A}$ from $\mathcal{P}$ to $\mathcal{Q}$. We thus compare the vector components at the same point, so that the same tangent space applies to them. The first notion of transport we will use is called "parallel transport." It amounts to just sliding the vector along the curve γ in sufficiently small steps that the change in the basis objects is negligible:



(Note that we have implicitly used this notion of transport throughout our physics careers to date. In a non-curved manifold, all points share the same tangent space, and parallel transport is trivial.) The transport will be "parallel" when the change associated with this notion of transport is zero.

To make this precise and quantitative, we need to define a mechanism to transport the vector between the two points. Let us assume a mathematical structure exists which transports as follows:

$$A^{\alpha}_{\rm PT}(\mathcal{P}\to\mathcal{Q}) = A^{\alpha}(\mathcal{P}) - \Pi^{\alpha}{}_{\beta\mu}dx^{\beta}A^{\mu}\ . \tag{7.4}$$

We then define our derivative operator D_β by comparing the vector field at $\mathcal{Q}$ to the vector field transported from $\mathcal{P}$ to $\mathcal{Q}$:

$$\begin{aligned} D_\beta A^\alpha &= \frac{A^\alpha(\mathcal{Q}) - A^\alpha_{\rm PT}(\mathcal{P}\to\mathcal{Q})}{dx^\beta} \\ &= \partial_\beta A^\alpha + \Pi^\alpha{}_{\beta\mu}A^\mu\ . \end{aligned} \tag{7.5}$$

The object we have called $\Pi^\alpha{}_{\beta\mu}$ is known as the "connection" — it connects separated points in the manifold, and allows us to compare them. We can further specify the properties of the connection by demanding it satisfy some additional properties. We will require that

$$\Pi^{\bar\alpha}{}_{\bar\beta\bar\mu}A^{\bar\mu} = \left(\frac{\partial x^{\bar\alpha}}{\partial x^\alpha}\right)\left(\frac{\partial x^\beta}{\partial x^{\bar\beta}}\right)\Pi^\alpha{}_{\beta\mu}A^\mu - \left(\frac{\partial^2 x^{\bar\alpha}}{\partial x^\beta \partial x^\mu}\right)\left(\frac{\partial x^\beta}{\partial x^{\bar\beta}}\right)A^\mu\ ; \tag{7.6}$$

this guarantees that $D_\beta A^\alpha$ transforms between representations as tensor components should, with the term involving the second derivative of the coordinate transformation matrix cancelling the analogous term that prevents the partial derivative from being tensorial.

We further require that $D_\gamma g_{\alpha\beta} = 0$ due to the equivalence principle: the metric has no derivative in a local Lorentz frame, so it cannot have a derivative in any frame. These properties tell us that the connection coefficient is in fact just the Christoffel symbol, and the derivative which follows from this transport operation is the covariant derivative:

$$\Pi^\alpha{}_{\beta\mu} = \Gamma^\alpha{}_{\beta\mu}\ , \qquad D_\beta = \nabla_\beta\ . \tag{7.7}$$

Suppose we transport a vector field along a curve parameterized by a quantity λ which increases as one moves along it[2]. The tangent vector along this curve has components $u^\alpha = dx^\alpha/d\lambda$. The vector field is parallel transported if

$$u^\beta \nabla_\beta A^\alpha \equiv \frac{DA^\alpha}{d\lambda} = 0 \; . \tag{7.8}$$

7.3 Parallel transport in the local Lorentz frame

The notion derivative operation $DA^\alpha/d\lambda$ describes how the vector field components A^α change as one moves along the curve γ. Even in the case of parallel transport,

$$\left.\frac{DA^\alpha}{d\lambda}\right|_{\rm PT} = 0 \; , \tag{7.9}$$

it should be emphasized that the vector components *do* change, since

$$u^\beta \nabla_\beta A^\alpha = u^\beta \partial_\beta A^\alpha + \Gamma^\alpha{}_{\beta\mu} u^\beta A^\mu \; . \tag{7.10}$$

Parallel transport can thus be regarded as defining an evolution for the components A^α that obeys the differential equation

$$\left.\frac{dA^\alpha}{d\lambda}\right|_{\rm PT} = -\Gamma^\alpha{}_{\beta\mu} u^\beta A^\mu \; . \tag{7.11}$$

Because within a local Lorentz frame first derivatives of the metric are zero, the Christoffel symbols are zero in that frame. This means that

$$\left.\frac{dA^\alpha}{d\lambda}\right|_{\rm PT} = 0 \qquad \text{within the LLF} \; . \tag{7.12}$$

Parallel transport thus corresponds to the components of a vector not changing within the LLF as we slide along the curve γ — hopefully, a very intuitively sensible correspondence.

7.4 Lie transport

Parallel transport is just one way to define transport of a vector or tensor along the curve γ. A second notion of transport comes by defining the event $\mathcal{P}$ as at coordinate x^α, the event $\mathcal{Q}$ as at coordinate $x^\alpha + dx^\alpha \equiv (x')^\alpha$. Since the curve γ has tangent vector $u^\alpha = d^\alpha/d\lambda$, we can further write $(x')^\alpha = x^\alpha + u^\alpha d\lambda$.

We now regard the shift from $\mathcal{P}$ to $\mathcal{Q}$ as a kind of coordinate transformation; let's apply this to a vector field $\vec{A}$:

$$\begin{aligned} A^\alpha_{\rm LT}(\mathcal{P}\to\mathcal{Q}) &= \text{components } A^\alpha \text{ ``transformed'' from coordinate } x^\alpha \text{ to } (x')^\alpha \\ &= \frac{\partial (x')^\alpha}{\partial x^\beta} A^\beta(\mathcal{P}) \\ &= (\delta^\alpha{}_\beta + (\partial_\beta u^\alpha) d\lambda)\, A^\beta(\mathcal{P}) \; , \end{aligned} \tag{7.13}$$

yielding

$$A^\alpha_{\rm LT}(\mathcal{P}\to\mathcal{Q}) = A^\alpha(\mathcal{P}) + (\partial_\beta u^\alpha) A^\beta(\mathcal{P}) d\lambda \; . \tag{7.14}$$

If the displacement $d\lambda$ along γ is small, we can also expand the vector field at event $\mathcal{Q}$ in a Taylor series:

$$\begin{aligned} A^\alpha(\mathcal{Q}) &= A^\alpha(x^\beta + dx^\beta) \\ &= A^\alpha(x^\beta) + dx^\beta (\partial_\beta A^\alpha) \\ &= A^\alpha(\mathcal{P}) + (u^\beta\, d\lambda)(\partial_\beta A^\alpha) \; . \end{aligned} \tag{7.15}$$

[2]We will discuss and more precisely define a particular λ parameter for certain very important curves in spacetime in an upcoming lecture.

We now define a derivative along the curve γ by comparing $A^\alpha(\mathcal{Q})$ with $A^\alpha_{\rm LT}(\mathcal{P}\to\mathcal{Q})$:

$$\begin{aligned}\mathcal{L}_{\vec{u}}A^\alpha &\equiv \frac{A^\alpha(\mathcal{Q})-A^\alpha_{\rm LT}(\mathcal{P}\to\mathcal{Q})}{d\lambda}\\ &= u^\beta(\partial_\beta A^\alpha)-A^\beta(\partial_\beta u^\alpha)\ .\end{aligned}\tag{7.16}$$

This construction is known as the *Lie derivative*, and tells us how the field $\vec{A}$ varies as one "flows" along the vector field $\vec{u}$. Lie derivatives are often encountered in fluid dynamics, since they express a very natural notion of how a quantity evolves as one moves along the flow lines associated with the motion of a fluid.

It's not hard to show that

$$\mathcal{L}_{\vec{u}}A^\alpha = u^\beta\nabla_\beta A^\alpha - A^\beta\nabla_\beta u^\alpha\ .\tag{7.17}$$

The connection coefficients which enter into the covariant derivative cancel out, reducing the expression in terms of covariant derivatives to one that only involves partial derivatives. However, the fact that it can be written using covariant derivatives tells us that the Lie derivative is itself tensorial: the components of the Lie derivative of some tensorial object transform between frames as a tensor.

If we repeat this exercise but apply the Lie derivative to a scalar, we find

$$\mathcal{L}_{\vec{u}}\Phi = u^\alpha\partial_\alpha\Phi = u^\alpha\nabla_\alpha\Phi\ .\tag{7.18}$$

For a 1-form, we find

$$\begin{aligned}\mathcal{L}_{\vec{u}}p_\alpha &= u^\beta\partial_\beta p_\alpha + p_\beta\partial_\alpha u^\beta\\ &= u^\beta\nabla_\beta p_\alpha + p_\beta\nabla_\alpha u^\beta\ .\end{aligned}\tag{7.19}$$

The generalization to general rank tensors is probably not surprising at this point:

$$\begin{aligned}\mathcal{L}_{\vec{u}}T^\alpha{}_\beta &= u^\mu\partial_\mu T^\alpha{}_\beta - T^\mu{}_\beta\partial_\mu u^\alpha + T^\alpha{}_\mu\partial_\beta u^\mu\\ &= u^\mu\nabla_\mu T^\alpha{}_\beta - T^\mu{}_\beta\nabla_\mu u^\alpha + T^\alpha{}_\mu\nabla_\beta u^\mu\ .\end{aligned}\tag{7.20}$$

In 8.962, we use the Lie derivative for one particularly important task: characterizing important symmetry and conservation principles associated with motion along certain trajectories in spacetime.

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Lecture 8
Application of the Lie derivative; tensor densities;
party tricks with the determinant of the metric

This lecture is the last one in which the focus is on developing mathematical tools. We will continue to develop new tensors that tell us about important quantities in spacetime, but after this lecture we will have all the calculational machinery we need to do this.

8.1 Lie derivatives and symmetries

The Lie derivative $\mathcal{L}_{\vec{u}}$ tells us about how a tensorial quantity changes as one moves along the curve whose tangent is $\vec{u}$. In particular, we describe a tensor as being *Lie transported* if $\mathcal{L}_{\vec{u}}(\text{tensor}) = 0$.

Imagine that such a curve exists in our manifold, and it is parameterized by a quantity λ that increases as we move along the curve; the tangent is defined as $u^\alpha = dx^\alpha/d\lambda$. We can change our coordinates such that $x^0 = \lambda$ on the curve, and such that $x^1 = x^2 = x^3 =$ constant on the curve. With such coordinates,

$$u^\alpha = \frac{dx^\alpha}{d\lambda} \doteq \delta^\alpha{}_0 \; , \tag{8.1}$$

and so in these coordinates $\partial_\mu u^\alpha = 0$ everywhere along this curve.

Now imagine that some tensor is Lie transported. The statement $\mathcal{L}_{\vec{u}}(\text{tensor}) = 0$ which defines Lie transport turns into

$$u^\alpha \partial_\alpha(\text{tensor}) + (\text{combinations of tensor indices contracted on derivatives of } u^\alpha) = 0 \; . \tag{8.2}$$

Since, in these coordinates, u^α has no non-zero derivatives, the equation of Lie transport further simplifies to

$$u^\alpha \partial_\alpha(\text{tensor}) = \frac{\partial(\text{tensor})}{\partial x^0} = 0 \; . \tag{8.3}$$

This means that the tensor *does not change* as we slide along the curve: it is independent of the coordinate x^0 that we introduced.

Recall a foundational principle of tensor analysis: if a tensorial equation holds in one representation, then it holds in *all* representations. If we find that our tensor is independent of some coordinate, then we have shown it is Lie transported in a representation akin to what we have sketched above. This means there must exist some vector $\vec{u}$ along which the tensor is Lie transported in *all* representations. Lie transport gives us a covariant, frame-independent way of characterizing a symmetry of a tensor field.

8.2 Killing vectors and Killing's equation

Suppose the tensor in question is the metric of spacetime. Let us imagine that there exists some vector field $\vec{\xi}$ along which the metric is Lie transported: $\mathcal{L}_{\vec{\xi}} g_{\alpha\beta} = 0$. From the discussion above, this tells us that there exists a coordinate x^0 such that $\partial g_{\alpha\beta}/\partial x^0 = 0$: the metric is constant with respect to that coordinate. The converse is also true: if the metric is constant with respect to some coordinate, then a vector $\vec{\xi}$ exists such that the metric is Lie transported along $\vec{\xi}$. (Proof: if $\partial(\text{tensor})/\partial y = 0$ for some coordinate y, then we can define the vector field $\vec{\xi}$ as the one which flows in the direction in which y increases, holding all other coordinates field. By the logic of the

above construction, this is nothing more than the Lie derivative along $\vec{\xi}$ in coordinates adapted to this flow, and as such holds in all representations.)

Let us expand the Lie derivative for this situation:

$$\mathcal{L}_{\vec{\xi}} g_{\alpha\beta} = 0 \tag{8.4}$$

expands to

$$\xi^\gamma \nabla_\gamma g_{\alpha\beta} + g_{\alpha\gamma}\nabla_\beta \xi^\gamma + g_{\gamma\beta}\nabla_\alpha \xi^\gamma = 0 \ . \tag{8.5}$$

But remember that the covariant derivative of the metric is zero. The first term on the left-hand side of (8.5) is zero; on the other two terms, we can move the metric inside the derivatives and lower the indices on the components of $\vec{\xi}$:

$$\boxed{\boxed{\nabla_\alpha \xi_\beta + \nabla_\beta \xi_\alpha = 0}} \tag{8.6}$$

This result is called *Killing's equation*, and $\vec{\xi}$ is known as a Killing vector[1]. We will use Killing vectors as tools for characterizing symmetries of spacetime; from your experience with Lagrangian mechanics, it should not surprise you that they also will play a role in developing a notion of quantities that are conserved as things move about in spacetime.

8.3 Tensor densities

Tensor "densities" are a further category of mathematical objects which will prove useful in our future work. These objects are best regarded as things that transform between different representations almost, but not quite, in the way that tensors transform. When we change representation, we find that they "miss" the correct transformation law by a factor that looks like the determinant of the matrix which effects the transformation.

One of the most important tensor densities is the Levi-Civita symbol. We treated this as a tensor when our focus was special relativity in inertial coordinates, but (as we'll see in a moment), this is rare example of a case where we can do this. Recall how this symbol was defined:

$$\begin{aligned} \tilde{\epsilon}_{\alpha\beta\gamma\delta} &= +1 \quad \text{for indices 0123 and even permutations} \\ &= -1 \quad \text{for odd permutations of 0123} \\ &= 0 \quad \text{for any index repeated.} \end{aligned} \tag{8.7}$$

(We write this with a tilde to indicate that the Levi-Civita symbol is *not* a tensor.)

To show that this is not a tensor, we quote a theorem: Given any 4×4 matrix M whose components are $M^\alpha{}_\mu$,

$$\tilde{\epsilon}_{\alpha\beta\gamma\delta} M^\alpha{}_\mu M^\beta{}_\nu M^\gamma{}_\rho M^\delta{}_\sigma = \tilde{\epsilon}_{\mu\nu\rho\sigma}|M| \ , \tag{8.8}$$

where $|M|$ is the determinant of the matrix M. Let's take in particular $M^\alpha{}_{\bar{\alpha}} = \partial x^\alpha / \partial x^{\bar{\alpha}}$. Doing so tells us what the Levi-Civita symbol is in the new representation:

$$\tilde{\epsilon}_{\bar{\alpha}\bar{\beta}\bar{\gamma}\bar{\delta}} = \left|\frac{\partial x^{\bar{\mu}}}{\partial x^\mu}\right| \tilde{\epsilon}_{\alpha\beta\gamma\delta} \left(\frac{\partial x^\alpha}{\partial x^{\bar{\alpha}}}\right)\left(\frac{\partial x^\beta}{\partial x^{\bar{\beta}}}\right)\left(\frac{\partial x^\gamma}{\partial x^{\bar{\gamma}}}\right)\left(\frac{\partial x^\delta}{\partial x^{\bar{\delta}}}\right) \ . \tag{8.9}$$

If the determinant of the transformation matrix were not present, then this would be tensorial[2]. Instead, we call this a "tensor density of weight 1."

[1] Named for a 19th century German mathematician, Wilhelm Killing, who did a lot of foundational work on Lie algebras independent of Lie, as well as work on non-Euclidean geometries. He was apparently a lovely person, despite his (in English) homicidal surname.

[2] It *is* tensorial if the determinant of the transformation matrix happens to be 1, which in fact is the case when the spacetime is the metric of special relativity in inertial coordinates. This is why it was acceptable for us to treat this symbol as a tensor when we first encountered it.

Another important tensor density is related to the metric. The metric is of course a tensor:

$$g_{\bar{\alpha}\bar{\beta}} = g_{\alpha\beta}\left(\frac{\partial x^\alpha}{\partial x^{\bar{\alpha}}}\right)\left(\frac{\partial x^\beta}{\partial x^{\bar{\beta}}}\right) . \tag{8.10}$$

Take the determinant of both sides of this:

$$\bar{g} = \left|\frac{\partial x^{\bar{\alpha}}}{\partial x^\alpha}\right|^{-2} g . \tag{8.11}$$

We use g, with no indices, for the determinant of the metric; $\bar{g}$ is the determinant of the metric after changing representation. The determinant of the metric is a tensor density of weight -2.

8.4 Converting tensor densities to tensors

The fact that tensor densities do not transform between representations like tensors makes them difficult to work with. Fortunately, if we have a tensor density of weight w, it is easy to convert it to a true tensor: we just multiply by certain powers of the determinant of the metric:

$$(\text{Tensor density of weight } w)|g|^{w/2} = \text{Tensor} . \tag{8.12}$$

Note the absolute value: g might be negative, and $w/2$ might be half integer. There's no harm in introducing an absolute value here; everything derived above remains true if we replace g by $|g|$.

Perhaps the most important example of a tensor density converted to a tensor is the Levi-Civita symbol. We make a tensor which forms covariant volume operators in this way:

$$\epsilon_{\alpha\beta\gamma\delta} \equiv \sqrt{|g|}\,\tilde{\epsilon}_{\alpha\beta\gamma\delta} . \tag{8.13}$$

You are in fact already familiar with the effect of this, though you likely came at the result by a different route. Consider the 3-volume element in spherical polar coordinates. The line element is given by $ds^2 = g_{ij}dx^i dx^j = dr^2 + r^2 d\theta^2 + r^2\sin^2\theta d\phi^2$. From this, we deduce that $g = g_{rr}g_{\theta\theta}g_{\phi\phi} = r^4\sin^2\theta$. Defining three principle basis vectors $d\mathbf{x}_1 = dr\mathbf{e}_r$, $d\mathbf{x}_2 = d\theta\mathbf{e}_\theta$, $d\mathbf{x}_3 = d\phi\mathbf{e}_\phi$, we find

$$dV^3 = \sqrt{g}\,\tilde{\epsilon}_{ijk}\,dx^i_1\,dx^j_2\,dx^k_3 = r^2\sin^2\theta\,dr\,d\theta\,d\phi . \tag{8.14}$$

Similar exercises allow us to work out 4-volume elements in given spacetimes, or the 3-volume associated with a particular constant value of some time coordinate in such a spacetime.

8.5 Party trick

The last section presents a fairly miscellaneous, but often quite useful, trick that uses the determinant of the metric. In particular, it turns out that a certain combination of Christoffel symbols can be computed very simply and effectively using g.

The Christoffel combination that is simplified is given by

$$\begin{aligned}
\Gamma^\mu{}_{\mu\alpha} &= g^{\mu\beta}\Gamma_{\beta\mu\alpha} \\
&= \frac{1}{2}g^{\mu\beta}\left(\partial_\mu g_{\alpha\beta} + \partial_\alpha g_{\beta\mu} - \partial_\beta g_{\mu\alpha}\right) \\
&= \frac{1}{2}g^{\mu\beta}\partial_\alpha g_{\mu\beta} .
\end{aligned} \tag{8.15}$$

Note that we *are* summing over μ here; this is not like the homework exercise you did on problem set 3 in which repeated indices were not summed. In going from line 1 to line 2, we are using the form of the Christoffel we worked out a few lectures ago; in going from line 2 to line 3 we use

the fact that the combination $(\partial_\mu g_{\alpha\beta} - \partial_\beta g_{\alpha\mu})$ is antisymmetric on exchange of μ and β, but the combination is contracted with the symmetric $g^{\mu\beta}$.

What we will now show is that

$$\frac{1}{2} g^{\mu\beta} \partial_\alpha g_{\mu\beta} = \frac{1}{\sqrt{|g|}} \partial_\alpha \left(\sqrt{|g|} \right) = \partial_\alpha \left(\ln \sqrt{|g|} \right) \ , \tag{8.16}$$

and thus

$$\Gamma^\mu{}_{\mu\alpha} = \frac{1}{\sqrt{|g|}} \partial_\alpha \left(\sqrt{|g|} \right) = \partial_\alpha \left(\ln \sqrt{|g|} \right) \ . \tag{8.17}$$

To show this, let's examine the following matrix algebra analysis. Consider some matrix M, and imagine the following variation:

$$\begin{aligned} \delta \ln(\det \mathsf{M}) &= \ln \left[\det \left(\mathsf{M} + \delta \mathsf{M} \right) \right] - \ln \left(\det \mathsf{M} \right) \\ &= \ln \left[\frac{\det \left(\mathsf{M} + \delta \mathsf{M} \right)}{\det \mathsf{M}} \right] \\ &= \ln \left[\det \left(\mathsf{I} + \mathsf{M}^{-1} \cdot \delta \mathsf{M} \right) \right] \ . \end{aligned} \tag{8.18}$$

On the last line, we used the fact that $(\det \mathsf{M})^{-1} = \det \mathsf{M}^{-1}$, as well as the fact that $(\det \mathsf{M})(\det \mathsf{N}) = \det (\mathsf{M} \cdot \mathsf{N})$. The I appearing on that line is the identity matrix.

We now introduce a useful identify: if $\boldsymbol{\epsilon}$ is a "small" matrix, then

$$\det \left(\mathsf{I} + \boldsymbol{\epsilon} \right) \simeq 1 + \mathrm{Tr}(\boldsymbol{\epsilon}) \ , \tag{8.19}$$

where $\mathrm{Tr}(\boldsymbol{\epsilon}) = g^{\alpha\beta} \epsilon_{\alpha\beta}$ is the trace of $\boldsymbol{\epsilon}$. Note that this quantity is a scalar.

So, under the assumption that $\delta \mathsf{M}$ is a very small variation,

$$\delta \ln(\det \mathsf{M}) = \ln \left[\det \left(\mathsf{I} + \mathsf{M}^{-1} \cdot \delta \mathsf{M} \right) \right] = \ln \left[1 + \mathrm{Tr} \left(\mathsf{M}^{-1} \cdot \delta \mathsf{M} \right) \right] = \mathrm{Tr} \left(\mathsf{M}^{-1} \cdot \delta \mathsf{M} \right) \ . \tag{8.20}$$

Now, set the matrix M to the metric components $g_{\alpha\beta}$: $\det \mathsf{M}$ becomes $|g|$, M^{-1} becomes $g^{\alpha\beta}$, and Eq. (8.20) becomes

$$\delta \ln |g| = \mathrm{Tr} \left(g^{\mu\beta} \delta g_{\beta\gamma} \right) = g^{\mu\beta} \delta g_{\mu\beta} \ . \tag{8.21}$$

Imagine that this variation arises from a small variation in the coordinates, δx^α; divide by this variation, take the limit, and find

$$\partial_\alpha \ln |g| = g^{\mu\beta} \partial_\alpha g_{\beta\mu} \ . \tag{8.22}$$

Combining this with our earlier result, we have

$$\Gamma^\mu{}_{\mu\alpha} = \frac{1}{2} g^{\mu\beta} \partial_\alpha g_{\mu\beta} = \partial_\alpha \ln \sqrt{|g|} = \frac{1}{\sqrt{|g|}} \partial_\alpha \sqrt{|g|} \ . \tag{8.23}$$

A particularly useful application of this party trick is to the divergence of a vector:

$$\begin{aligned} \nabla_\alpha A^\alpha &= \partial_\alpha A^\alpha + \Gamma^\alpha{}_{\alpha\beta} A^\beta \\ &= \partial_\alpha A^\alpha + \Gamma^\beta{}_{\beta\alpha} A^\alpha \\ &= \partial_\alpha A^\alpha + \frac{A^\alpha}{\sqrt{|g|}} \partial_\alpha \left(\sqrt{|g|} \right) \\ &= \frac{1}{\sqrt{|g|}} \partial_\alpha \left(\sqrt{|g|} A^\alpha \right) \ . \end{aligned} \tag{8.24}$$

What's really nice about this is that it only involves partial derivatives. This form enables a really lovely way to write Gauss's theorem that works very nicely even in arbitrary curved geometries:

$$\int_{V^4} \left(\nabla_\alpha A^\alpha \right) \sqrt{|g|} d^4x = \int_{V^4} \partial_\alpha \left(\sqrt{|g|} A^\alpha \right) d^4x = \oint_{\partial V^4} A^\alpha \sqrt{|g|} d\Sigma_\alpha \ . \tag{8.25}$$

(Here $d^4x = dx^0\,dx^1\,dx^2\,dx^3$ is the product of the differentials of our 4 spacetime coordinates, in whatever coordinate system is being used.)

Equation (8.25) can be quite valuable when dealing with vector-valued quantities that may be subject to some kind of conservation principle (e.g., number flux or charge/current density). Given that energy and momentum are expressed using the stress-energy energy tensor, and are subject to a nice integral conservation law in flat spacetime, one might wonder — can we generalize this to tensors? Unfortunately, there is no analogous result for the divergence of tensors. An attempt to construct such a generalization fails in two ways:

- First, when we expand the divergence of a tensor, the result is just not as amenable to a nice simplification except under very special circumstances:

$$\nabla_\alpha A^{\alpha\beta} = \partial_\alpha A^{\alpha\beta} + \Gamma^\alpha{}_{\alpha\gamma} A^{\gamma\beta} + \Gamma^\beta{}_{\alpha\gamma} A^{\alpha\gamma} \,. \tag{8.26}$$

  The second term on the right-hand side could be simplified using our result involving the determinant of the metric, but the third term cannot be simplified. The only circumstance when this simplifies is if the tensor $A^{\alpha\beta}$ is antisymmetric, in which case that final term is zero by symmetry-antisymmetry. This unfortunately is rarely useful.

- Even if the divergence can be simplified, *we cannot integrate it up* to get something useful. Consider the stress-energy tensor, $T^{\alpha\beta}$. The divergence of this tells us about conservation of energy and momentum. However, the resulting mathematical object, $\nabla_\alpha T^{\alpha\beta}$ is a 4-vector. In a curved geometry, we cannot "add up" 4-vectors from different regions of an integrand: each point in the integrand has its own tangent space, and so vector-valued contributions at one point cannot be simply combined with contributions at another point.

A corollary of this final point is that many global conservation laws simply do not work in curved spacetime. They may work *under certain circumstances*, but one has to be careful to understand what those circumstances are to make sure that the conservation law has been properly translated to the curved spacetime context.

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Lecture 9
Motion of an unforced body in curved spacetime

9.1 Motion in curved spacetime

How does a body, subject to no external *non-gravitational* forces, move in spacetime? For intuition, go back to special relativity in inertial coordinates. In this situation, a body is at rest in some set of frames, and moves with constant velocity in all other inertial frames. Its motion in these frames is simply a straight line:

$$x^\alpha = x_0^\alpha + u^\alpha \tau \ , \tag{9.1}$$

where τ is proper time along this trajectory. The Einstein equivalence principle tells us that (9.1) holds in a local Lorentz frame. Our challenge is to take this idea and translate it into a form that holds more generally.

What does moving in "simply a straight line" really mean? A more complete expression of this idea is that the trajectory at moment $\tau + d\tau$ moves in the same direction in which it was going at moment τ. The *tangent* to the trajectory remains constant over this interval, which means that the trajectory *parallel transports its tangent vector.*

As usual, imagine trajectories that are parameterized by some λ which grows along their worldlines. We define $u^\alpha = dx^\alpha/d\lambda$ as the tangent. If the tangent vector is itself parallel transported along the worldline, then it satisfies

$$u^\alpha \nabla_\alpha u^\beta = 0 \quad \longrightarrow \quad \frac{Du^\alpha}{d\lambda} = 0 \ . \tag{9.2}$$

(The second form of this equation introduces the notation $u^\alpha \nabla_\alpha \equiv D/d\lambda$. We will use this operator occasionally to mean "the covariant derivative along the curve whose tangent is $\vec{u}$.") Expanding the covariant derivative yields

$$u^\alpha \partial_\alpha u^\beta + \Gamma^\beta{}_{\alpha\gamma} u^\alpha u^\gamma = 0 \ , \tag{9.3}$$

which is typically written in the form

$$\frac{du^\beta}{d\lambda} + \Gamma^\beta{}_{\alpha\gamma} u^\alpha u^\gamma = 0 \qquad \text{or} \qquad \frac{d^2 x^\beta}{d\lambda} + \Gamma^\beta{}_{\alpha\gamma} \frac{dx^\alpha}{d\lambda}\frac{dx^\gamma}{d\lambda} = 0 \ . \tag{9.4}$$

Equation (9.4) is known as the *geodesic equation*, and the curves $x^\alpha(\lambda)$ which solve it are *geodesics.*

As a brief but occasionally useful aside, a more general form of the geodesic equation is to allow its normalization to change as it is transported:

$$\frac{Du^\alpha}{d\lambda^*} = \kappa(\lambda^*) u^\alpha \ . \tag{9.5}$$

As you will show on a homework exercise, we can always change our parameterization from λ^* to λ in such a way that the "normal" geodesic equation results. Define $v^\alpha = dx^\alpha/d\lambda$. As you will show on an upcoming problem set, if

$$\frac{d\lambda}{d\lambda^*} = \exp\left[\int_0^\lambda \kappa(\lambda^*) d\lambda^* \right] \ , \tag{9.6}$$

then $v^\alpha \nabla_\alpha v^\beta = 0$ The parameter conversion (9.6) puts the equation in the "standard" geodesic form, in which the tangent vector maintains its normalization as it is transported.

A choice of λ which maintains normalization (i.e., for which $Du^\alpha/d\lambda = 0$) is known as an *affine parameterization*. An affine parameterization corresponds to one which has uniformly spaced "tickmarks" in local Lorentz frames along the worldline. At least for timelike trajectories, a very natural choice for this is the proper time τ. As such, it is very common to see Eq. (9.4) written using τ rather than λ. Note that linear transformations leave affine parameterizations affine: if λ is an affine parameterization, then so is

$$\lambda' = a\lambda + b \, , \tag{9.7}$$

with a and b both constants. Such a reparameterization simply shifts the origin (choice of b), and changes the spacing of tickmarks in each local Lorentz frame (choice of a). The change of spacing is like changing units (e.g., from seconds to minutes).

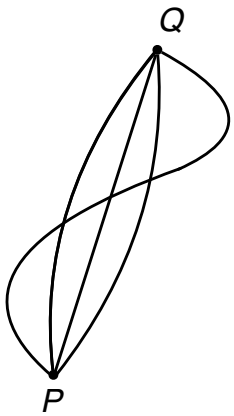
What about motion that *is* subject to an external force? In this case, we simply replace the 0 on the right-hand side of the geodesic equation with an acceleration. Consider a timelike trajectory for which proper time is an appropriate parameter. Our equation of motion in this case becomes

$$\frac{Du^\alpha}{d\tau} = \frac{du^\alpha}{d\tau} + \Gamma^\alpha{}_{\mu\nu} u^\mu u^\nu = a^\alpha \, . \tag{9.8}$$

The acceleration that you put in depends upon the physics of whatever is forcing your body's motion. For instance, if the body has a charge q, mass m, and is in an electromagnetic field with components $F^{\alpha\beta}$, the right-hand side is $(q/m)F^{\alpha\beta}u_\beta$.

9.2 Geodesics via extremization

There is a second route to deriving the geodesic equation, which can be regarded as a rigorous derivation of the idea that "the shortest path between two points is a straight line." Our variant of this to consider the proper time which elapses going from event $\mathcal{P}$ to event $\mathcal{Q}$. Consider every possible path that connects $\mathcal{P}$ to $\mathcal{Q}$, including paths which are physically implausible; a sample of some of these paths might look like



Our only major constraint will be that the paths must be everywhere timelike. Let us compute the proper time $\Delta\tau$ experienced by an observer who moves along one of these paths. We will apply a variational principle in order to find the trajectory for which the accumulated $\Delta\tau$ is *extremal*.

As an aside, you should be able to convince yourself that the extremum you wish to compute is the *maximum* of accumulated proper time, rather than a minimum. If this is not clear, consider a path that consists of multiple segments on which the observer moves at nearly light speed in

the frame which sees $\mathcal{P}$ and $\mathcal{Q}$ as they are drawn here. Such a path must include very large accelerations. Supposing that the observer survives these accelerations, the accumulated proper time on each segment will be very small thanks to time dilation. We can find a large number of such (highly accelerated) trajectories with accumulated proper time approaching zero, which is the minimum possible. The minimum of $\Delta\tau$ is thus not unique; the unique trajectory found by variational methods must be the maximum.

Consider each path to have an affine parameter λ that increases from $\mathcal{P}$ to $\mathcal{Q}$; the accumulated proper time for an observer who follows one of these paths is

$$\Delta\tau = \int_{\mathcal{P}}^{\mathcal{Q}} d\lambda \sqrt{-g_{\alpha\beta}\frac{dx^\alpha}{d\lambda}\frac{dx^\beta}{d\lambda}} \ . \tag{9.9}$$

This follows from the fact that $ds^2 = -d\tau^2$ within each local Lorentz frame adapted to this timelike observer.

We now will consider variations in this path, though holding the endpoints (events $\mathcal{P}$ and $\mathcal{Q}$) fixed. To facilitate this analysis, we define

$$f \equiv g_{\alpha\beta}\frac{dx^\alpha}{d\lambda}\frac{dx^\alpha}{d\lambda} \ , \tag{9.10}$$

and then consider variations in $\Delta\tau$ as we we vary the path:

$$\begin{aligned} \delta(\Delta\tau) &= \int_{\mathcal{P}}^{\mathcal{Q}} d\lambda\, \delta\left[(-f)^{1/2}\right] \\ &= -\frac{1}{2}\int_{\mathcal{P}}^{\mathcal{Q}} \frac{\delta f}{(-f)^{1/2}}\, d\lambda \ . \end{aligned} \tag{9.11}$$

Let us now make a choice for λ: we set $\lambda = \tau$ (proper time along that particular trajectory), so that $dx^\alpha/d\lambda$ is the 4-velocity u^α, and $f = g_{\alpha\beta}u^\alpha u^\beta = -1$. Our variational integral is then

$$\delta(\Delta\tau) = -\frac{1}{2}\int \delta f\, d\tau \ . \tag{9.12}$$

Our goal will be to find the trajectory for which this variation is zero; that will identify the trajectory which gives us an extremum in $\Delta\tau$. To facilitate this, let's make a further definition: we define

$$I = \frac{1}{2}\int_{\mathcal{P}}^{\mathcal{Q}} \left(g_{\alpha\beta}\frac{dx^\alpha}{d\tau}\frac{dx^\beta}{d\tau}\right) d\tau \tag{9.13}$$

as an action describing a body's motion from $\mathcal{P}$ to $\mathcal{Q}$; then, $\delta(\Delta\tau) = \delta I$.

We now examine how this action varies as we vary the trajectory, and look for the trajectory for which this variation is *stationary* (i.e, for which $\delta I = 0$ when $\delta x^\alpha \neq 0$). We take $x^\alpha \to x^\alpha + \delta x^\alpha$, $g_{\alpha\beta} \to g_{\alpha\beta} + \delta x^\gamma \partial_\gamma g_{\alpha\beta}$, and impose the boundary condition that $\delta x^\alpha = 0$ at the endpoints of each trajectory. Doing so, we find

$$\delta I = \frac{1}{2}\int_{\mathcal{P}}^{\mathcal{Q}} d\tau \left[\partial_\gamma g_{\alpha\beta}\frac{dx^\alpha}{d\tau}\frac{dx^\beta}{d\tau}\delta x^\gamma + g_{\alpha\beta}\frac{d(\delta x^\alpha)}{d\tau}\frac{dx^\beta}{d\tau} + g_{\alpha\beta}\frac{dx^\alpha}{d\tau}\frac{d(\delta x^\beta)}{d\tau}\right] \ . \tag{9.14}$$

Let us integrate the second term on the right-hand side by parts:

$$\begin{aligned} \frac{1}{2}\int_{\mathcal{P}}^{\mathcal{Q}} d\tau \left[g_{\alpha\beta}\frac{d(\delta x^\alpha)}{d\tau}\frac{dx^\beta}{d\tau}\right] &= -\frac{1}{2}\int_{\mathcal{P}}^{\mathcal{Q}} d\tau \left[g_{\alpha\beta}\frac{d^2x^\beta}{d\tau^2} + \frac{dg_{\alpha\beta}}{d\tau}\frac{dx^\beta}{d\tau}\right]\delta x^\alpha \\ &= -\frac{1}{2}\int_{\mathcal{P}}^{\mathcal{Q}} d\tau \left[g_{\alpha\beta}\frac{d^2x^\beta}{d\tau^2} + \partial_\gamma g_{\alpha\beta}\frac{dx^\beta}{d\tau}\frac{dx^\gamma}{d\tau}\right]\delta x^\alpha \ . \end{aligned} \tag{9.15}$$

To get the first line, we discarded a boundary term, using the fact that $\delta x^\alpha = 0$ at the limits $\mathcal{P}$ and $\mathcal{Q}$. For the second line, we used the fact that $d/d\tau = (dx^\gamma/d\tau)\partial_\gamma$. The third term on the right-hand side can be manipulated similarly:

$$\frac{1}{2}\int_{\mathcal{P}}^{\mathcal{Q}} d\tau \left[g_{\alpha\beta}\frac{dx^\alpha}{d\tau}\frac{d(\delta x^\beta)}{d\tau}\right] \;=\; -\frac{1}{2}\int_{\mathcal{P}}^{\mathcal{Q}} d\tau \left[g_{\alpha\beta}\frac{d^2x^\alpha}{d\tau^2} + \partial_\gamma g_{\alpha\beta}\frac{dx^\alpha}{d\tau}\frac{dx^\gamma}{d\tau}\right]\delta x^\beta \,. \tag{9.16}$$

Let us now put all the pieces together, and write $dx^\alpha/d\tau = u^\alpha$:

$$\begin{aligned}\delta I \;=\; \frac{1}{2}\int_{\mathcal{P}}^{\mathcal{Q}} d\tau \Big[& \partial_\gamma g_{\alpha\beta}u^\alpha u^\beta \delta x^\gamma \\ & -\partial_\gamma g_{\alpha\beta}u^\beta u^\gamma \delta x^\alpha - \partial_\gamma g_{\alpha\beta}u^\gamma u^\alpha \delta x^\beta \\ & -g_{\alpha\beta}\frac{du^\beta}{d\tau}\delta x^\alpha - g_{\alpha\beta}\frac{du^\alpha}{d\tau}\delta x^\beta\Big] \,.\end{aligned} \tag{9.17}$$

We will cycle the dummy indices in this formula in the following way in order to facilitate grouping the terms in a particularly convenient and useful way:

- First line: Leave the indices as is.
- Second line, left term: $\alpha \to \gamma$, $\beta \to \alpha$, $\gamma \to \beta$.
- Second line, right term: $\alpha \to \beta$, $\beta \to \gamma$, $\gamma \to \alpha$.
- Third line, left term: $\alpha \to \gamma$, $\beta \to \alpha$.
- Third line, right term: $\beta \to \gamma$.

The result of this exercise allows us to write

$$\delta I = -\int_{\mathcal{P}}^{\mathcal{Q}} d\tau \left[g_{\gamma\alpha}\frac{du^\alpha}{d\tau} + \frac{1}{2}\left(\partial_\alpha g_{\beta\gamma} + \partial_\beta g_{\gamma\alpha} - \partial_\gamma g_{\alpha\beta}\right)u^\alpha u^\beta\right]\delta x^\gamma \,. \tag{9.18}$$

We recognize the second term in the square brackets here as $\Gamma_{\gamma\alpha\beta}$, so

$$\delta I = -\int_{\mathcal{P}}^{\mathcal{Q}} d\tau \left[g_{\gamma\alpha}\frac{du^\alpha}{d\tau} + \Gamma_{\gamma\alpha\beta}u^\alpha u^\beta\right]\delta x^\gamma \,. \tag{9.19}$$

We require $\delta I = 0$ for any δx^γ. This is the case if the term in square brackets is zero; hitting this with $g^{\mu\gamma}$, we see that extremization leads us to

$$\frac{du^\mu}{d\tau} + \Gamma^\mu{}_{\alpha\beta}u^\alpha u^\beta = 0 \,. \tag{9.20}$$

This is the geodesic equation yet again.

9.3 Recap and yet another form

Geodesics generalize the notion of "straight" trajectories which describe inertial motion in special relativity to curves spacetime. The technique of parallel transport expresses the idea that since no force acts, the tangent to the trajectory moves parallel to itself (i.e., it is parallel transported). In special relativity, an accelerated trajectory always accumulates less proper time between two events than the inertial trajectory that connects these events. Extremization of accumulated proper time shows that geodesic motion expresses this concept in a curved spacetime. Geodesics thus generalize the notion of inertial motion for bodies moving in a curved spacetime.

It is useful to re-organize the geodesic equation once more by rewriting it in terms of 4-momentum, rather than 4-velocity:

$$u^\alpha \nabla_\alpha u^\beta = 0 \quad \rightarrow \quad p^\alpha \nabla_\alpha p^\beta = 0 \,, \tag{9.21}$$

which rewrites the geodesic equation in the form

$$m\frac{dp^\beta}{d\tau} + \Gamma^\beta{}_{\alpha\gamma} p^\alpha p^\gamma = 0 \,. \tag{9.22}$$

This becomes particularly useful when we define an interval of affine parameter using the rule $\Delta\lambda = \Delta\tau/m$. Using this, we then have

$$p^\alpha \equiv \frac{dx^\alpha}{d\lambda} \tag{9.23}$$

and

$$\frac{dp^\beta}{d\lambda} + \Gamma^\beta{}_{\alpha\gamma} p^\alpha p^\gamma = 0 \,. \tag{9.24}$$

The reason that this is useful is that we can take the limit $m \to 0$. Massless particles move on light-like trajectories in relativistic kinematics; the very notion of proper time does not make sense along such trajectories. However, if we imagine a limit of a massive body approaching the light-like trajectory, we can consider $m \to 0$ while $\Delta\tau \to 0$, but doing so in such a way that $\Delta\tau/m$ is always constant. Equation (9.24) is a form of the geodesic equation that is perfectly suited for studying light-like trajectories, provided we use (9.23) to relate the 4-momentum associated with the trajectory to its path in spacetime.

9.4 Things that are conserved along geodesics

Along with being a useful form for studying light-like trajectories, Eq. (9.24) is also useful for identifying quantities that are conserved along geodesics. Equation (9.24) can be rewritten $p^\alpha(\nabla_\alpha p^\beta) = 0$; because the metric commutes with the covariant derivative, we can further rewrite this

$$p^\alpha \left(\nabla_\alpha p_\beta\right) = 0 \,, \tag{9.25}$$

which expands to become

$$m\frac{dp_\beta}{d\tau} - \Gamma^\gamma{}_{\beta\alpha} p^\alpha p_\gamma = 0 \,, \tag{9.26}$$

or

$$m\frac{dp_\beta}{d\tau} = \Gamma_{\gamma\beta\alpha} p^\alpha p^\gamma = \frac{1}{2}\left(\partial_\beta g_{\alpha\gamma} + \partial_\alpha g_{\gamma\beta} - \partial_\gamma g_{\beta\alpha}\right) p^\alpha p^\gamma \,. \tag{9.27}$$

Look carefully at the last two terms in (9.27): the derivatives of the metric are antisymmetric on exchange of α and γ, but they are contracted with the symmetric combination $p^\alpha p^\gamma$. These two terms thus do not contribute to the right-hand side of (9.27), and we obtain the following very interesting form of the geodesic equation:

$$m\frac{dp_\beta}{d\tau} = \frac{1}{2}\partial_\beta g_{\alpha\gamma} p^\alpha p^\gamma \,. \tag{9.28}$$

"What is so interesting about this?" you may reasonably ask. Notice that, if the metric is independent of a particular x^β (so that $\partial_\beta g_{\alpha\gamma} = 0$ for some value of β), then p_β is *constant everywhere on the geodesic*:

$$\partial_\beta g_{\alpha\gamma} = 0 \quad \longrightarrow \quad \frac{dp_\beta}{d\tau} = 0 \quad \longrightarrow \quad p_\beta = \text{constant on geodesic trajectories.} \tag{9.29}$$

This should remind you of a result from Lagrangian mechanics: if a Lagrangian does not depend on a coordinate, then the momentum conjugate to that coordinate is constant.

A lecture back, we discussed Killing vectors, and one of our conclusions was that if the metric was independent of a particular coordinate (i.e., it did not vary along that coordinate), then there exists a Killing vector associated with this symmetry. Let us imagine that our spacetime admits a Killing vector $\vec{\xi}$. We can construct a scalar $p^\beta \xi_\beta$; let us see how this scalar evolves along a geodesic:

$$\begin{aligned} \frac{D(p^\beta \xi_\beta)}{d\lambda} &= p^\alpha \nabla_\alpha \left(p^\beta \xi_\beta \right) \\ &= \xi_\beta \left(p^\alpha \nabla_\alpha p^\beta \right) + p^\alpha p^\beta \nabla_\alpha \xi_\beta \ . \end{aligned} \tag{9.30}$$

The first term on the final line of Eq. (9.30) vanishes because p^α solves the geodesic equation — the term in parenthesis equals zero. In the second term, because $p^\alpha p^\beta$ is symmetric under exchange of α and β, we can rewrite $\nabla_\alpha \xi_\beta$ in a way that makes this symmetry manifestly clear:

$$\begin{aligned} \frac{D(p^\beta \xi_\beta)}{d\lambda} &= \frac{1}{2} p^\alpha p^\beta \left(\nabla_\alpha \xi_\beta + \nabla_\beta \xi_\alpha \right) \\ &= 0 \ . \end{aligned} \tag{9.31}$$

The final reduction is because $\vec{\xi}$ is a Killing vector, and thus satisfies Killing's equation, $\nabla_\alpha \xi_\beta + \nabla_\beta \xi_\alpha = 0$. We thus see that $p^\beta \xi_\beta$ is constant along any geodesic trajectory.

Two very important examples of this are as follows:

- Suppose that, in some choice of coordinates, $\partial_t g_{\alpha\beta} = 0$. We know that there exists a "timelike" Killing vector, $\vec{\xi}^T$, in this spacetime. We also know that p_0 is conserved along all trajectories; to tie this to special relativity intuition, we define the *conserved energy* along the geodesic as $E \equiv -p_0$. (To understand the minus sign, think about the special relativity limit: the "downstairs" component in that limit is $-E$.) We can write this as $E = -p^\beta \xi^T_\beta$, which is a coordinate invariant expression; the form $E = -p_0$ only holds in the coordinates for which $\partial_t g_{\alpha\beta} = 0$.

  - *Caution*: the conserved energy $E = -p_0$ is **not necessarily** the energy that is measured by an observer who measures properties of the geodesic whose 4-momentum is p^α! The measured energy can be found by invoking the Einstein equivalence principle: if we go into a local Lorentz frame around an observer whose 4-velocity is $\vec{u}$, they measure the energy along a geodesic whose 4-momentum is $\vec{p}$ to be $E_{\vec{u}} = -p_\alpha u^\alpha$. We will see a very specific example of the difference between a conserved energy along a geodesic and the energy someone measures associated with this geodesic in a few weeks. It is important to always think carefully about the specific, quantitative meaning of words like "energy" in the context of an analysis.

- Suppose that, in some choice of coordinates, $\partial_\phi g_{\alpha\beta} = 0$. This means that there is an "axial" Killing vector, $\vec{\xi}^\Phi$, in this spacetime. We know that p_ϕ is conserved along all trajectories; we call this the "axial angular momentum," and define $L_z \equiv p_\phi$. We can write this in coordinate invariant form $L_z = p^\beta \xi^\Phi_\beta$.

9.5 A geodesic in a specific spacetime

To wrap up this discussion, let's look at geodesics in a particularly interesting spacetime:

$$ds^2 = -(1 + 2\Phi)dt^2 + (1 - 2\Phi)(dx^2 + dy^2 + dz^2) \ . \tag{9.32}$$

(We will develop this spacetime quite soon as one of our first solutions once we've made the Einstein field equation.) The function Φ which appears here is required to be small ($\Phi \ll 1$). We imagine that it varies in space but not in time: $\Phi = \Phi(x, y, z)$.

Let us consider a slowly moving body in this spacetime: consider a body whose 4-velocity in these coordinates has components $u^\alpha \doteq (dt/d\tau, dx^i/d\tau)$, with $dt/d\tau \gg dx^i/d\tau$ for all i. (This is equivalent to saying that the spatial velocity components $v^i = dx^i/dt$ in the frame in which we have written down this spacetime are all much smaller than the speed of light.) The geodesic equation in this spacetime has the form

$$\frac{d^2x^\alpha}{d\tau^2} + \Gamma^\alpha{}_{\mu\nu}\frac{dx^\mu}{d\tau}\frac{dx^\nu}{d\tau} = 0 \; . \tag{9.33}$$

Because we are considering the slow motion limit, the second term in the geodesic equation is dominated by the terms with $\mu = \nu = 0$; the contributions related to the spatial 4-velocity components are much less important than the contributions from $dt/d\tau$. Our geodesic equation, to a high level of accuracy, can be written

$$\frac{d^2x^\alpha}{d\tau^2} + \Gamma^\alpha{}_{00}\left(\frac{dt}{d\tau}\right)^2 = 0 \; , \tag{9.34}$$

or

$$\frac{d^2x^\alpha}{dt^2} + \Gamma^\alpha{}_{00} = 0 \; . \tag{9.35}$$

Let's focus on the spatial components:

$$\frac{d^2x^i}{dt^2} + \Gamma^i{}_{00} = 0 \; . \tag{9.36}$$

The Christoffel symbol we need is given by

$$\begin{aligned} \Gamma^i{}_{00} &= \frac{1}{2}g^{i\alpha}\Gamma_{\alpha 00} \\ &= \frac{1}{2}g^{i\alpha}\left(\partial_t g_{0\alpha} + \partial_t g_{\alpha 0} - \partial_\alpha g_{00}\right) \\ &= -\frac{1}{2}g^{i\alpha}\partial_\alpha g_{00} \; . \end{aligned} \tag{9.37}$$

We have used the fact that the metric does not depend on t to eliminate the ∂_t terms[1]. Using $g^{i\alpha} = (1-2\Phi)^{-1}\delta^{i\alpha}$ plus the fact that $\Phi \ll 1$, we find

$$\begin{aligned} \Gamma^i{}_{00} &= -\frac{1}{2}(1-2\Phi)^{-1}\delta^{ij}\partial_j(1-2\Phi) \\ &= \delta^{ij}\partial_j\Phi + \mathcal{O}(\Phi^2) \; . \end{aligned} \tag{9.38}$$

Our equation of motion in this spacetime is thus

$$\frac{d^2x^i}{dt^2} = -\delta^{ij}\partial_j\Phi \; . \tag{9.39}$$

Geodesic motion in this spacetime says that the body moves through these coordinates with an acceleration given by the negative gradient of the function Φ that appears in the line element (9.32). This is *exactly* the law of motion that describes a body moving under the influence of a Newtonian gravitational potential Φ. We will in fact see quite soon that Eq. (9.32) is a key tool in making sure that the relativistic theory of gravity we develop includes and respects the Newtonian limit.

[1] Using this, we find the timelike component of the geodesic equation yields $d^2t/d\tau^2 = 0$. In other words, in this limit, coordinate time and proper time are the same up to a choice of origin and a choice of units. This is appropriate for a Newtonian limit.

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Lecture 10
Spacetime curvature

10.1 Quantifying curvature

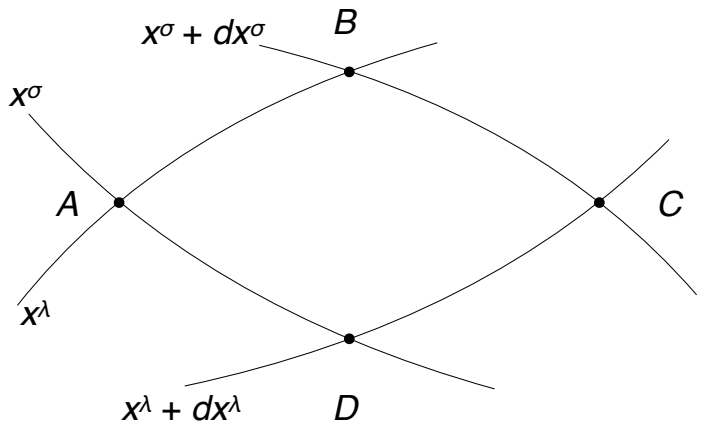
We have spent much of the past several lectures figuring out how describe tensors and mathematical operations using tensors in curved spacetime manifolds; we have described the motion of freely falling bodies moving through a curved spacetime. However, we have not yet precisely defined what curvature actually *is*. Today we rectify this.

Our concept of curvature centers on the idea that initially parallel trajectories do not remain parallel. This is an important concept to bear in mind (and in Lecture 11, we add some quantitative flesh to this conceptual skeleton), but for cleanly developing a notion of curvature there is an even better tool: take some vector and parallel propagate it around a closed, but finite-sized loop. If the manifold in which this vector moves around has curvature, then the resulting vector will point in a new direction after returning to its starting point (cf. the left-hand panel of the figure below, sketching a vector parallel propagated around a triangle, $A \to B \to C \to A$, that is embedded on the surface of a sphere). If the manifold is *flat*, then the vector will be unchanged when it returns to its starting point (cf. the right-hand panel of the figure below, sketching a vector parallel propagated around a triangle, $A \to B \to C \to A$, that is drawn on a flat surface).



(Such a parallel transport around a closed path is known as a holonomy; those of you interested in the more mathematical side of things can find lots of discussion and details of this operation.) When we are done, the vector has returned to its starting point, and so comparing the initial and final states can be done cleanly using just the components. The basis objects "live" in the same tangent space, and we don't have to worry about how they change during the propagation.

For our analysis, consider parallel transport of a vector with components V^α around the following "diamond," from $A \to B \to C \to D \to A$:



We need only consider two spacetime directions in this analysis. Label one with the index λ, the other with σ. (In the 4 dimensions that we consider for spacetime, there are of course two other; call them x^β and x^γ. We will hold them fixed throughout this analysis.) As we go along the leg from A to B, the coordinate x^λ is held constant, while x^σ increases to $x^\sigma + dx^\sigma$. As we go from B to C, x^λ increases to $x^\lambda + dx^\lambda$; the other is held fixed at $x^\sigma + dx^\sigma$. As we go from C to D, we decrease from $x^\sigma + dx^\sigma$ to x^σ; the other direction is held fixed at $x^\lambda + dx^\lambda$. Finally, returning along D to A, we decrease from $x^\lambda + dx^\lambda$ to x^λ, holding the other coordinate at x^σ.

Let's examine how V^α changes as we parallel transport around these paths. First go from A to B, which we will call "Leg I." We parallel transport with fixed x^λ, in the direction of increasing x^σ. In other words,

$$\nabla_{\vec{e}_\sigma} V^\alpha = 0 \longrightarrow \frac{\partial V^\alpha}{\partial x^\sigma} = -\Gamma^\alpha{}_{\sigma\mu} V^\mu \ . \tag{10.1}$$

Treating this as a differential equation for V^α, we find

$$V^\alpha(B) = V^\alpha_{\rm init} - \int_{\rm I} \Gamma^\alpha{}_{\sigma\mu} V^\mu \, dx^\sigma \ . \tag{10.2}$$

(The notation on the integral means "integrated along Leg I.")

Repeat this to go from B to C ("Leg II"):

$$V^\alpha(C) = V^\alpha(B) - \int_{\rm II} \Gamma^\alpha{}_{\lambda\mu} V^\mu \, dx^\lambda \ . \tag{10.3}$$

From C to D ("Leg III"):

$$V^\alpha(D) = V^\alpha(C) + \int_{\rm III} \Gamma^\alpha{}_{\sigma\mu} V^\mu \, dx^\sigma \ . \tag{10.4}$$

Note the sign flip, which is due to the fact that we are now integrating from $x^\sigma + dx^\sigma$ to x^σ.

Finally, from D back to A ("Leg IV"):

$$V^\alpha_{\rm final} = V^\alpha(C) + \int_{\rm IV} \Gamma^\alpha{}_{\lambda\mu} V^\mu \, dx^\lambda \ . \tag{10.5}$$

We are interested in seeing how the vector changes after going around this loop:

$$\delta V^\alpha \equiv V^\alpha_{\rm final} - V^\alpha_{\rm init} \ . \tag{10.6}$$

As emphasized above, by beginning and ending at the same event, it is easy for us to make this comparison — the initial and final vectors "live" in the same tangent space, so comparing components is sufficient to understand how the vector changes under this operation. Using what we derived above and organizing terms a bit, we have

$$\begin{aligned}\delta V^\alpha &= \int_{\rm IV} \Gamma^\alpha{}_{\lambda\mu} V^\mu \, dx^\lambda - \int_{\rm II} \Gamma^\alpha{}_{\lambda\mu} V^\mu \, dx^\lambda \\ &\quad + \int_{\rm III} \Gamma^\alpha{}_{\sigma\mu} V^\mu \, dx^\sigma - \int_{\rm I} \Gamma^\alpha{}_{\sigma\mu} V^\mu \, dx^\sigma \ . \end{aligned} \tag{10.7}$$

As we go through these contributions, keep in mind that Leg I is evaluated along constant x^λ; Leg III is evaluated along constant $x^\lambda + dx^\lambda$; Leg II is evaluated along constant $x^\sigma + dx^\sigma$; and Leg IV is evaluated along constant x^σ. With this in mind, it follows that

$$\int_{\rm III} \mathcal{I} \, dx^\sigma - \int_{\rm I} \mathcal{I} \, dx^\sigma = \delta x^\lambda \int_{\rm I} \frac{\partial \mathcal{I}}{\partial x^\lambda} dx^\sigma \ , \tag{10.8}$$

$$\int_{\rm IV} \mathcal{I} \, dx^\lambda - \int_{\rm II} \mathcal{I} \, dx^\lambda = -\delta x^\sigma \int_{\rm II} \frac{\partial \mathcal{I}}{\partial x^\sigma} dx^\lambda \tag{10.9}$$

for any integrand $\mathcal{I}$. (We are imaging that δx^λ and δx^σ are sufficiently small that terms of order $(\delta x)^2$ can be discarded; if you want more rigor, please replace $=$ with $\simeq$ and imagine taking a limit at the end of the calculation.) Using this, we can rewrite Eq. (10.7) as

$$\delta V^\alpha = \int_{x^\sigma}^{x^\sigma + dx^\sigma} \delta x^\lambda \frac{\partial}{\partial x^\lambda} \left(\Gamma^\alpha{}_{\sigma\mu} V^\mu \right) dx^\sigma - \int_{x^\lambda}^{x^\lambda + dx^\lambda} \delta x^\sigma \frac{\partial}{\partial x^\sigma} \left(\Gamma^\alpha{}_{\lambda\mu} V^\mu \right) dx^\lambda \ . \tag{10.10}$$

Expanding the derivatives, this becomes

$$\delta V^\alpha = \delta x^\lambda \delta x^\sigma \left[\partial_\lambda \Gamma^\alpha{}_{\sigma\mu} V^\mu - \partial_\sigma \Gamma^\alpha{}_{\lambda\mu} V^\mu + \Gamma^\alpha{}_{\sigma\mu} \partial_\lambda V^\mu - \Gamma^\alpha{}_{\lambda\mu} \partial_\sigma V^\mu \right] \ . \tag{10.11}$$

To reduce those terms involving the partial derivatives of the vector components, remember that they are parallel transported, and so we can write

$$\partial_\lambda V^\mu = -\Gamma^\mu{}_{\lambda_\nu} V^\nu \ , \qquad \partial_\sigma V^\mu = -\Gamma^\mu{}_{\sigma\nu} V^\nu \ . \tag{10.12}$$

Inserting this, we find

$$\delta V^\alpha = \delta x^\lambda \delta x^\sigma \left[\left(\partial_\lambda \Gamma^\alpha{}_{\sigma\mu} - \partial_\sigma \Gamma^\alpha{}_{\lambda\mu} \right) V^\mu + \left(\Gamma^\alpha{}_{\lambda\mu} \Gamma^\mu{}_{\sigma\nu} - \Gamma^\alpha{}_{\sigma\mu} \Gamma^\mu{}_{\lambda_\nu} \right) V^\nu \right] \ . \tag{10.13}$$

Relabel dummy indices on the last two terms to exchanging μ and ν; the final expression then becomes

$$\delta V^\alpha = R^\alpha{}_{\mu\lambda\sigma} \delta x^\lambda \delta x^\sigma V^\mu \ , \tag{10.14}$$

where we have introduced

$$\boxed{\boxed{R^\alpha{}_{\mu\lambda\sigma} = \partial_\lambda \Gamma^\alpha{}_{\sigma\mu} - \partial_\sigma \Gamma^\alpha{}_{\lambda\mu} + \Gamma^\alpha{}_{\lambda\nu} \Gamma^\nu{}_{\sigma\mu} - \Gamma^\alpha{}_{\sigma\nu} \Gamma^\nu{}_{\lambda\mu}}} \tag{10.15}$$

a quantity known as the *Riemann curvature tensor*. Despite being constructed out of Christoffel symbols, it is truly a tensor: when we change representations, the terms which cause the Christoffels to not transform in a tensorial manner all cancel out.

Notice that the first two contributions to this tensor have the heuristic form "two derivatives of the metric" (bearing in mind that the Christoffel symbol is one derivative of the metric). It is nicely consistent that a quantity which looks like the second derivative of the metric, which we earlier showed cannot be eliminated by going to a local Lorentz frame, is in fact a tensor. The

final two terms are of the form "nonlinear first derivative of the metric," a sign that any differential equations we make out of this are going to end up being nonlinear ones.

One can show the calculation we did is equivalent to looking at the commutator of partial derivatives[1]:

$$[\nabla_\lambda, \nabla_\sigma] V^\alpha = R^\alpha{}_{\mu\lambda\sigma} V^\mu \,, \tag{10.16}$$

$$[\nabla_\lambda, \nabla_\sigma] p_\alpha = -R^\mu{}_{\alpha\lambda\sigma} p_\mu \,. \tag{10.17}$$

10.2 Curvature coupling

The way in which we derived the Riemann curvature tensor gives us a flavor of the circumstances in which it matters: curvature tends to enter physics when we look at processes that occur over some kind of extended region. In our case, we had to go around a loop that enclosed a finite region (if we had gone $A \to B \to A$, there would have been no effect), and we ended up with an effect that is proportional to the product of the length of each side of our "parallel transport diamond." This is consistent with the idea that we can make spacetime look like special relativity up to $\mathcal{O}(\delta x)$ from some event, but we cannot do so at $\mathcal{O}(\delta x^2)$.

In our previous lecture, we found that geodesic motion generalizes the notion of free-fall for a body moving in a curved spacetime. The geodesic equation,

$$\frac{du^\alpha}{d\tau} + \Gamma^\alpha{}_{\mu\nu} u^\mu u^\nu = 0 \,, \tag{10.18}$$

has no curvature in it. We can now see that there's a more-or-less hidden assumption in our discussion of the idea that freely-falling bodies follow geodesic trajectories: we assume that these bodies are so small that the whole body "fits" into a local Lorentz frame. Strictly speaking, when we describe a body as moving along a geodesic, we idealize that body as a zero-size test particle. In reality, we expect that a body of finite size will "couple" to spacetime curvature. For example, the Earth is large enough that it couples to the Riemann curvature tensor associated with the spacetime metric produced by the Sun. This coupling corrects the leading description of the Earth's orbit as following a geodesic, and can be shown to have a small but non-negligible effect, especially when integrated up over many orbits. This in fact is how one describes the precession of the equinoxes in general relativity.

10.3 Symmetries and counting of independent components

The Riemann curvature tensor is a 4-index object, with each index taking n possible values in n dimensional space ($n = 4$ for spacetime, but it will be useful to leave this more general in the following discussion). This suggests naively that it has n^4 different components (256 in spacetime). However, we have not yet looked to see how many symmetries it has.

One symmetry can be seen quite simply: go back to our parallel propagation of the vector around our "curvature diamond," but do so in the opposite direction. Doing so exchanges the roles of λ and σ, and yields a change δV^α with opposite sign. This tells us that

$$R^\alpha{}_{\mu\lambda\sigma} = -R^\alpha{}_{\mu\sigma\lambda} \,. \tag{10.19}$$

Finding that the Riemann tensor is antisymmetric on the last two indices reduces the number of independent components from n^4 to $n^2 \cdot (n(n-1))/2$, which means from 256 to 96 in 4-dimensional spacetime. This is already a significant reduction, but we are just getting started.

[1] **Caution:** the commutator acting on a 1-form is written incorrectly in the 1st edition of Schutz, just in case you happen to have a copy of that text. This error has been corrected in later editions.

To go further, it is useful to lower the first index

$$R_{\alpha\mu\lambda\sigma} = g_{\alpha\nu} R^{\nu}{}_{\mu\lambda\sigma} \ , \tag{10.20}$$

and then expand all the terms in a local Lorentz frame. Within the LLF, all 1st derivatives of the metric vanish; however, 2nd derivatives of the metric do *not* vanish. This amounts to going into a representation in which we keep the $\partial\Gamma$ terms in Eq. (10.15), but we drop the Γ^2 terms. It should be emphasized that we do not *need* to do this. Going to the LLF is solely to expedite the calculation. We could do this exercise using Riemann in its full glory, though it is significantly more tedious to do so.

Expanding in the LLF, we find

$$\begin{aligned} \left(R_{\alpha\mu\lambda\sigma}\right)^{\rm LLF} &= g_{\alpha\nu} \left[\partial_\lambda \Gamma^{\nu}{}_{\sigma\mu} - \partial_\sigma \Gamma^{\nu}{}_{\lambda\mu}\right] \\ &= \partial_\lambda \Gamma_{\alpha\sigma\mu} - \partial_\sigma \Gamma_{\alpha\lambda\mu} \ . \end{aligned} \tag{10.21}$$

On the second line, we have used the fact that the metric commutes with partial derivatives in the LLF. (Note also that in this form the fact that $R_{\alpha\mu\lambda\sigma} = -R_{\alpha\mu\sigma\lambda}$ is manifestly obvious.) Inserting the definition of the Christoffels and gathering terms yields

$$\left(R_{\alpha\mu\lambda\sigma}\right)^{\rm LLF} = \frac{1}{2} \left(\partial_\lambda \partial_\mu g_{\alpha\sigma} - \partial_\lambda \partial_\alpha g_{\sigma\mu} - \partial_\sigma \partial_\mu g_{\alpha\lambda} + \partial_\sigma \partial_\alpha g_{\lambda\mu}\right) \ . \tag{10.22}$$

By staring at Eq. (10.22) for a little while, we notice the following symmetries:

- Antisymmetry on the last two indices: $R_{\alpha\mu\lambda\sigma} = -R_{\alpha\mu\sigma\lambda}$. We already deduced this one, but including it on this list for completeness. As already discussed, this symmetry reduces the number of independent components from n^4 to $n^3(n-1)/2$.

- Antisymmetry on the first two indices: $R_{\alpha\mu\lambda\sigma} = -R_{\mu\alpha\lambda\sigma}$. We replace the n^2 that are possible for the first two indices with $n(n-1)/2$, yielding $n^2(n-1)^2/4$. This further reduces the number of independent components to 36 for $n = 4$.

- Block symmetry of the first two with the last two: $R_{\alpha\mu\lambda\sigma} = R_{\lambda\sigma\alpha\mu}$. It's useful to think of the blocks $\alpha\mu$ and $\lambda\sigma$ as each having N possible values, with $N = n(n-1)/2$. This symmetry tells us that only $N(N+1)/2$ of them are independent; plugging in $N = n(n-1)/2$, we find that the number of components has been reduced to $n(n-1)(n^2-n+2)/8 = (n^4-2n^3+3n^2-2n)/8$, whose value is 21 when $n = 4$.

- Cyclic permutation of last three indices sums to zero: $R_{\alpha\mu\lambda\sigma} + R_{\alpha\lambda\sigma\mu} + R_{\alpha\sigma\mu\lambda} = 0$.

  - Another way to write this form is $R_{\alpha[\mu\lambda\sigma]} = 0$, where

    $$R_{\alpha[\mu\lambda\sigma]} = \frac{1}{3!} \left(R_{\alpha\mu\lambda\sigma} - R_{\alpha\lambda\mu\sigma} + R_{\alpha\lambda\sigma\mu} - R_{\alpha\mu\sigma\lambda} + R_{\alpha\sigma\mu\lambda} - R_{\alpha\sigma\lambda\mu}\right) \ , \tag{10.23}$$

    uses "total antisymmetrization" on the 3 indices μ, λ, and σ. By using the other symmetries we've established one can show that these two ways of writing this symmetry property of the Riemann tensor are completely equivalent.

  When 3 indices are completely antisymmetric, the number of independent ways of choosing them is $n(n-1)(n-2)/3!$. By folding in the other Riemann symmetries, we can in fact express this rule as $R_{[\alpha\mu\lambda\sigma]}$ (i.e., total antisymmetrization on all four indices); there are $n(n-1)(n-2)(n-3)/4! = (n^4-6n^3+11n^2-6n)/24$ totally antisymmetric arrangements of these indices. Bearing in that this counts up constraints on the number of independent components, we find that the number of components in the Riemann tensor is

$$N_{\rm Riemann}(n) = \frac{n^4-2n^3+3n^2-2n}{8} - \frac{n^4-6n^3+11n^2-6n}{24} = \frac{n^2(n^2-1)}{12} \ . \tag{10.24}$$

For $n = 4$, this takes the value 20: **exactly** the number of constraints we found that we could not satisfy at second-order in displacement when we developed the idea of the local Lorentz frame.

A few other values are also interesting:

- For $n = 1$, $N_{\rm Riemann} = 0$. This is no way to develop the "curvature diamond," and thus no meaningful holonomy operation, on a 1-dimensional manifold.
- For $n = 2$, $N_{\rm Riemann} = 1$. A single number characterizes the curvature of a 2-dimensional manifold. This can be regarded as related to the radius of curvature associated with a point. For the surface of a sphere, this quantity is constant everywhere, but it varies more generally (e.g., for an oblate or prolate spheroid, or a very bumpy spheroidal body).
- For $n = 3$, $N_{\rm Riemann} = 6$. Interestingly, this is exactly the same as the number of components in a symmetric 3×3 tensor. We will revisit this observation soon.

10.4 Riemann normal coordinates

With the Riemann tensor in hand, it is possible to actually write down the metric in a freely falling representation and show how the 20 components of Riemann allow us to build the metric that is locally flat, with quadratic corrections. Those interested in a full calculation of this representation are referred to Sec. 1.11 of the textbook "A Relativist's Toolkit" by Eric Poisson, one of the suggested supplemental texts for this course.

Let γ be a timelike geodesic trajectory through spacetime, and consider a LLF around this curve. Let t be proper time on this curve, and define Cartesian coordinates x^i in this frame, undefined up to a rotation, but with $x^i = 0$ on the curve. In the LLF,

$$g_{tt} = -1 - R_{tjtk}\, x^j x^k + \mathcal{O}(x^3) \,, \tag{10.25}$$

$$g_{tj} = -\frac{2}{3} R_{tijk}\, x^i x^k + \mathcal{O}(x^3) \,, \tag{10.26}$$

$$g_{jk} = \delta_{jk} - \frac{1}{3} R_{jikl}\, x^i x^l + \mathcal{O}(x^3) \,. \tag{10.27}$$

This shows exactly how the metric components look when we perform the transformation to the local Lorentz / freely falling frame.

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Lecture 11
More on spacetime curvature

11.1 Variants of the curvature tensor

The Riemann curvature tensor that we derived in the previous lecture is one of the most important quantities we have developed this term. Certain variants of this curvature are also important. The first such is found by constructing the *trace*:

$$R_{\mu\nu} \equiv R^{\alpha}{}_{\mu\alpha\nu} = g^{\alpha\beta} R_{\beta\mu\alpha\nu} \; . \tag{11.1}$$

This combination is known as the *Ricci* curvature tensor. Note the trace is taken on the first and third indices. A trace over the first and second indices would be identically zero, thanks to the antisymmetry of these indices; likewise a trace on the third and fourth is zero. By using the Riemann symmetries, you should be able to convince yourself that contracting on indices one and three yields the only unique trace of the Riemann curvature (up to a sign).

It is not hard to show that Ricci is symmetric on its two indices: expanding the definition of the Riemann tensor and constructing the trace yields

$$R_{\mu\nu} = \partial_\alpha \Gamma^{\alpha}{}_{\mu\nu} - \partial_\nu \Gamma^{\alpha}{}_{\alpha\mu} + \Gamma^{\alpha}{}_{\alpha\beta}\Gamma^{\beta}{}_{\nu\mu} - \Gamma^{\alpha}{}_{\nu\beta}\Gamma^{\beta}{}_{\alpha\mu} \; . \tag{11.2}$$

Recalling the fact that the Christoffel symbol is symmetric on its bottom two indices, we see that the first and third terms on the right-hand side of (11.2) are clearly symmetric. The fourth term is also clearly symmetric, noting that the indices α and β in this term are dummies. The only unclear term is the second one. We can put it into a form that makes things much clearer by using a result from a problem set exercise:

$$\partial_\nu \Gamma^{\alpha}{}_{\alpha\mu} = \partial_\nu \partial_\mu \left(\ln \sqrt{|g|} \right) \; . \tag{11.3}$$

This is clearly symmetric with respect to μ and ν, and so we see that Ricci tensor is indeed symmetric.

In 4-dimensional spacetime, the Ricci tensor has 10 independent components. Heuristically, you can think of it as "half" the 4-D Riemann tensor (which we recall has 20 independent components). The other half will be developed below. Interestingly, in 3-D space, Ricci has 6 independent components — exactly the number of components that Riemann has in 3 dimensions. In 3 dimensions, Riemann and Ricci are in an important fundamental sense equivalent to one another.

The trace of the Ricci tensor,

$$R \equiv R^{\mu}{}_{\mu} = g^{\mu\nu} R_{\mu\nu} \; , \tag{11.4}$$

is called the *Ricci scalar*. It is of course just one number (or more accurately one function, since it may vary over the manifold), and totally characterizes curvature in 2 dimensions (recall that Riemann has only 1 independent component in 2-D). An interesting quantity we can construct using the Ricci scalar and the Ricci tensor is a "trace-reversed" Ricci tensor:

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R \; . \tag{11.5}$$

Notice that $G^{\mu}{}_{\mu} = -R$; this can be shown using the fact that $g^{\mu\nu} g_{\mu\nu} = 4$ in 4-dimensional spacetime. We will discuss this tensor a lot more and give it a name in the very near future.

The final variant curvature tensor we will discuss is the totally trace-free part of the Riemann tensor: we take Riemann and subtract from it everything that can yield a non-zero trace. Carefully thinking about the ways in which this can be done, we end up with

$$C_{\alpha\mu\lambda\sigma} = R_{\alpha\mu\lambda\sigma} - \frac{2}{n-2}\left(g_{\alpha[\lambda}R_{\sigma]\mu} - g_{\mu[\lambda}R_{\sigma]\alpha}\right) + \frac{2}{(n-2)(n-1)}g_{\alpha[\lambda}g_{\sigma]\mu}R\,. \tag{11.6}$$

This quantity is known as the *Weyl* (pronounced like "vile" in English, named for the German mathematician Hermann Weyl) curvature tensor. The normalization of the terms that involve the metric are chosen because $g^{\mu\nu}g_{\mu\nu} = n$ in n dimensions; note that the Weyl tensor is only defined for $n \geq 3$. In the definition of $C_{\alpha\mu\lambda\sigma}$, we are using antisymmetrization on certain pairs of indices like so:

$$X_{\mu[\beta}Y_{\gamma]\nu} = \frac{1}{2}\left(X_{\mu\beta}Y_{\gamma\nu} - X_{\mu\gamma}Y_{\beta\nu}\right)\,. \tag{11.7}$$

The Weyl tensor has the same symmetries as the Riemann tensor, but thanks to the various traces that have been removed the number of independent components is equal to that of Riemann minus that of Ricci:

$$\begin{aligned} N_{\text{Weyl}}(n) &= N_{\text{Riemann}}(n) - N_{\text{Ricci}}(n) \\ &= \frac{n^2(n^2-1)}{12} - \frac{n(n+1)}{2} \\ &= \frac{n^4 - 7n^2 - 6n}{12}\,. \end{aligned} \tag{11.8}$$

The number of independent components is 10 for $n = 4$, accounting for the "other" 10 components of Riemann that did not end up in Ricci. It is zero for $n = 3$, consistent with the fact that both Riemann and Ricci have 6 independent components in this case.

11.1.1 A brief aside on the Weyl curvature

The Weyl curvature is important for some analyses because of the simple way that it behaves under what are called *conformal transformations*. Such transformations describe a local change of scale: if two events have a proper separation (interval) of ds, then the conformal rescaling is

$$ds \to d\tilde{s} \equiv \Omega\, ds\,. \tag{11.9}$$

The function Ω, which can vary from event to event in spacetime, is called the scale factor; it is a way of introducing a remapping of spacetime that adjusts the scale associated with the separation of various events. Changing the proper separation in this way is equivalent to rescaling the metric by the conformal factor squared, allowing us to define

$$\tilde{g}_{\mu\nu} = \Omega^2 g_{\mu\nu}\,. \tag{11.10}$$

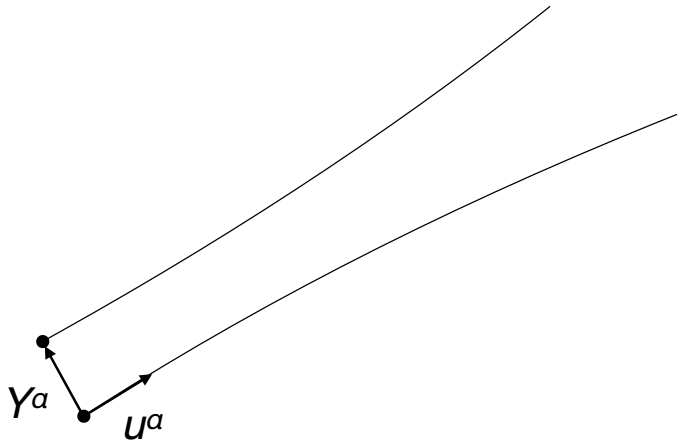
Conformal transformations of this kind are often useful because they do not affect *light cones*: if p^μ is null in the original metric, then it is also null in the conformally transformed metric: $\tilde{g}_{\mu\nu}p^\mu p^\nu = \Omega^2 g_{\mu\nu}p^\mu p^\nu = 0$. This can be particularly useful for understanding how or even whether different regions in a spacetime manifold are in causal contact with one another (i.e., if events in one part of the manifold can send a lightlike signal to events in another part of the manifold).

Unfortunately, conformal transformations tend to horribly mangle the spacetime's curvature tensors; see Appendix G of Carroll for discussion of how Riemann is affected. However, the Weyl curvature is *not* affected by such a transformation. As such, you will sometimes see Weyl called the "conformally invariant" curvature.

The Weyl curvature makes an important physical contrast with the Ricci curvature. We will soon see that the Ricci curvature is connected to the matter and energy content of spacetime. Weyl will then tell us about a spacetime's "vacuum" curvature.

11.2 Breakdown of parallelism: geodesic deviation

Our original motivating idea to lay out the notion of "curvature" was that initially parallel trajectories do not remain parallel: like lines of longitude followed from the equator (which both start on a trajectory that is 90° to the equator) to a pole, they eventually cross; or, they eventually diverge, depending on the detailed nature of the manifold's curvature. Here we make this notion precise.



11.2.1 Formal calculation

Consider a two-parameter family of geodesic trajectories in spacetime, $x^\alpha(\lambda, s)$. The parameter λ is affine parameter along each geodesic. This means that $u^\alpha(s) = (\partial x^\alpha/\partial\lambda)_s$ is the tangent along the geodesic with parameter s. (If we choose λ to be proper time, then u^α is the usual 4-velocity for this geodesic.) The parameter s tells us about the separation of neighboring geodesics. The vector[1] $Y^\alpha(\lambda) = (\partial x^\alpha/\partial s)_\lambda$ points from the event at affine parameter λ on the geodesic at s to the event at affine parameter λ on the geodesic at $s + ds$.

Before diving into our analysis, it is useful to note that

$$u^\beta \nabla_\beta Y^\mu = \frac{\partial^2 x^\mu}{\partial\lambda\partial s} + \Gamma^\mu{}_{\beta\nu} u^\beta Y^\nu \quad \text{and} \quad Y^\beta \nabla_\beta u^\mu = \frac{\partial^2 x^\mu}{\partial s\partial\lambda} + \Gamma^\mu{}_{\beta\nu} Y^\beta u^\nu \ .$$

Because partial derivatives commute, because the connection is symmetric on its downstairs indices, and because we can relabel dummy indices, we see that $u^\beta \nabla_\beta Y^\mu = Y^\beta \nabla_\beta u^\mu$. We will make use of this identity very soon.

[1] In an older version of my notes which might still be floating around, this vector was labeled ξ^α. I have changed from this older notation because $\vec{\xi}$ also denotes a Killing vector, and ξ^α will also be used in a future lecture to describe the generator of an infinitesimal coordinate transformation. The letter ξ is a bit overloaded.

Let us now evaluate the *covariant acceleration* of the separation vector Y^μ:

$$\begin{aligned}
\frac{D^2Y^\mu}{d\lambda^2} &\equiv u^\alpha\nabla_\alpha\left(u^\beta\nabla_\beta Y^\mu\right) \\
&= u^\alpha\nabla_\alpha\left(Y^\beta\nabla_\beta u^\mu\right) \\
&= \left(u^\alpha\nabla_\alpha Y^\beta\right)\left(\nabla_\beta u^\mu\right) + u^\alpha Y^\beta\nabla_\alpha\nabla_\beta u^\mu \\
&= \left(u^\alpha\nabla_\alpha Y^\beta\right)\left(\nabla_\beta u^\mu\right) + u^\alpha Y^\beta\left(\nabla_\beta\nabla_\alpha u^\mu + R^\mu{}_{\nu\alpha\beta}u^\nu\right) \\
&= \left(u^\alpha\nabla_\alpha Y^\beta\right)\left(\nabla_\beta u^\mu\right) + Y^\beta\nabla_\beta\left(u^\alpha\nabla_\alpha u^\mu\right) - \left(Y^\beta\nabla_\beta u^\alpha\right)\left(\nabla_\alpha u^\mu\right) + R^\mu{}_{\nu\alpha\beta}u^\nu u^\alpha Y^\beta \\
&= R^\mu{}_{\nu\alpha\beta}u^\nu u^\alpha Y^\beta \ .
\end{aligned}$$

The first line essentially just uses the definition $D/d\lambda \equiv u^\alpha\nabla_\alpha$ twice. To get to the second line, we use the identity we worked out above. To go to the third line, we expand the action of ∇_α. To go to the fourth line, we use the commutator rule $[\nabla_\alpha, \nabla_\beta]u^\mu = R^\mu{}_{\nu\alpha\beta}u^\nu$.

In going from the fourth to the fifth line, we use $u^\alpha\left(\nabla_\beta W_\alpha{}^\mu\right) = \nabla_\beta\left(u^\alpha W_\alpha{}^\mu\right) - \left(\nabla_\beta u^\alpha\right)W_\alpha{}^\mu$ (with $W_\alpha{}^\mu = \nabla_\alpha u^\mu$). We then simplify the fifth line by using the fact that the first and third terms on the right-hand side are equal but opposite in sign after using the identity and relabeling dummy indices. We also use the fact that, since u^μ is the tangent along a geodesic trajectory, and the quantity in parentheses in the second term is zero.

The final line gives us the covariant acceleration of the separation vector, and is known as the *equation of geodesic deviation*. Notice what it says: the separation of two nearby geodesics "accelerates" at a rate proportional to this separation and to the Riemann curvature of the spacetime in which these geodesics "live."

11.2.2 LLF frame calculation

We now look at this calculation in a somewhat less formal manner, considering how the various terms develop if we examine them in a particularly local Lorentz frame. Consider two nearby geodesics, both parameterized by affine parameter λ: Geodesic 1 follows the curve $x^\alpha(\lambda)$, and has tangent vector $u^\alpha = dx^\alpha/d\lambda$; geodesic 2 follows the curve $z^\alpha(\lambda)$, and has tangent vector $v^\alpha = dz^\alpha/d\lambda$. Let $Y^\alpha = z^\alpha - x^\alpha$ be the vector that points from events at parameter λ on geodesic 1 to events at λ on geodesic 2.

We expand all important elements of the geometry in a LLF centered on event A at $\lambda = \lambda_0$ on geodesic 1. Let A' be the corresponding event on geodesic 2. We assume that A and A' are sufficiently close to one another that their tangent vectors are parallel at $\lambda = \lambda_0$; we elaborate on what this means quantitatively below. Other important quantities we need are the metric and the connection at these two events:

$$g_{\mu\nu}\big|_A = \eta_{\mu\nu} \ , \qquad g_{\mu\nu}\big|_{A'} = \eta_{\mu\nu} \ ; \qquad \Gamma^\mu{}_{\alpha\beta}\big|_A = 0 \ , \qquad \Gamma^\mu{}_{\alpha\beta}\big|_{A'} = Y^\gamma\left(\partial_\gamma\Gamma^\mu{}_{\alpha\beta}\right)\big|_A \ . \tag{11.11}$$

Notice that the connection vanishes at event A, but it does *not* vanish at A'. The value given there is the leading correction, found by Taylor-expanding over the displacement Y^α.

The geodesic equation for geodesic 1, evaluated at event A, is

$$\left.\frac{d^2x^\mu}{d\lambda^2}\right|_A = 0 \ . \tag{11.12}$$

The corresponding equation for geodesic 2, evaluated at event A', is

$$\left.\frac{d^2z^\mu}{d\lambda^2}\right|_{A'} + \left(\Gamma^\mu{}_{\alpha\beta}v^\alpha v^\beta\right)\big|_{A'} = 0 \ . \tag{11.13}$$

Combining our different definitions and assumptions, Eq. (11.13) can be rewritten

$$\left.\frac{d^2 z^\mu}{d\lambda^2}\right|_{A'} = -Y^\gamma \left(\partial_\gamma \Gamma^\mu{}_{\alpha\beta} u^\alpha u^\beta\right)\big|_A \ . \tag{11.14}$$

Here we used $v^\alpha = u^\alpha$, which follows from our requirement that the tangents to geodesics 1 and 2 be parallel at $\lambda = \lambda_0$. This requires us to parallel transport from A' to A, but this transport is trivial[2] in the LLF.

Taking the difference between the geodesic equation on geodesic 2 and on geodesic 1, we find

$$\frac{d^2 Y^\mu}{d\lambda^2} = -\partial_\gamma \Gamma^\mu{}_{\alpha\beta} u^\alpha u^\beta Y^\gamma \ . \tag{11.15}$$

We drop the A and A' subscripts from here on; all quantities are now evaluated at A.

This acceleration equation uses $d/d\lambda = u^\alpha \partial_\alpha$ as its derivative operator. To write this in proper tensorial form, we need to express things in terms of $D/d\lambda \equiv u^\alpha \nabla_\alpha$. Begin by looking at the "velocity" equation for the separation vector:

$$\frac{DY^\mu}{d\lambda} = u^\alpha \nabla_\alpha Y^\mu = \frac{dY^\mu}{d\lambda} + \Gamma^\mu{}_{\alpha\beta} Y^\alpha u^\beta \ . \tag{11.16}$$

It is tempting to set the connection to zero since we are in the LLF. However, we need to take one more derivative to make an acceleration equation, and so we will wait until we have taken all derivatives before zeroing our connections. The fact that the connection has non-zero slope in the LLF is important.

Taking the next derivative, we construct our acceleration for the separation vector:

$$\begin{aligned}
\frac{D^2 Y^\mu}{d\lambda^2} &= u^\gamma \nabla_\gamma \left(\frac{dY^\mu}{d\lambda} + \Gamma^\mu{}_{\alpha\beta} Y^\alpha u^\beta\right) \\
&= \frac{d^2 Y^\mu}{d\lambda^2} + u^\gamma \Gamma^\mu{}_{\gamma\nu} \frac{dY^\nu}{d\lambda} + \left(u^\gamma \nabla_\gamma \Gamma^\mu{}_{\alpha\beta}\right) u^\beta Y^\alpha + \Gamma^\mu{}_{\gamma\nu} \left(u^\gamma \nabla_\gamma u^\beta\right) Y^\alpha + \Gamma^\mu{}_{\alpha\beta} u^\beta \left(u^\gamma \nabla_\gamma Y^\mu\right) \\
&= \frac{d^2 Y^\mu}{d\lambda^2} + \partial_\gamma \Gamma^\mu{}_{\alpha\beta} u^\beta u^\gamma Y^\alpha + O(\Gamma^2) \ .
\end{aligned} \tag{11.17}$$

On the final line of this equation, we have gone into the LLF near event A, we have used the fact that u^α is a geodesic, and we used the fact that $dY^\mu/d\lambda = v^\mu - u^\mu = 0$.

Finally, use the previous result we found for $d^2 Y^\mu/d\lambda^2$ to write this

$$\begin{aligned}
\frac{D^2 Y^\mu}{d\lambda^2} &= \partial_\gamma \Gamma^\mu{}_{\alpha\beta} u^\beta u^\gamma Y^\alpha - \partial_\gamma \Gamma^\mu{}_{\alpha\beta} u^\alpha u^\beta Y^\gamma \\
&= \left(\partial_\gamma \Gamma^\mu{}_{\alpha\beta} - \partial_\alpha \Gamma^\mu{}_{\beta\gamma}\right) u^\beta u^\gamma Y^\alpha \\
&= R^\mu{}_{\beta\gamma\alpha} u^\beta u^\gamma Y^\alpha \ .
\end{aligned} \tag{11.18}$$

In going from the first to the second line, we relabeled dummy indices; to go to the third line, we identified that combination of connection derivatives as the Riemann curvature in the LLF. The final result is identical to that found in the formal calculation, modulo a few different choices of dummy index labels.

This second version of the calculation highlights how it is that the second order terms in the metric (via the first order correction to the connection) cause nearby geodesics to deviate from one another. This is the hallmark of a *tidal* effect: the geodesics deviate because "free fall" means slightly different things to freely falling observers that are slightly separated from one another.

[2] At least to linear order in important quantities; if you do this transport and keep all terms, you will find an additional term of order Γ^2. We discard this term at the end of the calculation by our assumption that we are in the LLF.

11.3 A differential curvature rule: The Bianchi identity

Our final identity is one involving derivatives of the Riemann tensor. To derive it, recall the identity involving the commutator of covariant derivatives:

$$[\nabla_\lambda, \nabla_\sigma] p_\alpha = -R^\mu{}_{\alpha\lambda\sigma} p_\mu \ . \tag{11.19}$$

It is not hard to show that this generalizes to a higher rank tensor:

$$[\nabla_\lambda, \nabla_\sigma] M_{\alpha\beta} = -R^\mu{}_{\alpha\lambda\sigma} M_{\mu\beta} - R^\mu{}_{\beta\lambda\sigma} M_{\alpha\mu} \ . \tag{11.20}$$

With this established, let us examine two relations. The first is

$$\boxed{[\nabla_\alpha, \nabla_\beta] \nabla_\gamma p_\delta = -R^\mu{}_{\gamma\alpha\beta} \nabla_\mu p_\delta - R^\mu{}_{\delta\alpha\beta} \nabla_\gamma p_\mu} \tag{11.21}$$

The second requires some manipulation:

$$\begin{aligned} \nabla_\alpha [\nabla_\beta, \nabla_\gamma] p_\delta &= \nabla_\alpha \left(-R^\mu{}_{\delta\beta\gamma} p_\mu \right) \\ &= -p_\mu \nabla_\alpha R^\mu{}_{\delta\beta\gamma} - R^\mu{}_{\delta\beta\gamma} \nabla_\alpha p_\mu \\ &= -p^\mu \nabla_\alpha R_{\mu\delta\beta\gamma} - R^\mu{}_{\delta\beta\gamma} \nabla_\alpha p_\mu \ . \end{aligned} \tag{11.22}$$

On the final line, used the fact that the metric commutes with the covariant derivative to raise the index μ on the 1-form component and simultaneously lower it on the Riemann curvature. We finally use Riemann symmetry to swap some of the indices around on the first term on the right-hand side to write this equation

$$\boxed{\nabla_\alpha [\nabla_\beta, \nabla_\gamma] p_\delta = -p^\mu \nabla_\alpha R_{\beta\gamma\mu\delta} - R^\mu{}_{\delta\beta\gamma} \nabla_\alpha p_\mu} \tag{11.23}$$

Now, let's look at what happens when we antisymmetrize these equations on the indices α, β, and γ. Begin with the left-hand side of Eq. (11.20):

$$\begin{aligned} [\nabla_{[\alpha}, \nabla_\beta] \nabla_{\gamma]} p_\delta &= \frac{1}{3!} \Big([\nabla_\alpha, \nabla_\beta] \nabla_\gamma + [\nabla_\beta, \nabla_\gamma] \nabla_\alpha + [\nabla_\gamma, \nabla_\alpha] \nabla_\beta \\ &\qquad - [\nabla_\alpha, \nabla_\gamma] \nabla_\beta - [\nabla_\gamma, \nabla_\beta] \nabla_\alpha - [\nabla_\beta, \nabla_\alpha] \nabla_\gamma \Big) p_\delta \ . \end{aligned} \tag{11.24}$$

Expanding every one of these commutators, one finds that this can be written equivalently as

$$\begin{aligned} [\nabla_{[\alpha}, \nabla_\beta] \nabla_{\gamma]} p_\delta &= \frac{1}{3!} \Big(\nabla_\alpha [\nabla_\beta, \nabla_\gamma] + \nabla_\beta [\nabla_\gamma, \nabla_\alpha] + \nabla_\gamma [\nabla_\alpha, \nabla_\beta] \\ &\qquad - \nabla_\alpha [\nabla_\gamma, \nabla_\beta] - \nabla_\gamma [\nabla_\beta, \nabla_\alpha] - \nabla_\beta [\nabla_\alpha, \nabla_\gamma] \Big) p_\delta \\ &= \nabla_{[\alpha} [\nabla_\beta, \nabla_{\gamma]}] p_\delta \ . \end{aligned} \tag{11.25}$$

This is exactly the left-hand side of Eq. (11.22), antisymmetrized on these three indices. In other words, **antisymmetrizing the left-hand side of Eq. (11.20) on α, β, and γ is exactly the same as antisymmetrizing the left-hand side of Eq. (11.22).**

If this antisymmetrization makes the left-hand sides equal, then it must make the right-hand sides equal too. Let's look at what we get when we antisymmetrize the right-hand sides of (11.20) and (11.22) on these three indices and equate them to each other:

$$R^{\mu}{}_{[\gamma\alpha\beta]}\nabla_{\mu}p_{\delta} + R^{\mu}{}_{\delta[\alpha\beta}\nabla_{\gamma]}p_{\mu} = p^{\mu}\nabla_{[\alpha}R_{\beta\gamma]\mu\delta} + R^{\mu}{}_{\delta[\beta\gamma}\nabla_{\alpha]}p_{\mu} \ . \tag{11.26}$$

The first term on the right-hand side of (11.25) vanishes by virtue of the Riemann symmetry $R^{\mu}{}_{[\gamma\alpha\beta]} = 0$. The second term on the right-hand side of (11.25) is identical to the second term on the left-hand side of (11.25); some of the indices are ordered slightly differently, but because they are being antisymmetrized, we can cyclically permute them without changing the equation. These two terms thus cancel out. What remains is

$$p^{\mu}\nabla_{[\alpha}R_{\beta\gamma]\mu\delta} = 0 \ . \tag{11.27}$$

This must hold for any p^{μ}, so we conclude that

$$\boxed{\boxed{\nabla_{[\alpha}R_{\beta\gamma]\mu\delta} = 0}} \tag{11.28}$$

Using the antisymmetry on the first two indices, this can be written in the equivalent form

$$\boxed{\boxed{\nabla_{\alpha}R_{\beta\gamma\mu\delta} + \nabla_{\beta}R_{\gamma\alpha\mu\delta} + \nabla_{\gamma}R_{\alpha\beta\mu\delta} = 0}} \tag{11.29}$$

Equations (11.27) and (11.28) are both known as the *Bianchi identity*. The Bianchi identity is the last major ingredient that we need to turn our theory of spacetime curvature into a fully fledged theory of gravity.

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Lecture 12
Making a theory of gravity

12.1 Final prep: Revisiting the Bianchi identity

In the previous lecture, we derived the Bianchi identity for the Riemann tensor:

$$\nabla_\alpha R_{\beta\gamma\mu\nu} + \nabla_\beta R_{\gamma\alpha\mu\nu} + \nabla_\gamma R_{\alpha\beta\mu\nu} = 0 \; . \tag{12.1}$$

Let's hit this equation with $g^{\beta\mu}$. Because the metric has no covariant derivative, it commutes with all of the derivatives, and we find the following variant of this identity:

$$\nabla_\alpha R_{\gamma\nu} + \nabla^\mu R_{\gamma\alpha\mu\nu} - \nabla_\gamma R_{\alpha\nu} = 0 \; . \tag{12.2}$$

(Why the minus sign on the final term? The contraction is on indices 2 and 3, rather than indices 1 and 3 for the "standard" contraction from Riemann to Ricci. We thus get a minus sign from the associated Riemann symmetry.) Let's do this again, now using $g^{\gamma\nu}$:

$$\nabla_\alpha R - \nabla^\mu R_{\alpha\mu} - \nabla^\nu R_{\alpha\nu} = 0 \; . \tag{12.3}$$

Recognizing that μ and ν are dummy indices in those two terms and that $\nabla_\alpha = \nabla^\mu g_{\alpha\mu}$, we can reorganize this as

$$\nabla^\mu \left(R_{\alpha\mu} - \frac{1}{2} g_{\alpha\mu} R \right) = 0 \; . \tag{12.4}$$

The quantity in parenthesis is the as-yet-unnamed tensor $G_{\alpha\mu}$ which we briefly noted in the previous lecture as the "trace-reversed" Ricci curvature. We now see that it has a further interesting property, namely that its divergence vanishes:

$$\nabla^\mu G_{\mu\alpha} = 0 \; . \tag{12.5}$$

(Using its symmetry to swap indices around.) Divergence-free quantities tend to show up when we connect a form of mathematical analysis to physics. In anticipation of the possibility that this tensor might prove useful for us, we will call it the *Einstein curvature* tensor.

12.2 Making a theory of gravity: Guiding principles

We are now ready to put all of the ingredients we have developed over the previous 11 lectures together to make a relativistic theory of gravity. Two main ingredients go into this: first, using the principle of equivalence (in the form of the Einstein equivalence principle) to express laws of physics (particularly laws of motion) in a form appropriate to physics in a curved spacetime; and second, some sort of field equation that connects spacetime to sources of matter and energy.

We have already had some practice with using the equivalence principle. The key idea, sometimes called the "minimal coupling[1]" principle, is to begin with a law of physics that is valid in inertial coordinates in flat spacetime; to then use that form of the law in a local Lorentz frame in which, by the Einstein equivalence principle, it should hold as well; and then to rewrite the law in a covariant, tensorial form. We finally assert that this law holds in curved spacetime. An example

[1] In contrast to "curvature coupling," which can modify these laws when the physical process under discussion extends over a large enough region that processes couple to the spacetime's curvature tensors.

of this is how we describe force-free motion: in a freely falling frame, bodies that move unaffected by non-gravitational forces follow a trajectory describe by

$$\frac{d^2x^\mu}{d\tau^2} = 0 \,, \tag{12.6}$$

a "straight" line in the inertial coordinates. A more general statement of such motion is that the tangent vector is parallel transported along the trajectory, so the fully covariant formulation of becomes

$$u^\alpha \nabla_\alpha u^\beta = 0 \,. \tag{12.7}$$

Equations (12.6) and (12.7) are identical in a LLF. However, the second form is written in a fully tensorial, covariant form, so we assert that it describes motion more generally.

A second example, which will be quite relevant very shortly, is our law of local energy and momentum conservation. In special relativity and in inertial coordinates, we write this law

$$\partial_\mu T^{\mu\nu} = 0 \,. \tag{12.8}$$

To make this tensorial, we simply need to "upgrade" the derivative from partial to covariant:

$$\nabla_\mu T^{\mu\nu} = 0 \,. \tag{12.9}$$

12.3 The Newtonian limit

A principle we must hold in mind over all of these ideas is that the theory of gravity we develop must be compatible with a Newtonian limit. Kepler's laws, for example, are extremely well tested, and empirically describe the motion of bodies in our solar system to extraordinarily good accuracy. At least for non-relativistic motion, we know that Newtonian gravity works very well; our relativistic theory of gravity must incorporate this limit.

The equation of motion in Newtonian gravity is

$$\frac{d^2x^i}{dt^2} = -\nabla\Phi = -\delta^{ij}\partial_j\Phi \,. \tag{12.10}$$

This is very much *not* a covariant formulation (a specific time t and spatial coordinates x^i have been picked out), but our covariant equation of motion should reduce to this in an appropriate limit. We begin with the geodesic equation, which we write as a second-order equation for the trajectory $x^\mu(\tau)$ that a freely falling body follows:

$$\frac{d^2x^\mu}{d\tau^2} + \Gamma^\mu{}_{\alpha\beta}\frac{dx^\alpha}{d\tau}\frac{dx^\beta}{d\tau} = 0 \,. \tag{12.11}$$

The slow motion limit is defined by requiring that the timelike component of the 4-velocity be much larger than the spatial components:

$$\frac{dx^0}{d\tau} = \frac{dt}{d\tau} \gg \frac{dx^i}{d\tau} \qquad \text{for all } i. \tag{12.12}$$

(You should be able to convince yourself that this is equivalent to $v^i \ll c$ for all i.) With this simplification, we expect that the $\alpha = 0$, $\beta = 0$ indices contribute much more to the sum over dummy indices than any of the others and we write this

$$\frac{d^2x^\mu}{d\tau^2} + \Gamma^\mu{}_{00}\left(\frac{dt}{d\tau}\right)^2 = 0 \qquad \text{or} \qquad \frac{d^2x^\mu}{d\tau^2} = -\Gamma^\mu{}_{00}\left(\frac{dt}{d\tau}\right)^2 \,, \tag{12.13}$$

moving the Christoffel term to the right-hand side in anticipation of how we wish to present this result later.

The Christoffel symbol we must now compute is

$$\Gamma^{\mu}{}_{00} = \frac{1}{2} g^{\mu\nu} \left(\partial_t g_{\nu 0} + \partial_t g_{0\nu} - \partial_\nu g_{00} \right) \ . \tag{12.14}$$

To recover the Newtonian limit, we can take the metric to be time independent, yielding

$$\Gamma^{\mu}{}_{00} = -\frac{1}{2} g^{\mu\nu} \partial_\nu g_{00} \ . \tag{12.15}$$

We also imagine that the spacetime metric is *almost* that of special relativity, writing

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu} \ , \tag{12.16}$$

where all of the components of $h_{\mu\nu}$ are "small": all of its components are much less than 1. It's not hard to show that the inverse tensor corresponding to (12.16) is given by

$$g^{\mu\nu} = \eta^{\mu\nu} - h^{\mu\nu} + \mathcal{O}(h^2) \ , \tag{12.17}$$

where $h^{\mu\nu} = \eta^{\mu\alpha}\eta^{\nu\beta} h_{\alpha\beta}$. Since all the components of $h_{\mu\nu}$ are considered to be small, we will drop contributions $\mathcal{O}(h^2)$; this is why we raise indices on $h_{\mu\nu}$ with $\eta^{\mu\nu}$. Putting all this together yields for our Christoffel symbol

$$\Gamma^{\mu}{}_{00} = -\frac{1}{2} \eta^{\mu\nu} \partial_\nu h_{00} \ . \tag{12.18}$$

Our equation of motion then becomes

$$\frac{d^2 t}{d\tau^2} = 0 \qquad (\text{using } \partial_t h_{00} = 0) \ , \tag{12.19}$$

$$\frac{d^2 x^i}{d\tau^2} = \frac{1}{2} \delta^{ij} \partial_j h_{00} \left(\frac{dt}{d\tau} \right)^2 \qquad \text{or} \ ,$$

$$\frac{d^2 x^i}{dt^2} = \frac{1}{2} \delta^{ij} \partial_j h_{00} \ . \tag{12.20}$$

The equation for t tells us that in this limit a freely falling observer's proper time is the same as coordinate time, up to a linear transformation (i.e., change of origin, change of units). The equation governing x^i is the gravitational Newtonian equation of motion if

$$h_{00} = -2\Phi \qquad \text{or} \qquad g_{00} = -(1 + 2\Phi) \ . \tag{12.21}$$

We will hold this correspondence with the Newtonian limit in our heads as we develop a field equation for spacetime.

12.4 A field equation for relativistic gravity

The next, and arguably most crucial step of all, is to develop a field equation for spacetime: some kind of relation that tells us how much "gravity" arises per unit of "source." Based on what we've developed so far, our "gravity" must be upgraded from a Newtonian gravitational potential to something like the spacetime metric. But we need to think carefully about what to use for "source" in such a relationship.

As guidance, let's write down the field equation of Newtonian gravity:

$$\nabla^2 \Phi = \delta^{ij} \partial_i \partial_j \Phi = 4\pi G \rho \ . \tag{12.22}$$

There is a lot flawed about this equation from a relativistic perspective. Note that it is manifestly not covariant or tensorial: the right-hand side is ρ, which in Newtonian physics we have always taken to be mass density. In our $c = 1$ units, this is the same thing as energy density. Energy density, however, we now recognize as not a scalar, but rather as one component of the stress energy tensor: $\rho = T_{00}$. To make this tensorial, one hypothesis is that we should "upgrade" the right-hand side from something proportional to ρ to something proportional to $T_{\mu\nu}$.

On the left-hand side, we have spatial derivatives (not covariant: we've chosen time and space) acting on the Newtonian potential Φ. Our analysis of Newtonian motion shows us that the metric and Φ play similar roles as far as gravitational motion is concerned, so presumably we want something involving two derivatives of the metric. Heuristically, our proposed field equation has the structure

$$(\text{two derivatives of the spacetime metric}) = T_{\mu\nu} \ . \tag{12.23}$$

Two derivatives of the spacetime metric is what goes into making curvature tensors (along with a few nonlinear combinations with the metric, and of first derivatives of the metric). So a plausible hypothesis for the left-hand side is one of the curvature tensors that we developed. Clearly it cannot be the Riemann tensor — Riemann has too many indices, and too many independent components to equate to $T_{\mu\nu}$. Both Ricci and Einstein, however, fit. To choose which one, recall that the stress-energy tensor is divergence free: $\nabla^\mu T_{\mu\nu} = 0$. The tensor which goes on the left-hand side must be divergence free as well; this picks out the Einstein tensor as what we should put on the left-hand side:

$$G_{\mu\nu} = \kappa T_{\mu\nu} \ . \tag{12.24}$$

We have introduced a constant of proportionality κ because curvature and stress-energy have different dimensions; there must be some factor to have sure that their units are consistent. Our last task is to fix this constant.

One bit of sleight of hand helps to organize terms for this analysis. Recall that $G_{\mu\nu}$ is the "trace-reversed" Ricci tensor; you should be able to convince yourself that if you trace-reverse $G_{\mu\nu}$ you just get the Ricci tensor $R_{\mu\nu}$ back. We can trace-reverse any 2-index tensor by subtracting one half the metric times the trace of the tensor. Taking the trace reverse of both sides of Eq. (12.24) gives us a second way of writing our proposed field equation which will prove handy in the next calculation:

$$R_{\mu\nu} = \kappa \left(T_{\mu\nu} - \frac{1}{2} g_{\mu\nu} T \right) \ , \qquad \text{where} \qquad T = g^{\mu\nu} T_{\mu\nu} \ . \tag{12.25}$$

It should be emphasized that Eqs. (12.24) and (12.25) are essentially identical, differing only by a trace-reverse operation.

In order to fix the constant κ, we take everything to be static, and go to the weak-field limit in a particular coordinate representation which allows us to write $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$. The components of $h_{\mu\nu}$ are all taken to be small, meaning that we will discard any term of $\mathcal{O}(h^2)$. Let's take our source to be a static perfect fluid:

$$T_{\mu\nu} = (\rho + P) u_\mu u_\nu + P g_{\mu\nu} \simeq \rho u_\mu u_\nu \ . \tag{12.26}$$

We are using the idea that for a non-relativistic fluid, $\rho \gg P$. Bearing in mind that $\partial P / \partial \rho = c_s^2$, where c_s is the speed of sound, this is akin to saying that the speed of sound is much smaller than the speed of light, a reasonable condition to impose for the Newtonian limit.

Requiring the fluid to be static implies that $u^\mu \doteq (u^0, 0, 0, 0)$. To normalize the only non-zero velocity component, we enforce

$$g_{\mu\nu} u^\mu u^\nu = -1 \qquad \longrightarrow \qquad u^0 = 1 + \frac{1}{2} h_{00} + \mathcal{O}(h^2) \ . \tag{12.27}$$

Lowering the index so that we have the component we need in our stress-energy tensor yields

$$u_0 = g_{0\mu}u^\mu = (\eta_{00} + h_{00})u^0 = -1 + \frac{1}{2}h_{00} + \mathcal{O}(h^2) \; . \tag{12.28}$$

We now have enough pieces to generate at least one component of the proposed field equation: let's look at

$$R_{00} = \kappa \left(T_{00} - \frac{1}{2} g_{00} T \right) \; . \tag{12.29}$$

On the right-hand side, discarding terms of $\mathcal{O}(h^2)$, we have

$$T_{00} = \rho u_0 u_0 = \rho(1 - h_{00}) \; , \tag{12.30}$$

$$T = g^{\mu\nu} T_{\mu\nu} = \rho u^\mu u_\mu = -\rho \; , \tag{12.31}$$

so we find

$$T_{00} - \frac{1}{2} g_{\mu\nu} T = \frac{1}{2}\rho + \mathcal{O}(h) \; . \tag{12.32}$$

As we'll see, it is acceptable to truncate at this order on the right-hand side.

Let's now build the left-hand side:

$$R_{00} = R^\alpha{}_{0\alpha 0} = R^i{}_{0i0} \; . \tag{12.33}$$

We can reduce the sum over spacetime index α to a sum over spatial index i since $R^0{}_{000}$ is zero by Riemann symmetries. Inserting the definition of Riemann in terms of Christoffel derivatives and nonlinear Christoffel terms,

$$\begin{aligned} R_{00} &= \partial_i \Gamma^i{}_{00} - \partial_0 \Gamma^i{}_{0i} + \mathcal{O}(\Gamma^2) \\ &= \frac{1}{2}\partial_i \left[g^{i\mu} \left(\partial_0 g_{\mu 0} + \partial_0 g_{0\mu} - \partial_\mu g_{00} \right) \right] \\ &= -\frac{1}{2}\partial_i \left[\eta^{i\mu} \partial_\mu h_{00} \right] \\ &= -\frac{1}{2}\delta^{ij}\partial_i \partial_j h_{00} \\ &\equiv -\frac{1}{2}\nabla^2 h_{00} \; . \end{aligned} \tag{12.34}$$

In each step, we simplify by discarding terms that involve ∂_0 (nothing depends on time), by using the fact that derivatives of $g_{\mu\nu}$ become derivatives of $h_{\mu\nu}$ since $\eta_{\mu\nu}$ is a constant, and by keeping only terms that are linear in h.

Equating R_{00} to $T_{00} - (1/2)g_{00}T$, we find

$$\nabla^2 h_{00} = -\kappa\rho \; . \tag{12.35}$$

Recovering Newtonian free-fall motion in the limit requires that $h_{00} = -2\Phi$. Making this substitution, we see that this limit of our proposed field equation duplicates the Newtonian field equation if we put $\kappa = 8\pi G$. This allows us to at last write down a field equation for spacetime:

$$\boxed{\boxed{G_{\mu\nu} = 8\pi G T_{\mu\nu}}} \tag{12.36}$$

This is known as the *Einstein field equation*, often abbreviated as just "the Einstein equation" or "the Einstein equations," depending on whether you think of this as one tensor equation with 10 components, or 10 coupled equations[2].

[2]Note that if we had kept the $\mathcal{O}(h)$ term in Eq. (12.32), it would have changed the differential equation in such as way as to generate an $\mathcal{O}(h^2)$ correction to the solution. This is ignorable in a leading-order analysis; we will revisit issues like this when we look at solutions to this equation in more detail.

12.5 A few remarks on the Einstein field equation

The way we have derived the Einstein field equation here parallels how Einstein originally derived it: deduce that the source should be $T_{\mu\nu}$, infer that the metric plays the role of a potential in relativistic theory, and then use a divergence-free curvature as the quantity which is produced by this source. This is a rather ad hoc derivation. You might wonder — why do we make these particular choices? Could we not come up with other ways of combining tensors that is divergence free and covariant?

Indeed we can. In fact, one such modification is not hard to notice. Another divergence-free tensor is the metric itself (since *any* covariant derivative of the metric is zero). The metric has the wrong dimensions or units to be curvature, put perhaps we could introduce some scalar constant that corrects this. The result is a field equation of the form

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu} \ . \tag{12.37}$$

The constant scalar factor Λ introduced here is known as the *cosmological constant*. It's actually a matter of taste whether this term belongs on the left-hand side or the right-hand side of this field equation. On the left-hand side, it introduces a kind of mean curvature everywhere in spacetime. Moving it to the right-hand side, we can interpret it as a kind of stress-energy tensor:

$$T^{\Lambda}_{\mu\nu} = -\frac{\Lambda}{8\pi G} g_{\mu\nu} \ . \tag{12.38}$$

This can be thought of as a rather bizarre perfect fluid, one that has

$$\rho = \frac{\Lambda}{8\pi g} \ , \qquad P = -\rho \tag{12.39}$$

in a freely-falling frame. Such a stress-energy tensor arises in quantum field theory, and represents a form of vacuum energy, as it is isotropic and invariant to Lorentz transformations in the local Lorentz frame. We will set $\Lambda = 0$ in most of our analysis, though nonzero Λ is especially important and interesting in cosmology.

In the next lecture, we will re-derive (12.36) by working from an action principle. In doing so, we will see that Eq. (12.36) is, in a very well-defined way, the *simplest* theory of gravity that is compatible with general covariance and relativity. A particularly important aspect of the action approach will be that it shows how one can introduce modifications to our theory of gravity in a way that insures consistency with general covariance and relativity. One then wonders — which, if any, of such modifications is "right"? Ultimately that is a question that must be addressed by experiment and observation.

12.6 A word on units

Moving forward, we will use Newton's gravitational constant G to help keep track of gravitational terms. However, it is very common to set both G and c to 1 in many calculations[3]. When we set G and c to 1, we find that mass, length, and time are all measured in the same[4] units.

A useful way to keep terms straight in your head is to note that factors of G and c can be used to convert between different kinds of units. For example, the combination G/c^2 converts "normal" mass units to a length. A useful conversion factor in many astrophysical applications is

$$\frac{GM_\odot}{c^2} = 1.476\,\mathrm{km} \ , \tag{12.40}$$

[3]This in fact is the default in much of my research, so please watch in case I drop factors of G — it's so ingrained that I tend to forget once I've got some momentum at the blackboard.

[4]This contrasts with the system often used in quantum field theory: setting $\hbar$ and c to 1, we find that the dimensions of mass (and energy) are the *inverse* of the dimensions of length and time.

where $M_\odot \simeq 1.989 \times 10^{30}$ kg is the mass of the Sun. Another factor of c in the denominator converts masses to times:

$$\frac{GM_\odot}{c^3} = 4.924 \times 10^{-6}\,\mathrm{sec}\ . \tag{12.41}$$

When working in a problem dominated by the gravitational dynamics of a body with mass M, one often finds that important lengths in the problem scale with GM, or GM/c^2; important times in the problem scale with GM, or GM/c^3. Frequencies associated with gravitational dynamics tend to scale as $1/M$, or c^3/GM (a factor that comes up a lot when thinking about gravitational-wave sources, for instance).

The factor G/c^4 converts energy to length. This is an important one because the stress-energy tensor's components have the dimensions of energy density, or energy per length cubed. This means that $GT_{\mu\nu}/c^4$ has dimensions (length)$^{-2}$, which is exactly the dimensions we expect for *curvature*. The factor κ we introduced when developing the Einstein field equation is more completely written

$$G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}\ . \tag{12.42}$$

Notice that, in "normal" units, this factor is very small: $G/c^4 = 8.26 \times 10^{-45}\,\mathrm{m/J}$ — it takes a *lot* of energy density to generate spacetime curvature. This is another way of expressing the fact that gravity is the weakest of the fundamental forces of nature. But, because mass and energy only have one sign, gravity adds up and can have a significant effect.

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Lecture 13
Another approach to a theory of gravity

13.1 Field theories from Lagrangians

In the previous lecture, by assembling the various ingredients and tools we developed over the previous lectures and by insisting that we recover Newtonian gravity in an appropriate limit, we developed a field equation that relates the geometry of spacetime to the energy and momentum content of spacetime. This Einstein field equation is the core of Einstein's general theory of relativity; its presentation[1] is often taken to be the starting point of this subject. With this field equation in hand, we are ready to start developing solutions which describe gravity according to the principles we have been developing all semester, and to begin exploring how relativistic gravity differs from the Newtonian description.

Before doing so, it is useful to confront the fact that the derivation we used — though historically quite similar to how general relativity was first presented to the world — is somewhat dissatisfying. As we noted in the previous lecture, one could imagine other combinations of curvature tensors which reproduce the Newtonian limit and which satisfy the various principles which underlie relativistic physics. How do we choose?

As physicists, measurement and observation are the ultimate arbiters of how we assemble things. However, as *theoretical* physicists, we seek ways of organizing our model building to guide us toward plausible ways of understanding how spacetime *might* behave, subject to the various ideas or constraints we wish to put on the theory. The track we will follow in this lecture is one that Einstein's mathematician colleague David Hilbert originally developed, and published at nearly the same time in 1915 as Einstein published the field equations (though he always credited Einstein for the essential ideas that went into this analysis; they were in close contact during the years that general relativity was formulated). This approach is based on developing a Lagrangian density which describes gravitation.

Such an approach is now standard in much of field theory. The basic idea is to imagine that there is some function $\hat{\mathcal{L}}$ which characterizes the fields you wish to study, and that an action can be found by integrating this over all of spacetime:

$$S = \int d^4x \, \hat{\mathcal{L}} \, . \tag{13.1}$$

Since the action must be a Lorentz scalar, is not uncommon to write this

$$S = \int d^4x \sqrt{-g} \, \mathcal{L} \, , \tag{13.2}$$

including the factor $\sqrt{-g}$ needed to make a proper volume element in a spacetime $g_{\mu\nu}$. We require that this action be stationary with respect to variations in the fields, so schematically we require

$$\delta S = \int d^4x \left[\frac{\partial \hat{\mathcal{L}}}{\partial(\text{fields})} \right] \delta(\text{fields}) = 0 \, . \tag{13.3}$$

[1] A. Einstein, "Die Feldgleichungen der Gravitation," Sitzungsberichte der Preussischen Akademie der Wissenschaften, 844 (1915).

Enforcing

$$\frac{\partial \hat{\mathcal{L}}}{\partial(\text{fields})} = 0 \tag{13.4}$$

then yields Euler-Lagrange equations that describe the various fields that go into our theory.

To remind ourselves of how this works, let's imagine that $\mathcal{L}$ depends on scalar fields Φ^a and their derivatives $\partial_\mu \Phi^a$ in flat spacetime $\eta_{\mu\nu}$ (so that $\sqrt{-g} = 1$, and there is no distinction between $\mathcal{L}$ and $\hat{\mathcal{L}}$). Note that the superscript "a" just labels different fields, and is not intended to be a spacetime index. We put $\mathcal{L} = \mathcal{L}(\Phi^a, \partial_\mu \Phi^a)$, and in our variation we will treat the field and its derivatives as separate degrees of freedom.

We put $\Phi^a \to \Phi^a + \delta\Phi^a$, $\partial_\mu \Phi^a \to \partial_\mu \Phi^a + \partial_\mu(\delta\Phi^a)$, and so our variation in S becomes

$$\begin{aligned} \delta S &= \int d^4x \left[\frac{\partial \mathcal{L}}{\partial \Phi^a} \delta\Phi^a + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \Phi^a)} \partial_\mu(\delta\Phi^a) \right] \\ &= \int d^4x \left[\frac{\partial \mathcal{L}}{\partial \Phi^a} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \Phi^a)} \right) \right] \delta\Phi^a \ . \end{aligned} \tag{13.5}$$

On that last line, we have integrated by parts and assumed that there are no boundary terms, or that any such terms can be discarded. If the integral is over all of spacetime and we are working in the flat manifold of special relativity, then this is reasonable. In other circumstances, there may be subtle details that make this rather tricky; we will revisit this issue further below.

Requiring $\delta S = 0$, this analysis yields Euler-Lagrange equations for the field:

$$\frac{\partial \mathcal{L}}{\partial \Phi^a} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \Phi^a)} \right) = 0 \ . \tag{13.6}$$

Further progress requires that we use some Lagrangian density for our fields (often just called a "field Lagrangian" or "Lagrangian"). A common example is the massive scalar field: focusing on a single field so we can drop the label a, we write down a Lagrangian with a kinetic term and a potential term as follows:

$$\mathcal{L} = -\frac{1}{2} \eta^{\mu\nu} \left(\partial_\mu \Phi\right) \left(\partial_\nu \Phi\right) - \frac{1}{2} m^2 \Phi^2 \ . \tag{13.7}$$

The derivatives which go into the Euler-Lagrange equation are

$$\frac{\partial \mathcal{L}}{\partial \Phi} = -m^2 \Phi \ , \qquad \frac{\partial \mathcal{L}}{\partial(\partial_\mu \Phi)} = -\eta^{\mu\nu} \partial_\nu \Phi \ . \tag{13.8}$$

Plugging these in, the Euler-Lagrange equation yields

$$\Box \Phi - m^2 \Phi = 0 \ , \tag{13.9}$$

where $\Box = \eta^{\mu\nu} \partial_\mu \partial_\nu$ is the flat spacetime wave operator. This is simply the Klein-Gordon equation for a massive scalar field.

13.2 Constructing and varying an action for gravity: Version 1

We would like to develop a similar Lagrangian that is appropriate for a relativistic theory of gravity. Two main points guide our thinking:

- The action must be a scalar, so the Lagrangian must also be a scalar.
- To make sure that the equivalence principle is respected, this Lagrangian must be built out of curvature scalars so that it is not something that can be eliminated at any point simply by changing representation.

The suggests that the simplest possible action we can write down is

$$S = \frac{1}{16\pi G}\int d^4x\sqrt{-g}R = \frac{1}{16\pi G}\int d^4x\sqrt{-g}g^{\alpha\beta}R_{\alpha\beta}\ . \tag{13.10}$$

The factor of $1/G$ (which is c^4/G in "normal" units) insures that this final result has dimensions of action. The numerical factor, $1/16\pi$, is inserted more or less magically for later convenience. (We should really treat it as some undetermined number, and determine it later; I am skipping over these details because we've already done a version of this dance once.)

We will vary this action with respect to the metric. It turns out to be most convenient to do so using the inverse metric, $g^{\alpha\beta}$; we thus seek to compute

$$\delta S = \frac{1}{16\pi G}\int d^4x\frac{\partial}{\partial g^{\alpha\beta}}\left[\sqrt{-g}g^{\alpha\beta}R_{\alpha\beta}\right]\delta g^{\alpha\beta}\ . \tag{13.11}$$

Let us examine variations in that integrand:

$$\delta\left(\sqrt{-g}g^{\alpha\beta}R_{\alpha\beta}\right) = \left(\delta\sqrt{-g}\right)R + \sqrt{-g}\,\delta g^{\alpha\beta}\,R_{\alpha\beta} + \sqrt{-g}g^{\alpha\beta}\,\delta R_{\alpha\beta}\ . \tag{13.12}$$

The first term in (13.12) can be found using some of the tricks associated with the determinant of the metric we discussed in an earlier lecture; we find that

$$\delta\sqrt{-g} = -\frac{1}{2}\sqrt{-g}g_{\alpha\beta}\delta g^{\alpha\beta} \tag{13.13}$$

and so

$$\left(\delta\sqrt{-g}\right)R = -\frac{1}{2}\sqrt{-g}g_{\alpha\beta}\,R\,\delta g^{\alpha\beta}\ . \tag{13.14}$$

The second term of (13.12) is already in a form that is proportional to $\delta g^{\alpha\beta}$, and so requires no further massaging.

The third term, however, requires some work. To find the variation in the Ricci tensor, begin by looking at variations in Riemann:

$$\delta R^{\mu}{}_{\alpha\nu\beta} = \partial_\nu\delta\Gamma^{\mu}{}_{\beta\alpha} - \partial_\beta\delta\Gamma^{\mu}{}_{\nu\alpha} + \delta\Gamma^{\mu}{}_{\nu\gamma}\Gamma^{\gamma}{}_{\beta\alpha} + \Gamma^{\mu}{}_{\nu\gamma}\delta\Gamma^{\gamma}{}_{\beta\alpha} - \delta\Gamma^{\mu}{}_{\beta\gamma}\Gamma^{\gamma}{}_{\nu\alpha} - \Gamma^{\mu}{}_{\beta\gamma}\delta\Gamma^{\gamma}{}_{\nu\alpha}\ . \tag{13.15}$$

Although Christoffel symbols are not tensors, *differences* in Christoffel symbols are tensors: the pieces of the transformation rule that break the pattern needed to define a quantity as a tensor cancel out when we look at such a quantity. This means that $\delta\Gamma^{\mu}{}_{\beta\alpha}$ is tensorial, and we can sensibly define its covariant derivative:

$$\nabla_\nu\delta\Gamma^{\mu}{}_{\beta\alpha} = \partial_\nu\delta\Gamma^{\mu}{}_{\beta\alpha} + \delta\Gamma^{\gamma}{}_{\beta\alpha}\Gamma^{\mu}{}_{\nu\gamma} - \delta\Gamma^{\mu}{}_{\gamma\alpha}\Gamma^{\gamma}{}_{\nu\beta} - \delta\Gamma^{\mu}{}_{\beta\gamma}\Gamma^{\gamma}{}_{\nu\alpha}\ . \tag{13.16}$$

Using this expression, we can rewrite (13.15) using covariant derivatives rather than partial derivatives. Doing so, a miracle occurs: all the terms of the form $\delta\Gamma\,\Gamma$ cancel out, and we are left with

$$\delta R^{\mu}{}_{\alpha\nu\beta} = \nabla_\nu\delta\Gamma^{\mu}{}_{\beta\alpha} - \nabla_\beta\delta\Gamma^{\mu}{}_{\nu\alpha}\ . \tag{13.17}$$

Contracting on indices 1 and 3, we get the variation in Ricci:

$$\delta R_{\alpha\beta} = \nabla_\mu\delta\Gamma^{\mu}{}_{\beta\alpha} - \nabla_\beta\delta\Gamma^{\mu}{}_{\mu\alpha}\ . \tag{13.18}$$

We are not done: we still need to examine the variation in the Christoffel symbol. With some effort, one can show that

$$\delta\Gamma^{\mu}{}_{\alpha\beta} = \frac{1}{2}\left[\nabla_\gamma\left(g_{\alpha\nu}g_{\beta\lambda}g^{\mu\gamma}\delta g^{\nu\lambda}\right) - \nabla_\alpha\left(g_{\beta\gamma}\delta g^{\mu\gamma}\right) - \nabla_\beta\left(g_{\alpha\gamma}\delta g^{\mu\gamma}\right)\right]\ . \tag{13.19}$$

(Note that, although $\nabla_\alpha g_{\mu\nu} = 0$, we cannot assume $\nabla_\alpha \delta g^{\mu\nu} = 0$.) Using this, one can finally assemble

$$g^{\alpha\beta}\delta R_{\alpha\beta} = \nabla_\alpha \nabla_\beta \left(g^{\alpha\beta} g_{\mu\nu} \delta g^{\mu\nu} - \delta g^{\alpha\beta} \right) \equiv \nabla_\alpha v^\alpha \ . \tag{13.20}$$

This final form recognizes that the expression we at last landed on can be thought of as the covariant divergence of a vector field v^α whose form can be read out of Eq. (13.20).

Putting these three terms together, the variational principle we seek to enforce takes the form

$$\begin{aligned} \delta S &= \frac{1}{16\pi G} \int d^4x \sqrt{-g} \left[\left(R_{\alpha\beta} - \frac{1}{2} g_{\alpha\beta} R \right) \delta g^{\alpha\beta} + \nabla_\alpha v^\alpha \right] \\ &= \frac{1}{16\pi G} \int d^4x \sqrt{-g} \left[G_{\alpha\beta} \delta g^{\alpha\beta} + \nabla_\alpha v^\alpha \right] \ . \end{aligned} \tag{13.21}$$

The first term inside the square brackets looks pretty much just like what we want. The second term does not. Because this term is in the form of a divergence, one can imagine invoking the divergence theorem and converting it into a boundary term. We presumably could then discard this term by requiring the variation to vanish on the boundary.

For now, this argument is fine; however, it is at best somewhat glib. We can set variations in the metric to zero at the boundary, but we cannot set both variations of the metric *and* its derivative to zero — variations in the metric's derivative are not part of our variational scheme. To handle this term rigorously, we need to examine boundary regions in our analysis very carefully. This requires that we carefully define how to compute the contribution to curvature that arises for a 3-dimensional "slice" of spacetime carved out of our 4-dimensional manifold. (Or, we need to modify our variational scheme; such a modification is introduced later in these lecture notes.)

Students who wish to examine how to handle the boundary in (much) more detail are invited to carefully examine the discussion in Appendix E of Wald's textbook. For our purposes, we take the glib assertion that we can discard the $\nabla_\alpha v^\alpha$ term to be good enough[2] for the purposes of 8.962. Enforcing $\delta S = 0$, Eq. (13.21), then yields $G_{\alpha\beta} = 0$ — the Einstein field equation with no source.

13.3 Putting in a source

More generally, we can write our action for everything — matter, fields, spacetime — in the form

$$S = \int d^4x \sqrt{-g} \left(\frac{R}{16\pi G} + \mathcal{L}_{M/F} \right) \ , \tag{13.22}$$

where $\mathcal{L}_{M/F}$ is a Lagrangian for all matter or fields that live in spacetime. We then define the stress-energy tensor to be

$$T_{\alpha\beta} = -\frac{2}{\sqrt{-g}} \frac{\partial(\sqrt{-g}\mathcal{L}_{M/F})}{\partial g^{\alpha\beta}} \ . \tag{13.23}$$

The somewhat funny factors of -2 and $\sqrt{-g}$ are included to account for the terms which arise from varying the term which gives us a properly tensorial volume element. At this stage, we recognize the factor $1/16\pi$ that we included earlier as what was necessary in order for our field equation to have the correct Newtonian limit.

Carroll works out an example of how this variational principle duplicates the stress-energy tensor for a scalar field; see discussion from Eq. (4.76) until the end of Sec. 4.3 in his textbook. We work through another illustrative example here: consider

$$\mathcal{L}_{M/F} = -\frac{1}{4} F^{\mu\nu} F_{\mu\nu} \ . \tag{13.24}$$

[2] Vladimir Braginsky (`https://en.wikipedia.org/wiki/Vladimir_Braginsky`) used to attend meetings of the group in which I was a grad student, and once memorably advised me in a wonderfully thick Russian accent that "Perfect is enemy of Good Enough."

This describes the electromagnetic field. Put it inside the action integral, and write it as

$$S = \int d^4x\sqrt{-g}\, g^{\alpha\mu} g^{\beta\nu}\, F_{\alpha\beta} F_{\mu\nu} \ . \tag{13.25}$$

We now vary the spacetime metric, but leave the term $F_{\alpha\beta}F_{\mu\nu}$ unvaried. The variation we need is

$$\delta\left[\sqrt{-g}\, g^{\alpha\mu} g^{\beta\nu}\right] = -\frac{1}{2}\sqrt{-g}(g_{\sigma\rho}\delta g^{\sigma\rho}) g^{\alpha\mu} g^{\beta\nu} + \sqrt{-g} g^{\alpha\mu}\delta g^{\beta\nu} + \sqrt{-g} g^{\beta\nu}\delta g^{\alpha\mu} \ . \tag{13.26}$$

Applying this variation to the action integral, we have

$$\delta S = -\frac{1}{4}\int d^4x\sqrt{-g}\left[F_{\alpha\beta}F^{\alpha}{}_{\nu}\,\delta g^{\beta\nu} + F_{\alpha\beta}F_{\mu}{}^{\beta}\,\delta g^{\alpha\mu} - \frac{1}{2}F^{\mu\nu}F_{\mu\nu} g_{\sigma\rho}\delta g^{\sigma\rho}\right] \ . \tag{13.27}$$

We cycle dummy indices in each term in order to pull out a common factor of $\delta g^{\alpha\beta}$ and to facilitate combining some of the terms under the integrand: in the first term in square brackets under the integrand of (13.27), we take $\alpha \to \mu$, $\beta \to \alpha$, $\nu \to \beta$; in the second term, we take $\mu \to \beta$, $\beta \to \mu$; and in the third, we take $\sigma \to \alpha$, $\rho \to \beta$. This yields

$$\delta S = -\frac{1}{4}\int d^4x\sqrt{-g}\left[F_{\mu\alpha}F^{\mu}{}_{\beta} + F_{\alpha\mu}F_{\beta}{}^{\mu} - \frac{1}{2}g_{\alpha\beta}F_{\mu\nu}F^{\mu\nu}\right]\delta g^{\alpha\beta} \ . \tag{13.28}$$

Finally using the antisymmetry of the electromagnetic field tensor, we write this

$$\delta S = -\frac{1}{2}\int d^4x\sqrt{-g}\left[F_{\mu\alpha}F^{\mu}{}_{\beta} - \frac{1}{4}g_{\alpha\beta}F_{\mu\nu}F^{\mu\nu}\right]\delta g^{\alpha\beta} \equiv -\frac{1}{2}\int d^4x\sqrt{-g}\left[T^{\rm EM}_{\alpha\beta}\right]\delta g^{\alpha\beta} \ . \tag{13.29}$$

On the final line, we used the definition (13.23) to recognize that the quantity in square brackets in Eq. (13.29) is the electromagnetic stress-energy tensor.

13.4 Constructing and varying an action for gravity: Version 2

One of the reasons that the boundary term got us into difficulty in the variation we studied above is that we imagine the "field" that we vary to be the metric of spacetime. Doing so, we can impose a requirement that it not vary on the boundary of some region. This requirement does not constrain derivatives of the metric.

What if we treat the metric and its derivatives as independent degrees of freedom, akin to the variation of the scalar field that we examined at the beginning of these notes? More concretely, imagine that S depends *separately* on the metric and on the connection, and that we *separately* vary these two quantities. We write

$$S = \frac{1}{16\pi G}\int d^4x\sqrt{-g}\, R(\Gamma) = \frac{1}{16\pi G}\int d^4x\sqrt{-g} g^{\alpha\beta}\, R_{\alpha\beta}(\Gamma) \ . \tag{13.30}$$

The notation $R(\Gamma)$, $R_{\alpha\beta}(\Gamma)$ indicates somewhat schematically that the Ricci tensor and Ricci scalar are functions of the connection. We now imagine that the tensor $R_{\alpha\beta}$ *does not* change as we vary the metric. We also do not assume that the metric and the covariant derivative are necessarily compatible, i.e. we do not assume that $\nabla_\alpha g_{\mu\nu} = 0$. That would assume a particular value for the connection, and we wish to leave the connection unspecified until the variational principle guides us to its value.

With this idea in mind, let us revisit the variation. First, vary the action with respect to the metric. Begin by writing

$$\delta_g S = \int d^4x\left[\delta_g(\sqrt{-g})R + \sqrt{-g}R_{\alpha\beta}\,\delta g^{\alpha\beta}\right] \ . \tag{13.31}$$

We expand that first variation using

$$\delta_g\left(\sqrt{-g}\right) = -\frac{1}{2}\sqrt{-g}g_{\alpha\beta}\delta g^{\alpha\beta}\ , \tag{13.32}$$

as we worked out before. Nothing else varies here, since the Ricci tensor depends on the connection Γ rather than the metric g. With this, the variation of the action with the metric becomes

$$\begin{aligned}\delta_g S &= \frac{1}{16\pi G}\int d^4x\sqrt{-g}\left[R_{\alpha\beta}-\frac{1}{2}Rg_{\alpha\beta}\right]\delta g^{\alpha\beta} \\ &= \frac{1}{16\pi G}\int d^4x\sqrt{-g}\,G_{\alpha\beta}\,\delta g^{\alpha\beta}\ .\end{aligned} \tag{13.33}$$

Requiring that this vanish for any variation in the metric yields the vacuum field equation, $G_{\alpha\beta}=0$. Including other matter or fields and following the procedure described in the previous section to produce the stress energy tensor yields the complete field equation, $G_{\alpha\beta}=8\pi G T_{\alpha\beta}$.

Now consider varying the connection. Begin with the fact that, as discussed above, the variation in the Ricci curvature may be written

$$\delta R_{\alpha\beta} = \nabla_\mu \delta\Gamma^\mu{}_{\beta\alpha} - \nabla_\beta\delta\Gamma^\mu{}_{\mu\alpha}\ . \tag{13.34}$$

In what follows, let us explicitly indicate that we are integrating over some 4-volume V^4. We will require that the variation in the connection vanish on the boundary of the 4-volume V^4, i.e. that $\delta\Gamma^\mu{}_{\alpha\beta}\to 0$ on ∂V^4. The variation in the action upon varying the connection is given by

$$\delta_\Gamma S = \frac{1}{16\pi G}\int_{V^4} d^4x\left[\nabla_\mu\delta\Gamma^\mu{}_{\beta\alpha}-\nabla_\beta\delta\Gamma^\mu{}_{\mu\alpha}\right]\sqrt{-g}g^{\alpha\beta}\ . \tag{13.35}$$

Because variations in the connection are tensorial, we can expand the covariant derivative in the usual way. Let us do so, writing

$$\nabla_\mu\delta\Gamma^\mu{}_{\beta\alpha} = \partial_\mu\delta\Gamma^\mu{}_{\beta\alpha} + \text{terms of the form } \delta\Gamma\,\Gamma\ . \tag{13.36}$$

A similar form describes the other covariant derivative. Although the terms of the form $\delta\Gamma\,\Gamma$ are only indicated here schematically, it is important to get them right for the analysis which follows. I won't fill in all the algebra, which though straightforward is somewhat tedious, but just outline the procedure to be followed.

The terms which involve partial derivatives can now be integrated by parts. Analysis of the one we have written out proceeds as follows:

$$\int_{V^4} d^4x\,\partial_\mu\delta\Gamma^\mu{}_{\beta\alpha}\,g^{\alpha\beta}\sqrt{-g} = \int_{\partial V^4}\delta\Gamma^\mu{}_{\beta\alpha}\,g^{\alpha\beta}\,d\Sigma_\mu - \int_{V^4} d^4x\,\delta\Gamma^\mu{}_{\beta\alpha}\,\partial_\mu\left(g^{\alpha\beta}\sqrt{-g}\right)\ . \tag{13.37}$$

Note that in the first integral, we've absorbed a factor of $\sqrt{-g}$ into the definition of the "surface" element $d\Sigma_\mu$. The boundary term contributes nothing by virtue of the fact that the variation in the connection vanishes on ∂V^4; what remains is an integral of the variation against the partial derivative of the (inverse) metric times the determinant factor.

Doing a similar exercise for the other covariant derivative, putting all terms together, gathering together all the terms of the form $\delta\Gamma\,\Gamma$, and finally relabeling dummy indices, an apparent miracle occurs: *the whole thing can be written as variations in the connection linked to* **covariant** *derivatives of* $g^{\alpha\beta}\sqrt{-g}$. In particular, what we find is

$$\begin{aligned}\delta_\Gamma S &= \frac{1}{16\pi G}\int_{V^4} d^4x\left[\delta\Gamma^\mu{}_{\mu\alpha}\nabla_\beta\left(g^{\alpha\beta}\sqrt{-g}\right)-\delta\Gamma^\mu{}_{\beta\alpha}\nabla_\mu\left(g^{\alpha\beta}\sqrt{-g}\right)\right] \\ &= \frac{1}{16\pi G}\int_{V^4} d^4x\,\delta\Gamma^\mu{}_{\nu\alpha}\left[\delta^\nu{}_\mu\nabla_\beta\left(g^{\alpha\beta}\sqrt{-g}\right)-\delta^\nu{}_\beta\nabla_\mu\left(g^{\alpha\beta}\sqrt{-g}\right)\right]\ .\end{aligned} \tag{13.38}$$

On the second line, we've used the Kronecker delta and relabeled dummy indices to pull out a common factor of the variation in the connection.

This variation is stationary if the term in brackets vanishes, which is the case if

$$\nabla_\mu \left(g^{\alpha\beta}\sqrt{-g} \right) = 0 \, . \tag{13.39}$$

This condition is in fact met if our connection coefficients are equal to the Christoffel synbols. This demonstrates that this *Palatini variation* eliminates the problematic boundary term that arises when we vary only the metric, and yields the Christoffel connection as a choice compatible with the variation[3].

13.5 Advantage of the variational approach

Both of the variational frameworks we discussed yield the Einstein field equations, as does the "historical" approach discussed in the previous lecture. So what do we gain by going through all this additional analysis?

As emphasized earlier, the properties of any fundamental interaction such as gravity are ultimately ones that must be determined empirically; but, as theorists we seek a framework that allows us to "catalog" what properties may exist given particular assumptions about the properties this interaction must satisfy. The Einstein-Hilbert Lagrangian, $\mathcal{L} = R/(16\pi G)$, is the *simplest* one which is fully covariant and assembled only from curvature tensors. Since varying this action yields the Einstein field equation, this tells us that the theory of gravity arising from this field equation — general relativity — is in a very fundamental sense the simplest possible theory of gravity compatible with covariance and the equivalence principle.

One could imagine other possibilities, though. For example, what happens if we choose

$$\mathcal{L} = \frac{1}{16\pi G}\left(R - \frac{\alpha}{R} \right) \tag{13.40}$$

as our gravitational Lagrangian? Heuristically, this choice indicates that we expect things to look "just like" general relativity when $R \gg \sqrt{\alpha}$, but have very different behavior when $R \lesssim \sqrt{\alpha}$. This Lagrangian has introduced a particular curvature scale at which we expect gravitational dynamics to change character. One can show[4] that the field equations produced by this Lagrangian are

$$G_{\mu\nu} + \frac{\alpha}{R^2}\left[R_{\mu\nu} + \frac{1}{2} g_{\mu\nu} R \right] + \alpha \left[g_{\mu\nu}\nabla_\sigma\nabla^\sigma - \nabla_{(\mu}\nabla_{\nu)} \right] R^{-2} = 8\pi G T_{\mu\nu} \, . \tag{13.41}$$

This describes a theory of gravity for which the dynamics are identical to those of general relativity at "large curvature," but things are quite different for "small curvature." This Lagrangian was proposed as a way to explain the accelerated expansion of the universe. It was eventually falsified for being incompatible with measurements of the precession of orbits in our solar system — a textbook example of how to modify a theory in a way that is compatible with important physical principles, then empirically testing whether those modifications fit what we measure or not.

One can also imagine a Lagrangian that involves higher-order curvature terms. The most general such form is

$$\mathcal{L} = \frac{1}{16\pi G}\left(R + \alpha_1 R^2 + \alpha_2 R_{\mu\nu}R^{\mu\nu} + \alpha_3 R_{\mu\sigma\nu\rho}R^{\mu\sigma\nu\rho} \right) \, . \tag{13.42}$$

There is some theoretical prejudice that $\alpha_{1,2,3} \sim \ell_{\rm P}^2 \sim \hbar$, where $\ell_{\rm P}$ is the *Planck length*. A fair amount of literature and work these days examines the possibility of *Lovelock gravity*, a proposal that these curvature terms enter the Lagrangian in the form

$$\mathcal{L} = \frac{1}{16\pi G}\left[R + \alpha\left(R^2 - 4R_{\mu\nu}R^{\mu\nu} + R_{\mu\sigma\nu\rho}R^{\mu\sigma\nu\rho} \right] \right) \, . \tag{13.43}$$

[3] Interestingly, there are other choices that could be made to solve $\nabla_\mu(g^{\alpha\beta}\sqrt{-g}) = 0$, though discussing them goes well beyond our needs. See J. T. Wheeler, arXiv:2201.00938 for further discussion.

[4] See S. M. Carroll et al, Phys. Rev. D **70**, 043528 (2004).

Such a choice arises because the term after α is used to compute the Euler-Poincaré characteristic of curved manifolds, and is known as the Gauss-Bonnet term.

One can also imagine that gravity is not purely a consequence of spacetime geometry, but that some field (often taken to be a scalar) exists which couples to curvature. The gravity Lagrangian is then given by

$$\mathcal{L}_g = \frac{1}{16\pi G} f(\Phi) R \,, \tag{13.44}$$

which is sometimes usefully thought of as the coupling constant G become a quantity that depends on the scalar field:

$$\mathcal{L}_g = \frac{1}{16\pi G(\Phi)} R \,. \tag{13.45}$$

This must be supplemented by a Lagrangian describing the dynamics of the scalar field:

$$\mathcal{L}_\Phi = h(\Phi) \nabla^\alpha \nabla_\alpha \Phi - V(\Phi) \,, \tag{13.46}$$

with the principles of your *scalar-tensor* theory providing the functions $f(\Phi)$ and $h(\Phi)$, as well as the potential $V(\Phi)$. Note that such theories in general violate the equivalence principle — different bodies may fall at different rates depending on how they couple to Φ. Further discussion can be found in Sec. 4.8 of Carroll.

From our standpoint, the take-away message is that this framework gives us a way of developing a description of gravity which goes beyond general relativity, but does so in a methodical way that builds into the analysis from the start whatever principles you wish your theory of gravity to incorporate. For the purpose of 8.962, we will henceforth focus solely on the unadorned theory of general relativity; but the tools exist to examine how things differ if we deviate from this surprisingly simple starting point.

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Lecture 14
Solving the Einstein field equations: Weak field solutions

14.1 Solving the Einstein field equation: General considerations

For the remainder of 8.962, we will focus on solving the Einstein field equation $G_{\mu\nu} = 8\pi G T_{\mu\nu}$ and studying the properties of its solutions (including the cosmological constant in our analysis for one particular class of solutions).

When this equation is regarded as a statement about the geometry of spacetime and how it relates to the flow and density of energy and momentum in this spacetime, terms like "simple" and "beautiful" tend to be used. Indeed, as we saw in Lecture 13, Einstein's general relativity is in a very real sense the simplest possible theory of gravity that respects general covariance and the principle of equivalence, arising from the simplest Lagrangian that can be constructed from curvature. The result is a straightforward equivalence between a divergence-free tensor that describes spacetime curvature and the flow of energy and momentum through spacetime.

On the other hand, if we regard the field equations as differential equations that specify the spacetime metric — which is often what we want for the purpose of physical analysis — then they are a rather horribly complicated mess. Consider the situation for a perfect fluid. On the left-hand side, we have the Einstein tensor, which we can regard as a differential operator acting on the metric of spacetime:

$$G_{\mu\nu} = \mathcal{D}^2_{\mu\nu}\left[g_{\alpha\beta}\right] \ . \tag{14.1}$$

In this schematic form, $\mathcal{D}^2_{\mu\nu}$ is a 2nd-order, nonlinear partial differential operator. On the right, we have

$$T_{\mu\nu} = (\rho + P)u_\mu u_\nu + P g_{\mu\nu} \ . \tag{14.2}$$

The spacetime metric thus appears explicitly on the right-hand side, as well as under the operator on the left-hand side. It is also present *implicitly* on the right-hand side via the requirement that the four velocity be properly normalized: $u^\mu u^\nu g_{\mu\nu} = -1$.

So as a differential equation for the metric, the Einstein field equation sets us up for what appears likely to be a rather challenging analysis. To make progress, we will implement three approaches to solving this equation:

1. *"Weak" gravity.* Here we imagine that spacetime is not very different (a statement that must be made quantitative) from that of special relativity, at least in some coordinate systems, and we will linearize the field equations in that deviation from $\eta_{\mu\nu}$. Such an analysis can be regarded as the first order in a perturbative expansion. We will discuss in schematic terms how one can iterate this expansion to higher order in the last week of this course.

2. *Impose a symmetry.* By imposing a symmetry, the complexity of the operator $\mathcal{D}^2_{\mu\nu}$ and of any source terms is greatly reduced. The coupled nonlinear nastiness that remains is one that can be handled without too much trouble; this is how some of the most important solutions in general relativity have been constructed. By adding perturbations around these symmetric but exact solutions, we can do a tremendous amount of useful analysis. (Indeed, a rather large chunk of your lecturer's career is based on studying perturbations around a particular exact solution.)

3. *Just deal with it: Code up the equations and attack.* Numerical analysis can be a very powerful tool, provided we understand its limitations and can assess whether our solutions are converging to a description of the correct physics. Especially since about 2005, numerical methods for solving the Einstein field equations have been particularly productive; but it took decades of foundational work to understand how to cast the equations into a form that was amenable to this method of building solutions. We will discuss work in this vein in the last week of this course.

We begin by studying the linearized field equations.

14.2 The "weak-field" limit: Linearized gravity

The weak-field limit is defined by the idea that spacetime is *nearly* flat. We choose coordinates such that

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu} \ , \tag{14.3}$$

and in which the components describing the deviation from flatness are small enough that we can linearize with negligible error: $||h_{\mu\nu}|| \ll 1$, so terms $\mathcal{O}(h^2)$ can be neglected. Whether this linearization is accurate or not is ultimately a matter of measurement precision. It is conceivable that measurements which agree with the predictions of linearized theory will cease to agree if measurement techniques are significantly improved. In this part of 8.962, we take the ansatz that the linear approximation is acceptable, and develop this limit of general relativity in detail.

Coordinates in which one can write the spacetime metric in the form (14.3) are known as "nearly Lorentz coordinates." One can think of them as coordinates that hold in a convenient freely falling frame and are then extended to cover the entire manifold of (nearly flat) spacetime. Note that we have borrowed some of the ideas and results from this analysis in earlier sections of this course, particularly in making sure that our field equation included a Newtonian limit. Our goal now is to put this framework from which we have been borrowing onto solid ground.

Because the spacetime metric is a tensor, it of course is subject to the general rule for changing representation:

$$g_{\bar{\mu}\bar{\nu}} = \frac{\partial x^\alpha}{\partial x^{\bar{\mu}}} \frac{\partial x^\beta}{\partial x^{\bar{\nu}}} g_{\alpha\beta} \ . \tag{14.4}$$

When the spacetime is flat, we picked out a special category of representation changes, the Lorentz transformations, as playing an important role. One thing that makes the Lorentz transformations special is that the metric has the same representation in all Lorentz frames:

$$\eta_{\bar{\mu}\bar{\nu}} = \Lambda^\alpha{}_{\bar{\mu}} \Lambda^\beta{}_{\bar{\nu}} \eta_{\alpha\beta} \ , \tag{14.5}$$

and both $\eta_{\bar{\mu}\bar{\nu}}$ and $\eta_{\alpha\beta}$ are represented by the matrix $\mathrm{diag}(-1, 1, 1, 1)$.

The Lorentz transformation really only has physical meaning in special relativity — in general relativity, it is not generally applicable except within the confines of a local Lorentz frame. However, in the case of nearly flat spacetimes, we can extend this concept in a useful way. Begin by applying the Lorentz transformation to a nearly flat spacetime metric:

$$\begin{aligned} g_{\bar{\mu}\bar{\nu}} &= \Lambda^\alpha{}_{\bar{\mu}} \Lambda^\beta{}_{\bar{\nu}} \left(\eta_{\alpha\beta} + h_{\alpha\beta}\right) \\ &= \eta_{\bar{\mu}\bar{\nu}} + \Lambda^\alpha{}_{\bar{\mu}} \Lambda^\beta{}_{\bar{\nu}} h_{\alpha\beta} \\ &\equiv \eta_{\bar{\mu}\bar{\nu}} + h_{\bar{\mu}\bar{\nu}} \ . \end{aligned} \tag{14.6}$$

The background flat spacetime is unchanged by the Lorentz transformation, but the "perturbation" $h_{\alpha\beta}$ transforms just like any tensor field in special relativity. In this circumstance, we call this transformation a *background* Lorentz transformation, since we are taking advantage of the fact that the flat background to our perturbation respects Lorentz invariance.

The relationship $h_{\bar{\mu}\bar{\nu}} = \Lambda^{\alpha}{}_{\bar{\mu}}\Lambda^{\beta}{}_{\bar{\nu}}h_{\alpha\beta}$ tells us that a useful fiction is to regard $h_{\alpha\beta}$ as simply a tensor field that lives in flat spacetime. This is a fiction because in fact the spacetime is curved, but it is a useful one for us because it gives us a calculational "narrative" that is very similar to how we describe many field theories in special relativity.

Another operation that we need to work out is the form of the inverse metric in linearized theory. We begin by raising indices:

$$\begin{aligned} h^{\alpha\beta} &= g^{\alpha\mu}g^{\beta\nu}h_{\mu\nu} \\ &= \eta^{\alpha\mu}\eta^{\beta\nu}h_{\mu\nu} + \mathcal{O}(h^2) \ . \end{aligned} \tag{14.7}$$

We always find that indices related to spacetime tensors are raised and lowered using the background metric to linear order, which is consistent with the idea that we can treat $h_{\mu\nu}$ as a tensor field in a flat background. With this in hand, let us calculate the inverse metric:

$$\begin{aligned} \delta^{\alpha}{}_{\gamma} &= g^{\alpha\beta}g_{\beta\gamma} \\ &= (\eta^{\alpha\beta} + m^{\alpha\beta})\,(\eta_{\beta\gamma} + h_{\beta\gamma}) \ . \end{aligned} \tag{14.8}$$

The first line here is just the definition of the metric inverse $g^{\alpha\beta}$. On the second line, we have introduced a tensor $m^{\alpha\beta}$ whose elements are of the same order as the elements of $h_{\alpha\beta}$. We leave it undetermined in order to see what value it must have in order to provide the correct inverse metric. Expanding, we have

$$\delta^{\alpha}{}_{\gamma} + h^{\alpha}{}_{\gamma} + m^{\alpha}{}_{\gamma} + \mathcal{O}(h^2) = \delta^{\alpha}{}_{\gamma} \ , \tag{14.9}$$

which tells us that

$$h^{\alpha}{}_{\gamma} = -m^{\alpha}{}_{\gamma} \tag{14.10}$$

to linear order. Therefore,

$$g^{\alpha\beta} = \eta^{\alpha\beta} - h^{\alpha\beta} \tag{14.11}$$

to linear order in the deviation from $\eta_{\alpha\beta}$. This is a form that we used earlier without careful justification in our analysis of the Newtonian limit, although this result should intuitively remind you of the functional expansion $1/(1+x) \simeq 1 - x$ for small x.

14.3 Gauge transformations in linearized gravity

Consider next making a coordinate shift, defined as

$$x^{\alpha'} = x^{\alpha} + \xi^{\alpha}(x^{\beta}) \ . \tag{14.12}$$

It must be acknowledged that this is not the best notation — this form abuses the index notation somewhat. Calling the new coordinate $(x')^{\alpha}$ rather than $x^{\alpha'}$ would be more consistent with the way we use indices in relativity. However, it is best to live with this inconsistency. Our goal here is not to introduce a covariant relationship, but simply to indicate how two coordinate systems are related to one another. In particular, one of the reasons that this somewhat irritatingly inconsistent notation is commonly used is that we wish to introduce the coordinate transformation matrix

$$L^{\alpha'}{}_{\beta} = \delta^{\alpha}{}_{\beta} + \partial_{\beta}\xi^{\alpha} \ . \tag{14.13}$$

We imagine that both the new and the old coordinates are nearly inertial, and require the matrix of partials to be very small: $||\partial_{\alpha}\xi^{\beta}|| \ll 1$. We call this an "infinitesimal" coordinate transformation. Using these properties, it is simple to show that the inverse transformation is given by

$$L^{\alpha}{}_{\beta'} = \delta^{\alpha}{}_{\beta} - \partial_{\beta}\xi^{\alpha} + \mathcal{O}[(\partial\xi)^2] \ . \tag{14.14}$$

Let us now examine how the metric changes under this coordinate transformation:

$$\begin{aligned} g_{\mu'\nu'} &= L^{\alpha}{}_{\mu'} L^{\beta}{}_{\nu'} g_{\alpha\beta} \\ &= (\delta^{\alpha}{}_{\mu} - \partial_\mu \xi^\alpha)(\delta^{\beta}{}_{\nu} - \partial_\nu \xi^\beta)(\eta_{\alpha\beta} + h_{\alpha\beta}) \\ &= \eta_{\mu\nu} + h_{\mu\nu} - \partial_\mu \xi_\nu - \partial_\nu \xi_\mu + \mathcal{O}(h\,\partial\xi) \; . \end{aligned} \tag{14.15}$$

The effect of this coordinate transformation is to change the perturbation around flat spacetime:

$$h_{\mu\nu} \to h_{\mu\nu} - \partial_\mu \xi_\nu - \partial_\nu \xi_\mu \; . \tag{14.16}$$

This should remind you of a gauge transformation in electrodynamics, in which one adjusts the potential $A_\mu \to A_\mu - \partial_\mu \Lambda$. We thus call this operation a gauge transformation in linearized gravity.

Just as the gauge transformation in electrodynamics changes potentials but leaves the field unchanged, the linearized gravity gauge transformation has no effect on curvature. It is simple to show that the Riemann tensor is given by

$$R_{\mu\alpha\nu\beta} = \frac{1}{2} \left(\partial_\alpha \partial_\nu h_{\mu\beta} + \partial_\mu \partial_\beta h_{\alpha\nu} - \partial_\alpha \partial_\beta h_{\mu\nu} - \partial_\mu \partial_\nu h_{\alpha\beta} \right) \tag{14.17}$$

in linearized theory; just expand the formula given in terms of Christoffel symbols, discarding the nonlinear terms and using the fact that we raise and lower indices with the flat spacetime metric (which commutes with partial derivatives since it is a constant). If one changes the metric according to (14.16), then Riemann is apparently shifted by

$$\begin{aligned} \delta R_{\mu\alpha\nu\beta} &= \frac{1}{2} \Big(-\partial_\alpha \partial_\nu \partial_\mu \xi_\beta - \partial_\alpha \partial_\nu \partial_\beta \xi_\mu - \partial_\mu \partial_\beta \partial_\alpha \xi_\nu - \partial_\mu \partial_\beta \partial_\nu \xi_\alpha \\ &\qquad + \partial_\mu \partial_\nu \partial_\alpha \xi_\beta + \partial_\alpha \partial_\beta \partial_\nu \xi_\mu + \partial_\alpha \partial_\beta \partial_\mu \xi_\nu + \partial_\mu \partial_\nu \partial_\eta \xi_\alpha \Big) \; . \end{aligned} \tag{14.18}$$

Because partial derivatives commute, one can see that $\delta R_{\mu\alpha\nu\beta} = 0$. We will take advantage of this property to adjust our gauge in such a way that curvature tensors, written as derivatives of the metric, take a particularly convenient form for our analysis.

14.4 The linear gravity field equation and its leading solution

Having already built Riemann in order to examine its behavior under gauge transformations, let's build the various contractions that we need for the field equation. We start with the Ricci tensor:

$$\begin{aligned} R_{\alpha\beta} &= \eta^{\mu\nu} R_{\mu\alpha\nu\beta} \\ &= \frac{1}{2} \left(\partial_\alpha \partial^\mu h_{\mu\beta} + \partial_\beta \partial^\mu h_{\mu\alpha} - \partial_\alpha \partial_\beta h - \Box h_{\alpha\beta} \right) \; . \end{aligned} \tag{14.19}$$

Here $\Box = \partial^\mu \partial_\mu = \eta^{\mu\nu} \partial_\mu \partial_\nu$ is the wave operator in flat spacetime, and we have introduced $h = h^\mu{}_\mu = \eta^{\mu\nu} h_{\mu\nu}$, the trace of the metric perturbation.

The Ricci scalar follows simply from this:

$$R = \eta^{\alpha\beta} R_{\alpha\beta} = \partial^\alpha \partial^\mu h_{\mu\alpha} - \Box h \; . \tag{14.20}$$

Finally, the linearized Einstein tensor is given by

$$\begin{aligned} G_{\alpha\beta} &= R_{\alpha\beta} - \frac{1}{2} \eta_{\alpha\beta} R \\ &= \frac{1}{2} \left(\partial_\alpha \partial^\mu h_{\mu\beta} + \partial_\beta \partial^\mu h_{\mu\alpha} - \partial_\alpha \partial_\beta h - \Box h_{\alpha\beta} + \eta_{\alpha\beta} \Box h - \eta_{\alpha\beta} \partial^\mu \partial^\nu h_{\mu\nu} \right) \; . \end{aligned} \tag{14.21}$$

The presence of the trace h makes this expression somewhat unwieldy. Recalling that $G_{\alpha\beta}$ is a trace-reversed version of the Ricci curvature, it is worth seeing what happens if we build it using a trace-reversed version of the metric perturbation. We define

$$\bar{h}_{\alpha\beta} = h_{\alpha\beta} - \frac{1}{2}\eta_{\alpha\beta}h \; ; \tag{14.22}$$

notice that $\eta^{\alpha\beta}\bar{h}_{\alpha\beta} = h - 2h = -h$. We then write $h_{\alpha\beta} = \bar{h}_{\alpha\beta} + \frac{1}{2}\eta_{\alpha\beta}h$, and substitute this into the Einstein tensor (14.21). The result is

$$\begin{aligned} G_{\alpha\beta} &= \frac{1}{2}\Big(\partial_\alpha\partial^\mu\bar{h}_{\mu\beta} + \frac{1}{2}\eta_{\mu\beta}\partial_\alpha\partial^\mu h + \partial_\beta\partial^\mu\bar{h}_{\mu\alpha} + \frac{1}{2}\eta_{\mu\alpha}\partial_\beta\partial^\mu h - \partial_\alpha\partial_\beta h - \Box\bar{h}_{\alpha\beta} \\ &\quad -\frac{1}{2}\eta_{\alpha\beta}\Box h + \eta_{\alpha\beta}\Box h - \eta_{\alpha\beta}\partial^\mu\partial^\nu\bar{h}_{\mu\nu} - \frac{1}{2}\eta_{\alpha\beta}\eta_{\mu\nu}\partial^\mu\partial^\nu h\Big) \; . \end{aligned} \tag{14.23}$$

Carefully examining this expression, we see that all the terms involving the trace h cancel out: the second, fourth, and fifth terms on the top line of the right-hand side sum to zero, as do the first, second, and fourth expressions on the bottom line of the right-hand side. What remains is

$$G_{\alpha\beta} = \frac{1}{2}\left(\partial_\alpha\partial^\mu\bar{h}_{\mu\beta} + \partial_\beta\partial^\mu\bar{h}_{\mu\alpha} - \eta_{\alpha\beta}\partial^\mu\partial^\nu\bar{h}_{\mu\nu} - \Box\bar{h}_{\alpha\beta}\right) \; . \tag{14.24}$$

It should be emphasized that this trick is really nothing more than algebraic sleight of hand; however, it is no less useful for being essentially a bit of clever rearranging.

Notice that in what remains all the terms that are left except the very last term inside the parentheses involve divergences of the metric perturbation: the first three terms are each of the form $\partial^\mu\bar{h}_{\mu\nu}$ (or something involving a different index than ν). Just as in electromagnetic theory, we can use our gauge freedom to clean this up. We would like to set

$$\partial^\mu\bar{h}_{\mu\nu} = 0 \; , \tag{14.25}$$

a total of 4 conditions. The quantities ξ_ν which generate the infinitesimal coordinate transformation provide us with 4 four free functions to do this. Writing

$$h^{\rm new}_{\mu\nu} = h^{\rm old}_{\mu\nu} - \partial_\mu\xi_\nu - \partial_\nu\xi_\mu \; , \tag{14.26}$$

we trace reverse and find that

$$\bar{h}^{\rm new}_{\mu\nu} = \bar{h}^{\rm old}_{\mu\nu} - \partial_\mu\xi_\nu - \partial_\nu\xi_\mu + \eta_{\mu\nu}\partial^\alpha\xi_\alpha \; . \tag{14.27}$$

Take the divergence of this:

$$\begin{aligned} \partial^\mu\bar{h}^{\rm new}_{\mu\nu} &= \partial^\mu\bar{h}^{\rm old}_{\mu\nu} - \Box\xi_\nu - \partial_\nu\partial^\mu\xi_\mu + \eta_{\mu\nu}\partial^\mu\partial^\alpha\xi_\alpha \\ &= \partial^\mu\bar{h}^{\rm old}_{\mu\nu} - \Box\xi_\nu \; . \end{aligned} \tag{14.28}$$

(To simplify, we used the metric to lower the index on the last term and recognized that μ and α are dummy indices on the final two terms on the top line.) This tells us that if we choose our gauge generators to satisfy

$$\Box\xi_\nu = \partial^\mu\bar{h}^{\rm old}_{\mu\nu} \; , \tag{14.29}$$

then the metric in our new gauge will be divergence free. Solutions of (14.29) always exist as long as $\bar{h}^{\rm old}_{\mu\nu}$ is well behaved, and this metric perturbation will always be well behaved in the weak gravity applications to which we apply this framework. We call this gauge "Lorenz gauge," following the

essentially identical gauge choice used in electrodynamics to make the potential 4-vector divergence free. When we are in Lorenz gauge, the Einstein tensor is given by

$$G_{\alpha\beta} = -\frac{1}{2}\Box\bar{h}_{\alpha\beta} \ , \tag{14.30}$$

and we find the linearized Einstein field equation takes the form

$$\boxed{\Box\bar{h}_{\alpha\beta} = -16\pi G T_{\alpha\beta}} \tag{14.31}$$

This is starkly reminiscent of the sourced Maxwell's equation written in terms of the potential in Lorenz gauge, and very similar techniques can be used to construct its solution.

14.5 Static solutions

Let us begin by considering a perfect fluid source that is static ($u^t = 1 + \mathcal{O}(h)$, $u^i = 0$) and non-relativistic ($\rho \gg P$, corresponding to $c_{\rm sound} \ll c$). We can write the stress energy tensor

$$T_{\alpha\beta} \simeq \rho u_\alpha u_\beta \ . \tag{14.32}$$

Thanks to the static condition, the only non-negligible component is $T_{00} = \rho + \mathcal{O}(h)$. The only non-trivial field equation component is then

$$\Box\bar{h}_{00} = -16\pi G\rho \ . \tag{14.33}$$

Including the $\mathcal{O}(h)$ correction would lead to corrections to the solution $\bar{h}_{00}$ we find which are themselves second-order in h.

Since the source is static, the differential operator simplifies to

$$\nabla^2\bar{h}_{00} = -16\pi G\rho \ . \tag{14.34}$$

Solutions to this equation have a form reminiscent of a Coulomb potential. In the case of gravity, we recognize it as the equation for the Newtonian potential Φ (modulo a factor -4), and deduce that the solution is

$$\bar{h}_{00} = -4\Phi \ ; \qquad \text{all other } \bar{h}_{\mu\nu} = 0 \ . \tag{14.35}$$

To get the metric perturbation, we trace reverse again:

$$h_{\mu\nu} = \bar{h}_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}\bar{h} \ , \tag{14.36}$$

where the trace $\bar{h} = \eta^{\mu\nu}\bar{h}_{\mu\nu} = \eta^{00}\bar{h}_{00} = (-1)(-4\Phi) = 4\Phi$. This tells us that

$$\begin{aligned} h_{00} &= -4\Phi - \frac{1}{2}(-1)(4\Phi) \\ &= -2\Phi \ , \end{aligned} \tag{14.37}$$

$$\begin{aligned} h_{11} = h_{22} = h_{33} &= 0 - \frac{1}{2}(1)(4\Phi) \\ &= -2\Phi \ ; \end{aligned} \tag{14.38}$$

all off-diagonal components of $h_{\mu\nu} = 0$ in this case. We see that

$$ds^2 = -(1+2\Phi)dt^2 + (1-2\Phi)(dx^2 + dy^2 + dz^2) \ . \tag{14.39}$$

This is the line element we used (without justification) to explore the Newtonian limit.

On problem set 7, you will examine what happens when the body spins rigidly about an axis. Just as the static problem leads to a potential that is a lot like the Coulomb potential governing the electric field arising from the density field ρ, the rotating problem looks a lot like what we get for the magnetic potential arising from a density flow $\mathbf{J} = \rho\mathbf{v}$. The solution should remind you quite a bit of problems you have already seen in an electromagnetic content, albeit with some signs flipped, and various prefactors not quite the same. It should also be noted that linearity of these field equations means that we can add any *homogeneous* solution of the field equation; i.e., any solution of (14.31) for which $T_{\alpha\beta} = 0$. Modulo some subtleties having to do with gauge, such solutions can represent radiation. Hold that thought! — we throw away those solutions right now because they are not static (our current focus), but will be a major focus of material we discuss very soon.

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Lecture 15
Solving the Einstein field equations: Dynamical weak field solutions

15.1 Solving the linearized Einstein equation for dynamical sources

When sources are not dynamical, then the linearized Einstein field equation

$$\Box\bar{h}_{\alpha\beta} = -16\pi G T_{\alpha\beta} \tag{15.1}$$

simplifies significantly: one can find a frame in which time derivatives on the left-hand side do not contribute (this is typically a frame in which a localized source is at rest in the coordinates), and so the wave operator $\Box$ simplifies to a Laplacian ∇^2. Such an equation can be solved using techniques imported directly from studies of Maxwell's equations. In particular, the leading order solution for non-relativistic sources ($\rho \gg P$) is just Newtonian gravity for a distribution of mass/energy density ρ, and can be solved as a multipolar expansion describing how ρ is distributed in space. Subleading solutions (such as you explore on problem set 7) correct the Newtonian description with gravitational contributions that describe how the flow of this mass/energy density contributes to spacetime. The entire problem bears a striking resemblance to how one solves for electric and magnetic fields from a stationary distribution of charges and currents.

How does one solve Eq. (15.1) when sources are dynamical? For example, what if one considers a source in which the mass/energy distribution is highly disturbed from equilibrium and undergoes oscillations? Or, a source consisting of two bodies in orbit about one another? There is no frame in which the time derivatives can be neglected in this case, and one must solve Eq. (15.1) in a manner which reflects the source's dynamics.

Our tool for beginning this investigation takes advantage of the fact that (15.1) is a linear equation — double the amount of source on the right-hand side, and we double[1] the amount of resulting field on the left-hand side. Linear differential equations can often be solved using the technique of Green's functions. If you have not seen this technique before, I recommend the discussion in the textbook by Arfken[2]. Briefly, imagine that you need to find the particular solution to a differential equation

$$\mathcal{D}f(t,\mathbf{x}) = s(t,\mathbf{x}) \ , \tag{15.2}$$

where $\mathcal{D}$ is a linear differential operator, $s(t,\mathbf{x})$ is a source that depends on space and time in the chosen coordinate system, and $f(t,\mathbf{x})$ is the resulting field arising from this source. The detailed form of the operator $\mathcal{D}$ is not important for this brief overview, other than that it must be linear in the field f.

The idea of the Green's function is to recognize that much of the challenge in this analysis comes from the fact that the source may be complicated. However, because the equation is linear, we can replace the source with something simpler and build a more complicated source using superposition. In particular, let us replace the source with delta functions:

$$s(t,\mathbf{x}) \to \delta(t-t')\delta^{(3)}(\mathbf{x}-\mathbf{x}') \ . \tag{15.3}$$

[1] This somewhat glib description neglects the role of homogeneous solutions to (15.1), i.e., solutions for no source at all, $\Box\bar{h}_{\alpha\beta} = 0$. The amount of homogeneous solution in the problem is determined by boundary conditions; though very important to get this right, we can set this aside for the moment.

[2] G. B. Arfken, *Mathematical Methods for Physicists*, Sec. 16.5–16.6 in the 3rd edition. Apparently this book is now up to edition 7; the location of the discussion of Green's functions may be different in newer versions.

The primed coordinates $(t', \mathbf{x}')$ label the "source point," the location of the delta source. Our goal is to compute how much field arises at field point $(t, \mathbf{x})$ arisesdue to this delta spike of source at the source point. We call this solution the Green's function $G(t, \mathbf{x}; t', \mathbf{x}')$, and assert that our equation can now be written

$$\mathcal{D}G(t, \mathbf{x}; t', \mathbf{x}') = \delta(t - t')\delta^{(3)}(\mathbf{x} - \mathbf{x}') \; . \tag{15.4}$$

Suppose we have constructed the Green's function corresponding to our operator $\mathcal{D}$. It should be emphasized that this construction may take some effort; but imagine we have put in this effort, and we have $G(t, \mathbf{x}; t', \mathbf{x}')$ in hand for our problem. The solution for our field is then given by

$$f(t, \mathbf{x}) = \int dt' \int d^3x' G(t, \mathbf{x}; t', \mathbf{x}') s(t', \mathbf{x}') \; . \tag{15.5}$$

We can think of the Green's function $G(t, \mathbf{x}; t', \mathbf{x}')$ as providing the amount of field at $(t, \mathbf{x})$ that arises per unit of source at $(t', \mathbf{x}')$. This solution is essentially a convolution: we treat the source as a superposition of many delta functions, so by linearity the resulting field is a superposition of many Green's functions.

To prove that Eq. (15.5) gives the correct solution, simply act on it using the operator $\mathcal{D}$:

$$\begin{aligned} \mathcal{D}f(t, \mathbf{x}) &= \int dt' \int d^3x' \mathcal{D}G(t, \mathbf{x}; t', \mathbf{x}') s(t', \mathbf{x}') \\ &= \int dt' \int d^3x' \delta(t - t')\delta^{(3)}(\mathbf{x} - \mathbf{x}') s(t', \mathbf{x}') \\ &= s(t, \mathbf{x}) \; . \end{aligned} \tag{15.6}$$

On the first line, we use the fact that the operator $\mathcal{D}$ only acts on functions of the unprimed coordinates, so it commutes with integration over the primed coordinates, and thus hits the Green's function but not the source. We then use the definition of the Green's function, and are left with an integral that is trivial to perform. Thus, if we can develop the Green's function for our differential operator, then we can easily construct the particular solution that solves our equation.

Fortunately, the Green's function for the wave equation is very well known: the solution to

$$\Box G(t, \mathbf{x}; t', \mathbf{x}') = \delta(t - t')\delta^{(3)}(\mathbf{x} - \mathbf{x}') \tag{15.7}$$

is

$$G(t, \mathbf{x}; t', \mathbf{x}') = -\frac{\delta\left[t' - (t - |\mathbf{x} - \mathbf{x}'|)\right]}{4\pi|\mathbf{x} - \mathbf{x}'|} \; . \tag{15.8}$$

This *radiative Green's function* is derived in detail in, for example, J. D. Jackson, *Classical Electrodynamics*, Sec. 6.6 (2nd edition); we provide a sketch of this derivation in the next section of these notes. Notice the form

$$t - |\mathbf{x} - \mathbf{x}'| \equiv t_{\rm ret} \tag{15.9}$$

that enters the time dependence. The name $t_{\rm ret}$ we assign means that this is *retarded time*: it reflects the fact that information about the source at time t' and location $\mathbf{x}'$ only reaches an observer at $\mathbf{x}$ following an interval $|\mathbf{x} - \mathbf{x}'|$ (remember our $c = 1$ units). Our wave operator[3] describes functions which propagate information about sources across spacetime at the speed of light.

The general solution to Eq. (15.1) is thus given by

$$\begin{aligned} \bar{h}_{\alpha\beta}(t, \mathbf{x}) &= \int dt' \int d^3x' \left(-\frac{\delta\left[t' - (t - |\mathbf{x} - \mathbf{x}'|)\right]}{4\pi|\mathbf{x} - \mathbf{x}'|}\right)\left(-16\pi G T_{\alpha\beta}(t', \mathbf{x}')\right) \\ &= 4G \int d^3x' \frac{T_{\alpha\beta}(t_{\rm ret}, \mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|} \; . \end{aligned} \tag{15.10}$$

[3]Because the wave operator is second order in time derivatives, it is agnostic about whether this information propagates forward or backwards in time; both solutions work with the ∂_t^2 part of the $\Box$ operator. We discard an *advanced* solution that depends on $t_{\rm adv} = t + |\mathbf{x} - \mathbf{x}'|$ on the grounds that information propagating from the future to the past is inconsistent with the ordering of events. In essence, we enforce a boundary condition that information can only propagate from the past to the future.

15.2 Aside: The radiative Green's function

Our goal will be to use the solution (15.10) to characterize and understand solutions to the linearized Einstein equation. Before doing this, we briefly discuss where the radiative Green's function comes from, for the benefit of students who have not yet encountered this in their coursework.

We need two ingredients for this analysis:

- The Laplace operator acting on $1/|\mathbf{x}-\mathbf{x}'|$ generates the Dirac delta function:

$$\nabla^2\left(\frac{1}{|\mathbf{x}-\mathbf{x}'|}\right) = -4\pi\,\delta^{(3)}(\mathbf{x}-\mathbf{x}')\ . \tag{15.11}$$

  This relationship is discussed in many electrodynamics textbooks; the discussion in Griffiths[4] is typically clear and readable if you have not seen this result previously.

- The Fourier transform pair: if $f(t)$ is some function of time, then we define its Fourier transform

$$\tilde{f}(\omega) = \int_{-\infty}^{\infty} f(t)e^{i\omega t}dt\ . \tag{15.12}$$

  Note that this implies that the Fourier transform of $df/dt = i\omega\tilde{f}(\omega)$. The Fourier transform operation is inverted by

$$f(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \tilde{f}(\omega)e^{-i\omega t}d\omega\ . \tag{15.13}$$

  By combining the forward and reverse Fourier transforms, one can show that

$$\delta(t'-t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{i\omega(t'-t)}d\omega\ . \tag{15.14}$$

We begin by computing the Green's function for the *inhomogeneous Helmholtz equation*:

$$\left(\nabla^2 + k^2\right)G(\mathbf{x};\mathbf{x}') = \delta^{(3)}(\mathbf{x}-\mathbf{x}')\ . \tag{15.15}$$

For notational simplicity, let us define $\boldsymbol{\zeta} \equiv \mathbf{x}-\mathbf{x}'$, $\zeta = |\mathbf{x}-\mathbf{x}'|$. The Green's function which solves this can only depend on ζ, and must be spherically symmetric about $\boldsymbol{\zeta} = 0$. This allows us to rewrite this equation as

$$\frac{1}{\zeta}\frac{d^2}{d\zeta^2}\left(\zeta G\right) + k^2 G = \delta^{(3)}(\boldsymbol{\zeta})\ . \tag{15.16}$$

This form of the equation makes it clear that

$$G = -\frac{e^{\pm ik\zeta}}{4\pi\zeta}\ . \tag{15.17}$$

That this solution holds is clear for $\zeta > 0$. Exactly at $\zeta = 0$, we invoke the first Dirac delta identity and see that the solutions are equivalent at this point[5]. Returning to the original parameterization, we see that

$$\left(\nabla^2 + k^2\right)G(\mathbf{x};\mathbf{x}') = \delta^{(3)}(\mathbf{x}-\mathbf{x}') \tag{15.18}$$

is solved by

$$G(\mathbf{x};\mathbf{x}') = -\frac{e^{\pm ik|\mathbf{x}-\mathbf{x}'|}}{4\pi|\mathbf{x}-\mathbf{x}'|}\ . \tag{15.19}$$

[4] D. J. Griffiths, *Introduction to Electrodynamics*, Sec. 1.5.3, 4th edition.

[5] A more careful analysis would examine a very small region surrounding $\boldsymbol{\zeta} = 0$ and show that integrating the left-hand side of (15.16) over such a region returns a non-zero value consistent with with integrating the right-hand side of this equation over the same region. Griffith's discussion of Eq. (15.11) is an excellent prototype of such a calculation.

Note the ambiguous sign in the argument of the exponential.

Turn now to the wave equation:

$$\Box G(t,\mathbf{x};t',\mathbf{x}') = (-\partial_t^2 + \nabla^2)G = \delta(t-t')\delta^{(3)}(\mathbf{x}-\mathbf{x}') \; . \tag{15.20}$$

Take the Fourier transform of both sides. On the left-hand side, $G(t,\mathbf{x};t',\mathbf{x}') \to \tilde{G}(\omega,\mathbf{x};t',\mathbf{x}')$, $\partial_t^2 G \to -\omega^2\tilde{G}$; on the right-hand side, $\delta(t-t') \to e^{i\omega t'}$. The equation becomes

$$\left(\nabla^2 + \omega^2\right)\tilde{G} = e^{i\omega t'}\delta^{(3)}(\mathbf{x}-\mathbf{x}') \; . \tag{15.21}$$

This is the inhomogeneous Helmholtz equation with $k = \omega$, and with an extra factor of $e^{i\omega t'}$ attached to the right-hand side. We can simply lift the solution from our earlier discussion, finding

$$\tilde{G}(\omega,\mathbf{x};t',\mathbf{x}') = -\frac{e^{i\omega(t'\pm|\mathbf{x}-\mathbf{x}'|)}}{4\pi|\mathbf{x}-\mathbf{x}'|} \; . \tag{15.22}$$

We conclude by computing the inverse Fourier transform:

$$\begin{aligned}
G(t,\mathbf{x};t',\mathbf{x}') &= \frac{1}{2\pi}\int_{-\infty}^{\infty} d\omega\, e^{-i\omega t}\tilde{G}(\omega,\mathbf{x};t',\mathbf{x}') \\
&= \frac{1}{2\pi}\int_{-\infty}^{\infty} d\omega\, e^{-i\omega t}\left[-\frac{e^{i\omega(t'\pm|\mathbf{x}-\mathbf{x}'|)}}{4\pi|\mathbf{x}-\mathbf{x}'|}\right] \\
&= -\frac{1}{4\pi|\mathbf{x}-\mathbf{x}'|}\left\{\frac{1}{2\pi}\int_{-\infty}^{\infty}\exp\left[i\omega\left(t'-\left(t\mp|\mathbf{x}-\mathbf{x}'|\right)\right)\right]d\omega\right\} \\
&= -\frac{\delta\left[t'-(t\mp|\mathbf{x}-\mathbf{x}'|)\right]}{4\pi|\mathbf{x}-\mathbf{x}'|} \\
&= -\frac{\delta\left(t'-t_{\rm ret/adv}\right)}{4\pi|\mathbf{x}-\mathbf{x}'|} \; .
\end{aligned} \tag{15.23}$$

The advanced solution, $t' = t_{\rm adv} = t+|\mathbf{x}-\mathbf{x}'|$, describes information propagating backwards in time. It arises, as mentioned in a footnote on a previous page, because the wave equation, being second order in time, is agnostic about whether information propagates from the past to the future or vice versa. We are not agnostic about time's direction, and enforce the causal boundary condition by rejecting the advanced solution. The radiative Green's function we use is thus given by

$$\boxed{\boxed{G(t,\mathbf{x};t',\mathbf{x}') = -\frac{\delta\left(t'-t_{\rm ret}\right)}{4\pi|\mathbf{x}-\mathbf{x}'|}}} \tag{15.24}$$

where $t_{\rm ret} = t - |\mathbf{x}-\mathbf{x}'|$.

15.3 The general solution of the linearized Einstein field equation

It should be emphasized that the solution we have derived to the linearized Einstein field equation,

$$\bar{h}_{\alpha\beta}(t,\mathbf{x}) = 4G\int d^3x'\,\frac{T_{\alpha\beta}(t_{\rm ret},\mathbf{x}')}{|\mathbf{x}-\mathbf{x}'|} \; , \tag{15.25}$$

is *exact*, at least within the confines of the linearized approximation to general relativity. We made a specific gauge choice in order to write the field equation in a form which is amenable to our use of the radiative Green's function. This does not limit our solution in the slightest. Equation (15.25) can be used to find the spacetime for any stress-energy tensor, provided the resulting deviation from flat spacetime is weak enough that h^2 terms can be neglected.

This solution has an interesting character: the field at event $(t, \mathbf{x})$ depends upon the source at $(t_{\rm ret}, \mathbf{x}')$, meaning that the dynamics enters our solution with a retardation factor that reflects the finite speed with which information propagates. And, the solution has a factor that reflects a "one over distance" kind of falloff. These are exactly the behaviors that describe radiative potentials in electrodynamics. It looks like all ten components of $\bar{h}_{\alpha\beta}$ describe a radiative solution that propagates out from a source!

This might initially seem like a very interesting and physically reasonable outcome, but with a little thought you should be concerned. Suppose some aspect of the source reflects a static mass M centered at $\mathbf{x}'$. We expect that to generate metric components with contributions proportional to $GM/|\mathbf{x} - \mathbf{x}'|$, which are not radiative.

For some intuition about what might be happening here, consider an electromagnetic potential whose components in some frame are given by

$$\begin{aligned} A^0 &= \frac{q}{r} + \frac{q\omega R \sin(\mathbf{k}\cdot\mathbf{x} - \omega t)}{r} , \\ A^i &= \frac{qk^i R \sin(\mathbf{k}\cdot\mathbf{x} - \omega t)}{r} + \frac{qx^i R \cos(\mathbf{k}\cdot\mathbf{x} - \omega t)}{r^3} . \end{aligned} \tag{15.26}$$

If, while walking down a dark alley, someone shoved this potential in your face and demanded that you describe what physical situation this corresponded to, you would probably guess that it is a Coulomb point charge plus a plane electromagnetic wave. Once you are safe from the physicist mugger, you would want to compute the components of the field tensor. Doing so, you discover[6] the components have the following values:

$$F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu = \begin{pmatrix} 0 & -x & -y & -z \\ x & 0 & 0 & 0 \\ y & 0 & 0 & 0 \\ z & 0 & 0 & 0 \end{pmatrix} \times \frac{q}{r^3} . \tag{15.27}$$

This potential describes an electric field $\mathbf{E} = q\mathbf{x}/r^3$ and a magnetic field $\mathbf{B} = 0$: it is simply a Coulomb point charge at the origin. There is *no* radiation in this problem at all.

The potential that the mugger shoved in your face was nothing but the usual Coulomb potential, written in a gauge that made it seem radiative. Indeed, to generate this, I started with $A^0 = q/r$, $A^i = 0$ and applied a gauge change $A^\mu \to A^\mu - \partial^\mu \Lambda$ with $\Lambda = qR\cos(\mathbf{k}\cdot\mathbf{x} - \omega t)/r$.

15.4 Gauge invariant characterization of linearized theory

The moral of this story is that when we work with quantities like the metric, the gauge we use can obscure the physics if we are not careful. This motivates a way of understanding which aspects of a spacetime are truly radiative in all gauges, and which are not (at least when we linearize around a flat background spacetime). The following discussion is a brief synopsis of a framework that was presented in E. E. Flanagan and S. A. Hughes, gr-qc/0501041 (available at `arXiv.org`, and linked to the 8.962 website); this presentation in turn was based on ideas we learned from E. Bertschinger, which performed a similar analysis to perturbations of cosmological spacetimes. The end result is that we will find that the degrees of freedom represented by the 10 independent components of the spacetime perturbation $h_{\alpha\beta}$ describe 4 degrees of freedom governed by Poisson-like equations (and thus are not radiative) and 2 degrees of freedom governed by a wave equation. These represent 2 polarizations of gravitational radiation, and will be the main topic of our next lecture. (The remaining 4 degrees of freedom are purely gauge in nature.)

Begin by considering $h_{\mu\nu}$ as a tensor field in a flat background spacetime. We pick inertial coordinates, meaning that we select a time coordinate that we will focus on for this analysis, as

[6] If you check this, don't forget that $\partial^t = \eta^{00}\partial_t = -\partial/\partial t$.

well as 3 space directions. The 10 components of $h_{\mu\nu}$ then break up into 3 subgroups when we consider how they behave under rotations:

$$\begin{aligned} h_{\mu\nu} \quad \mapsto \quad & h_{tt} \equiv -2\phi \qquad \text{(a scalar)} \\ & h_{ti} \qquad \text{(a 3-vector)} \\ & h_{ij} \qquad \text{(a } 3\times 3 \text{ symmetric tensor)} \,. \end{aligned} \tag{15.28}$$

We can break these down further. Consider h_{ti} first. Any 3-vector can be written as a divergence-free function plus the gradient of a scalar, so we write

$$h_{ti} = \beta_i + \partial_i \gamma \,, \qquad \partial_i \beta_i = 0 \,. \tag{15.29}$$

This represents a total 3 degrees of freedom (1 for the function γ, 3 for the functions β_i, plus one constraint imposed by $\partial_i \beta_i = 0$). Note that in this analysis, the placement of spatial indices is immaterial — our time coordinate is fixed, so the spatial metric is just the Kronecker delta. (We modify the summation convention in this situation to sum over repeated indices no matter how they are positioned.) Including the free function ϕ, we have accounted for 4 degrees of freedom in the spacetime metric so far.

Extending this logic to h_{ij}, we find that the most general form of this 3×3 symmetric tensor is

$$h_{ij} = h_{ij}^{\mathrm{TT}} + \frac{1}{3} H \delta_{ij} + \partial_{(i}\epsilon_{j)} + (\partial_i \partial_j - \frac{1}{3}\delta_{ij}\nabla^2)\lambda \,. \tag{15.30}$$

Let's go through the quantities we have introduced here and examine their properties:

- The function H is a scalar under rotations, and represents 1 degree of freedom. We put $H = \delta^{ij} h_{ij}$, so that it is the trace of this 3×3 tensor. This means there is no contribution to the trace from any other terms in Eq. (15.30).

- The function λ is also a scalar under rotations, and represents 1 degree of freedom. It represents a trace-free double gradient of a scalar. The operator which acts upon it is defined to have zero trace since the trace is already bound up in the function H.

- The functions ϵ_j behave as a 3-vector under rotations. In order that it have no trace, we require that $\partial_i \epsilon_i = 0$. This means that this quantity represents 3 free functions plus 1 constraint, for a total of 2 degrees of freedom.

- The functions h_{ij}^{TT} describe the remaining divergence-free, trace-free degrees of freedom in h_{ij}. The 6 functions of h_{ij}^{TT} are subject to 3 constraints to make this quantity divergence free, $\partial_i h_{ij}^{\mathrm{TT}} = 0$, plus 1 constraint to make it trace free, $\delta^{ij} h_{ij}^{\mathrm{TT}} = 0$. These functions thus represent 2 degrees of freedom. The label "TT" indicates that these represent the *transverse* and *trace-free* parts of h_{ij}. "Trace-free" is hopefully clear; the reason why divergenceless corresponds to transverse will be made clear in the next lecture.

These choices sum to a total of 6 degrees of freedom. Including the 3 degrees of freedom associated with γ and β_i and the 1 associated with ϕ, we see that all 10 components of $h_{\mu\nu}$ are accounted for in the degrees of freedom provided by this decomposition.

Before we start working with this metric, we should account for our gauge freedom: we can change the metric by $h_{\mu\nu} \to h_{\mu\nu} - \partial_\mu \xi_\nu - \partial_\nu \xi_\mu$. Doing this has no effect on the curvature tensors, and thus no effect on the Einstein field equations. We write our gauge generator

$$\begin{aligned} \xi_\alpha \quad &\doteq \quad (\xi_t, \xi_i) \\ &\doteq \quad (A, B_i + \partial_i C) \qquad \text{with} \quad \partial_i B_i = 0 \,. \end{aligned} \tag{15.31}$$

The functions A and C are both scalars; B_i is a divergence-free vector. When we use this form of the generator to change gauge, we find that the functions we introduced to decompose the metric change as follows:

$$\begin{aligned}
\phi &\rightarrow \phi + \partial_t A \,, \\
\beta_i &\rightarrow \beta_i - \partial_t B_i \,, \\
\gamma &\rightarrow \gamma - A - \partial_t C \,, \\
H &\rightarrow H - 2\nabla^2 C \,, \\
\lambda &\rightarrow \lambda - 2C \,, \\
\epsilon_i &\rightarrow \epsilon_i - 2B_i \,, \\
h_{ij}^{\rm TT} &\rightarrow h_{ij}^{\rm TT} \,.
\end{aligned} \tag{15.32}$$

Notice that the TT contribution to h_{ij} is gauge invariant; these two degrees of freedom have the same form in *all* gauges. If we "stare" at these functions for a little while, we "notice" that the following combinations of metric functions are totally gauge invariant[7]:

$$\begin{aligned}
\Phi &= \phi + \partial_t \gamma - \frac{1}{2}\partial_t^2 \lambda \,, \\
\Theta &= \frac{1}{3}\left(H - \nabla^2 \lambda\right) \,, \\
\Psi_i &= \beta_i - \frac{1}{2}\partial_t \epsilon_i \,, \quad \partial_i \Psi_i = 0 \,.
\end{aligned} \tag{15.33}$$

These quantities, plus $h_{ij}^{\rm TT}$, represent the gauge-invariant, physical degrees of freedom in the metric $h_{\mu\nu}$. Notice that there are only 6 such degrees of freedom; in the 10 independent components of $h_{\mu\nu}$, 4 are purely gauge degrees of freedom. As we'll see when we talk in more depth about the dynamics of the Einstein field equations in general, this is consistent with the fact that the 10 Einstein field equations ($G_{\alpha\beta} = 8\pi G T_{\alpha\beta}$) are subject to 4 constraints due to the contracted Bianchi identity ($\nabla^\alpha G_{\alpha\beta} = 0$).

We would now like to see what this decomposition of spacetime implies for the Einstein field equations. Because, as shown in Lecture 14, curvature tensor components do not depend on the gauge choice in linearized theory, we should be able to write any curvature tensor using only the gauge-invariant functions Φ, Θ, Ψ_i, and $h_{ij}^{\rm TT}$. We will take advantage of this fact to write out the components of the Einstein tensor in this form. Before doing so, we first introduce a decomposition of the stress-energy tensor similar to what we used in the metric:

$$\begin{aligned}
T_{tt} &= \rho \,, \\
T_{ti} &= S_i + \partial_i S \,, \\
T_{ij} &= P\delta_{ij} + \sigma_{ij} + \partial_{(i}\sigma_{j)} + \left(\partial_i\partial_j - \frac{1}{3}\nabla^2\right)\sigma \,.
\end{aligned} \tag{15.34}$$

These quantities are subject to the constraints

$$\begin{aligned}
\partial_i S_i = 0 \,, \qquad & \partial_i \sigma_i = 0 \,, \\
\partial_i \sigma_{ij} = 0 \,, \qquad & \delta^{ij}\sigma_{ij} = 0 \,.
\end{aligned} \tag{15.35}$$

Several of these terms we have introduced have clear physical meaning: ρ is an energy density, P is a pressure, σ_{ij} is an anisotropic stress. However, some of these are merely rearranging terms in

[7] Scare quotes acknowledging that "stare" and "notice" somewhat understate the challenge of this process. However, it really is just a matter of gathering together functions in such a way that the pieces A, B_i, C which go into the shifted metric functions do not appear in the combinations that we assemble.

a way that is convenient. For instance, T_{ti} is a the flow of energy; $S_i + \partial_i S$ just rewrites this flow using functions with mathematically convenient properties. When we enforce $\nabla^\alpha T_{\alpha\beta} = 0$, we find that

$$\begin{aligned} \nabla^2 S &= \partial_t \rho \,, \\ \nabla^2 \sigma &= -\frac{3}{2} P + \frac{3}{2} \partial_t S \,, \\ \nabla^2 \sigma_i &= 2 \partial_t S_i \,. \end{aligned} \tag{15.36}$$

This amplifies the message that some of the quantities we introduced in the decomposition of the stress-energy tensor are not independent. We can freely specify ρ, P, S_i, and σ_{ij}, a total of 6 functions describing the source; the remaining 4 functions S, σ, and σ_i which complete the source are determined by the behavior of the 6 we can freely specify.

With some labor, we develop the ten components of the Einstein tensor and write the results using the gauge-invariant metric functions we worked out. The result of this exercise is

$$\begin{aligned} G_{tt} &= -\nabla^2 \Theta \,, \\ G_{ti} &= -\frac{1}{2} \nabla^2 \Psi_i - \partial_i \partial_t \Theta \,, \\ G_{ij} &= -\frac{1}{2} \Box h_{ij}^{\rm TT} - \partial_{(i} \Psi_{j)} - \frac{1}{2} \partial_i \partial_j \left(2\Phi + \Theta \right) + \delta_{ij} \left(\frac{1}{2} \nabla^2 (2\Phi + \Theta) - \partial_t^2 \Theta \right) \,. \end{aligned} \tag{15.37}$$

Enforcing $G_{\alpha\beta} = 8\pi G T_{\alpha\beta}$ yields the following 6 differential equations governing the gauge-invariant degrees of freedom in spacetime:

$$\nabla^2 \Phi = 4\pi G \left(\rho + 3P - \partial_t S \right) \,, \tag{15.38}$$

$$\nabla^2 \Theta = -8\pi G \rho \,, \tag{15.39}$$

$$\nabla^2 \Psi_i = -16\pi G S_i \,, \tag{15.40}$$

$$\Box h_{ij}^{\rm TT} = -16\pi G \sigma_{ij} \,. \tag{15.41}$$

This exercise confirms that, although the solution we derived earlier appears to be one in which the metric perturbation is entirely radiative, this is quite misleading. No more than *two* components of the spacetime — those described by the transverse, traceless piece of the metric, $h_{ij}^{\rm TT}$ — satisfy a wave equation in all gauges. Therefore, only those two components truly characterize radiative components of the spacetime. Four of the remaining components of the spacetime are governed by Poisson-type equations, and are often called the "longitudinal degrees of freedom" associated with spacetime. The last four components represent gauge degrees of freedom.

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Lecture 16
Gravitational radiation

16.1 Lesson of the gauge-invariant characterization of metric components

In the previous lecture, we showed that at linear order in deviations from flat spacetime, the Einstein field equations can be reduced to a single wave equation plus several Poisson-type equations:

$$\nabla^2\Phi = 4\pi G\left(\rho + 3P - \partial_t S\right) , \tag{16.1}$$

$$\nabla^2\Theta = -8\pi G\rho , \tag{16.2}$$

$$\nabla^2\Psi_i = -16\pi G S_i , \tag{16.3}$$

$$\Box h_{ij}^{\rm TT} = -16\pi G\sigma_{ij} . \tag{16.4}$$

See the previous lecture for precise definitions connecting the fields Φ, Θ, and Ψ_i to the metric $h_{\alpha\beta}$, and to see how the source terms that appear on the right-hand side of these equations connect to the stress-energy tensor components $T_{\alpha\beta}$.

Our focus in this lecture is to understand the nature of the radiative part of this solution. The transverse, traceless spatial metric components $h_{ij}^{\rm TT}$ describe a propagating, radiative spacetime contribution in all gauges, and thus truly describe a radiative component to the gravitational interaction. Radiation is a starkly non-Newtonian aspect of relativistic gravity, but an absolutely necessary one: *any* relativistic field theory must include radiation in order to connect distant fields to the behavior of a dynamical source, and this radiation must propagate with finite speed. Our particular goals are to understand the nature of $h_{ij}^{\rm TT}$ in terms of quantities that we can observe, and to see how one computes $h_{ij}^{\rm TT}$ given a source.

16.2 A purely radiative solution

To begin, we simply write down a metric that has $\Phi = \Theta = \Psi_i = 0$, and choose $h_{ij}^{\rm TT} = h_{ij}^{\rm TT}(t-z)$ — radiation propagating in the z direction. Our solution takes the form

$$h_{\alpha\beta} \doteq \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & h_{xx}^{\rm TT} & h_{xy}^{\rm TT} & 0 \\ 0 & h_{yx}^{\rm TT} & h_{yy}^{\rm TT} & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} . \tag{16.5}$$

Because this must be symmetric, $h_{xy}^{\rm TT} = h_{yx}^{\rm TT}$; because we demand that it be trace-free, $\delta^{ij}h_{ij}^{\rm TT} = 0$, we have $h_{xx}^{\rm TT} = -h_{yy}^{\rm TT}$. We will return to other aspects of this solution (e.g., why all components of the form $h_{\alpha z}$ are zero) later in this lecture.

What do freely falling bodies experience in this spacetime? Freely falling bodies follow geodesics, so we naturally start by looking at geodesics in the spacetime (16.5): we examine

$$\frac{du^\alpha}{d\tau} + \Gamma^\alpha{}_{\mu\nu}u^\mu u^\nu = 0 , \tag{16.6}$$

taking the bodies to be initially at rest in these coordinates, $u^\alpha(\tau = 0) \doteq (1, 0, 0, 0)$, and computing the connection coefficient from (16.5):

$$\begin{aligned} \Gamma^\alpha{}_{00} &= \eta^{\alpha\beta}\Gamma_{\beta 00} \\ &= \frac{1}{2}\eta^{\alpha\beta}\left(\partial_t h^{\rm TT}_{\beta 0} + \partial_t h^{\rm TT}_{0\beta} - \partial_\beta h^{\rm TT}_{00}\right) \\ &= 0\,. \end{aligned} \tag{16.7}$$

This means that $du^\alpha/d\tau = 0$ at $\tau = 0$; integrating this up, we find that the body has $u^\alpha\dot{(}1, 0, 0, 0)$ for all time.

Does this mean that this radiative solution has no effect? No! — what it means is that we've calculated the wrong thing. Geodesics tell us about the trajectory of a body through a specified coordinate system. We see that a body remains at rest *with respect to these coordinates* as the radiation passes by. But we have tremendous freedom in general relativity to adjust our coordinates, and it could very well be that the coordinates are themselves moving in some way.

Indeed, we know that we can make the metric have a representation that looks just like special relativity in the vicinity of any event. To detect an effect involving the radiation encoded in Eq. (16.5), we should examine some kind of *proper* quantity — i.e., something that we can measure, with value that is independent of the representation — and it should be something that involves enough of a spatial extent that it can "taste" the curvature corrections which provide the leading correction to the $\eta_{\alpha\beta}$ representation of the metric in a local Lorentz frame. Let us examine two bodies that are separated by a displacement L in the x direction:



Let's compute the time takes for a light pulse to travel from A to B. The light pulse follows a null geodesic, so

$$0 = -\left(\frac{dt}{d\lambda}\right)^2 + (1 + h^{\rm TT}_{xx})\left(\frac{dx}{d\lambda}\right)^2\,. \tag{16.8}$$

Rearranging, this gives us an equation for how much time dt elapses as the light travels through a spatial interval dx:

$$\begin{aligned} dt &= \sqrt{1 + h^{\rm TT}_{xx}}\,dx \\ &\simeq \left(1 + \frac{h^{\rm TT}_{xx}}{2}\right)dx\,. \end{aligned} \tag{16.9}$$

We've taken the positive root because light pulse travels in the direction of increasing x. If we examined light traveling in the opposite direction, or if the light reflected at $x = L$ and returning to $x = 0$, we would take the opposite sign. On the second line, we've used the fact that all the elements of $h_{\alpha\beta}$ are small (we are using linearized theory, after all) and so we use the binomial expansion. Integrating this up using the fact that A and B are located at fixed coordinate values $x = 0$ and $x = L$ respectively, the time it takes for light to travel from A to B is

$$\begin{aligned} T_{A\to B} &= \int_0^L \left(1 + \frac{h^{\rm TT}_{xx}}{2}\right)dx \\ &= L + \frac{1}{2}\int_0^L h^{\rm TT}_{xx}\,dx\,. \end{aligned} \tag{16.10}$$

This is enough to show us that the time of arrival of light pulses varies with the amplitude of this component of the radiative solution $h^{\rm TT}_{xx}$. By making use of a very well-defined time standard (and monochromatic light itself defines such a time standard), Eq. (16.10) defines a way in which the imprint of such *gravitational radiation* or *gravitational waves* can be measured.

Note that $h^{\rm TT}_{xx}$ is a function of t and z. As such, one might imagine that it pops out of the integral. However, an integral with respect to x is in a sense also an integral with respect to t, since the light is traveling along a trajectory with $dx/dt = 1 + \mathcal{O}(h)$. Because of this, the function $h^{\rm TT}_{xx}$ in general is not constant inside the integral. This means that without more information about the nature of the wave amplitude $h^{\rm TT}_{xx}$, we cannot simplify (16.10) further. However, there are some important circumstances in which further manipulation can be done; you will explore such a circumstance on problem set 8.

16.3 Geodesic deviation and gravitational radiation

Both body A and body B are freely falling bodies that follow geodesics. Because they are spatially separated from one another in the radiative spacetime (16.5), free fall for body A is a little different from free fall for body B. This shows up in the fact that the proper separation, as measured by light travel time, changes proportional to the amplitude of the radiation. A nice way to describe two nearby geodesics is to examine their geodesic separation. Let us take the two bodies to be close enough that, at least initially, they have the same 4-velocity:

$$u^\alpha_A \simeq u^\alpha_B \doteq (1,0,0,0) + \mathcal{O}(h) \; . \tag{16.11}$$

Slight differences at the level of h in their 4-velocities will couple to produce $\mathcal{O}(h^2)$ differences in the analysis that follows, which we can neglect. Taking the separation vector of these two bodies to be given by Y^α, the equation of geodesic deviation is given by

$$\frac{D^2 Y^\alpha}{d\tau^2} = R^\alpha{}_{\mu\nu\beta} u^\mu u^\nu Y^\beta \; . \tag{16.12}$$

Inserting our 4-velocities, expanding the various derivatives, and using one of the Riemann symmetries, this becomes

$$\frac{d^2 Y^i}{dt^2} = -R^i{}_{0j0} Y^j + \mathcal{O}(h^2) \; . \tag{16.13}$$

For our radiative perturbation, the only unique non-zero Riemann components are

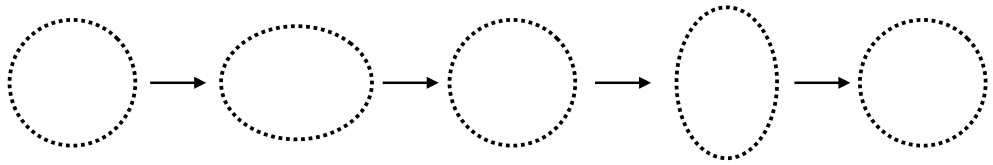
$$\begin{aligned} R^x{}_{0x0} &= -\frac{1}{2}\partial_t^2 h^{\rm TT}_{xx} \; , & (16.14)\\ R^y{}_{0y0} &= -\frac{1}{2}\partial_t^2 h^{\rm TT}_{yy} \\ &= \frac{1}{2}\partial_t^2 h^{\rm TT}_{xx} \; , & (16.15)\\ R^y{}_{0x0} &= -\frac{1}{2}\partial_t^2 h^{\rm TT}_{xy} \\ &= R^x{}_{0y0} & (16.16) \end{aligned}$$

All other components are either zero, or are related by either a Riemann symmetry or the fact that $\partial_z = -\partial_z$ when it acts on the metric (thanks to the fact that the metric depends on $t - z$). Using these forms, the equation of geodesic deviation becomes

$$\begin{aligned} \frac{d^2 Y^x}{dt^2} &= \frac{1}{2}\partial_t^2 h^{\rm TT}_{xx} Y^x + \frac{1}{2}\partial_t^2 h^{\rm TT}_{xy} Y^y & (16.17)\\ \frac{d^2 Y^y}{dt^2} &= \frac{1}{2}\partial_t^2 h^{\rm TT}_{xy} Y^x - \frac{1}{2}\partial_t^2 h^{\rm TT}_{xx} Y^y & (16.18)\\ \frac{d^2 Y^z}{dt^2} &= 0 \; . & (16.19) \end{aligned}$$

Let's assume that $Y^i = Y^i_0 + \delta Y^i$, with Y^i_0 a constant and δY^i having a magnitude $\mathcal{O}(h)$. Consider 2 limits:

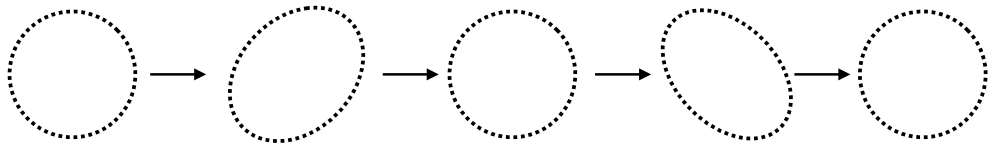
- First, imagine that $h^{\rm TT}_{xx}$ is a sinusoid, and $h^{\rm TT}_{xy} = 0$. Imagine a ring of test masses in the (x, y) plane. As this gravitational wave passes by, the separation vector components $(\delta Y^x, \delta Y^y)$ oscillate with the following pattern:



time increases to the right

We call this the "plus" polarization: it tidally stretches and squeezes along orthogonal axes that are aligned with the x and y axes of our coordinates. We denote this polarization h_+.

- Next, the opposite limit: $h^{\rm TT}_{xy}$ is a sinusoid, $h^{\rm TT}_{xx} = 0$. The separation vector now undergoes the following pattern as time goes on:



time increases to the right

This pattern is the "cross" polarization, since it tidally stretches and squeezes along orthogonal axes aligned at 45° to our coordinates' x and y axes. We denote this polarization $h_\times$.

Recall that $h^{\rm TT}_{ij}$ has only two independent degrees of freedom: the 6 components of the symmetric tensor are constrained by 3 conditions to make it divergence free, plus 1 trace-free condition. The polarizations h_+ and $h_\times$ completely account for these two independent degrees of freedom.

16.4 Computing gravitational radiation from a source

We conclude this lecture by computing radiation given a source. To do this, we are going to take advantage of what we learned from our characterization of the gauge-invariant degrees of freedom in the linearized metric: we compute $h_{\alpha\beta}$ by some convenient method, and then from this project out the $h^{\rm TT}_{ij}$ pieces of this tensor. The result is guaranteed to be the gauge-invariant, physical radiation content of the spacetime that we construct.

A particularly convenient starting point is the solution we already wrote down using the radiative Green's function for the linearized Einstein equation in Lorenz gauge: the field equation

$$\Box \bar{h}_{\alpha\beta} = -16\pi G T_{\alpha\beta} \tag{16.20}$$

has the solution

$$\bar{h}_{\alpha\beta}(t, \mathbf{x}) = 4G \int d^3x' \frac{T_{\alpha\beta}(t - |\mathbf{x} - \mathbf{x}'|, \mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|} \,, \tag{16.21}$$

where $\mathbf{x}$ denotes a field point (the place where we measure $h_{\alpha\beta}$) and $\mathbf{x}'$ is the source point (a point within the source whose stress-energy tensor is $T_{\alpha\beta}$) that we integrate over.

When we evaluate a source of this kind, we are typically interested in a situation in which $|\mathbf{x} - \mathbf{x}'| \gg$ the size of the source. For example, in many astrophysical applications, important sources are tens to hundreds of kilometers in size, and the source may be tens of millions or tens of billions of *light years* from the field point. Noting that 1 light year is about 10^{13} kilometers, we see that the "$\gg$" written above if anything understates the extreme disparity between these two scales. We can thus treat $|\mathbf{x} - \mathbf{x}'|$ as simply r, the distance to the source, and our solution to very good approximation may be written

$$\bar{h}_{\alpha\beta} = \frac{4G}{r} \int d^3x' \, T_{\alpha\beta}(t - r; \mathbf{x}') \; . \tag{16.22}$$

A more careful expansion uses the fact that $1/|\mathbf{x} - \mathbf{x}'|$ can be written as a sum over Legendre polynomials, or over spherical harmonics. Inserting this expansion allows one to define a multipole expansion, and leads to higher-order corrections to the form that we are about to derive.

For us, only the spatial components are important, so we reduce (16.22) to

$$\bar{h}_{ij} = \frac{4G}{r} \int d^3x' \, T_{ij}(t - r; \mathbf{x}') \; . \tag{16.23}$$

We now take advantage of an Easter egg that was hidden very early on in his course: you showed the "tensor virial theorem" on problem set 2,

$$\int d^3 \, x' \, T_{ij} = \frac{1}{2} \frac{d^2}{dt^2} \int d^3x' \, T_{00}(t - r; \mathbf{x}') x'_i x'_j \; . \tag{16.24}$$

Applying this to Eq. (16.23), we have

$$\begin{aligned} \bar{h}_{ij} &= \frac{2G}{r} \frac{d^2}{dt^2} \int d^3x' \, T_{00}(t - r; \mathbf{x}') x'_i x'_j \\ &= \frac{2G}{r} \frac{d^2 I_{ij}}{dt^2} \; . \end{aligned} \tag{16.25}$$

We now need to pull from this the transverse and traceless metric contribution.

Let's consider the transverse condition first. Far from the source, we have $\Box \bar{h}_{ij} = 0$, which suggests that we should expand our solution in plane waves:

$$\bar{h}_{jl} = A_{jl} e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)} \; . \tag{16.26}$$

The amplitude A_{jl} has no dependence on t or $\mathbf{x}$, though it may depend on ω or $\mathbf{k}$. A more complete presentation would write this as an integral over ω and $\mathbf{k}$; here we just examine things mode by mode[1], which is sufficient for our purposes.

Our "transverse" condition is that $\partial_j \bar{h}_{jl} = 0$. Let's enforce this:

$$\partial_j \bar{h}_{jl} = i k_j A_{jl}(\omega, \mathbf{k}) e^{i(\mathbf{k} \cdot x - \omega t)} = 0 \; . \tag{16.28}$$

This condition is equivalent to requiring that

$$k_j \bar{h}_{jl} = 0 \; . \tag{16.29}$$

[1] Such an expansion writes the waveform

$$\bar{h}_{jl} = \int d\omega \, d^3k \, A_{jl}(\omega, \mathbf{k}) e^{i(\mathbf{k} \cdot x - \omega t)} \; . \tag{16.27}$$

Everything given above carries over to this integral's integrand. Thanks to the linearity of this limit of general relativity, it is simple to combine each mode to assemble this integrated solution.

In other words, the wave has no components parallel to the direction of propagation. (This is why the form that I wrote down when we first began to explore radiative solutions had no z components — recall that it described a wave propagating along the z direction.) To guarantee that our solution satisfies this condition, define the projection tensor

$$P_{ij} = \delta_{ij} - n_i n_j \ , \qquad \text{where} \quad n_i = k_i/\sqrt{|\mathbf{k}|^2} \ . \tag{16.30}$$

Recalling some exercises from problem set 1 (problem 4 in particular), we recognize this as a tensor that projects tensor components into a subspace that is orthogonal to $\mathbf{n}$. The quantity $\mathbf{n}$ is in turn a unit vector (or unit 1-form) along the direction of propagation. It's also worth recalling from this homework exercise that the projection tensor P_{ij} can be regarded as the metric for geometric objects which live in that subspace orthogonal to $\mathbf{n}$. Using this, our transverse metric perturbation is

$$\bar{h}^{\rm T}_{ij} = \bar{h}_{lk} P_{li} P_{kj} \ . \tag{16.31}$$

Finally, we need to remove the trace. The general formula to make a general 2-index tensor[2] K_{ab} trace free is

$$K^{\rm TF}_{ab} = K_{ab} - \frac{1}{N}\left(g^{cd}K_{cd}\right) g_{ab} \ , \tag{16.32}$$

where g_{ab} is the metric describing the manifold in which the tensors K_{ab} and $K^{\rm TF}_{ab}$ reside, and N is the dimension of this manifold. (Note that $g^{ab}g_{ab} = N$.) For our current problem, the metric is P_{ij}, and $N = 2$. Applying this, and using the fact that raising and lowering indices is not meaningful in these spatial coordinates, we have

$$\bar{h}^{\rm TT}_{ij} = \bar{h}^{\rm T}_{ij} - \frac{1}{2}\bar{h}^{\rm T}_{mn} P_{mn} P_{ij} \ . \tag{16.33}$$

However, we also have

$$\bar{h}^{\rm T}_{mn} P_{mn} P_{ij} = \bar{h}_{lk} P_{lm} P_{kn} P_{mn} P_{ij} \ . \tag{16.34}$$

Using problem 4 on problem set 1 again,

$$P_{lm} P_{kn} P_{mn} P_{ik} = P_{lk} P_{ij} \ . \tag{16.35}$$

We thus finally have

$$\begin{aligned} \bar{h}^{\rm TT}_{ij} &= \bar{h}_{lk} P_{li} P_{kj} - \frac{1}{2}\bar{h}_{lk} P_{lk} P_{ij} \\ &= h^{\rm TT}_{ij} \ . \end{aligned} \tag{16.36}$$

In the final equality, we remove the overbar that denotes trace reversal. This is because once we've removed the trace, there's no distinction between the transverse-traceless metric perturbation $h^{\rm TT}_{ij}$ and its trace-reversed version $\bar{h}^{\rm TT}_{ij}$. Trace reversal has no effect when the trace is zero. (Though see the next section for a caveat when we apply this logic to realistic problems.) Using the solution we worked out earlier for $h^{\rm TT}_{ij}$, we have

$$\boxed{\boxed{h^{\rm TT}_{ij} = \frac{2G}{r}\frac{d^2 I_{lk}}{dt^2}\left(P_{li}P_{kj} - \frac{1}{2}P_{lk}P_{ij}\right)}} \tag{16.37}$$

The result is known as the *quadrupole formula* for the amplitude of gravitational radiation; it places a role in gravitational radiation theory quite similar to the role played by the dipole formula for the electromagnetic potential in electrodynamics.

[2] We use a and b for our indices because we are being agnostic about the nature of the geometry in which this tensor lives. It could be 4-dimensional spacetime; it could be N-dimensional space.

16.5 A subtlety regarding trace removal

A key step in the calculation above was equating $h_{ij}^{\rm TT}$ to $\bar{h}_{ij}^{\rm TT}$. This equality was justified by the idea that when the trace has been removed there is no difference between the two quantities. Our take-away slogan was that trace reversing has no effect when the trace is zero.

Strictly speaking, this is at best somewhat glib, and requires more careful justification. In particular, note that there are two notions of "trace" at play in this calculation, essentially because we are working with two different notions of "metric." We define $\bar{h}_{\alpha\beta} = h_{\alpha\beta} - (h/2)\eta_{\alpha\beta}$, with $h \equiv \eta^{\alpha\beta}h_{\alpha\beta}$. The trace here is defined using the *spacetime* metric $\eta_{\alpha\beta}$. By contrast, we define the traceless nature of the field $h_{ij}^{\rm TT}$ by requiring that $\delta^{ij}h_{ij}^{\rm TT} = 0$. This field has zero trace with respect to the *spatial* metric δ_{ij}.

So if the "trace" has been removed can we really say that $\bar{h}_{ij}^{\rm TT} = h_{ij}^{\rm TT}$? If our spacetime consists of nothing but flat spacetime and gravitational radiation, then yes. However, if there are other sources of gravity that contribute to the gauge-invariant fields Φ and θ that we defined in Lecture 15, then the answer is no. Those fields affect the notion of trace that enters the definition of $\bar{h}_{\alpha\beta}$, and are completely absent from $h_{ij}^{\rm TT}$. Aspects of these two non-radiative gauge-invariant fields get mixed into $\bar{h}_{ij}^{\rm TT}$, so the equality asserted above cannot be strictly correct.

The reason why the equality nonetheless works very well is because of aspects of the theory of gravitational radiation that we will discuss in Lecture 17. A gravitational wave is strictly speaking only defined in a context in which we can split aspects of spacetime that vary on timescales and lengthscales, τ and λ, that are much shorter than the timescales and lengthscales, $\mathcal{T}$ and $\mathcal{L}$, on which the "background" varies. In this context, the fields Φ and Θ (which recall satisfy Poisson-time equations) will always vary only very slowly with time (if at all), and will only vary on very long lengthscales.

Physics is ultimately about measurement, and when we measure the properties of curved spacetime we do so over a set of lengthscales and timescales that are ultimately determined by our measuring apparatus. When we focus on the timescale and lengthscales that correspond to radiation, then we indeed discover that $\bar{h}_{ij}^{\rm TT} = h_{ij}^{\rm TT}$; the "contaminating" pieces of Φ and Θ that affect the trace $h = \eta^{\alpha\beta}h_{\alpha\beta}$ and thereby enter into $\bar{h}_{ij}$ do not affect the radiative aspects of our solution on the lengthscales and timescales that characterize gravitational radiation.

An application of this principle can be seen in how ground-based gravitational-wave antenna like LIGO and Virgo operate. The gravitational waves that they probe describe spacetime oscillations at frequencies of tens to thousands of Hertz. *No other* gravitational phenomenon near the Earth's surface varies on similar time scales, so even if the Poisson-type fields "contaminate" our notion of the trace-free metric term as we have constructed it above, those fields do not vary on timescales that these antennas can probe. The strongest time-varying non-radiative gravitational term which the detectors have to worry about is the contribution to spacetime from the Moon, which varies on much longer timescales due to the Moon's orbit and the rotation of the Earth. (Of course, many *non*-gravitational phenomena oscillate in the antenna's sensitive band, generating noise. As such, it is hugely important to shield the test masses which probe for gravitational radiation from these noise sources. As detectors get more sensitive, local gravity may start to become a source of noise: future detectors are expected to be limited at the lowest frequencies by gravitational coupling to seismic modes in the ground on which the detectors are built. See https://arxiv.org/abs/gr-qc/9806018 for detailed discussion.)

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Lecture 17
Gravitational radiation on a curved background; radiative backreaction

17.1 Gravitational radiation in a general background spacetime: Conceptual foundation

Our detailed discussion of gravitational waves so far defines these waves as "small" perturbations about a flat background spacetime. A lot of useful results and formalism grow out of this analysis, but this definition is quite restrictive. In the general case, we imagine that waves are propagating on some reasonably well characterized but *non*-flat background:

$$g_{\alpha\beta} = \hat{g}_{\alpha\beta} + h_{\alpha\beta} \ . \tag{17.1}$$

The background $\hat{g}_{\alpha\beta}$ can now be something that varies in space and time; the perturbation $h_{\alpha\beta}$ will typically be small, though there may be circumstances where it is not as small as we imagined in our previous discussion.

Within this framework, how do we even define what part of the spacetime is "wave" and what part is "background"? Local measurements only measure some surrogate of the total spacetime metric $g_{\alpha\beta}$. Based on what we have laid out so far, a measurement does not enable use to clearly distinguish what is background from what is radiation. How do we distinguish between the radiative aspects of spacetime and its non-radiative, background aspects?

The trick to separating "radiation" from "background" is to use the idea that the radiation and background vary on very different scales. We imagine that the radiation varies on timescales τ and lengthscales λ that are far shorter than the timescales $\mathcal{T}$ and lengthscales $\mathcal{L}$ which describe the background. Derivatives associated with radiation will thus introduce scalings $\sim 1/\tau$ and $\sim 1/\lambda$ into the analysis[1]; derivatives associated with the background introduce scalings $\sim 1/\mathcal{T}$ and $\sim 1/\mathcal{L}$. A useful analogy is to think of water waves propagating on an ocean. If one imagines oneself to be an ant riding a cork that floats on the water's surface, you just follow the water's behavior, and are agnostic about whether you are in a wave, are following the curvature of the Earth, or are being affected by some local structure in the water. But if we zoom out and examine this cork from a larger perspective, we can use a separation of scales to tell whether the cork is rising and falling due to a passing wave, or is being affected by the curvature of the Earth or a local structure. Waves are associated with small scale, rapid oscillations; aspects of the dynamics related to larger-scale, long-duration structures are not impacted in such a small scale, rapid way. Our goal now is to understand how to similarly separate scales in our discussion of the dynamics of spacetime.

17.2 Gravitational radiation in a general background spacetime: Technical details

We now develop the concepts laid out above. We assume that the background varies on long scales $(\mathcal{T}, \mathcal{L})$. To set some context, it is useful to put numbers to these quantities: in a frame attached to the surface of the earth, $\mathcal{L}$ is on the order of $\sqrt{R_{\rm Earth}^3/GM_{\rm Earth}} \simeq 2.4 \times 10^8\,{\rm km}$, the typical

[1] In units with $c = 1$, $\tau = \lambda$. We maintain a distinction because there is no reason that the long scales are connected in a similar way, so we will often want to separately compare timescales and lengthscales.

length scale[2] associated with the Earth's spacetime curvature. In this frame, $\mathcal{T}$ is on the order of several hours, the timescale over which the Earth's rotation changes the orientation of this frame with respect to the solar system (thus changing the magnitude of tidal effects from the Moon and the Sun). Any gravitational wave will need to vary on short timescales (τ, λ) that are much smaller than $(\mathcal{T}, \mathcal{L})$.

We will implement this idea by defining an *averaging* operation: Imagine there exists a scale $L \sim$ (several)λ, $T \sim$ (several)τ, where "several" is small enough that $L \ll \mathcal{L}$, $T \ll \mathcal{T}$. Our background is then given by averaging spacetime over this scale, and the wave by subtracting this average from the total:

$$\hat{g}_{\alpha\beta} \equiv \langle g_{\alpha\beta} \rangle \ , \qquad h_{\alpha\beta} = g_{\alpha\beta} - \hat{g}_{\alpha\beta} \ . \tag{17.2}$$

A version of this averaging procedure was introduced in a paper by Dieter Brill and James Hartle[3], and adapted to the form that we use here by R. A. Isaacson[4]. The averaging consists of integrating the metric (or any other tensor) against a weighting function as follows:

$$\langle g_{\alpha\beta}(x^\mu) \rangle = \int d^4x' \, g_{\alpha'\beta'}[(x')^\mu] \, f^{\alpha'\beta'}_{\alpha\beta}[x^\mu, (x')^\mu] \ . \tag{17.3}$$

Note that when we implement this procedure, we are necessarily taking a somewhat coarse-grained view of the nature of spacetime, or at least of the background spacetime. Again, this is reminiscent of the water-wave metaphor: we can only sensibly define the action of the wave if we move away from viewing the water in the immediate vicinity of the floating cork and cast our view out on a scale similar to the wavelength of the waves themselves.

The weighting function $f^{\alpha'\beta'}_{\alpha\beta}[x^\mu, (x')^\mu]$ we have introduced here allows one to transport tensorial quantities from an event $(x')^\mu$ to an event x^μ. It is taken to be peaked when $x^\mu = (x')^\mu$, it has a width of order L, T, and is normalized such that

$$\int d^4x' \, f^{\alpha'\beta'}_{\alpha\beta}[x^\mu, (x')^\mu] = \delta^{\alpha'}{}_{\alpha} \delta^{\beta'}{}_{\beta} \ . \tag{17.4}$$

As shown in detail by Isaacson, this averaging procedure "washes out" oscillations on a scale $\lambda < L$, $\tau < T$, but does not have a large impact on features which vary on the long scales $\mathcal{L}, \mathcal{T}$. The output of this procedure is in fact a tensor up to errors on the order of the ratio of scales. In other words, when one changes representation, one finds that $\hat{g}_{\alpha\beta}$ follows the tensor transformation rule up to errors $\mathcal{O}[(\tau/\mathcal{T}), (\lambda/\mathcal{L})]$.

17.2.1 Christoffel symbols and curvature tensors for perturbation to a curved background

With this in mind, let's now develop the mathematical machinery which describes gravitational radiation in this framework. Writing $g_{\alpha\beta} = \hat{g}_{\alpha\beta} + h_{\alpha\beta}$, we will develop the connection and curvature tensors, expanding all quantities to linear order in the wave amplitude h. We begin by noting that

$$g^{\alpha\beta} = \hat{g}^{\alpha\beta} - h^{\alpha\beta} \ , \qquad \text{where} \quad h^{\alpha\beta} = \hat{g}^{\alpha\mu} \hat{g}^{\beta\nu} h_{\mu\nu} \ . \tag{17.5}$$

[2] We use the idea that the components of the Riemann tensor in a Cartesian-like coordinate representation have dimensions $1/(\text{length})^2$, and have defined this length as $\sqrt{1/(\text{typical Riemann component})}$.

[3] D. B. Brill and J. B. Hartle, Phys. Rev. **135**, 271 (1964); a link to this paper will be posted to the 8.962 webpage.

[4] R. A. Isaacson, Phys. Rev. **166**, 1272 (1968); a link to this paper will be posted to the 8.962 webpage.

It is simple to verify this by evaluating $g^{\alpha\mu}g_{\mu\beta}$ and discarding terms of $\mathcal{O}(h^2)$. Using this, let's assemble the Christoffel connection:

$$\begin{aligned}
\Gamma^{\alpha}{}_{\mu\nu} &= \frac{1}{2}g^{\alpha\beta}\left(\partial_\mu g_{\nu\beta} + \partial_\nu g_{\beta\mu} - \partial_\beta g_{\mu\nu}\right) \\
&= \frac{1}{2}\left(\hat{g}^{\alpha\beta} - h^{\alpha\beta}\right)\left(\partial_\mu \hat{g}_{\nu\beta} + \partial_\nu \hat{g}_{\beta\mu} - \partial_\beta \hat{g}_{\mu\nu} - \partial_\mu h_{\nu\beta} - \partial_\nu h_{\beta\mu} + \partial_\beta h_{\mu\nu}\right) \\
&= \hat{\Gamma}^{\alpha}{}_{\mu\nu} - h^{\alpha\beta}\hat{g}_{\beta\gamma}\hat{\Gamma}^{\gamma}{}_{\mu\nu} + \frac{1}{2}\hat{g}^{\alpha\beta}\left(\partial_\mu h_{\nu\beta} + \partial_\nu h_{\beta\mu} - \partial_\beta h_{\mu\nu}\right) + \mathcal{O}(h^2)\ . \qquad (17.6)
\end{aligned}$$

Here, $\hat{\Gamma}^{\alpha}{}_{\mu\nu}$ means a Christoffel connection defined in the background spacetime. Let us define $\hat{\nabla}_\mu$ as a covariant derivative taken with respect to the background. We can replace the partial derivatives in the above expression with $\hat{\nabla}_\mu$ at the cost of introducing additional background Christoffels coupling to the perturbation. Doing so and organizing terms, we find that the result can be written

$$\Gamma^{\alpha}{}_{\mu\nu} = \hat{\Gamma}^{\alpha}{}_{\mu\nu} + \delta\Gamma^{\alpha}{}_{\mu\nu}\ , \qquad (17.7)$$

where

$$\delta\Gamma^{\alpha}{}_{\mu\nu} = \frac{1}{2}\hat{g}^{\alpha\beta}\left(\hat{\nabla}_\mu h_{\nu\beta} + \hat{\nabla}_\nu h_{\beta\mu} - \hat{\nabla}_\beta h_{\mu\nu}\right)\ . \qquad (17.8)$$

Continuing in this fashion, it is straightforward to show that the Riemann tensor takes the form

$$R^{\alpha}{}_{\beta\gamma\delta} = \hat{R}^{\alpha}{}_{\beta\gamma\delta} + \delta R^{\alpha}{}_{\beta\gamma\delta}\ , \qquad (17.9)$$

where $\hat{R}^{\alpha}{}_{\beta\gamma\delta}$ is assembled from $\hat{g}_{\alpha\beta}$ in the usual way, but where we find

$$\delta R^{\alpha}{}_{\beta\gamma\delta} = \hat{\nabla}_\gamma \delta\Gamma^{\alpha}{}_{\beta\delta} - \hat{\nabla}_\delta \delta\Gamma^{\alpha}{}_{\beta\gamma}\ . \qquad (17.10)$$

17.2.2 Generalizing gauge transformations to a curved background

Before moving on, an important tool for what is to come is the notion of an infinitesimal coordinate transformation on a curved background, which will generalize the notion of a gauge transformation to such a background. We begin with a spacetime whose metric we represent as some background $\hat{g}_{\alpha\beta}$ plus a perturbation $h_{\alpha\beta}$; these are all expressed in terms of the coordinate x^μ. We wish to change to a new coordinate y^μ which is related to x^μ by an infinitesimal shift ξ^μ:

$$y^\mu = x^\mu + \xi^\mu \quad \rightarrow \quad x^\mu = y^\mu - \xi^\mu(y^\gamma)\ . \qquad (17.11)$$

(This second form is a somewhat more convenient form for the analysis we need to perform.) This coordinate transformation changes the metric according to

$$g_{\alpha\beta}(y^\mu) = g_{\gamma\delta}(x^\mu)\frac{\partial x^\gamma}{\partial y^\alpha}\frac{\partial x^\delta}{\partial y^\beta}\ . \qquad (17.12)$$

Expand this expression:

$$\begin{aligned}
g_{\alpha\beta}(y^\mu) &= \left(\hat{g}_{\gamma\delta} + h_{\gamma\delta}\right)\left(\delta^{\gamma}{}_{\alpha} - \partial_\alpha \xi^\gamma\right)\left(\delta^{\delta}{}_{\beta} - \partial_\beta \xi^\delta\right) \\
&= \hat{g}_{\alpha\beta} + h_{\alpha\beta} - \hat{g}_{\gamma\beta}\partial_\alpha\xi^\delta - \hat{g}_{\alpha\delta}\partial_\beta\xi^\gamma + \mathcal{O}[h\,\partial\xi, (\partial\xi)^2] \\
&= \hat{g}_{\alpha\beta} + h_{\alpha\beta} - \hat{g}_{\gamma\beta}(\hat{\nabla}_\alpha\xi^\gamma - \xi^\delta\hat{\Gamma}^{\gamma}{}_{\alpha\delta}) - \hat{g}_{\alpha\delta}(\hat{\nabla}_\beta\xi^\delta - \xi^\gamma\hat{\Gamma}^{\delta}{}_{\beta\gamma}) \\
&= \hat{g}_{\alpha\beta} + h_{\alpha\beta} - \hat{\nabla}_\alpha\xi_\beta - \hat{\nabla}_\beta\xi_\alpha + \xi^\gamma(\hat{\Gamma}_{\beta\alpha\gamma} + \hat{\Gamma}_{\alpha\beta\gamma}) \\
&= \hat{g}_{\alpha\beta} + h_{\alpha\beta} - \hat{\nabla}_\alpha\xi_\beta - \hat{\nabla}_\beta\xi_\alpha + \xi^\gamma\partial_\gamma\hat{g}_{\alpha\beta}\ . \qquad (17.13)
\end{aligned}$$

On going from the fourth to the fifth line, we've used the definition of the Christoffel symbol. On that last line, the background in this expression is currently expressed as a function of the old coordinates, x^γ. We want it to appear as a function of the new coordinates, y^μ:

$$\begin{aligned}\hat{g}_{\alpha\beta}(x^\mu) &= \hat{g}_{\alpha\beta}(y^\mu - \xi^\mu) \\ &= \hat{g}_{\alpha\beta}(y^\mu) - \xi^\mu \partial_\mu \hat{g}_{\alpha\beta} \ . \end{aligned} \tag{17.14}$$

Note, we could do this expansion for the perturbation as well: $h_{\alpha\beta}(x^\mu) = h_{\alpha\beta}(y^\mu - \xi^\mu) = h_{\alpha\beta}(y^\mu) - \xi^\mu \partial_\mu h_{\alpha\beta}$. This means that $h_{\alpha\beta}(x^\mu) = h_{\alpha\beta}(y^\mu)$ plus terms that are of order "small squared." We neglect these "small squared" contributions.

Substituting the result for $\hat{g}_{\alpha\beta}(x^\mu)$ into $g_{\alpha\beta}(y^\mu)$ yields

$$g_{\alpha\beta}(y^\mu) = \hat{g}_{\alpha\beta} + h_{\alpha\beta} - \hat{\nabla}_\alpha \xi_\beta - \hat{\nabla}_\beta \xi_\alpha \ , \tag{17.15}$$

from which we read off

$$h_{\alpha\beta} \to h_{\alpha\beta} - \hat{\nabla}_\alpha \xi_\beta - \hat{\nabla}_\beta \xi_\alpha \ . \tag{17.16}$$

Equation (17.16) generalizes the notion of a gauge transformation to a curved background.

17.2.3 The Einstein tensor for a perturbation to a curved background

We now are at last ready to assemble the Einstein tensor. For simplicity, we will do this in vacuum; this will give us an equation that describes how radiation propagates on a curved background. Making this simplification allows us to set $\hat{R}_{\mu\nu} = \hat{G}_{\mu\nu} = 0$; it is straightforward but tedious to lift this approximation, and the resulting equations are significantly more complicated to work with. As you will see, what results from this already simplified analysis is complicated enough for now.

The piece we need to start this analysis comes from the Riemann term $\delta R^\alpha{}_{\mu\beta\nu}$. Writing this out in terms of the perturbation, we find that

$$\delta R^\alpha{}_{\mu\beta\nu} = \frac{1}{2}\left(\hat{\nabla}_\beta \hat{\nabla}_\mu h^\alpha{}_\nu + \hat{\nabla}_\beta \hat{\nabla}_\nu h^\alpha{}_\mu - \hat{\nabla}_\beta \hat{\nabla}^\alpha h_{\mu\nu} - \hat{\nabla}_\nu \hat{\nabla}_\mu h^\alpha{}_\beta - \hat{\nabla}_\nu \hat{\nabla}_\beta h^\alpha{}_\mu + \hat{\nabla}_\nu \hat{\nabla}^\alpha h_{\mu\beta}\right) \ . \tag{17.17}$$

Contracting on α and β yields

$$\delta R_{\mu\nu} = \frac{1}{2}\left(\hat{\nabla}_\alpha \hat{\nabla}_\mu h^\alpha{}_\nu + \hat{\nabla}_\alpha \hat{\nabla}_\nu h^\alpha{}_\mu - \hat{\Box} h_{\mu\nu} - \hat{\nabla}_\nu \hat{\nabla}_\mu h\right) \ . \tag{17.18}$$

Note that the final two terms inside parentheses on the right-hand side of (17.17) sum to zero when we contract on α and β. We have introduced $\hat{\Box} = \hat{g}^{\mu\nu}\hat{\nabla}_\mu \hat{\nabla}_\nu$, a covariant wave operator on the background spacetime. The Ricci scalar corresponding to this is given by

$$\delta R = \hat{g}^{\mu\nu} R_{\mu\nu} = -\hat{\Box} h + \hat{\nabla}_\alpha \hat{\nabla}_\beta h^{\beta\alpha} \ . \tag{17.19}$$

With all this in hand, we finally assemble the Einstein tensor[5]:

$$\begin{aligned}\delta G_{\mu\nu} &= \delta R_{\mu\nu} - \frac{1}{2}\hat{g}_{\mu\nu}\delta R \\ &= -\frac{1}{2}\hat{\Box} h_{\mu\nu} + \frac{1}{2}\hat{g}_{\mu\nu}\hat{\Box} h - \frac{1}{2}\hat{\nabla}_\nu \hat{\nabla}_\mu h + \frac{1}{2}\hat{\nabla}_\alpha \hat{\nabla}_\mu h^\alpha{}_\nu + \frac{1}{2}\hat{\nabla}_\alpha \hat{\nabla}_\nu h^\alpha{}_\mu - \frac{1}{2}\hat{g}_{\mu\nu}\hat{\nabla}_\alpha \hat{\nabla}_\beta h^{\beta\alpha} \ . \end{aligned} \tag{17.20}$$

[5]Note that if we were not in a vacuum, $\delta G_{\mu\nu}$ would contain an additional term proportional to $h_{\mu\nu}\hat{R}$. This is an example of how this problem becomes more complicated as we go beyond this simple first exposition of these ideas.

Just as when we first assembled the analog of this tensor with a flat background, the result is somewhat unwieldy. We can clean it up significantly by replacing the metric perturbation with its trace-reversed version: we put

$$\bar{h}_{\mu\nu} = h_{\mu\nu} - \frac{1}{2} h\, \hat{g}_{\mu\nu} \ , \qquad \text{where} \quad h \equiv \hat{g}^{\alpha\beta} h_{\alpha\beta} \ . \tag{17.21}$$

Using this, we substitute $h_{\mu\nu} = \bar{h}_{\mu\nu} + (h/2)\hat{g}_{\mu\nu}$ and use the fact that $\hat{\nabla}_\alpha \hat{g}_{\mu\nu} = 0$. The result is

$$\delta G_{\mu\nu} = -\frac{1}{2}\hat{\Box}\bar{h}_{\mu\nu} + \frac{1}{2}\hat{\nabla}_\alpha\hat{\nabla}_\mu \bar{h}^\alpha{}_\nu + \frac{1}{2}\hat{\nabla}_\alpha\hat{\nabla}_\nu \bar{h}^\alpha{}_\mu - \frac{1}{2}\hat{g}_{\mu\nu}\hat{\nabla}_\alpha\hat{\nabla}_\beta \bar{h}^{\beta\alpha} \ . \tag{17.22}$$

This is similar in form to what we found when we developed this tensor on a flat background. However, in that case, all of the derivatives which appeared after the wave operator were partial derivatives. We were able to rearrange all of them so that they took the form of a divergence of the metric perturbation. We then selected a gauge such that the divergence vanished, and found that the wave had a much simpler form.

Our derivatives are now covariant rather than partial, so when we exchange their order, we find a coupling to the background Riemann tensor. Let's examine this for the two terms in (17.22) that are the most problematic:

$$\begin{aligned} \hat{\nabla}_\alpha\hat{\nabla}_\mu \bar{h}^\alpha{}_\nu &= \hat{\nabla}_\mu\hat{\nabla}_\alpha \bar{h}^\alpha{}_\nu + \hat{R}^\alpha{}_{\beta\alpha\mu}\bar{h}^\beta{}_\nu - \hat{R}^\beta{}_{\nu\alpha\mu}\bar{h}^\alpha{}_\beta \\ &= \hat{\nabla}_\mu\hat{\nabla}_\alpha \bar{h}^\alpha{}_\nu - \hat{R}_{\beta\nu\alpha\mu}\bar{h}^{\alpha\beta} \\ &= \hat{\nabla}_\mu\hat{\nabla}_\alpha \bar{h}^\alpha{}_\nu - \hat{R}_{\alpha\mu\beta\nu}\bar{h}^{\alpha\beta} \ . \end{aligned} \tag{17.23}$$

We used the fact that $\hat{R}^\alpha{}_{\beta\alpha\nu} = \hat{R}_{\beta\nu}$, which is zero by our assumption that the spacetime is vacuum. We also used a Riemann symmetry to arrange the indices on the final line. Repeating this exercise for the other term of this form yields

$$\hat{\nabla}_\alpha\hat{\nabla}_\nu \bar{h}^\alpha{}_\mu = \hat{\nabla}_\nu\hat{\nabla}_\alpha \bar{h}^\alpha{}_\mu - \hat{R}_{\alpha\mu\beta\nu}\bar{h}^{\alpha\beta} \ . \tag{17.24}$$

To get this form, we used Riemann symmetry, the symmetry of the metric, and relabeled dummy indices. Our expression for the Einstein tensor now becomes

$$\delta G_{\mu\nu} = -\frac{1}{2}\hat{\Box}\bar{h}_{\mu\nu} - \hat{R}_{\alpha\mu\beta\nu}\bar{h}^{\alpha\beta} + \frac{1}{2}\hat{\nabla}_\mu\hat{\nabla}_\alpha \bar{h}^\alpha{}_\nu + \frac{1}{2}\hat{\nabla}_\nu\hat{\nabla}_\alpha \bar{h}^\alpha{}_\nu - \frac{1}{2}\hat{g}_{\mu\nu}\hat{\nabla}_\alpha\hat{\nabla}_\beta \bar{h}^{\beta\alpha} \ . \tag{17.25}$$

As we justify in the next subsection, we can choose our gauge such that $\hat{\nabla}_\alpha \bar{h}^\alpha{}_\nu = 0$; we can further set gauge such that the metric perturbation is trace free. This choice is known as "TT" gauge. In TT gauge, there is no difference between $\bar{h}_{\mu\nu}$ and $h_{\mu\nu}$, and our perturbed Einstein equation yields the following equation for the propagation of gravitational waves on a curved background:

$$\boxed{\boxed{\hat{\Box} h_{\mu\nu} + 2\hat{R}_{\alpha\mu\beta\nu}h^{\alpha\beta} = 0}} \tag{17.26}$$

Equation (17.26) describes how gravitational waves propagate over a curved background. In addition to having dynamics determined by the wave operator $\hat{\Box}$, we see that radiation "scatters" off of spacetime curvature. (Note that the presentation of this equation in Flanagan and Hughes 2005 has a minus sign error in front of the Riemann term.)

17.2.4 Lorenz gauge and TT gauge

To get our final wave equation, we asserted that we could put the metric perturbation into a gauge that is divergence free. Let's examine this condition more closely. Under a change of gauge, we change from $h^{\rm old}_{\mu\nu}$ to

$$h^{\rm new}_{\mu\nu} = h^{\rm old}_{\mu\nu} - \hat{\nabla}_\mu \xi_\nu - \hat{\nabla}_\nu \xi_\mu \ . \tag{17.27}$$

Applying this to the trace-reversed perturbation, we find

$$\bar{h}^{\rm new}_{\mu\nu} = \bar{h}^{\rm old}_{\mu\nu} - \hat{\nabla}_\mu \xi_\nu - \hat{\nabla}_\nu \xi_\mu + \hat{g}_{\mu\nu}\hat{\nabla}^\alpha \xi_\alpha \ . \tag{17.28}$$

Let's take the covariant background divergence of this:

$$\hat{\nabla}^\mu \bar{h}^{\rm new}_{\mu\nu} = \hat{\nabla}^\mu \bar{h}^{\rm old}_{\mu\nu} - \hat{\Box}\xi_\nu - \hat{\nabla}^\mu \hat{\nabla}_\nu \xi_\mu + \hat{\nabla}_\nu \nabla^\alpha \xi_\alpha \ . \tag{17.29}$$

Let's focus on that final term, $\hat{\nabla}_\nu \nabla^\alpha \xi_\alpha = \hat{\nabla}_\nu \nabla^\mu \xi_\mu = \hat{\nabla}_\nu \hat{\nabla}_\mu \xi^\mu$ (relabeling the dummy index). If we could exchange the order of derivatives, this would cancel perfectly with the preceding term, leaving us with a fairly simple condition. However, doing such an exchange introduces a factor of the Riemann tensor:

$$\begin{aligned} \hat{\nabla}_\nu \hat{\nabla}_\mu \xi^\mu &= \hat{\nabla}_\mu \hat{\nabla}_\nu \xi^\mu + \hat{R}^\mu{}_{\gamma\nu\mu}\xi^\gamma \\ &= \hat{\nabla}_\mu \hat{\nabla}_\nu \xi^\mu - \hat{R}_{\gamma\nu}\xi^\gamma \\ &= \hat{\nabla}_\mu \hat{\nabla}_\nu \xi^\mu \\ &= \hat{\nabla}^\mu \hat{\nabla}_\nu \xi_\mu \ . \end{aligned} \tag{17.30}$$

In going from the first to the second lines, we use the fact that the Riemann tensor which enters this expression is contracted on indices 1 and 4, yielding the (negative) Ricci tensor. But Ricci is zero by our simplifying assumption that the background is a vacuum solution. So, in fact we *can* exchange those derivatives, and we find

$$\hat{\nabla}^\mu \bar{h}^{\rm new}_{\mu\nu} = \hat{\nabla}^\mu \bar{h}^{\rm old}_{\mu\nu} - \hat{\Box}\xi_\nu \ . \tag{17.31}$$

We can thus develop a curved spacetime Lorenz gauge, allowing us to set all divergences of $\bar{h}_{\mu\nu}$ to zero by choosing gauge generators such that

$$\hat{\Box}\xi_\nu = \hat{\nabla}\bar{h}^{\rm old}_{\mu\nu} \ . \tag{17.32}$$

Solutions of this exist everywhere that the background satisfies the vacuum Einstein equations. If the background were not vacuum, this condition would be modified by a coupling of the gauge generator to the background Ricci tensor, which implies a coupling the stress-energy tensor; we comment on such situations in the next subsection.

There isn't just one Lorenz gauge: if ξ_ν is the generator of Lorenz gauge, and χ_ν satisfies $\hat{\Box}\chi_\nu = 0$, then $\xi_\nu + \chi_\nu$ is also a generator of Lorenz gauge. Let us assume that $\bar{h}^{\rm L}_{\mu\nu}$ is a trace-reversed metric perturbation that satisfies Lorenz gauge. Let's change gauge once again using a generator χ_ν that satisfies $\hat{\Box}\chi_\nu = 0$:

$$\bar{h}^{\rm new}_{\mu\nu} = \bar{h}^{\rm L}_{\mu\nu} - \hat{\nabla}_\mu \chi_\nu - \hat{\nabla}_\nu \chi_\mu + \hat{g}_{\mu\nu}\hat{\nabla}^\alpha \chi_\alpha \ . \tag{17.33}$$

Let's take the trace of this:

$$\bar{h}^{\rm new} = \bar{h}^{\rm L} - 2\hat{\nabla}^\nu \chi_\nu + 4\hat{\nabla}^\alpha \chi_\alpha = \hat{h}^{\rm L} + 2\hat{\nabla}^\nu \chi_\nu \ . \tag{17.34}$$

This means that if we choose these generators such that

$$\hat{\nabla}^\nu \chi_\nu = -\frac{1}{2}\bar{h}^{\rm L} \ , \tag{17.35}$$

then the trace of the new metric is zero. Choosing this gauge means that there is no difference between $\bar{h}_{\mu\nu}$ and $h_{\mu\nu}$. This defines the TT gauge.

17.2.5 Considerations on a non-vacuum background

At several points in the above discussion, we used the fact that $T_{\mu\nu} = 0$ and thus $R_{\mu\nu} = 0$ in order to simplify the equations. By inspection it is clear that if we could not set these terms to zero then the discussion would become at a minimum more complicated. How do we handle this situation?

The key is to use the separation of scales again. Regions which are non vacuum vary on long length and time scales; the radiation varies on much shorter scales. By taking the Fourier transform of relevant quantities and examining how they behave in frequency and wavenumber space, one can develop useful relationships which are similar to those we have developed here, though with some modifications. As long as the separation of scales is valid, these modifications are not too difficult to develop and work with.

17.3 Energy content of gravitational waves: The Isaacson stress-energy tensor

We now have a good prescription for generating gravitational waves, and for studying how they evolve as they propagate out across spacetime, nicely analogous to our framework for studying the generation and propagation of electromagnetic radiation. One should recall that an important aspect of electromagnetic radiation is that it backreacts on a source, carrying away energy. Does gravitational radiation likewise carry away energy? This was quite a hotly debated topic in the mid 20th century (with some holdouts continuing to debate the issue into the late 1980s). The fact that general relativity can be a complicated nonlinear mess makes deducing the answer not exactly straightforward. On the one hand, one could argue that in a freely falling, locally Lorentzian representation, spacetime always just looks like special relativity. There's nothing there in this frame, so how can it carry energy?

Richard Feynman put forward one of the first clear physical arguments[6] that this radiation must carry energy: Imagine a very long stick on which there are two rings, free to slide up and down the stick, but with a small amount of friction. Place those rings on opposite ends of the stick, and allow the whole apparatus to float in free-fall (say in a high orbit about the Earth). The center of mass of the stick follows a geodesic; the rest of the stick follows along since the stick's molecules are bound to that center of mass. The two rings follow geodesics of their own, modulo a slight frictional coupling to the stick. Now, a gravitational wave comes by. It has no significant impact on the stick; the intermolecular forces within the stick effectively oppose the squeezing and stretching that the wave exerts. But by the equation of geodesic deviation, the rings slide back and forth, responding to the waves as they float in free fall. They rub a bit against the stick, and because of the slight frictional coupling generate a little bit of heat. The gravitational wave has thus heated up the stick; this wave must therefore carry energy.

The concerns about a freely-falling frame are not misplaced, however; note that a key part of Feynman's argument is that we need an extended object so that the impact of a gravitational wave can be discerned. Indeed, key to making a notion of energy carried by these waves work is that we will need to examine important quantities over some finite-sized region of spacetime. We also will need to go to second order in perturbation theory, since all first-order terms vanish when we average over such a region. This analysis was first done by Isaacson; our discussion is a synopsis of what Isaacson developed. We begin by writing the metric of spacetime

$$g_{\alpha\beta} = \hat{g}_{\alpha\beta} + \varepsilon h_{\alpha\beta} + \varepsilon^2 j_{\alpha\beta} \ . \tag{17.36}$$

We have here introduced an order-counting parameter ε. The value of this parameter is 1; however, its powers allow us to keep track of the order in perturbation theory that we are studying.

[6] This is discussed in Daniel Kennefick's book *Traveling at the Speed of Thought* (Princeton University Press, 2007; see pages 134-136). This book goes through the travails of the history of gravitational waves, and contains a lot of fascinating and occasionally sordid details.

Let us focus on the vacuum Einstein equation, setting $T_{\alpha\beta} = 0$. We write

$$\begin{aligned} 0 &= G_{\alpha\beta}\left[\hat{g}_{\mu\nu} + \varepsilon h_{\mu\nu} + \varepsilon^2 j_{\mu\nu}\right] \\ &= G^{(0)}_{\alpha\beta}\left[\hat{g}_{\mu\nu}\right] + \varepsilon G^{(1)}_{\alpha\beta}\left[h_{\mu\nu}; \hat{g}_{\mu\nu}\right] + \varepsilon^2 G^{(1)}_{\alpha\beta}\left[j_{\mu\nu}; \hat{g}_{\mu\nu}\right] + \varepsilon^2 G^{(2)}_{\alpha\beta}\left[h_{\mu\nu}; \hat{g}_{\mu\nu}\right] \ . \end{aligned} \tag{17.37}$$

The meaning of these terms is as follows:

$G^{(0)}_{\alpha\beta}\left[\hat{g}_{\mu\nu}\right]$	$\rightarrow$	"Standard" Einstein tensor
$G^{(1)}_{\alpha\beta}\left[h_{\mu\nu}; \hat{g}_{\mu\nu}\right]$	$\rightarrow$	1st-order correction
$G^{(1)}_{\alpha\beta}\left[j_{\mu\nu}; \hat{g}_{\mu\nu}\right]$	$\rightarrow$	Operator for 1st-order correction acting on the 2nd-order piece of metric
$G^{(2)}_{\alpha\beta}\left[h_{\mu\nu}; \hat{g}_{\mu\nu}\right]$	$\rightarrow$	Messy 2nd-order correction to the Einstein tensor.

The last piece, $G^{(2)}_{\alpha\beta}[h_{\mu\nu}; \hat{g}_{\mu\nu}]$, introduces many terms of the schematic form (suppressing indices) $h\hat{\nabla}\hat{\nabla}h$ or $(\hat{\nabla}h)(\hat{\nabla}h)$. Details can be found in Isaacson's paper; I will write out the key terms at a relevant point below.

Key for us is that we require the Einstein field equation to hold at each order in ε:

- $\mathcal{O}(1)$: $G^{(0)}_{\alpha\beta}[\hat{g}_{\mu\nu}] = 0$. This is a statement that $\hat{g}_{\mu\nu}$ is a vacuum solution to the field equation.
- $\mathcal{O}(\varepsilon)$: $G^{(1)}_{\alpha\beta}[h_{\mu\nu}; \hat{g}_{\mu\nu}] = 0$: This reproduces the analysis we did leading to the wave equation $\hat{\Box} h_{\mu\nu} + 2R_{\alpha\mu\beta\nu}h^{\alpha\beta} = 0$.
- $\mathcal{O}(\varepsilon^2)$: $G^{(1)}_{\alpha\beta}[j_{\mu\nu}; \hat{g}_{\mu\nu}] = -G^{(2)}_{\alpha\beta}[h_{\mu\nu}; \hat{g}_{\mu\nu}]$. The 2nd order perturbation obeys a *sourced* wave equation, with terms of order h^2 acting as this source.

In other words, terms that are quadratic in the wave amplitude act as a source for the 2nd-order corrections to spacetime.

Let us use the notion of averaging to separate the 2nd-order contributions as follows:

$$j_{\mu\nu} = \langle j_{\mu\nu} \rangle + \Delta j_{\mu\nu} \ . \tag{17.38}$$

The averaged term $\langle j_{\mu\nu} \rangle$ varies on the long scales $(\mathcal{T}, \mathcal{L})$; the oscillation $\Delta j_{\mu\nu}$ varies on the short scales (τ, λ). This suggests that we regroup terms in the metric according to how they vary on different scales:

$$g_{\mu\nu} = \left[\hat{g}_{\mu\nu} + \varepsilon^2 \langle j_{\mu\nu} \rangle\right]_{\rm L} + \left[\varepsilon h_{\mu\nu} + \varepsilon^2 \Delta j_{\mu\nu}\right]_{\rm S} \ . \tag{17.39}$$

The subscripts "L" and "S" denote terms that vary on the long and short scales, respectively. Let's apply the Brill-Hartle averaging to the $\mathcal{O}(\varepsilon^2)$ Einstein tensor; doing so yields

$$\left\langle G^{(1)}_{\alpha\beta}\left[j_{\mu\nu}; \hat{g}_{\mu\nu}\right] \right\rangle = -\left\langle G^{(2)}_{\alpha\beta}\left[h_{\mu\nu}; \hat{g}_{\mu\nu}\right] \right\rangle \ . \tag{17.40}$$

Here we can use a trick make possible by the Brill-Hartle averaging procedure: when we average the second derivative of a tensor function, the result is very close to the second derivative of the average of the function:

$$\left\langle \partial^2 f_{\mu\nu} \right\rangle = \partial^2 \left\langle f_{\mu\nu} \right\rangle + \mathcal{O}\left[(\tau/\mathcal{T})^2, (\lambda/\mathcal{L})^2\right] \ . \tag{17.41}$$

Using this, our averaged Einstein equation becomes

$$G^{(1)}_{\alpha\beta}\left[\langle j_{\mu\nu} \rangle ; \hat{g}_{\mu\nu}\right] = -\left\langle G^{(2)}_{\alpha\beta}\left[h_{\mu\nu}; \hat{g}_{\mu\nu}\right] \right\rangle \ . \tag{17.42}$$

Because the form on the left-hand side is linear in $\langle j_{\mu\nu}\rangle$, we can modify it further, writing

$$G_{\alpha\beta}\left[\hat{g}_{\mu\nu}+\varepsilon^2\langle j_{\mu\nu}\rangle\right]=-\left\langle G^{(2)}_{\alpha\beta}\left[h_{\mu\nu};\hat{g}_{\mu\nu}\right]\right\rangle\ . \tag{17.43}$$

This is a *very* interesting representation of the Einstein field equation. It is saying that the Einstein tensor for the long-lengthscale varying metric terms is governed by a source that looks like an averaged 2nd-order term involving the gravitational wave field $h_{\mu\nu}$. It suggests that we should introduce a definition:

$$T^{\rm GW}_{\alpha\beta}=-\frac{1}{8\pi G}\left\langle G^{(2)}_{\alpha\beta}\left[h_{\mu\nu};\hat{g}_{\mu\nu}\right]\right\rangle\ . \tag{17.44}$$

A straightforward but lengthy calculation (see Isaacson for details) shows that

$$T^{\rm GW}_{\alpha\beta}=\frac{1}{32\pi G}\left\langle\hat{\nabla}_\alpha\bar{h}_{\mu\nu}\hat{\nabla}_\beta\bar{h}^{\mu\nu}-\frac{1}{2}\hat{\nabla}_\alpha\bar{h}\hat{\nabla}_\beta\bar{h}-\hat{\nabla}_\alpha\bar{h}_{\beta\gamma}\hat{\nabla}_\mu\bar{h}^{\mu\gamma}-\hat{\nabla}_\beta\bar{h}_{\alpha\gamma}\hat{\nabla}_\mu\bar{h}^{\mu\gamma}\right\rangle\ . \tag{17.45}$$

Putting this into TT gauge simplifies things tremendously:

$$\boxed{\boxed{T^{\rm GW}_{\alpha\beta}=\frac{1}{32\pi G}\left\langle\hat{\nabla}_\alpha h^{\rm TT}_{\mu\nu}\hat{\nabla}_\beta h^{\mu\nu}_{\rm TT}\right\rangle}} \tag{17.46}$$

This result is known as the Isaacson stress-energy tensor.

17.4 The power carried by quadrupolar gravitational waves

Let's examine what this tensor tells us about the energy flux carried by quadrupolar gravitational waves in a nearly flat region. By working in this region and choosing nearly flat coordinates, $\hat{\nabla}_\alpha\to\partial_\alpha$, and $h^{\rm TT}_{\mu\nu}\to h^{\rm TT}_{ij}$. We look at T_{00}, which tells us about the flux of energy in the timelike direction:

$$T_{00}=\frac{dE}{dA\,dt}=\frac{1}{32\pi G}\left\langle\partial_t h^{\rm TT}_{ij}\partial_t h^{\rm TT}_{ij}\right\rangle\ . \tag{17.47}$$

The total rate of energy carried by these waves can be found by integrating over a large sphere:

$$\frac{dE}{dt}=r^2\int d\Omega\,T_{00}\ . \tag{17.48}$$

We insert

$$h^{\rm TT}_{ij}=\frac{2G}{r}\frac{d^2I_{kl}}{dt^2}\left(P_{ki}P_{lj}-\frac{1}{2}P_{kl}P_{ij}\right)\ , \tag{17.49}$$

where the projection tensors are given by $P_{ij}=\delta_{ij}-n_in_j$. Performing these integrals is facilitated by using the identities

$$\int d\Omega\,n_in_j=\frac{4\pi}{3}\delta_{ij}\ ,\quad \int d\Omega\,n_in_jn_k=0\ ,\quad \int d\Omega\,n_in_jn_kn_l=\frac{4\pi}{15}\left(\delta_{ij}\delta_{kl}+\delta_{ik}\delta_{jl}+\delta_{il}\delta jk\right)\ . \tag{17.50}$$

(You can demonstrate the validity of these identities by choosing a particular coordinate system in which, $n_x=\sin\theta\cos\phi$, $n_y=\sin\theta\sin\phi$, $n_z=\cos\theta$. It is then straightforward to integrate these forms up.) Putting all these pieces together yields

$$\boxed{\boxed{\frac{dE}{dt}=\frac{G}{5}\left\langle\frac{d^3I_{ij}}{dt^3}\frac{d^3I_{ij}}{dt^3}\right\rangle}} \tag{17.51}$$

This result is also known as the quadrupole formula, though it is more properly the quadrupole power formula (in contrast to our earlier quadrupole amplitude formula). Using it, one can derive

the rate at which a binary system's members fall toward one another due to the backreaction of gravitational wave emission. The result you get works describes quite well the characteristics of many of the sources that are measured by detectors like LIGO today. (There are corrections to this because multipoles higher than the quadrupole also radiate; and, it can be tricky to properly define and compute the multipole moments enter these radiation formulas. On problem set 8, you will compute what you get if you compute the quadrupole moment using a weak-field description of the source.)

17.5 More on the energy and momentum carried by gravitational waves

This section is based quite heavily on Chapters 6 and 12 of *Gravitation* by Eric Poisson and Clifford M. Will, especially sections 6.1, which introduces and discusses the Landau-Lifshitz pseudotensor, and section 12.2, which discusses how to compute radiative backreaction. Interested students are directed to these sections of that book for further discussion.

This Isaacson stress-energy tensor derived above is a nice conceptual tool, particularly for demonstrating that the notion of (for example) energy carried by gravitational waves can be made consistent with the principle of equivalence only by introducing a separation of lengthscales and timescales. One must be able to unambiguously separate the oscillatory, short-scale behavior associated with a wave from the slowly varying, long-scale behavior associated with a background. If such a separation can be made, then the Isaacson tensor provides an object that is rigorously a tensor, at least up to errors that scale as the ratio of lengthscales. This tensor demonstrates that notions of quantities like "energy carried by a wave" must be quadratic in a wave's amplitude, and must incorporate a notion of averaging over some finite region in order to wash out the small-scale oscillations. What's particularly nice is that, because it is a tensor (up least up to errors that scale as the ratio of lengthscales or ratio or timescales), it makes it unambiguously clear that a rigorous notion of energy and momentum carried by gravitational waves exists.

Although I like the Isaacson tensor as a conceptual tool, it turns out not to be such a useful tool for many practical computations. There are two key issues which make it not so useful in practice:

- First, it turns out to not be adequate for computing the angular momentum carried by radiation. A careful calculation of angular momentum radiated according to the Isaacson tensor would yield zero for most components, because the leading terms which appear to contribute to this calculation in fact cancel out upon integrating over the surface of a large sphere. One needs to include terms to higher order in $1/R$ (where R is distance from the radiating source) in order to recover subleading terms that do not vanish upon integration; this cannot be done with Isaacson's tensor.

- Second, although the Isaacson tensor tells us about the "energy" and "momentum" carried by radiation, it does not define what that "energy" or "momentum" actually **is**. Suppose for example you compute the total flux of energy carried by gravitational waves, $dE^{\rm GW}/dt$. You might imagine that you can use conservation of energy to balance this against the "energy" associated with a source, asserting that $d(E^{\rm source} + E^{\rm GW})/dt = 0$. Enforcing this equation, you would learn how the source energy changes, and hope to infer from this change how the system evolves in response.

  The problem is that there is no unique way to define $E^{\rm source}$: one can define "the" energy of a gravitational system in multiple ways in general relativity. Any and indeed, typically *all*, of these definitions may be acceptable and useful, depending on your analysis. The Isaacson tensor does not tell us which notion of "energy" is actually being carried off by gravitational waves. There are some limits for which this this energy is unambiguous (such as the weak-field example you will study in problem set 8), but it is ambiguous in general.

So, how do we do an analysis of this type in a more general situation? The trick is define *a* notion of energy (and momentum and angular momentum) which can be applied both to your system as well as the radiation, and to use that notion consistently throughout the calculation. Such a calculation is *not* covariant, but as long as everything is done consistently, that is fine. One can obtain covariant results that unambiguously describe measurable quantities provided one uses the non-covariant ingredients in the analysis in a fully self consistent way.

One very useful way of defining such quantities is to use what is called the *Landau-Lifshitz pseudotensor*, a quantity defined in the classic text *The Classical Theory of Fields* by Landau and Lifshitz. Interested students are directed to Chapters 6 and 12 of the text by Poisson and Will for a very readable discussion to get further details; those with particular stamina and algebraic fortitude are directed to Landau and Lifshitz itself for detailed proofs of many of the statements I will sketch below.

Landau and Lifshitz reformulate general relativity in an interesting way such that the field equation becomes

$$\partial_\mu \partial_\nu H^{\alpha\mu\beta\nu} = 16\pi G(-g)(T^{\alpha\beta} + t_{\rm LL}^{\alpha\beta}) \; . \tag{17.52}$$

Here g is the determinant of the metric $g_{\alpha\beta}$. Notice that only *partial* derivatives appear in this equation. The four-index field on the left-hand side is given by

$$H^{\alpha\mu\beta\nu} = \mathfrak{g}^{\alpha\beta}\mathfrak{g}^{\mu\nu} - \mathfrak{g}^{\alpha\nu}\mathfrak{g}^{\beta\mu} \; , \tag{17.53}$$

where the so-called "gothic metric" is $\mathfrak{g}^{\alpha\beta} \equiv \sqrt{-g}g^{\alpha\beta}$. The field $H^{\alpha\mu\beta\nu}$ is designed so that it has the same symmetries under index exchange as the Riemann tensor.

This reformulation of the field equation is *exact*: no approximation has been introduced. Its main "deficit," so to speak, is that it is not a covariant formulation: the quantities on both sides are tensor densities (note the factors of g) rather than tensors. This is not a flaw, but it's a characteristic that we must bear in mind as we interpret its results. It means (for example) that if we change representations, we must be careful how we interpret the components of terms in this equation, or of quantities that we derive from these terms.

In this formulation of general relativity, the two-index field $t_{\rm LL}^{\alpha\beta}$ is known as the Landau-Lifshitz pseudotensor. It is built from a fairly lengthy sequence of products of partial derivatives of the gothic metric, contracted by various combinations of the "normal" metric. See Eq. (6.5) of Poisson and Will for details. One reason why this is nice is that one can show rather simply that

$$\partial_\beta \left[(-g) \left(T^{\alpha\beta} + t_{\rm LL}^{\alpha\beta} \right) \right] = 0 \; . \tag{17.54}$$

Proof: Take the ∂_β derivative of Eq. (17.52). The derivative operators are symmetric under exchange of β and ν; the field $H^{\alpha\mu\beta\nu}$ is antisymmetric under this exchange. By contracting an antisymmetric field with a symmetric operator, the left-hand side is zero; therefore, the right-hand side must be zero as well.

This derivative of the stress-energy tensor and the Landau-Lifshitz pseudotensor equation expresses local conservation of energy. Here, the stress energy tensor $T^{\alpha\beta}$ as usual describes the flow of 4-momentum through spacetime. The Landau-Lifshitz pseudotensor can then be interpreted — though be *very* cautious about this interpretation — as something like energy and momentum associated with gravity. The reason to be cautious is that this formulation is quite specific to the reformulation of the field equations developed by Landau and Lifshitz. Also, because these are pseudotensors, the notion of "gravitational energy" at some location can be made to vary arbitrarily by changing coordinates. Indeed, one can *always* make $t_{\rm LL}^{\alpha\beta} = 0$ simply by going to the nearly inertial representation of a freely-falling frame. If anyone corners you at a cocktail party[7] and

[7] Or, perhaps more realistically, in an elevator at an APS meeting.

begins expounding on their method for uniquely defining gravitational energy, smile politely and move away as quickly as possible[8].

With all these caveats in mind, the Landau-Lifshitz pseudotensor turns out to be an excellent tool for many applications in which one is interested in understanding how gravitational waves backreact on a source. For example, the flux of energy carried by gravitational waves can be written

$$\frac{dE}{dt} = \oint_{\partial V} (-g) t_{\rm LL}^{0k} dS_k \ , \tag{17.55}$$

where the integral is taken over a large sphere bounding the volume V that surrounds a source, and where dS_k is an (outward directed) area element on that sphere. When evaluated in nearly flat spacetime far from a source, this yields results that agree with the Isaacson formula. Importantly, the Landau-Lifshitz formulation allows us to define what E actually means: the total E contained in a volume V is given by

$$E = \int_V (-g) \left(T^{00} + t_{\rm LL}^{00} \right) d^3x' = \frac{1}{16\pi} \oint_{\partial V} \partial_j H^{0j0k} dS_k \ . \tag{17.56}$$

The second form invokes Eq. (17.52) and uses integration by parts, assuming that V is large enough that a surface term's contribution is negligible. (In practice, one takes V to be all of space, so the bounding surface ∂V is a sphere of infinite radius.) In this formulation, we can compute E from the above integral, and use the equation for dE/dt to infer how E changes due to the backreaction of gravitational-wave emission. Similar formulas allow us to define the momentum and angular momentum of a system, and to infer how these quantities change due to backreaction.

The abundance of tools to describe how gravitational radiation carries energy and backreacts on a system is symptomatic of the fact that general relativity's coordinate freedom means we cannot uniquely define notions of "the" energy associated with gravitational systems. This should be regarded as a feature of the theory; understanding the conceptual nuances associated with defining a system's "energy" and the rate at which this "energy" changes typically yields dividends.

For the purposes of 8.962, the major takeaway is that one should be aware that there exist different approaches to problems like backreaction, and one has freedom to develop and pick tools which are well adapted to your problem. Landau and Lifshitz's reformulation of general relativity is quite elegant, and defines really useful tools, but experience has shown that it works best when the gothic metric $\mathfrak{g}^{\alpha\beta}$ can be regarded as a fairly small deviation from $\eta^{\alpha\beta}$. In this context, it makes it possible to define an iterative scheme for analyzing such a deviation. This approach is an important foundation for post-Newtonian and post-Minkowskian studies of gravitating systems. When applied to systems for which $\mathfrak{g}^{\alpha\beta}$ is not a small deviation from $\eta^{\alpha\beta}$, this reformulation is quite cumbersome and difficult to use. Studying the strong-field spacetimes of black holes in this framework, for example, is rather nightmarish.

[8] Whatever you do, do *not* attempt to correct their misunderstandings. Noble as your impulse may be, nothing good can come of it.

Reading a distant clock in a Robertson-Walker spacetime

During some of the Zoom lectures on cosmology, the question has come up as to how we know that a cosmologically distant clock is seen by us to "run slow." We all seem to "know" that an interval of time Δt on a distant clock should be measured by us here on Earth as an interval $\Delta t_z = (1+z)\Delta t$. How can we show this?

To begin, let us write the Robertson-Walker line element as follows:

$$ds^2 = -dt^2 + a^2(t) R_0^2 \left[d\chi^2 + S_k(\chi) d\Omega^2 \right] \ .$$

Note that t is proper time for *all* comoving observers in this spacetime: in these coordinates, the 4-velocity of a comoving observer has components $u^\alpha \doteq (dt/d\tau, d\chi/d\tau, d\theta/d\tau, d\phi/d\tau) = (1,0,0,0)$. During some of our Zoom discussions, I may have said something about the nature of the time coordinate that contradicted this. If so, I was incorrect. In fact, whenever a metric has $g_{tt} = -1$ and $g_{ti} = 0$, the coordinate t is the proper time of a comoving observer in that spacetime.

Next, imagine a clock at radial coordinate χ_e. Photons are emitted from the clock at $t = t_{e1}$ and at $t = t_{e2}$, carrying a picture of the clock's face at these moments. The emission times of these photons bound a time interval $\Delta t_e = t_{e2} - t_{e1}$. The photons then propagate radially inward to $\chi = 0$, and are measured at times t_{m1} and t_{m2}. Our goal is to compute the time interval $\Delta t_m = t_{m2} - t_{m1}$ measured at $\chi = 0$.

Both photons travel along null geodesics for which $0 = -dt^2 + a^2(t) R_0^2 \, d\chi^2$, or

$$R_0 \, d\chi = -dt/a(t) \ .$$

(Choosing the minus sign for the inward radial trajectory.) For the first photon, we have

$$0 - R_0 \chi_e = -\int_{t_{e1}}^{t_{m1}} \frac{dt}{a(t)} \ .$$

For the second photon,

$$0 - R_0 \chi_e = -\int_{t_{e2}}^{t_{m2}} \frac{dt}{a(t)} \ .$$

Since χ is a comoving radial coordinate, both of these photons are emitted from $\chi = \chi_e$ and are measured at the same $\chi = 0$. The left-hand sides of these equations are thus identical. We can therefore equate their right-hand sides to each other and rearrange:

$$\begin{aligned} 0 &= \int_{t_{e2}}^{t_{m2}} \frac{dt}{a(t)} - \int_{t_{e1}}^{t_{m1}} \frac{dt}{a(t)} \\ &= \int_{t_{m1}}^{t_{m2}} \frac{dt}{a(t)} - \int_{t_{e1}}^{t_{e2}} \frac{dt}{a(t)} \ . \end{aligned}$$

On the second line, we've used the fundamental theorem of calculus to rearrange the limits of integration. If we assume that both Δt_e and Δt_m are short compared to the timescale over which the scale factor $a(t)$ changes, then we can write

$$0 = \frac{\Delta t_m}{a(t_m)} - \frac{\Delta t_e}{a(t_e)}$$

[using $a(t_e) \equiv a(t_{e1}) \simeq a(t_{e2})$, and likewise for $a(t_m)$]. Taking the time of measurement to be now, so that $a(t_m) = 1$, we find

$$\Delta t_m = \frac{\Delta t_e}{a(t_e)} \equiv [1 + z(t_e)] \Delta t_e \ .$$

We measure a time interval Δt_e on a cosmologically distant clock redshifted to $\Delta t_m = (1+z)\Delta t_e$.